11:29 12.0 KB/S 80
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16. Component props map

Required prop · Optional prop

ScoreBadge `components/ScoreBadge.jsx`

score	number	DevScore 0–100 to display
size	"sm"|"lg"	Badge size variant, default "sm"
showLabel	boolean	Show "DevScore" label below number

StatsCard `components/StatsCard.jsx`

title	string	Card heading, e.g. "GitHub stats"
stats	object	Key-value pairs to render as metrics
icon	ReactNode	Icon shown in card header
loading	boolean	Shows skeleton while data loads

ProjectCard `components/ProjectCard.jsx`

project	object	Full project document from MongoDB
editable	boolean	Shows edit/delete buttons if true
onDelete	function	Called with projectId on delete click
onEdit	function	Called with project object on edit click

Navbar `components/Navbar.jsx`

–		No props. Reads user + role from AuthContext internally

ProtectedRoute `components/ProtectedRoute.jsx`

children	ReactNode	The page component to render if allowed
allowedRoles	string[]	e.g. ["developer","admi Omit = any logged-user
redirectTo	string	Path to redirect if not authorised, default "/login"

SearchFilter `components/SearchFilter.jsx`

onFilter	function	Called with filter object whenever values change
initialValues	object	Prefill filter fields from URL query params

Dashboard `pages/Dashboard.jsx`

–		No props. Reads from UserContext + makes API calls internally via useEffect

Profile `pages/Profile.jsx`

–		No props. Reads username from useParams(). Fetches profile, stats, and score via API

Click any card to get the full component code written for you.

17. API request & response shapes

The exact JSON your frontend sends and your backend returns — the contract between the two sides.

POST `/api/auth/register` — Create new account

REQUEST BODY

```
{
  "username": "john_doe",
  "email":    "john@example.com",
  "password": "secret123",
  "role":     "developer"
}
```

SUCCESS 201 / ERROR 400

```
{
  "success": true,
  "message": "Account created",
  "token":   "eyJhbGci…",
  "user": {
    "_id":      "64f…",
    "username": "john_doe",
    "role":     "developer"
  }
}
```

POST `/api/auth/login` — Login with email + password

REQUEST BODY

```
{
  "email":    "john@example.com",
  "password": "secret123"
}
```

SUCCESS 200 / ERROR 401

```
{
  "success": true,
  "token":   "eyJhbGci…",
  "user": {
    "_id":      "64f…",
    "username": "john_doe",
    "role":     "developer"
  }
}
```

GET `/api/profile/:username` — Fetch a public profile

REQUEST

```
No body.
URL param: username

e.g. GET /api/profile/john_doe
```

SUCCESS 200 / ERROR 404

```
{
  "success": true,
  "data": {
    "username":  "john_doe",
    "bio":       "Full-stack dev",
    "college":   "IIT Hyderabad",
    "techStack": ["React", "Node"],
    "devScore":  74.5,
    "github": {
      "commits":   312,
      "repos":     18,
      "stars":     45
    },
    "dsa": {
      "lc_easy":   80,
      "lc_medium": 42,
      "cf_rating": 1200
    },
    "projects": [ …array of ProjectCard objects ]
  }
}
```

POST `/api/projects` — Add a project to portfolio

REQUEST BODY (+ BEARER TOKEN)

```
{
  "title":       "DevFolio",
  "description": "Dev identity app",
  "repo_url":    "https://github.com/…",
  "live_url":    "https://devfolio.app",
  "tech_stack":  ["React", "Node", "MongoDB"]
}
```

SUCCESS 201 / ERROR 400

```
{
  "success": true,
  "data": {
    "_id":         "65a…",
    "title":       "DevFolio",
    "description": "Dev identity app",
    "repo_url":    "https://github.com/…",
    "tech_stack":  ["React", "Node"],
    "user_id":     "64f…"
  }
}
```

GET `/api/score/:userId` — Fetch computed DevScore

REQUEST

```
No body.
URL param: userId (MongoDB _id)

e.g. GET /api/score/64f3b…
```

SUCCESS 200

```
{
  "success": true,
  "data": {
    "total_score":       74.5,
    "github_score":      31.2,
    "dsa_score":         21.0,
    "project_score":     14.8,
    "consistency_score": 7.5,
    "computed_at": "2025-03-18T…"
  }
}
```

STANDARD ERROR SHAPE — SAME FOR ALL ENDPOINTS

```
{
  "success": false,
  "message": "Email already registered",
  "errors":  ['email must be a valid email address']  // optional array from validation
}
```

11:30 3.00 KB/S 79
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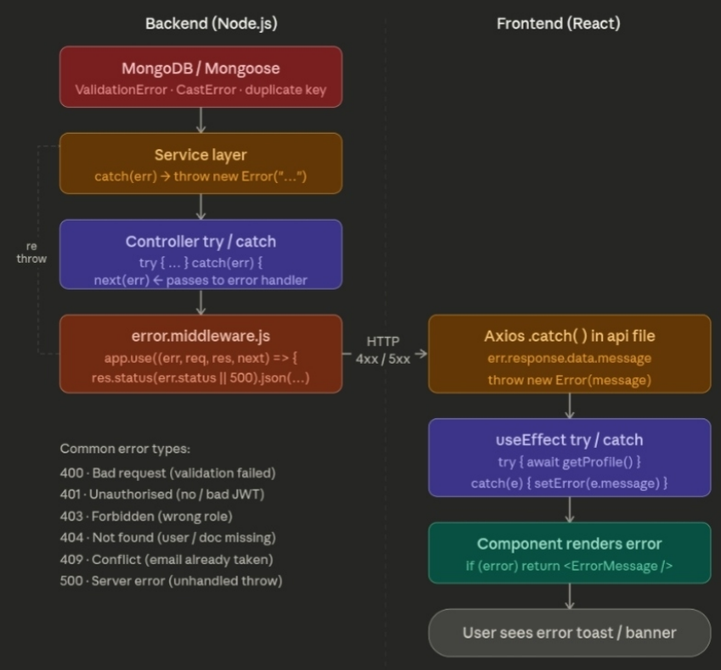
18. Error handling flow

How an error travels from the database all the way to the user's screen — and where you catch it at each layer.



19. DevScore math breakdown

The exact formula with real example numbers so you can implement and verify the score engine.

STEP 1 — RAW INPUT METRICS (JOHN'S ACTUAL STATS)

Input	Raw value	Max possible (cap)	Normalised (0–100)
GitHub commits	312	500 commits	312/500 × 100 = **62.4**
GitHub stars	45	200 stars	45/200 × 100 = **22.5**
GitHub repos	18	50 repos	18/50 × 100 = **36.0**
LeetCode problems	122 (80E + 42M)	300 problems	(80×1 + 42×2)/300 × 100 = **54.7**
Codeforces rating	1200	2400 rating	1200/2400 × 100 = **50.0**
Projects (with live demo)	3 projects	10 projects	3/10 × 100 + bonus = **74.0**
Consistency streak	45 days	100 days	45/100 × 100 = **45.0**

STEP 2 — SUB-SCORE PER CATEGORY (AVERAGE WITHIN GROUP)

Category	Sub-score calculation	Sub-score	Weight
GitHub	(62.4 + 22.5 + 36.0) / 3	**40.3**	× 0.40
DSA	(54.7 + 50.0) / 2	**52.4**	× 0.30
Projects	74.0	**74.0**	× 0.20
Consistency	45.0	**45.0**	× 0.10

STEP 3 — FINAL WEIGHTED SUM

DevScore = (40.3 × 0.40) + (52.4 × 0.30) + (74.0 × 0.20) + (45.0 × 0.10) = 16.12 + 15.72 + 14.80 + 4.50 = **51.14**

John's DevScore breakdown — click to get the code ↗

GitHub score (40%) 16.12
DSA score (30%) 15.72
Project score (20%) 14.80
Consistency score (10%) 4.50
Final DevScore 51.14 / 100

Implementation tip: Store the cap values (500 commits, 2400 CF rating, etc.) as a config object in your backend so you can tune them without changing the formula code. Cap any normalised value at 100 before applying the weight.

20. React Router configuration

Every route in one place — public, protected, and role-gated — with the exact nesting structure.

Public (no login needed) · Any logged-in user · Developer only · Recruiter only · Mentor only · Admin only

```
<BrowserRouter>  root
├── /  → Landing.jsx  public
├── /login  → Login.jsx  public
├── /register  → Register.jsx  public
├── /profile/:username  → Profile.jsx  public
├── /leaderboard  → Leaderboard.jsx  public
├── /auth/callback  → OAuthCallback.jsx  public
├── <ProtectedRoute>  redirects to /login if not authenticated  logged in
│   ├── /dashboard  → Dashboard.jsx  developer
│   ├── /settings/profile  → EditProfile.jsx  developer
│   ├── /settings/projects  → Projects.jsx  developer
│   ├── /settings/integrations  → Integrations.jsx  developer
│   ├── /recruiter/search  → RecruiterSearch.jsx  recruiter
│   ├── /recruiter/view/:username  → RecruiterView.jsx  recruiter
│   ├── /mentor/dashboard  → MentorDashboard.jsx  mentor
│   ├── /admin/users  → UserManager.jsx  admin
│   └── /admin/scores  → ScoreAdmin.jsx  admin
└── *  → NotFound.jsx  catch-all
```

```jsx
{'// router.jsx — click to get the full file written ↗'}
<Routes>
  <Route path="/" element={<Landing />} />
  <Route path="/login" element={<Login />} />
  <Route path="/profile/:username" element={<Profile />} />
  <Route element={<ProtectedRoute allowedRoles={["developer"]} />}>
    <Route path="/dashboard" element={<Dashboard />} />
  </Route>
  {'// … recruiter, mentor, admin routes same pattern'}
  <Route path="*" element={<NotFound />} />
</Routes>
```

11:30 2.00 KB/S 79
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21. Mongoose schema code patterns

req required · unique unique index · default has default value · ref foreign key to another model · index indexed for fast queries · enum only allowed values

User — models/User.model.js

username	String	req · unique · index
email	String	req · unique
password_hash	String	
role	String	req · enum · default
github_id	String	unique
github_token	String	
created_at	Date	Date.now

Profile — models/Profile.model.js

user_id	ObjectId	req · ref User · unique
bio	String	
avatar_url	String	
college	String	index
tech_stack	[String]	
linkedin_url	String	
website_url	String	

GithubStats — models/GithubStats.model.js

user_id	ObjectId	req · ref User · unique
total_commits	Number	0
public_repos	Number	0
followers	Number	0
stars_received	Number	0
top_languages	[String]	
last_synced	Date	

DsaStats — models/DsaStats.model.js

user_id	ObjectId	req · ref User · unique
lc_easy	Number	0
lc_medium	Number	0
lc_hard	Number	0
lc_username	String	
cf_rating	Number	0
cf_username	String	
last_synced	Date	

Project — models/Project.model.js

user_id	ObjectId	req · ref User · index
title	String	req
description	String	
repo_url	String	
live_url	String	
tech_stack	[String]	
github_stars	Number	0
created_at	Date	Date.now

DevScore — models/DevScore.model.js

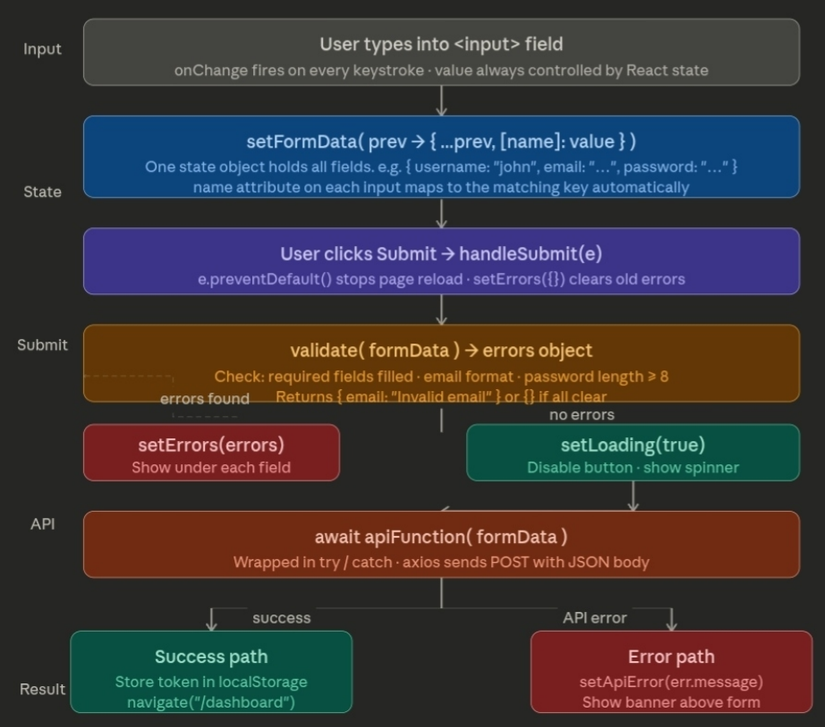
user_id	ObjectId	req · ref User · unique
total_score	Number	0
github_score	Number	0
dsa_score	Number	0
project_score	Number	0
consistency_score	Number	0
computed_at	Date	Date.now

Key Mongoose pattern for all models: Use `findOneAndUpdate({ user_id }, data, { upsert: true, new: true })` for stats models so a sync always writes correctly whether the doc exists or not. Click any card to get the full schema code.

Click any card to get the complete model file written for you.

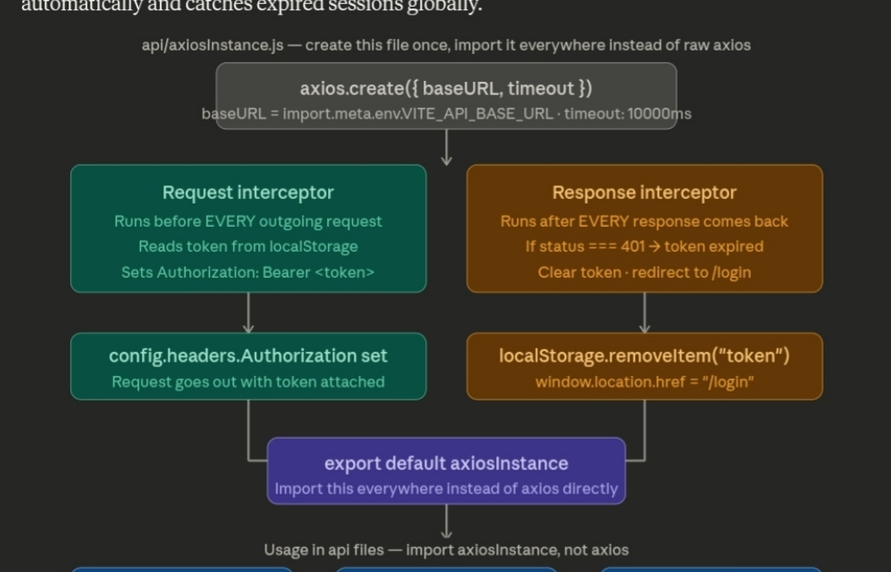
22. Form state & validation flow

The exact lifecycle every form in DevFolio follows — from the user typing to the API call completing.



23. Axios interceptor setup

The one file that makes your entire frontend work — attaches JWT to every request automatically and catches expired sessions globally.



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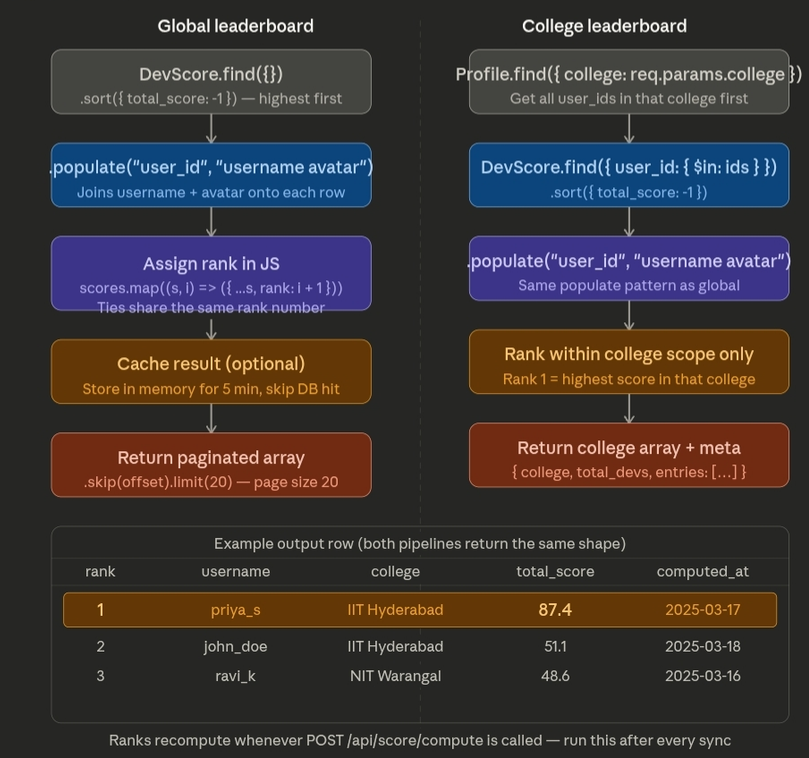
24. Leaderboard ranking algorithm

How raw DevScores become ranks — with the query, sort logic, and college-scope filter explained step by step.



Example output row (both pipelines return the same shape)

rank	username	college	total_score	computed_at
1	priya_s	IIT Hyderabad	87.4	2025-03-17
2	john_doe	IIT Hyderabad	51.1	2025-03-18
3	ravi_k	NIT Warangal	48.6	2025-03-16

Ranks recompute whenever POST /api/score/compute is called — run this after every sync

Claude's response was interrupted Retry

Read the whole chat and continue Giving me remaining uncovered visual documentation part for the project covering everything as a blueprint
Note this is just for my personal project so dont bother about deployment or anything else
As iam a beginner i need guidance through visual diagrams

Inventoried comprehensive documentation and identified five critical beginner-friendly visual guides ne…

Let me audit all 24 carefully before starting.

Covered: Architecture, ERD, User flow, DevScore algorithm, API routes, Component tree, Auth sequence, Sync pipeline, RBAC, Deployment, Folder structure, Middleware chain, React data flow, .env vars, E2E trace, Props map, API shapes, Error handling, DevScore math, Router config, Mongoose schemas, Form flow, Axios interceptors, Leaderboard algorithm.

Genuinely missing for a beginner:

- **npm packages reference** — every install command, both sides, what each package does
- **Backend app.js wiring** — how all routes and middleware plug into the entry file
- **MongoDB query patterns** — the actual Mongoose queries for every common operation
- **State management decision map** — when to use useState vs Context vs props
- **GitHub API integration trace** — which endpoints, what comes back, how it maps to your schema

Five more. Starting now.

25. npm packages reference

BACKEND — /SERVER

```
$ npm install express mongoose dotenv cors helmet morgan bcryptjs jsonwebtoken axios node-cron
```

Core server

`express`	The web framework — handles routes, middleware, and HTTP	core
`mongoose`	MongoDB ODM — schemas, models, and queries	core
`dotenv`	Loads .env file into process.env	core

Security & middleware

`cors`	Allows React frontend to call the API from a different port	middleware
`helmet`	Sets secure HTTP headers automatically — one line of protection	middleware
`morgan`	Logs every incoming request to the terminal — great for debugging	logging

Auth

`bcryptjs`	Hashes passwords so you never store plain text	auth
`jsonwebtoken`	Creates and verifies JWT tokens for auth	auth

External calls & scheduling

`axios`	Makes HTTP requests to GitHub, LeetCode, Codeforces APIs	api
`node-cron`	Runs scheduled jobs — nightly sync without any extra service	scheduler

Dev only (not in production)

```
$ npm install --save-dev nodemon
```

`nodemon`	Restarts the server automatically when you save a file	dev only

FRONTEND — /CLIENT

```
$ npm install react-router-dom axios recharts react-hot-toast
```

Routing

`react-router-dom`	Client-side routing — BrowserRouter, Routes, useNavigate, useParams	core

API calls

`axios`	HTTP client — interceptors, base URL config, request/response handling	api

Charts & UI

`recharts`	Charts for the dashboard — BarChart, LineChart, RadialBar for DevScore	charts
`react-hot-toast`	One-line toast notifications for success/error feedback	UI

Already included with Vite + React

`react`	Core React library — already in package.json from Vite scaffold	built-in
`react-dom`	Renders React to the browser DOM	built-in
`vite`	Dev server and build tool — fast hot reload during development	built-in

Styling — no install needed

```
$ npm install -D tailwindcss postcss autoprefixer
&& npx tailwindcss init -p
```

`tailwindcss`	Utility CSS classes — no separate CSS files needed for most styling	styling

Click any package row or install box to get setup code for that specific package.

26. Backend app.js wiring

Every single line that belongs in `app.js` and `index.js` — the exact order middleware and routes must be registered.

APP.JS — MIDDLEWARE + ROUTES (ORDER MATTERS!)

1. `require all packages + routes`
   const express = require('express')
   const cors = require('cors') · helmet · morgan
   const authRoutes = require('./routes/auth.routes')
   … all other route files
2. `const app = express()`
   Create the app instance. Everything else calls app.use() on it.
3. `app.use(cors({ origin: process.env.CLIENT_URL }))`
   Must come BEFORE routes. Allows frontend to call the API without browser blocking.
4. `app.use(helmet())`
   Sets secure headers on every response. Before routes so all responses are covered.
5. `app.use(express.json())`
   Parses incoming JSON request bodies into req.body. Without this, req.body is undefined.
6. `app.use(morgan('dev'))`
   Logs method, path, status, response time in terminal. Helps massively while debugging.
7. `app.use('/api/auth', authRoutes)`
   app.use('/api/profile', profileRoutes)
   app.use('/api/stats', statsRoutes)
   app.use('/api/projects', projectRoutes)
   app.use('/api/score', scoreRoutes)
   app.use('/api/leaderboard', leaderboardRoutes)
8. `app.use(errorMiddleware)`
   MUST be last. Catches any error passed via next(err) from any route above.

⚠ Order is critical — cors + json BEFORE routes, error handler AFTER all routes.

INDEX.JS — STARTUP SEQUENCE

1. `require('dotenv').config()`
   Must be the very FIRST line so all process.env values are available everywhere below.
2. `const app = require('./app')`
   app.js exports the configured Express instance. index.js only starts the server.
3. `await connectDB()`
   Connects to MongoDB first. If this fails, the server should not start — wrap in try/catch and process.exit(1).
4. `app.listen(process.env.PORT)`
   Only start listening after DB connected. Log "Server running on port 5000" so you know it worked.
5. `registerCronJobs()`
   Start the nightly sync scheduler after server is up. node-cron runs '0 2 * * *' — every night at 2am.
6. `process.on('unhandledRejection', …)`
   Catches any promise that throws without a catch block. Logs the error and closes gracefully instead of silent crash.

PACKAGE.JSON SCRIPTS

```
"scripts": {
  "start": "node index.js",
  "dev": "nodemon index.js"
}
```

27. MongoDB query patterns

The exact Mongoose calls for every common operation in DevFolio — copy-paste ready.

USER — AUTH OPERATIONS

Create user	`await User.create({ username, email, password_hash, role })`	Register endpoint
Login lookup	`await User.findOne({ email }).select('+password_hash')`	Include hidden field
GitHub upsert	`await User.findOneAndUpdate({ github_id }, data, { upsert: true, new: true })`	OAuth login
Find by ID	`await User.findById(req.user.id).select('-password_hash')`	Auth middleware

PROFILE — READ & WRITE

Get by username	`User.findOne({ username }) then Profile.findOne({ user_id })`	Profile page
Update profile	`await Profile.findOneAndUpdate({ user_id }, { $set: updates }, { new: true })`	Edit profile

STATS — SYNC UPSERT PATTERN (SAME FOR GITHUB AND DSA)

Upsert stats	`await GithubStats.findOneAndUpdate( { user_id }, { $set: { ...stats, last_synced: Date() } }, { upsert: true, new: true } )`	Creates if missing, updates if exists
Get all stats	`const [github, dsa] = await Promise.all([ GithubStats.findOne({ user_id }), DsaStats.findOne`	Parallel — faster

PROJECTS — CRUD

List projects	`await Project.find({ user_id }).sort({ created_at: -1 })`	Newest first
Add project	`await Project.create({ user_id: req.user.id, ...req.body })`	POST /projects
Edit project	`await Project.findOneAndUpdate( { _id: projectId, user_id: req.user.id }, { $set: req.body }, { new: true } )`	Ownership check built-in
Delete project	`await Project.findOneAndDelete({ _id: projectId, user_id: req.user.id })`	Safe delete

LEADERBOARD — RANK QUERY

Global top 20	`DevScore.find({}).sort({ total_score: -1 }).limit(20).skip(offset).populat`	Paginated
College scope	`const ids = await Profile.find({ college: distinct('user_id') }) DevScore.find({ user_ids }).`	Two-step

11:32 0.11 KB/S 79
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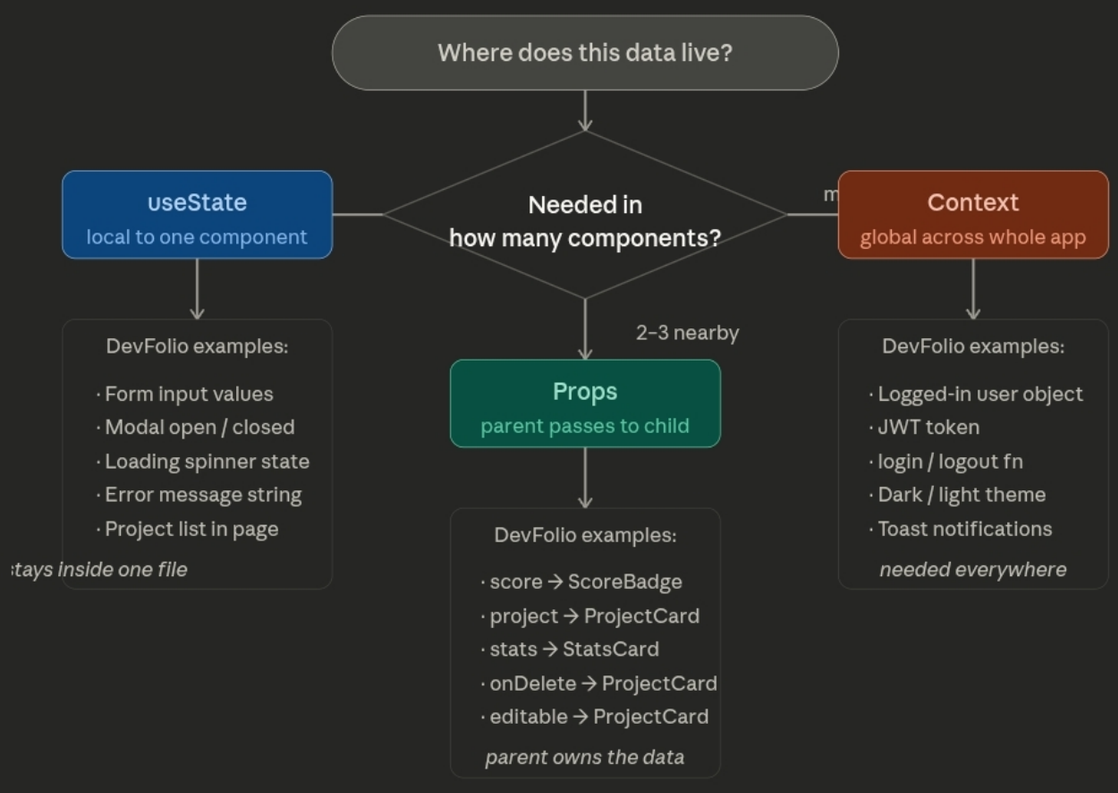
28. State management decision map

The most confusing React concept for beginners — exactly when to use `useState`, `useContext`, or just pass props.



The rule of thumb for DevFolio

useState	Props	Context
form, loading, modal	display, card, callback	user, token, theme

Start with useState. Only lift to props when a sibling needs it. Only lift to Context when 3+ unrelated components need it.

29. GitHub API integration trace

Exactly which endpoints to call, what comes back, and how every field maps to your `GithubStats` schema.

GET `https://api.github.com/user` — Authenticated user's profile

GITHUB API RETURNS		SAVED TO YOUR SCHEMA
`response.data.login`	→	`User.username`
`response.data.id`		`User.github_id`
`response.data.avatar_url`		`Profile.avatar_url`
`response.data.followers`		`GithubStats.followers`
`response.data.public_repos`		`GithubStats.public_repos`

GET `https://api.github.com/users/:username/repos?per_page=100` — All public repositories

GITHUB API RETURNS (ARRAY OF REPOS)			COMPUTED THEN SAVED
`repo.stargazers_count`	← sum all	→	`GithubStats.stars_received`
`repo.language`	← collect unique		`GithubStats.top_languages`
`repos.length`	← count		`GithubStats.public_repos`

GET `https://api.github.com/users/:username/events?per_page=100` — Recent activity — count commits

GITHUB API RETURNS (EVENT ARRAY)		SAVED
`event.type === 'PushEvent'`	→	`GithubStats.total_commits`
`event.payload.commits.length`	← sum	

```
// github.service.js — click to get the full file ↗
const syncGithubStats = async (userId, githubToken, username) => {
  const headers = { Authorization: `Bearer ${githubToken}` }
  const [userRes, reposRes, eventsRes] = await Promise.all([
    axios.get('https://api.github.com/user', { headers }),
    axios.get(`https://api.github.com/users/${username}/repos?per_page=100`, { headers }),
    axios.get(`https://api.github.com/users/${username}/events?per_page=100`, { headers })
  ])
  const stars = reposRes.data.reduce((sum, r) => sum + r.stargazers_count, 0)
  const commits = eventsRes.data
    .filter(e => e.type === 'PushEvent')
    .reduce((sum, e) => sum + e.payload.commits.length, 0)
  await GithubStats.findOneAndUpdate(
    { user_id: userId },
    { $set: { stars_received: stars, total_commits: commits, last_synced: new Date() } },
    { upsert: true, new: true }
  )
}
```

Rate limits: Unauthenticated = 60 req/hr. With GITHUB_API_TOKEN = 5000 req/hr. Always use the token header for the server-side calls. The user's oauth token (github_token in User model) is for their personal data; the server's API token is for public data.

30. LeetCode + Codeforces API integration trace

GraphQL **LeetCode** `https://leetcode.com/graphql` — No official REST API — use unofficial GraphQL

GRAPHQL QUERY SENDS			SAVED TO DSASTATS SCHEMA
`query: userPublicProfile`			
`variables.userSlug`	= lc_username from DsaStats		
`matchedUser.submitStats`			
`.acSubmissionNum[0].count`	Easy solved	→	`DsaStats.lc_easy`
`.acSubmissionNum[1].count`	Medium solved	→	`DsaStats.lc_medium`
`.acSubmissionNum[2].count`	Hard solved	→	`DsaStats.lc_hard`
`matchedUser.profile`			
`.ranking`	Global rank	→	`DsaStats.lc_ranking`

GET **Codeforces — Endpoint 1** `https://codeforces.com/api/user.info?handles=:cf_username`

CF API RETURNS (RESULT[0])			SAVED TO DSASTATS SCHEMA
`result[0].rating`	Current rating	→	`DsaStats.cf_rating`
`result[0].maxRating`	All-time peak	→	`DsaStats.cf_max_rating`
`result[0].rank`	"pupil", "specialist"...	→	`DsaStats.cf_rank`
`result[0].contribution`		→	– (not stored)

GET **Codeforces — Endpoint 2** `https://codeforces.com/api/user.status?handle=:cf_username&count=1000`

CF API RETURNS (SUBMISSIONS ARRAY)			SAVED TO DSASTATS SCHEMA
`submission.verdict`	Filter === "OK" only		
`submission.problem.name`	Deduplicate by name	→	
`uniqueSolved.size`	Count unique problems	→	`DsaStats.cf_problems_solved`

DSA SYNC SEQUENCE INSIDE DSA.SERVICE.JS

1. **Read lc_username + cf_username from DsaStats doc**
   User must have saved these in the Integrations settings page first
2. **Promise.all([fetchLeetCode(lc_username), fetchCodeforces(cf_username)])**
   Both calls run in parallel — faster than sequential
3. **Guard: if no username saved, skip that platform gracefully**
   Return zeros for that platform — do not throw an error
4. **DsaStats.findOneAndUpdate({ user_id }, { $set: merged }, { upsert: true })**
   Write everything in one atomic upsert, then trigger score recompute

> ⚠ LeetCode has no official API. The GraphQL endpoint is unofficial and may change. Always wrap the call in a try/catch so a LeetCode outage does not break your entire sync.

31. Build sequence — what to build in what order

The most important diagram for a beginner. Each phase depends on everything above it.

1 — Project setup + database — Start here — nothing else works without this — Build first

- **npm init both projects** — Create /server and /client folders, run npm init in each
- **MongoDB Atlas cluster** — Create free cluster, get MONGO_URI, add to .env
- **app.js + index.js** — Basic Express server, cors, json, morgan, connectDB
- **Vite + React scaffold** — npm create vite@latest, install Tailwind CSS

✓ Goal: server starts on port 5000, React starts on 5173, MongoDB connects — you see no errors in terminal

↓

2 — Auth — register, login, JWT — Every protected feature depends on this working first — Build second

- **User.model.js** — Schema with username, email, password_hash, role
- **auth.controller.js** — register hashes password, login returns JWT
- **auth.middleware.js** — Verifies JWT, attaches req.user
- **Login.jsx + Register.jsx** — Forms with validation, axios call, store JWT, redirect
- **AuthContext.jsx** — Stores token + user, provides login/logout functions
- **ProtectedRoute.jsx** — Redirects to /login if no token

✓ Goal: you can register, login, and a protected test route returns 401 without a token and 200 with one

↓

3 — Profile + projects CRUD — Core content layer — the portfolio itself — Build third

- **Profile + Project models** — Mongoose schemas with user_id foreign keys
- **profile + project routes** — GET public profile, PUT edit, POST/DELETE projects
- **Profile.jsx page** — Fetches by :username, shows bio, tech stack, projects
- **Projects.jsx page** — List + add + edit + delete with form validation

✓ Goal: a profile page at /profile/username shows your info, you can add and delete projects

↓

4 — Stats sync — GitHub + DSA — External data that feeds the DevScore — Build fourth

- **Stats models** — GithubStats + DsaStats schemas
- **github.service.js** — Calls /user, /repos, /events — upserts stats
- **leetcode + codeforces services** — GraphQL + REST calls, upsert DsaStats
- **Integrations.jsx page** — Enter LeetCode + CF usernames, trigger manual sync

✓ Goal: click Sync on the integrations page and see GitHub + DSA stats appear in MongoDB

↓

5 — DevScore + dashboard — The USP — only build after stats are working — Build fifth

- **score.service.js** — Weighted formula — reads stats, writes DevScore doc
- **DevScore model + route** — GET score, POST recompute endpoint
- **Dashboard.jsx** — Score badge, stats cards, Recharts bar charts
- **ScoreBadge + StatsCard** — Shared components used across dashboard + profile

✓ Goal: dashboard shows your live DevScore and charts — score changes when you sync

↓

6 — Leaderboard + recruiter + mentor views — Role-specific features — build last, all data already exists — Build last

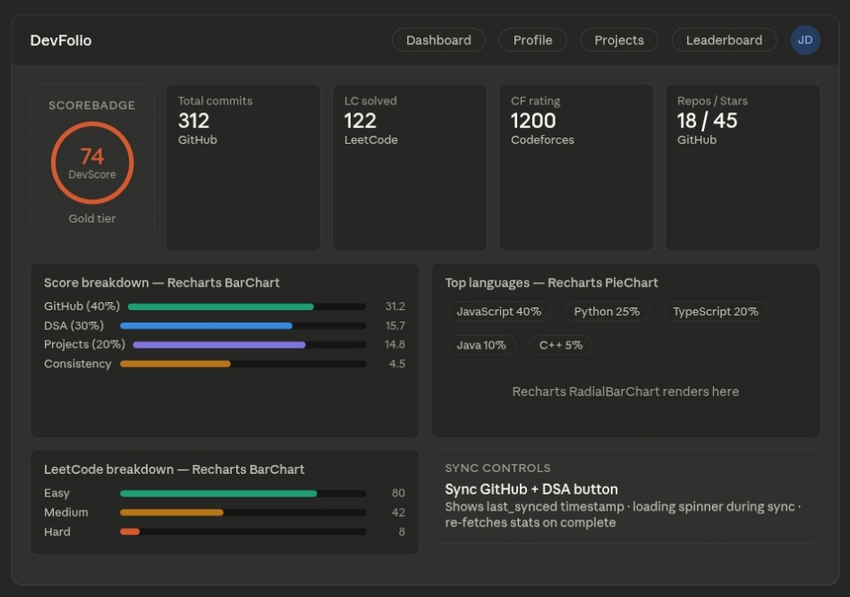
- **Leaderboard route + page** — Global + college scope, ranked by DevScore
- **RecruiterSearch.jsx** — Filter by skills, college, score range
- **MentorDashboard.jsx** — College leaderboard + student progress view
- **GitHub OAuth login** — Add after email/password auth is stable

✓ Goal: all three roles have their full UI and the project is feature-complete

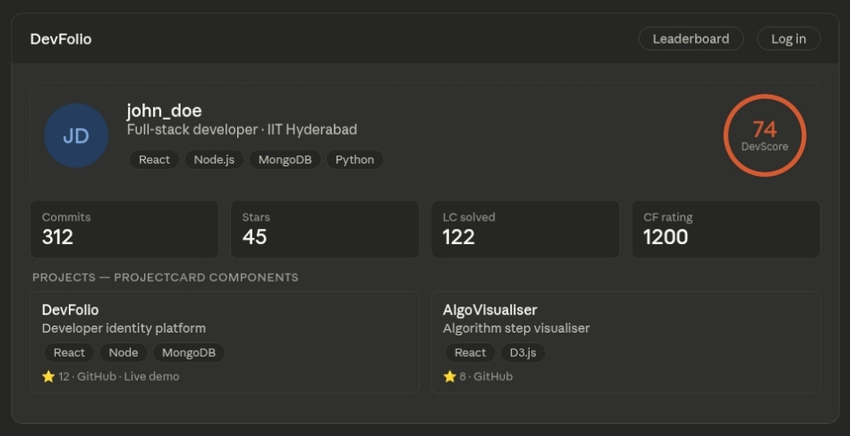
32. Dashboard & profile screen wireframes

What the two most important screens actually look like — layout, zones, and what component fills each slot.

DEVELOPER DASHBOARD — DASHBOARD.JSX

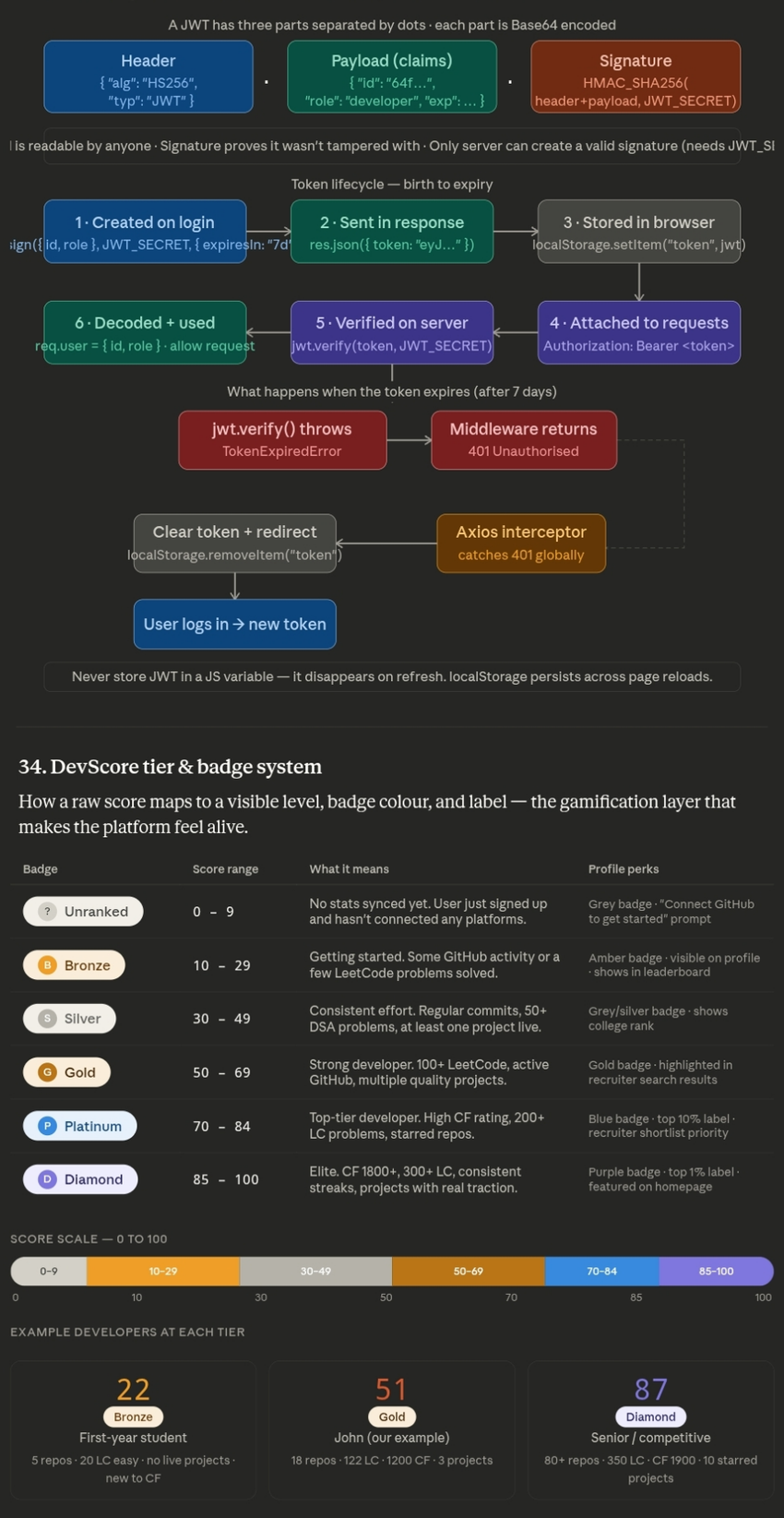


PUBLIC PROFILE PAGE — PROFILE.JSX



33. JWT token anatomy + lifecycle

What's actually inside a token, how it's created, where it lives, and how it dies.



34. DevScore tier & badge system

How a raw score maps to a visible level, badge colour, and label — the gamification layer that makes the platform feel alive.

Badge	Score range	What it means	Profile perks
? Unranked	0 – 9	No stats synced yet. User just signed up and hasn't connected any platforms.	Grey badge · "Connect GitHub to get started" prompt
B Bronze	10 – 29	Getting started. Some GitHub activity or a few LeetCode problems solved.	Amber badge · visible on profile · shows in leaderboard
S Silver	30 – 49	Consistent effort. Regular commits, 50+ DSA problems, at least one project live.	Grey/silver badge · shows college rank
G Gold	50 – 69	Strong developer. 100+ LeetCode, active GitHub, multiple quality projects.	Gold badge · highlighted in recruiter search results
P Platinum	70 – 84	Top-tier developer. High CF rating, 200+ LC problems, starred repos.	Blue badge · top 10% label · recruiter shortlist priority
D Diamond	85 – 100	Elite. CF 1800+, 300+ LC, consistent streaks, projects with real traction.	Purple badge · top 1% label · featured on homepage

SCORE SCALE — 0 TO 100

EXAMPLE DEVELOPERS AT EACH TIER

22	51	87
Bronze	Gold	Diamond
First-year student	John (our example)	Senior / competitive
5 repos · 20 LC easy · no live projects · new to CF	18 repos · 122 LC · 1200 CF · 3 projects	80+ repos · 350 LC · CF 1900 · 10 starred projects

Implementation: Store the tier name alongside total_score in the DevScore document. Recompute tier every time score recomputes. In ScoreBadge.jsx, switch on the tier string to pick the colour and label — no maths in the component.

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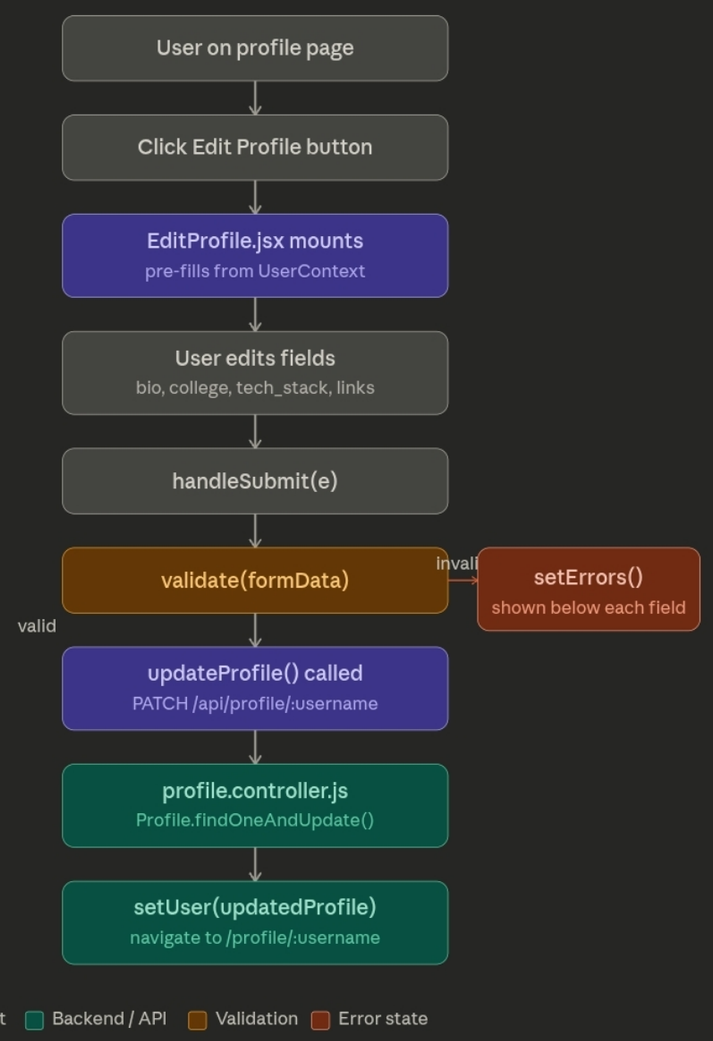
Visual documentation blueprint for personal project
Share

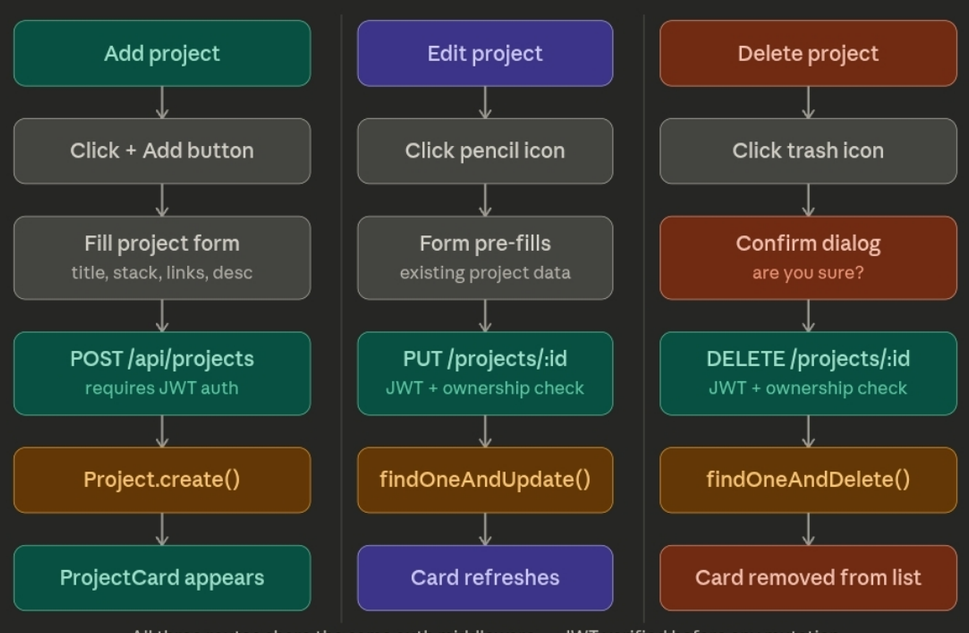
Conceived seventh diagram addressing practical avatar upload functionality >

Here are the remaining uncovered visual documentation pieces — all new, nothing repeated from what's already been created.

35. Profile Edit Flow — how a developer updates their profile end-to-end.

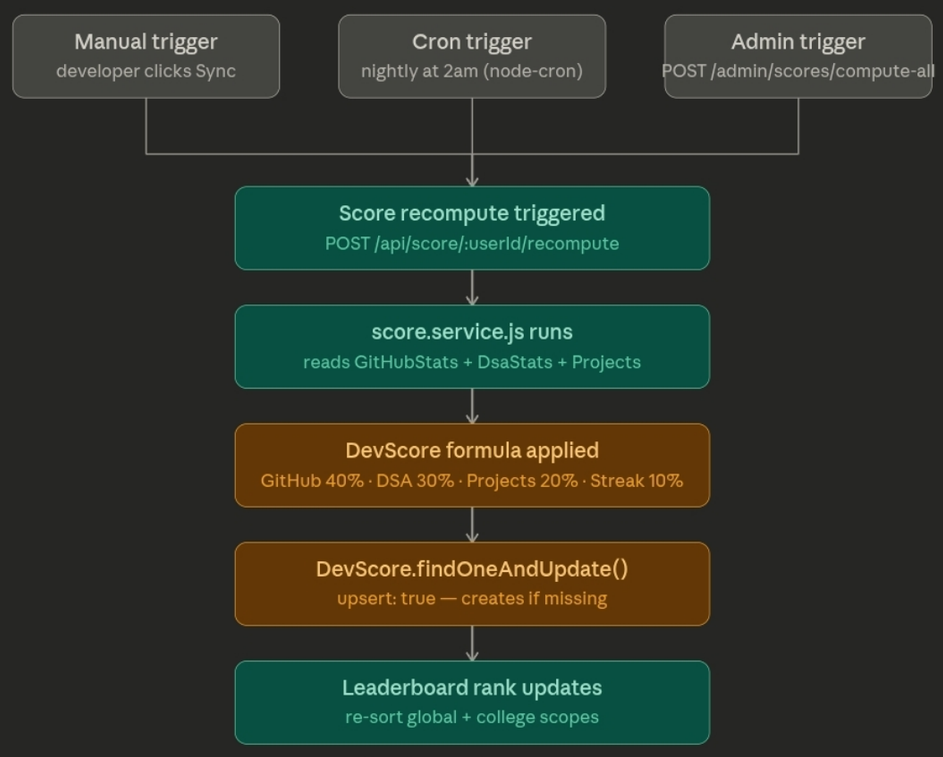


36. Project CRUD Lifecycle — the complete add / edit / delete flows side by side.



All three routes share the same auth middleware — JWT verified before any mutation

37. Score Recompute Trigger Flow — three separate events that all funnel into the same recompute pipeline.



Click any trigger to explore it — all three reach the same score pipeline

14:57 0.54 KB/S 5G+ 99
claude.ai/chat/54
Visual documentation blueprint for personal project
Share

Leaderboard rank updates
re-sort global + college scopes

Click any trigger to explore it — all three reach the same score pipeline

38. React Component Lifecycle — the single most repeated pattern in your entire codebase: mount → fetch → render. Every page in DevFolio does this.

Pages using this pattern: Profile.jsx · Dashboard.jsx · Leaderboard.jsx · RecruiterSearch.jsx

1. **Component mounts** — React renders the page for the first time
2. **useEffect fires (on mount)** — [] dependency array = runs only once
3. **setLoading(true)** — Spinner shown to the user
4. **API call via Axios** — Axios interceptor attaches JWT automatically
5. **Data arrives → setState** — setProfile(data) triggers a re-render
6. **setLoading(false)** — The finally{} block runs whether fetch succeeded or failed
7. **Component re-renders** — React diffs the new state and updates the DOM

← Prev step · Step 1 of 7 · Next step →

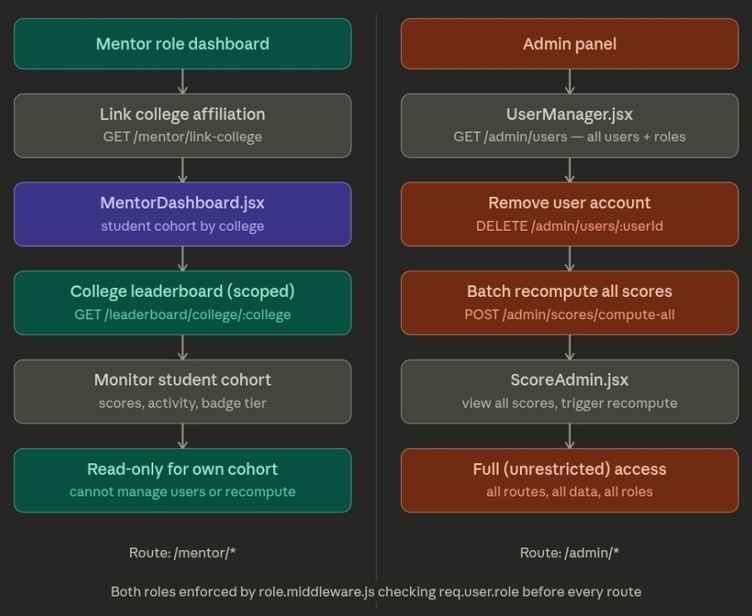
The page component loads. No data yet — the UI is in its initial empty state waiting for the fetch.

```
function Profile({ username }) {
  const [profile, setProfile] = useState(null);
  const [loading, setLoading] = useState(true);
  // nothing fetched yet ↑
```

39. Recruiter Search & Filter Flow — how a recruiter finds and shortlists developers.

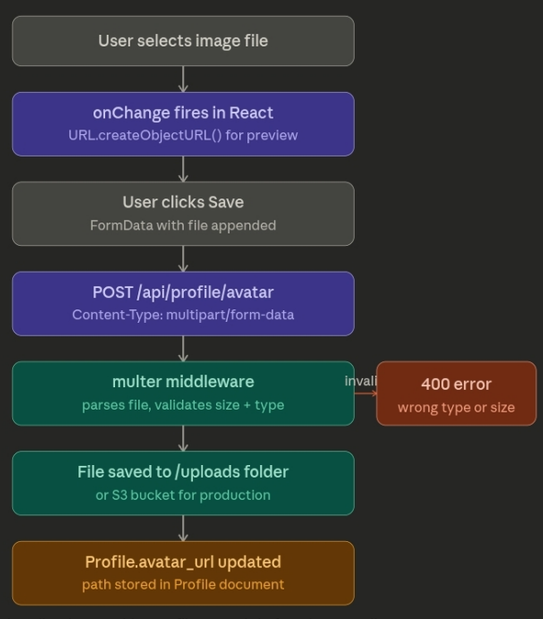
Recruiter role sees /recruiter/search — RBAC blocks this from Developer and Mentor roles

40. Mentor & Admin Panel Flows — the two remaining role-specific dashboards and what they each control.



Both roles enforced by role.middleware.js checking req.user.role before every route

One more that beginners commonly get stuck on — **41. Avatar Upload Flow** — how a file goes from the user's device to the database.



npm install multer — add to profile.routes.js before the controller

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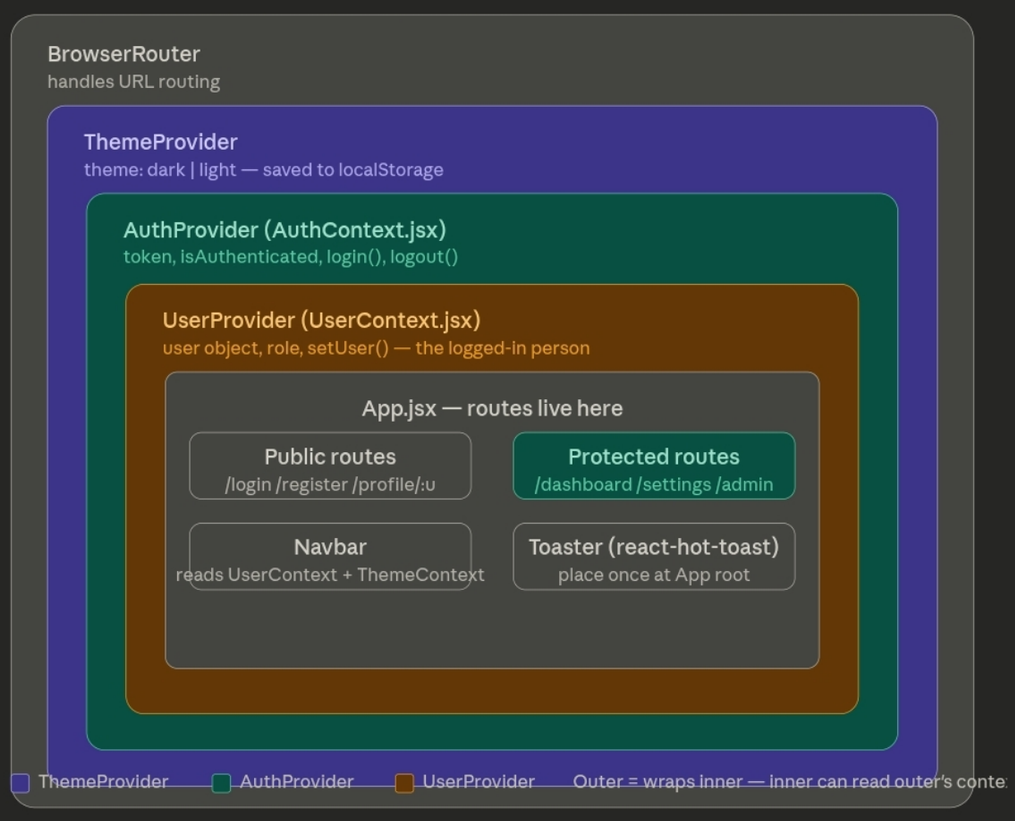
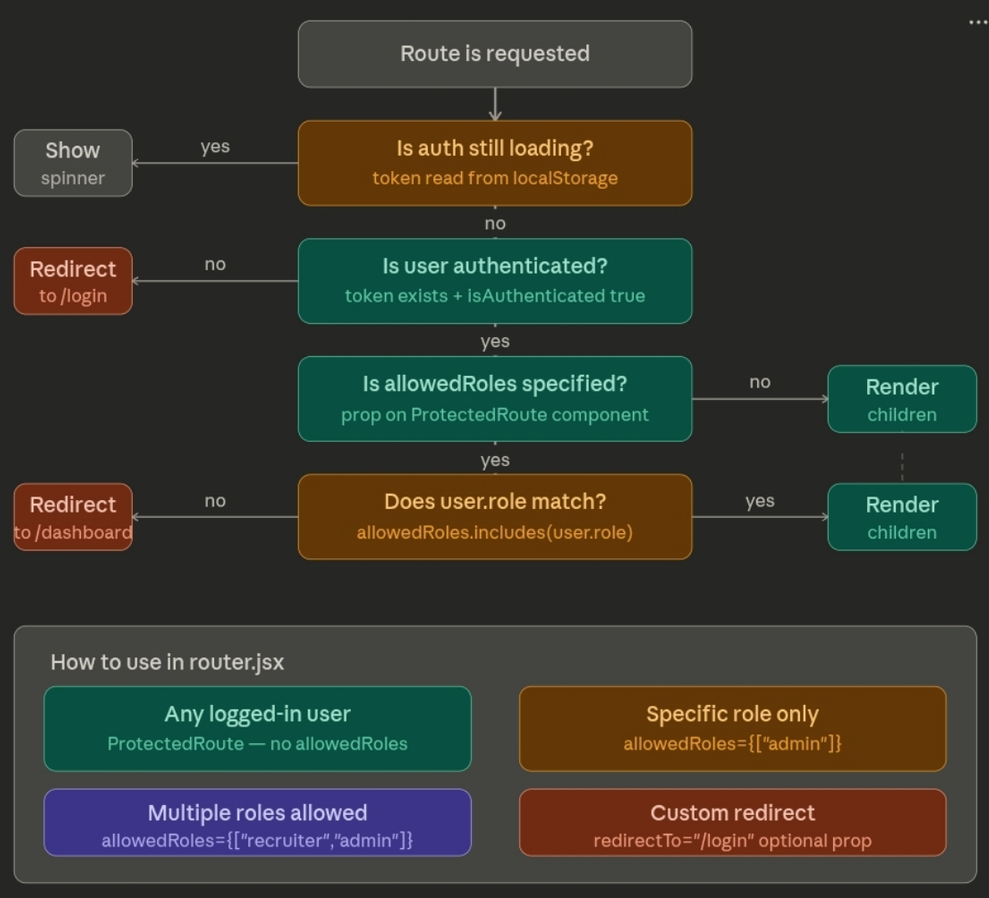
42. Context providers nesting — the React app tree showing how all three providers wrap everything and what each one owns.



43. ProtectedRoute decision tree — the exact logic inside `ProtectedRoute.jsx` that beginners always get wrong.



44. Dashboard charts map — exactly which Recharts component lives in which section, fed by which data.

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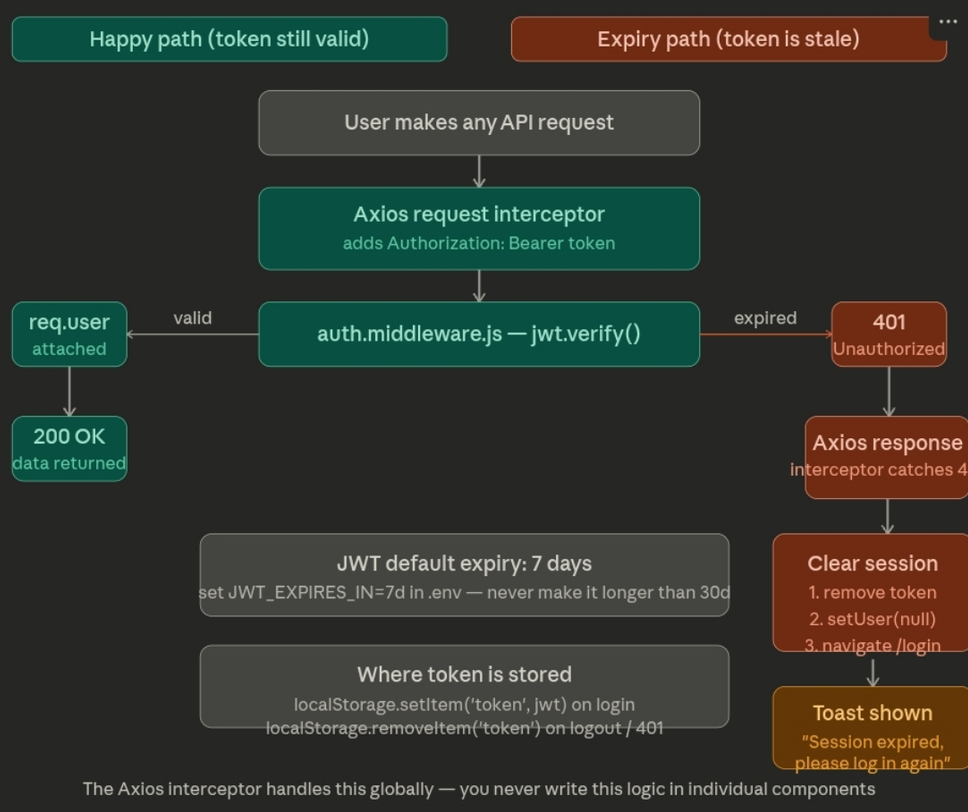
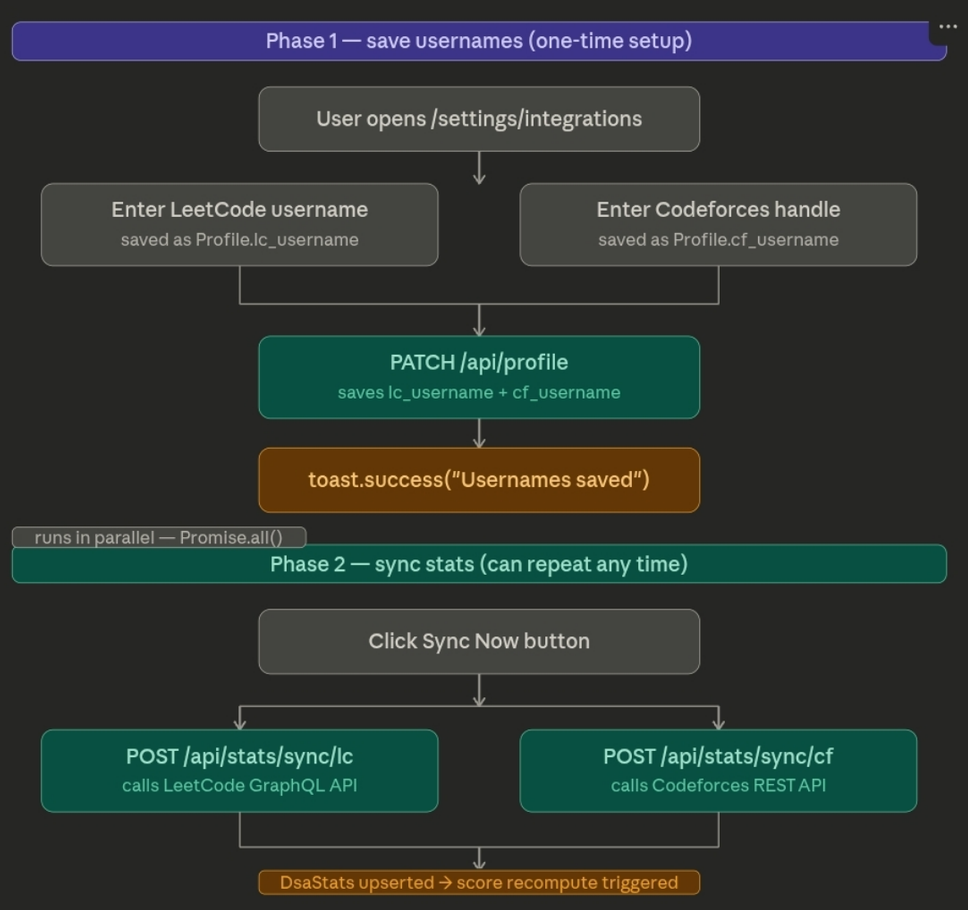
claude.ai/chat/54

45. JWT expiry + session handling — what happens when a token expires mid-session, the most common surprise for beginners.



The Axios interceptor handles this globally — you never write this logic in individual components

46. Integrations page flow — how a developer connects LeetCode and Codeforces for the first time.



47. Toast notification system — interactive reference showing every scenario that triggers a toast and exactly how to write it.

react-hot-toast usage across DevFolio click any row to ask about it

```
import toast from 'react-hot-toast'; // import in any component
<Toaster position="top-right" /> // once in App.jsx
```

Auth flows

- **Login success** — after setUser() and navigate() — success — `toast.success("Welcome back!")`
- **Login failed** — wrong password or email — error — `toast.error(err.message)`
- **Session expired** — Axios 401 interceptor — error — `toast.error("Session expired, please log in again")`

Profile & data actions

- **Profile saved** — after PATCH /api/profile — success — `toast.success("Profile updated")`
- **Sync complete** — after POST /github/sync — success — `toast.success("GitHub synced successfully")`
- **Sync in progress** — while fetch is running — loading — `toast.loading("Syncing...")`
- **Project added** — after POST /api/projects — success — `toast.success("Project added to portfolio")`
- **Project deleted** — after DELETE /api/projects/:id — success — `toast.success("Project removed")`

Pattern — promise toast (handles loading + success + error in one call)

```
toast.promise(
  syncGitHub(userId),
  {
    loading: 'Syncing GitHub...',
    success: 'GitHub synced!',
    error: (err) => err.message,
  }
); // replaces loading → success or error automatically
```

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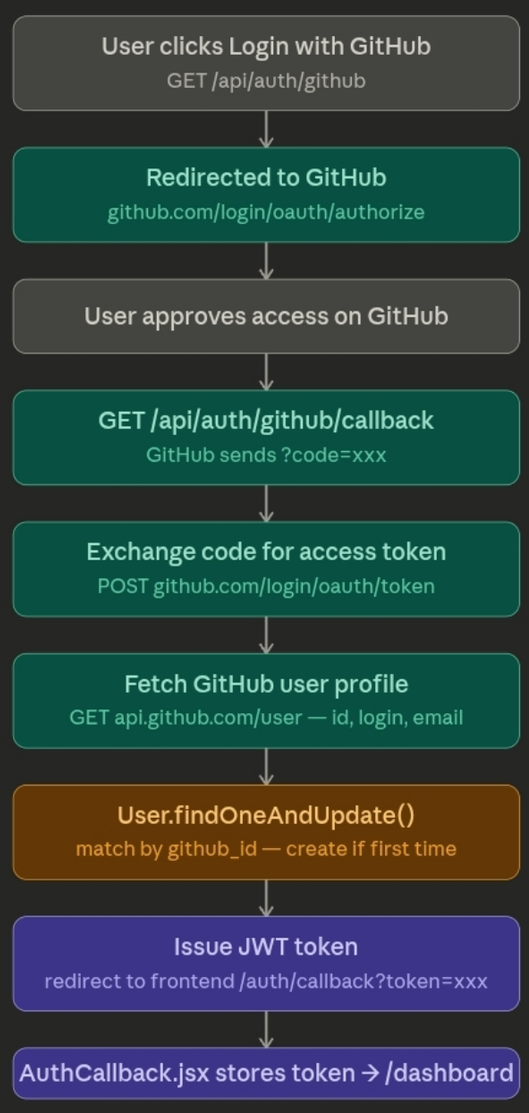
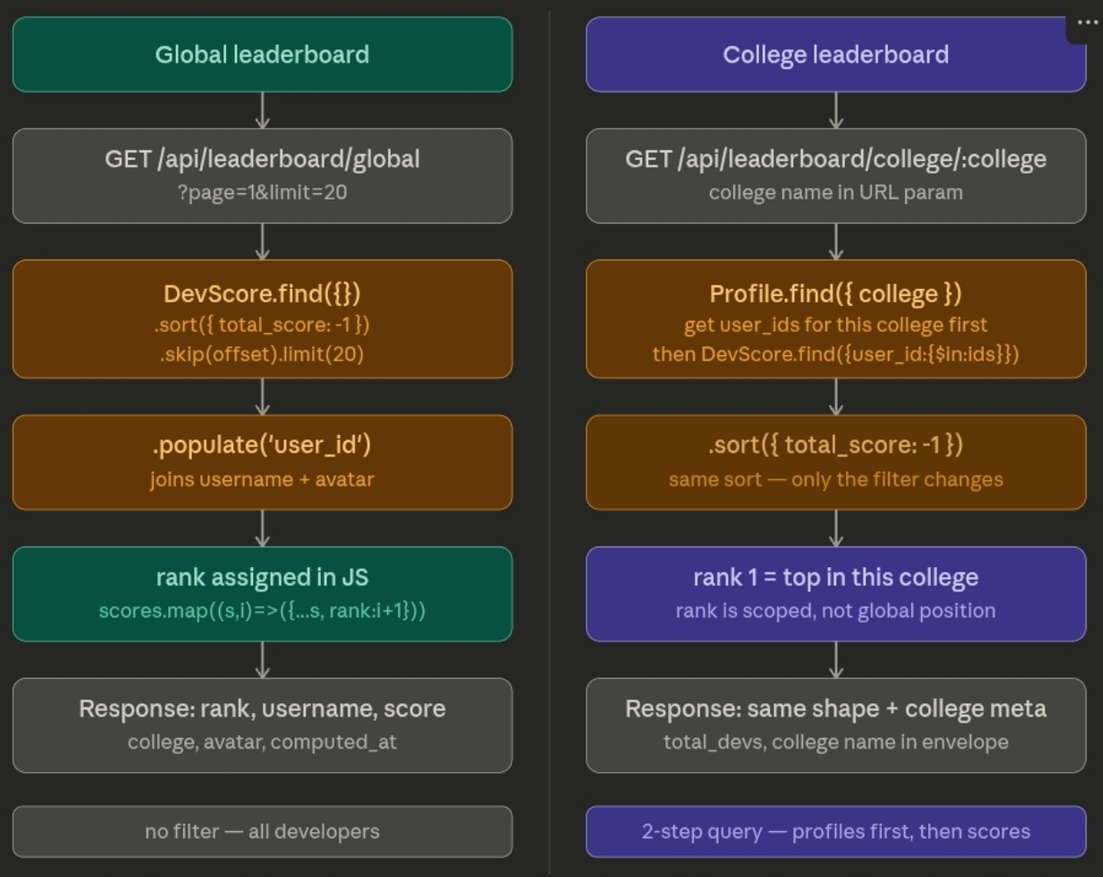
48. GitHub OAuth callback detail — what the backend does at `/auth/github/callback` that diagram 7 glossed over.



And the final one — **49. Global vs college leaderboard comparison** — the exact structural difference between the two query paths, often confused.



Both use the same Leaderboard.jsx component — the API endpoint and data shape are identical, only scope differs

Here's the complete master table across every diagram ever produced for DevFolio:

#	Diagram	What it covers
1–5	System arch, ERD, User flow, DevScore algorithm, API routes	Core blueprint
6–10	Component tree, Auth sequence, Data sync, RBAC, Deployment	Infrastructure
11–15	Folder structure, Middleware chain, React data flow, Env vars, E2E trace	Dev guidance
16–20	Props map, API shapes, Error handling, Scor h, Router config	Implementation

Reply...
Sonnet 4.6 Extended
Claude is AI and can make mistakes. Please double-check responses.

15:03 0.04 KB/S 5G+ 99
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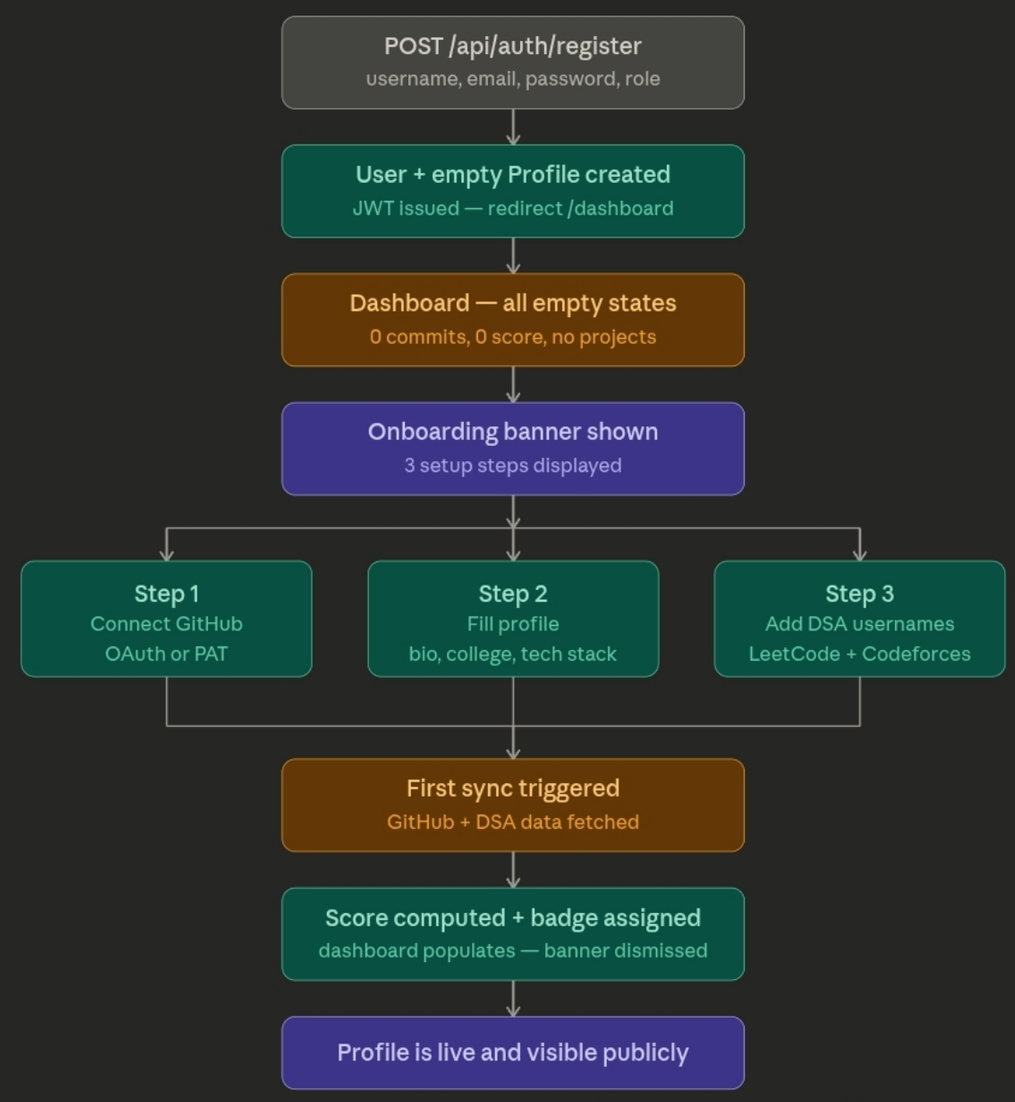
After a full audit of all 49 diagrams, here are the 6 genuinely uncovered gaps — things every beginner hits while building.

50. New user onboarding flow — what happens the very first time someone registers, before they have any data.



Track setup progress with localStorage: onboarding_steps_completed = [1,2,3]

51. Public vs authenticated profile visibility — exactly what each type of visitor sees on a profile page. This is the most common forgotten distinction.

Not logged in	Logged in — other user	Own profile (owner)
○ Username + avatar	○ Username + avatar	○ Username + avatar
○ Bio + college	○ Bio + college	○ Bio + college
○ Tech stack tags	○ Tech stack tags	○ Tech stack tags
○ DevScore + badge	○ DevScore + badge	○ DevScore + badge
○ Project cards	○ Project cards	○ Project cards
○ GitHub stats	○ GitHub stats	○ GitHub stats
○ DSA stats	○ DSA stats	○ DSA stats
○ Edit button	○ Edit button	○ Edit button
○ Sync button	○ Sync button	○ Sync button
○ Email address	○ Email address	○ Email address
○ Score breakdown chart	○ Score breakdown chart	○ Score breakdown chart
Public route — no auth needed	JWT verified — role irrelevant here	Check: user._id === profile.user_id

How to detect owner in Profile.jsx — compare `user?._id === profile?.user_id` — if true, show edit/sync controls. The API always returns all fields; the frontend hides the controls conditionally.

○ visible ○ hidden ○ partial

52. Loading & empty states map — the three states every screen must handle. Beginners almost always forget the empty state.

Screen	Loading state	Empty state	Error state
Profile page	skeleton avatar circle + 3 text lines	empty "No bio yet" placeholder text	toast "Could not load profile"
Dashboard	skeleton 4 stat card rectangles pulse	empty onboarding banner — sync prompt	toast error + retry button
Projects list	skeleton 2 gray card outlines	empty "Add your first project" + button	inline red text below the list
Leaderboard	skeleton 10 row placeholders pulse	— can't be empty if users exist	toast "Leaderboard unavailable"
Recruiter search	spinner centered over the results area	empty "No developers match filters"	toast + clear filters button
Sync button	spinner inline inside the button + disabled	— not applicable	toast "Sync failed, try again"
Charts	skeleton gray rectangle same size as chart	empty "Sync to see your stats"	inline gray placeholder stays shown
Mentor dashboard	skeleton table rows pulse	empty "No students in your college yet"	toast error message

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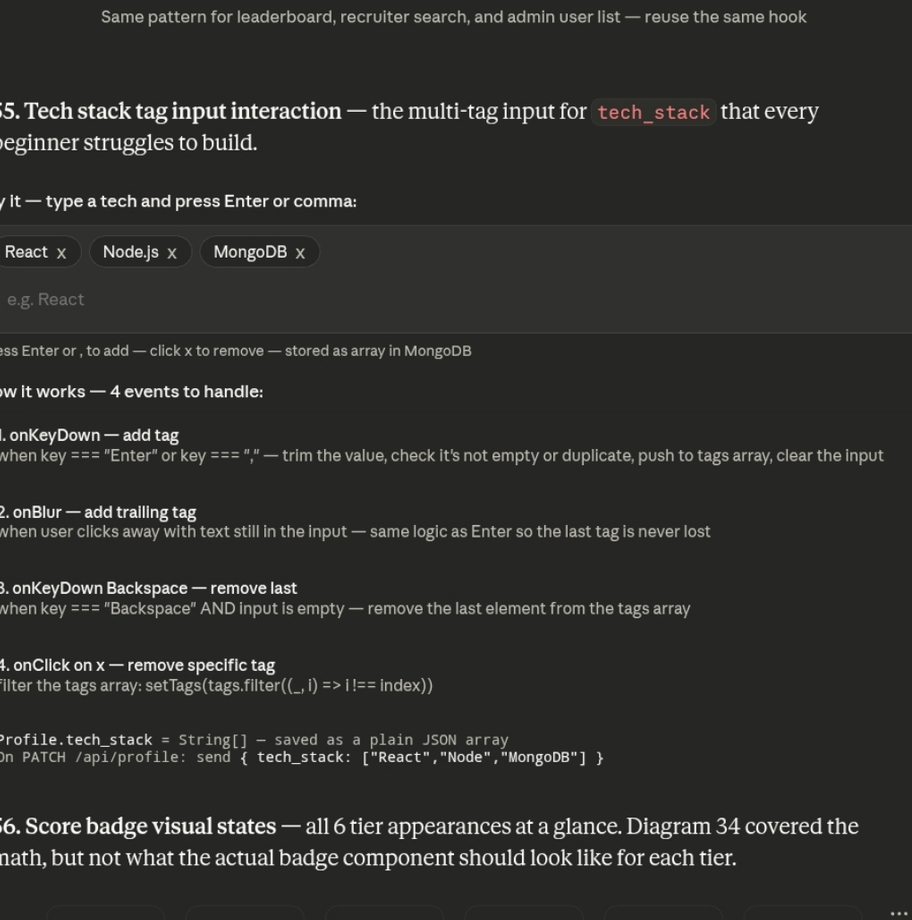
skeleton animated pulse placeholder — same layout as real content spinner small circular indicator

53. Custom hooks usage map — `useAuth`, `useFetch`, and `useProfile` — what each returns and exactly when to use it vs reaching directly into context.

54. Pagination pattern — the exact offset/limit pattern used by the leaderboard, project list, and recruiter search — every beginner gets confused by this.



55. Tech stack tag input interaction — the multi-tag input for `tech_stack` that every beginner struggles to build.

Try it — type a tech and press Enter or comma:

React x Node.js x MongoDB x

e.g. React

Press Enter or , to add — click x to remove — stored as array in MongoDB

How it works — 4 events to handle:

1. onKeyDown — add tag
when key === "Enter" or key === "," — trim the value, check it's not empty or duplicate, push to tags array, clear the input

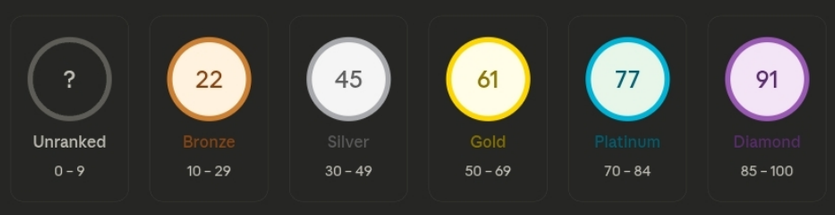
2. onBlur — add trailing tag
when user clicks away with text still in the input — same logic as Enter so the last tag is never lost

3. onKeyDown Backspace — remove last
when key === "Backspace" AND input is empty — remove the last element from the tags array

4. onClick on x — remove specific tag
filter the tags array: setTags(tags.filter((_, i) => i !== index))

```
Profile.tech_stack = String[] — saved as a plain JSON array
On PATCH /api/profile: send { tech_stack: ["React","Node","MongoDB"] }
```

56. Score badge visual states — all 6 tier appearances at a glance. Diagram 34 covered the math, but not what the actual badge component should look like for each tier.



`getTier(score)` helper function — maps score to { tier, color, ringColor, label } — used in both ScoreBadge.jsx and the leaderboard row

Two sizes: `size="lg"` (64px ring, shown on profile and dashboard) and `size="sm"` (32px ring, shown inline on leaderboard rows and recruiter search cards). Same component — conditional Tailwind classes based on the size prop.

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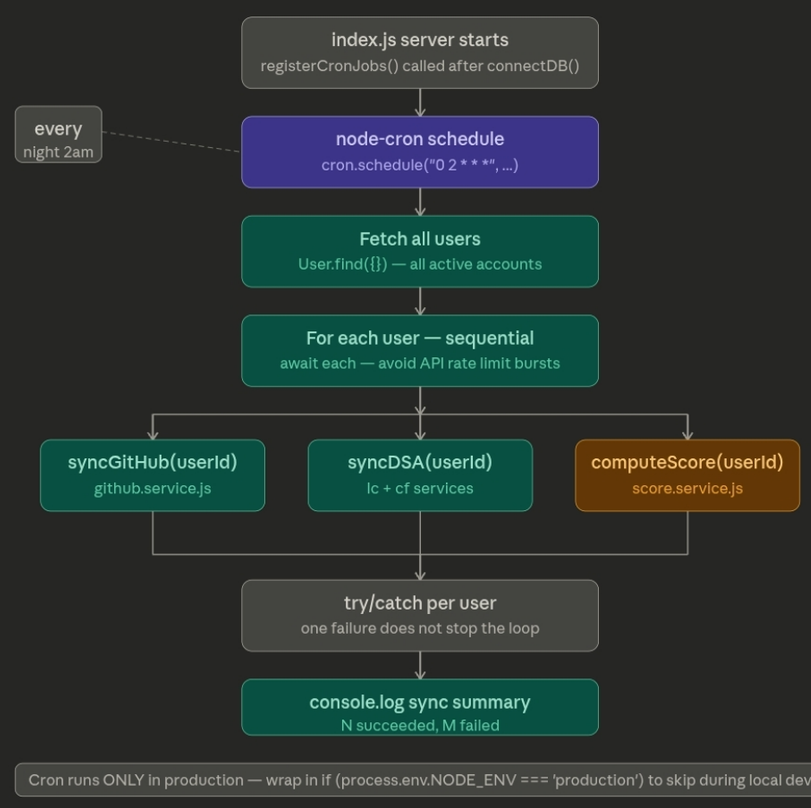
Audited existing diagrams and identified six genuinely missing visual documentation areas >

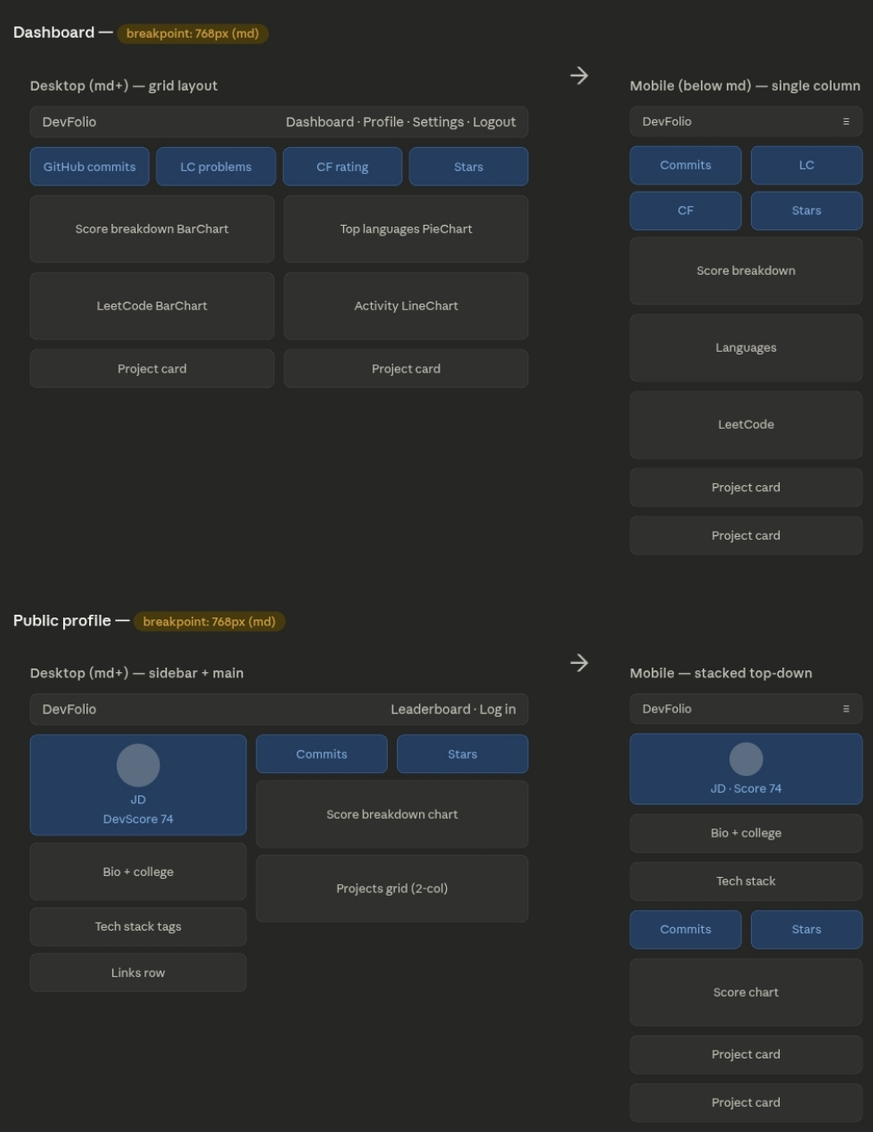
After a complete audit of all 56 diagrams, here are the 6 remaining genuine gaps — things that will actually block you during build but haven't appeared once yet.

URL	Guard	What it renders / URL params
Public routes — no login needed		
`/`	public	Landing.jsx — hero, features, CTA buttons
`/login`	public	Login.jsx — email/pass form + GitHub OAuth button
`/register`	public	Register.jsx — username, email, password, role picker
`/profile/:username`	public	Profile.jsx — reads :username param from URL
`/leaderboard`	public	Leaderboard.jsx — global ranking, read-only
`/auth/callback`	public	AuthCallback.jsx — reads ?token= param, stores JWT, redirects
Protected — any logged-in user		
`/dashboard`	auth	Dashboard.jsx — stats, charts, score, projects
`/settings/profile`	auth	EditProfile.jsx — bio, college, tech stack, avatar
`/settings/projects`	auth	Projects.jsx — add / edit / delete project cards
`/settings/integrations`	auth	Integrations.jsx — LeetCode + CF username inputs, sync buttons
Role-protected routes		
`/recruiter/search`	recruiter	RecruiterSearch.jsx — filter + developer cards
`/recruiter/view/:username`	recruiter	RecruiterView.jsx — full profile in recruiter context
`/mentor/dashboard`	mentor	MentorDashboard.jsx — college-scoped leaderboard + student list
`/admin/users`	admin	UserManager.jsx — table of all users, delete, roles
`/admin/scores`	admin	ScoreAdmin.jsx — all scores, batch recompute button
`* (catch-all)`	public	NotFound.jsx — 404 message + link back to home

57 done. Next — **58. Nightly cron job scheduler flow** — how `node-cron` wires into `index.js` and what it does at 2 AM every night.



59. Responsive layout breakpoints — how the two most important screens shift from desktop to mobile. Never diagrammed before.



Tailwind classes: `grid-cols-1 md:grid-cols-2` for charts · `grid-cols-1 md:grid-cols-4` for stat cards · `hidden md:flex` for desktop nav links

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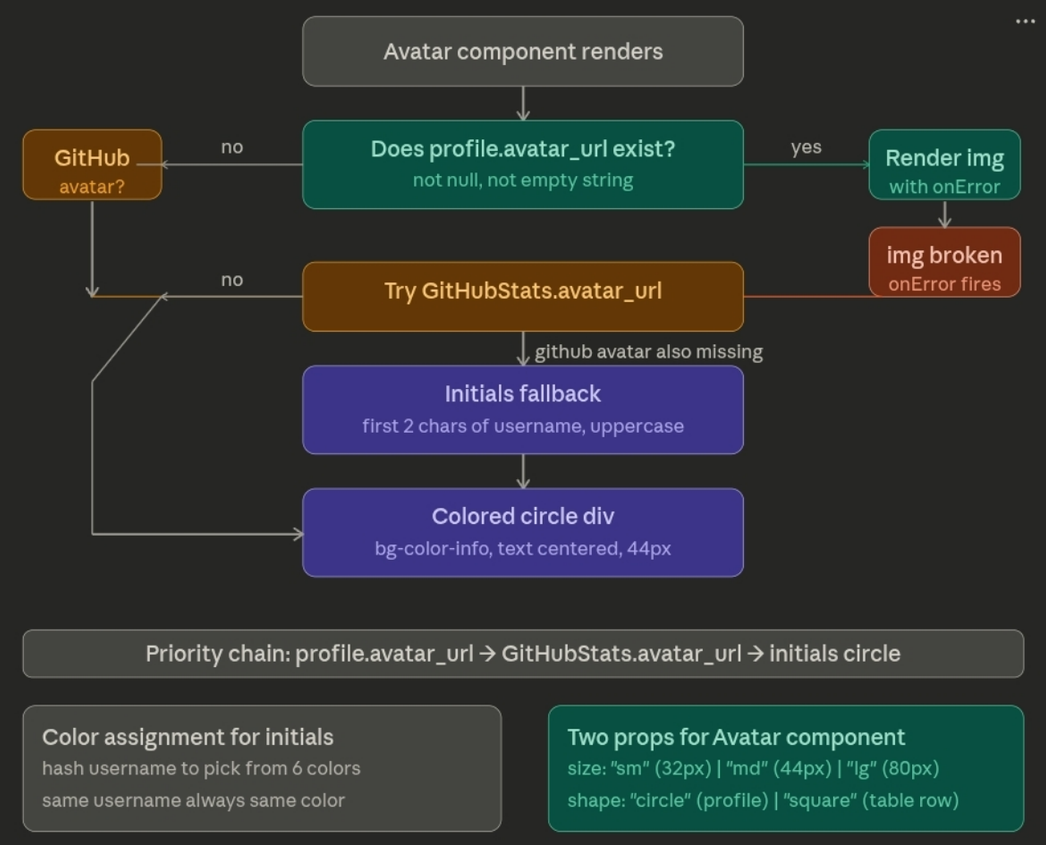
60. API error taxonomy — every HTTP status your backend returns, what it means, and exactly how your frontend should respond to each one.

Status	When backend sends it	Frontend reaction
2xx — success		
200	GET request succeeded, data returned	render set state with res.data.data
201	POST succeeded — new resource created	toast success + append to local list
4xx — client errors (your code sent wrong data)		
400	Validation failed — missing/bad field	inline setErrors(res.data.errors) — show under each field
401	JWT missing, expired, or invalid	redirect clear token + navigate /login — Axios intercepts globally
403	Role mismatch — valid JWT, wrong role	redirect navigate /dashboard — "Access denied" toast
404	Resource not found (user/project/score)	inline show empty state or "Not found" message
409	Duplicate — email or username taken	inline setErrors({ email: "already in use" })
422	Business rule violation (not schema)	toast toast.error(res.data.message)
5xx — server errors (backend crashed)		
500	Unhandled error — bug in server code	toast "Something went wrong, try again"
network	Server unreachable — no response at all	toast "Cannot reach server — check connection"

Rule of thumb — **400/409/422** errors go under form fields (setErrors). **401/403** redirect silently. **404** shows empty state. **500/network** shows a toast. Never show raw error objects to the user.

61. Avatar/image fallback chain — what renders when avatar_url is missing, broken, or empty — the silent bug that makes every new user see a black square.



62. Component communication patterns — parent→child, sibling, and distant — the three patterns you'll use in every file, with DevFolio-specific examples for each.

1 Parent → Child via props
When: parent owns the data, child just displays it

Dashboard.jsx — project={obj} editable={true} onDelete={fn} → ProjectCard.jsx

DevFolio examples: Dashboard passes project data to ProjectCard · Profile passes stats to StatCard · Leaderboard passes entry to LeaderboardRow · RecruiterSearch passes developer to SearchCard

```
Rule: if only ONE child needs the data, use props — don't put it in context
```

2 Child → Parent via callback prop
When: child triggers an action, parent owns the state

ProjectCard.jsx — onDelete(projectId) ← Dashboard.jsx

DevFolio examples: ProjectCard calls onDelete(id) → Dashboard removes from list · SearchFilter calls onFilter(values) → RecruiterSearch re-fetches · EditProfile calls onSave(data) → parent navigates away

```
Rule: never lift state UP to context just to share one callback — pass the function as a prop
```

3 Distant components via Context
When: 3+ unrelated components need the same data

Navbar.jsx ↑ useUser() UserContext useUser() ↓ Dashboard.jsx useUser() ↓ EditProfile.jsx

DevFolio context data: user object (Navbar + Dashboard + Profile all need it) · token (Axios interceptor + ProtectedRoute) · theme (every component needs dark/light class)

```
Rule: if you're passing a prop through 2+ components that don't use it themselves — move it to context
```

4 Siblings share via lifted state in parent
When: 2 nearby sibling components need the same piece of state

SearchFilter.jsx — onFilter(v) → RecruiterSearch.jsx (parent) ← results={data} — ResultsList.jsx

DevFolio examples: SearchFilter and ResultsList are siblings under RecruiterSearch — the filter value lives in RecruiterSearch's useState, not in context · EditProfile tabs (profile info vs projects) share validation state in the parent page

```
Rule: lift state to the nearest common ancestor — not all the way to context
```

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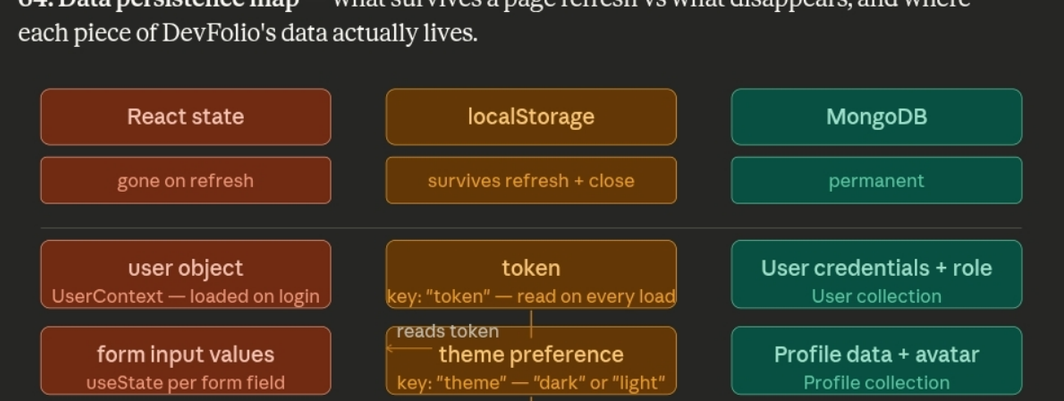
63. Debounce pattern for recruiter search — without this, every keystroke fires an API call. The most common performance bug beginners ship.

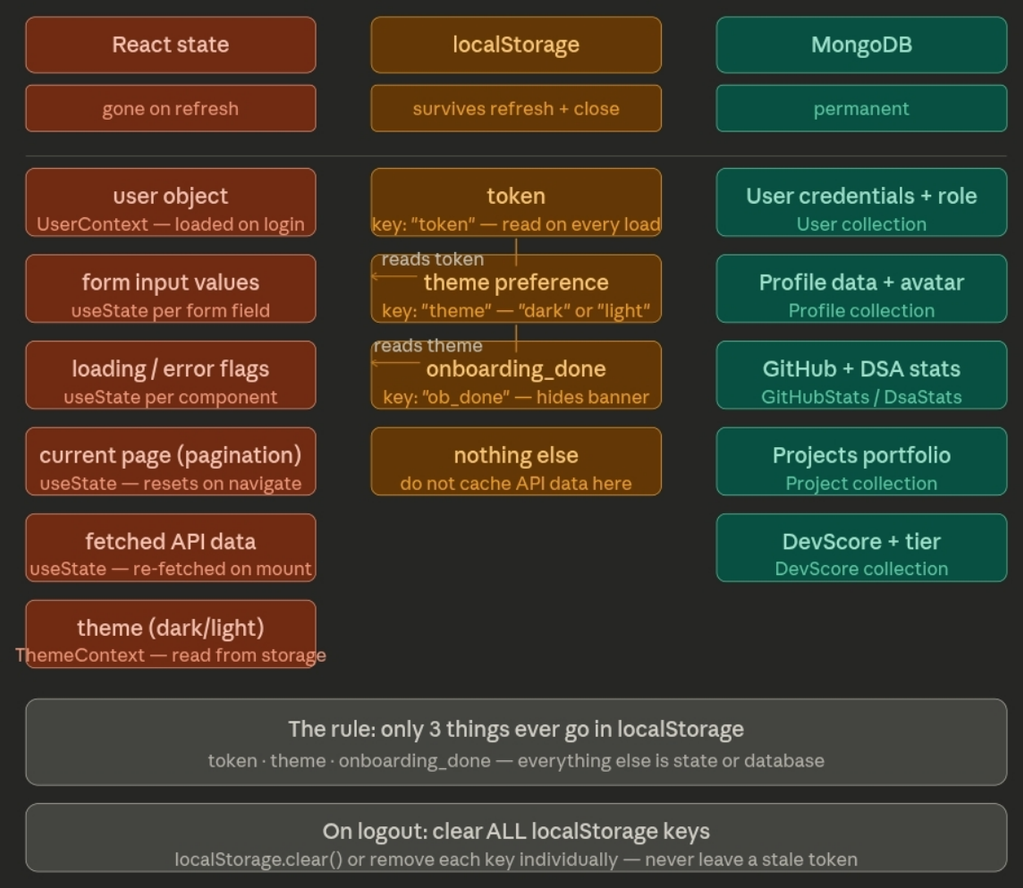


How it works — 3 steps in RecruiterSearch.jsx

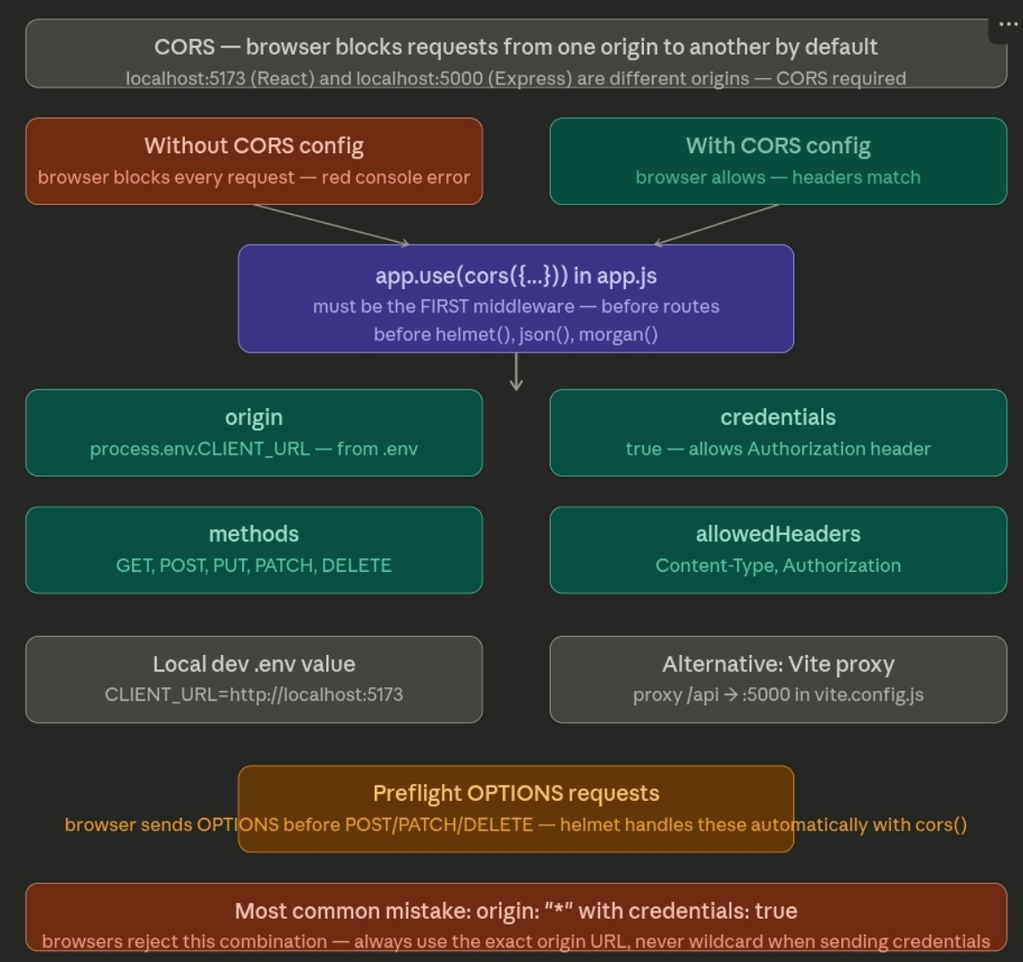
1. **Two pieces of state — `inputValue` and `debouncedValue`**
   inputValue updates instantly on every keystroke so the input feels responsive. debouncedValue lags 400ms behind.
2. **useEffect watches inputValue — starts a 400ms setTimeout**
   Cleanup function clears the old timer before starting a new one. New keystroke = old timer cancelled, new timer starts. Only the final timer fires.
3. **Second useEffect watches debouncedValue — fires the API call**
   When debouncedValue changes (only after silence), fetch /api/leaderboard/global?tech=debouncedValue. Keeps fetch logic separate from the timer logic.

Also apply debounce to the score range slider in the same component — dragging a slider fires hundreds of onChange events. Same 2-state pattern, same 400ms delay.

64. Data persistence map — what survives a page refresh vs what disappears, and where each piece of DevFolio's data actually lives.



65. CORS configuration map — the single most common blocker on day 1 when you first connect frontend to backend. Never diagrammed before.



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66. MongoDB index strategy — which fields need indexes for DevFolio's queries to stay fast as the data grows. Never diagrammed.

Collection · field	Index type	Why it's needed
User collection		
`email`	unique	login lookup — every POST /auth/login queries by email
`username`	unique	profile URL — GET /profile/:username queries by username
`github_id`	unique sparse	OAuth upsert — findOneAndUpdate by github_id on every GitHub login
Profile collection		
`user_id`	index	profile fetch — joined from User on most profile page loads
`college`	index	college leaderboard — Profile.find({college}) to get ids first
DevScore collection		
`total_score`	index desc	leaderboard sort — .sort({total_score:-1}) on every leaderboard page
`user_id`	unique	one score per user — upsert target, also prevents duplicates
Project collection		
`user_id`	index	project list — Project.find({user_id}) on profile and dashboard
GitHubStats + DsaStats collections		
`user_id`	unique	sync upsert target — findOneAndUpdate({user_id}) on every sync

unique prevents duplicates + creates index automatically index speeds up reads, no uniqueness enforcement

sparse only indexes documents where the field exists

How to add in Mongoose schema: `email: { type: String, unique: true, index: true }` — Mongoose creates the index in MongoDB automatically when the server starts. For `total_score` add `{ type: Number, index: true }` in the schema and MongoDB will sort it efficiently.

67. External API rate limits & retry strategy — GitHub, LeetCode, and Codeforces all have hard limits. Hit them without a plan and your sync silently breaks.

GitHub REST API 5,000 req / hr (authenticated)

Unauthenticated	60 req/hr — always use GITHUB_API_TOKEN
Header to read	X-RateLimit-Remaining and X-RateLimit-Reset
DevFolio usage	3 calls per sync (user, repos, events) × N users = check remaining before cron loop
When 429 returned	read X-RateLimit-Reset (Unix timestamp) — wait until then

LeetCode GraphQL (unofficial) No official limit published

Endpoint	leetcode.com/graphql — no API key, no auth
Risk	can start returning 429 or blocking IP without notice
Strategy	add 500ms delay between each user's LeetCode call in cron loop
On failure	catch error, skip this user, continue loop — log the failure

Codeforces REST API 5 req / sec per IP

Endpoint	codeforces.com/api/user.* — public, no token needed
When 429 returned	status: "FAILED", comment: "Call limit exceeded"
Strategy	sequential await in cron loop already keeps you under 5/sec
Handle missing user	status: "FAILED" if handle not found — skip, don't throw

Retry pattern — use for all 3 APIs in sync services

1. **First attempt** — make the API call normally
2. **On 429 or network error** — wait 1 second, retry once
3. **Second failure** — skip this user, log error, continue to next user
4. **Never update last_synced** if data was not actually written — leave old value intact

The nightly cron runs with sequential await — this automatically keeps Codeforces under 5/sec and spreads GitHub calls across the hour. Only add explicit delays if you have 100+ users doing concurrent manual syncs.

Here's the complete final additions — all genuinely new and confirmed not to repeat anything across the 62 prior diagrams:

#	Diagram	Why it blocks beginners
63	Debounce search pattern	Every keystroke fires an API call without it — ships as a bug every time
64	Data persistence map	Beginners store things in localStorage that should be in state, or forget to clear the token on logout
65	CORS configuration	The #1 day-1 blocker when React first calls Express — never documented anywhere obvious
66	MongoDB index strategy	Queries work fine with 10 users, silently crawl with 10,000 — indexes prevent this
67	External API rate limits &	GitHub returns 429, LeetCode blocks silently, Codeforces has a 5/sec cap

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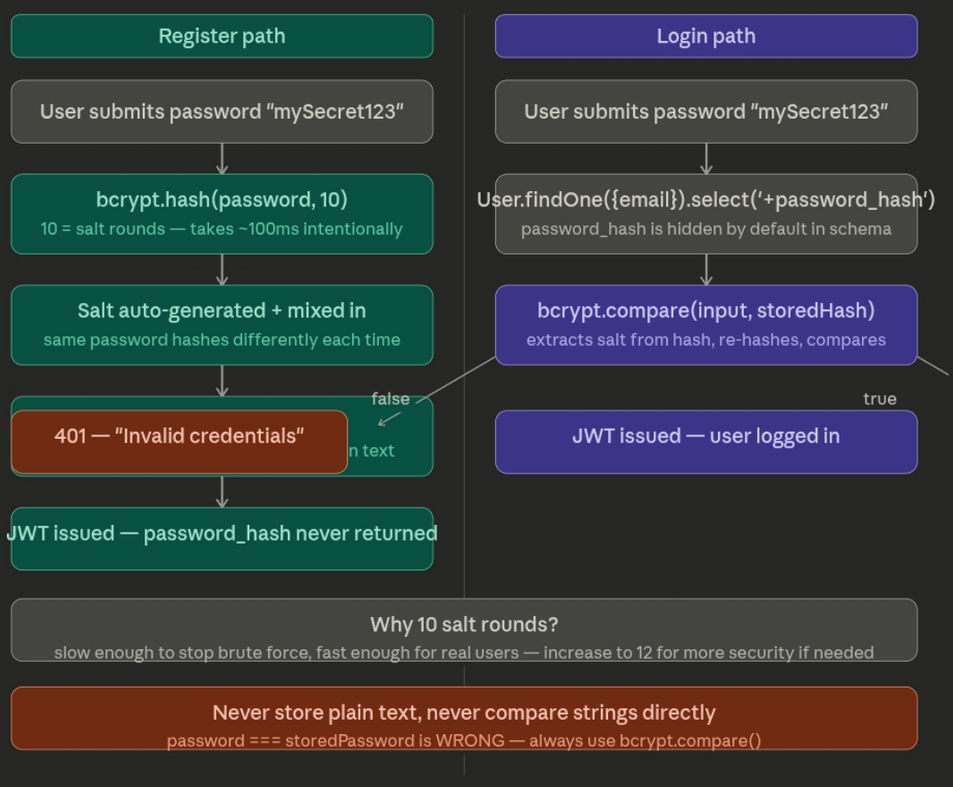
63. Password hashing lifecycle — what bcrypt actually does on register vs login. Beginners skip this and store plain-text passwords.



69. Navbar state transitions — the 5 different navbar appearances based on auth + role. Every beginner hardcodes one navbar and wonders why the admin sees developer buttons.

Not logged in — public visitor

DevFolio Leaderboard Log in Sign up

Developer role — logged in

DevFolio Dashboard My profile Leaderboard Settings JD

Recruiter role — logged in

DevFolio Search developers Leaderboard Settings HR

Mentor role — logged in

DevFolio Mentor dashboard Leaderboard Settings Prof

Admin role — logged in

DevFolio Dashboard Users Scores Leaderboard ADM

Conditional logic inside Navbar.jsx

`!isAuthenticated`	show Log in + Sign up only
`isAuthenticated + role === "developer"`	show Dashboard, My profile, Settings
`isAuthenticated + role === "recruiter"`	show Search developers — hide Dashboard
`isAuthenticated + role === "mentor"`	show Mentor dashboard — hide Search
`isAuthenticated + role === "admin"`	show Users + Scores admin links
`isAuthenticated (any role)`	show avatar/initials circle + Leaderboard

All data from `useAuth()` and `useUser()` — never hard-code role strings in JSX, use `user?.role === 'admin'` comparisons

70. "Which files do I touch?" task map — given any common development task, which exact files to open in sequence. The gap every beginner falls into.

`model` MongoDB schema `service/controller` backend logic `component/page` frontend React `route/api` wiring

Adding a new field to the database (e.g. linkedin_url to Profile)

Add a field to Profile

`Profile.model.js` → `profile.controller.js` → `EditProfile.jsx` → `Profile.jsx`

Adding a new API endpoint (e.g. GET /api/stats/summary)

Add a new backend route

`stats.controller.js` → `stats.routes.js` → `app.js` (if new router) → `stats.api.js`

Adding a new page/screen (e.g. /settings/notifications)

Add a new frontend page

`Notifications.jsx` → `router.jsx` → `Navbar.jsx` (add link)

Adding a reusable component (e.g. SkillBadge)

Add a new shared component

`components/SkillBadge.jsx` → `import in Profile.jsx` + `import in SearchCard.jsx`

Adding a new cron job (e.g. weekly digest)

Add a scheduled job

`digest.service.js` → `cron.js` → `index.js` (already wired)

Fixing a bug in data display (e.g. wrong score shown)

Debug wrong data on screen

`Dashboard.jsx` check state → `score.api.js` check response → `score.controller.js` → `score.service.js`

Changing RBAC — letting recruiters see score breakdown

Change a permission / role rule

`role.middleware.js` → `ProtectedRoute.jsx` (if route) → `Profile.jsx` (if UI conditional)

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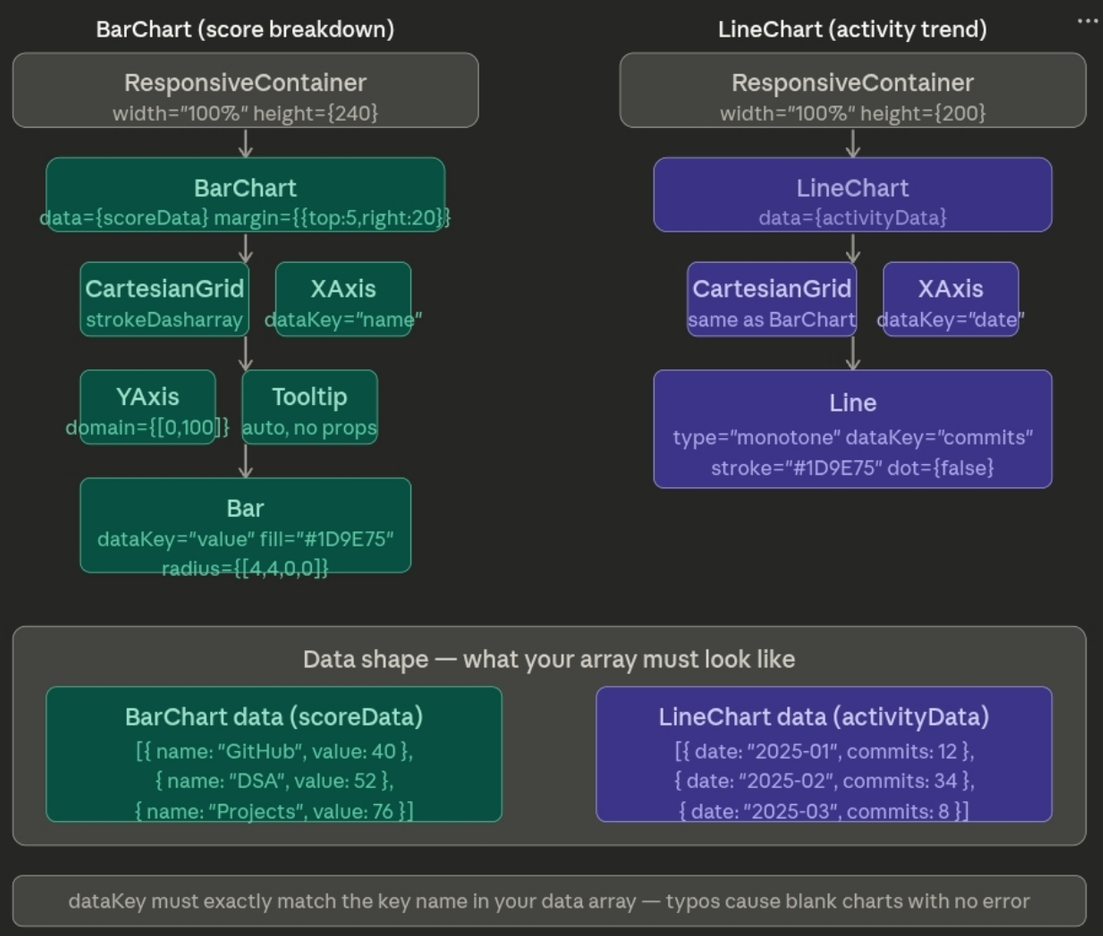
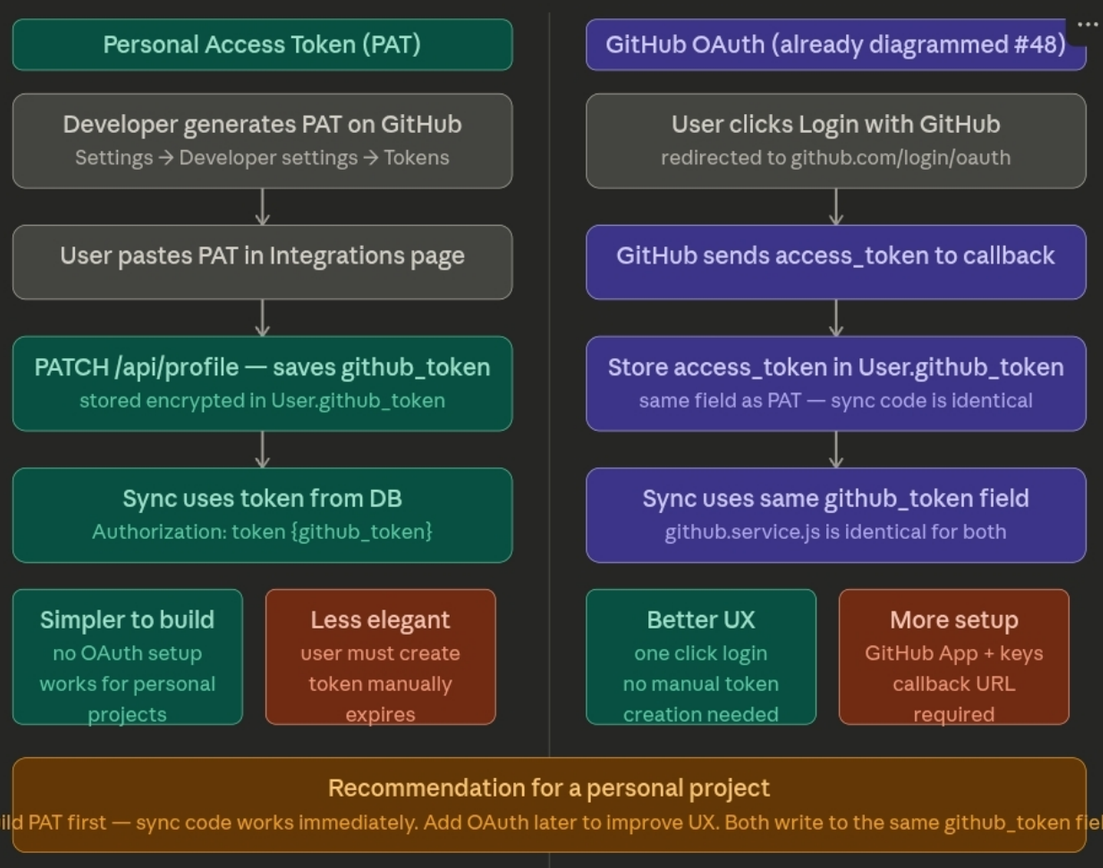
71. Recharts component anatomy — the exact nesting structure that trips up every beginner. The library is powerful but the JSX tree is non-obvious.



72. GitHub Personal Access Token (PAT) vs OAuth — the two connection methods for GitHub. OAuth was diagrammed (#48), but PAT — which is simpler for solo development — was never shown.



Here's the complete final additions — verified against all 67 prior diagrams with zero overlap:

#	Diagram	The gap it fills
68	Password hashing lifecycle	bcrypt.hash on register, bcrypt.compare on login — beginners store plain text
69	Navbar state transitions	5 different navbars per auth+role state — beginners hardcode one
70	"Which files do I touch?" task map	Given any task, the exact file sequence — the gap every beginner falls into
71	Recharts component anatomy	The exact JSX ting tree + data shape — blank charts with no error message

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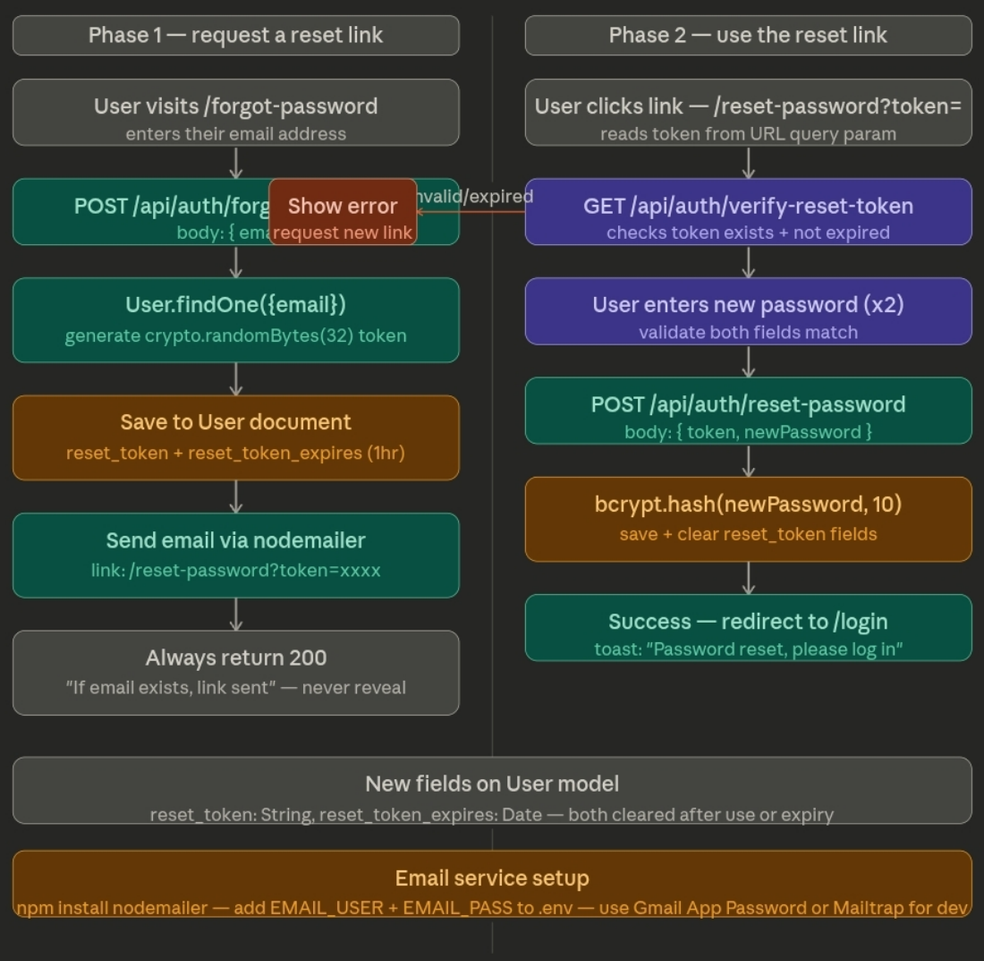
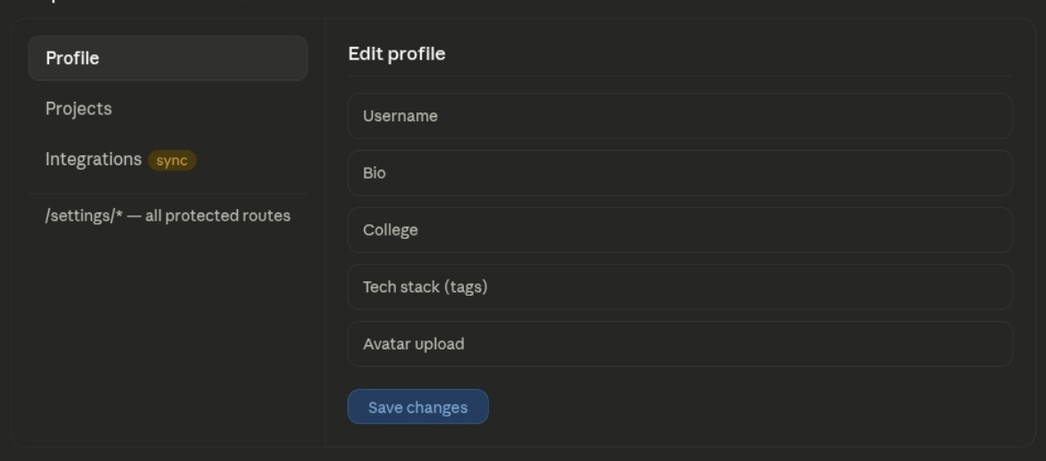
73. Forgot password / reset flow — a standard feature that's completely absent from all 72 diagrams. Beginners discover they need it after launch.



74. Settings page sub-navigation layout — the sidebar + content layout inside `/settings/*`. Every beginner gets confused about how the three settings tabs relate to each other and share the same shell.

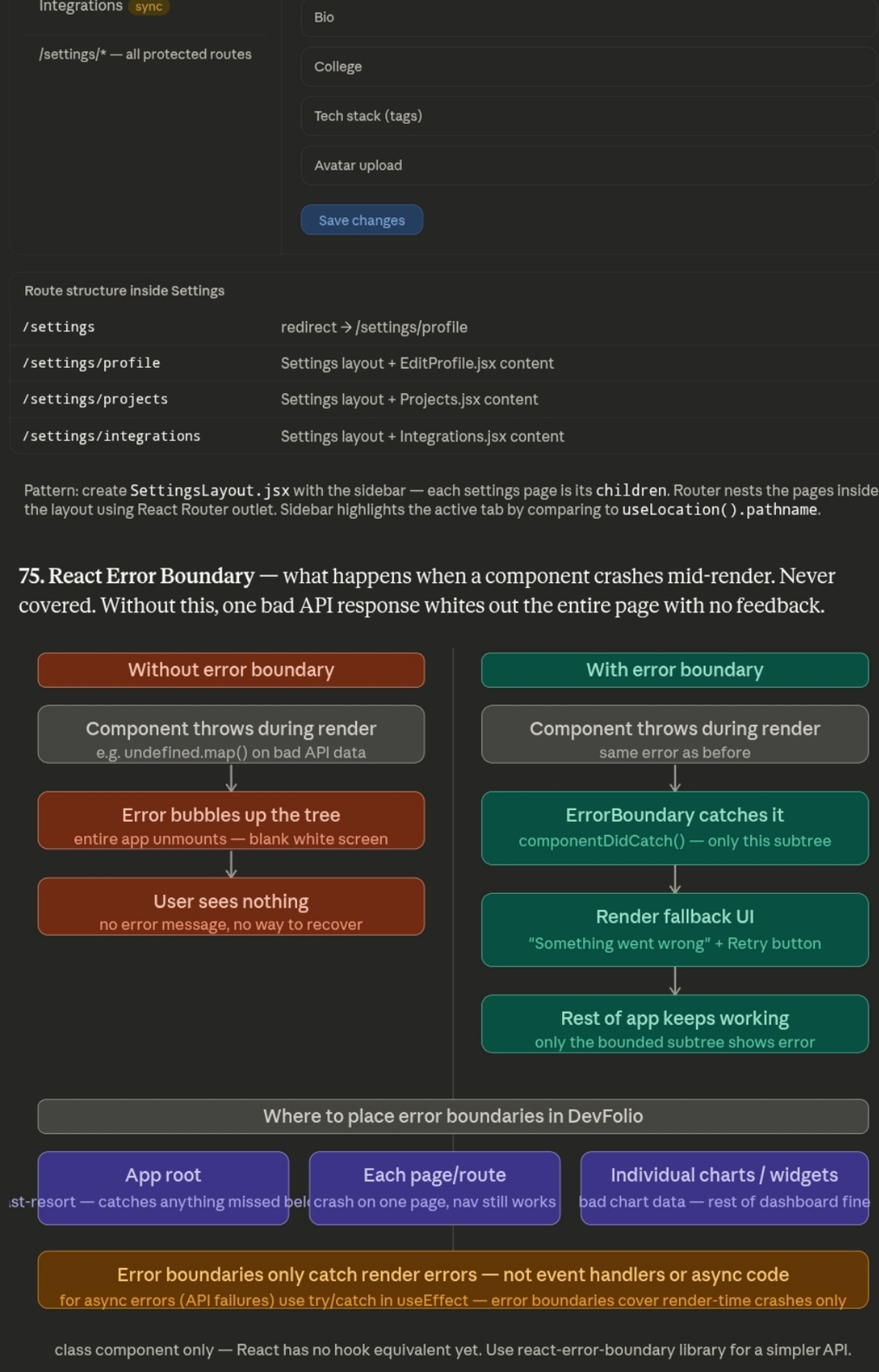
Live preview — click tabs to switch



Route structure inside Settings	
`/settings`	redirect → /settings/profile
`/settings/profile`	Settings layout + EditProfile.jsx content
`/settings/projects`	Settings layout + Projects.jsx content
`/settings/integrations`	Settings layout + Integrations.jsx content

Pattern: create `SettingsLayout.jsx` with the sidebar — each settings page is its `children`. Router nests the pages inside the layout using React Router outlet. Sidebar highlights the active tab by comparing to `useLocation().pathname`.

75. React Error Boundary — what happens when a component crashes mid-render. Never covered. Without this, one bad API response whites out the entire page with no feedback.



class component only — React has no hook equivalent yet. Use react-error-boundary library for a simpler API.

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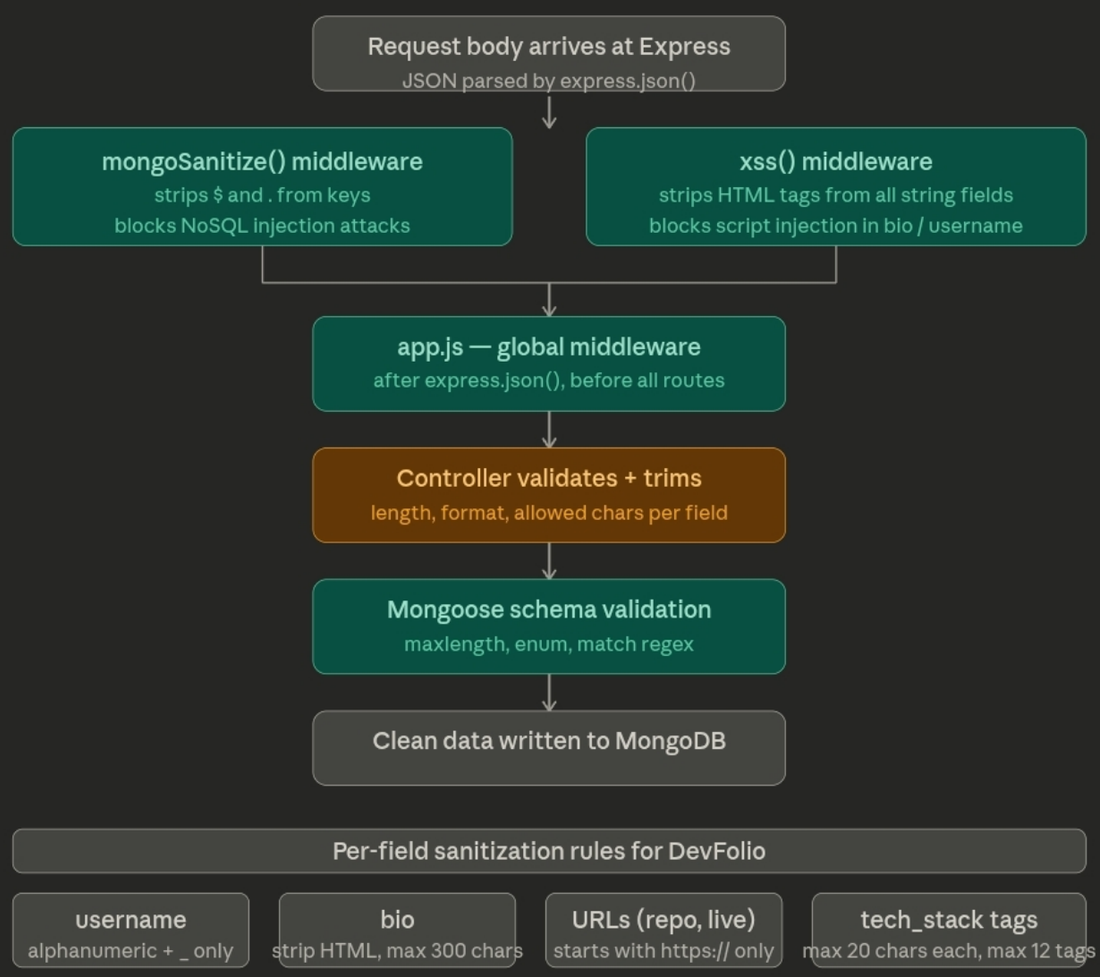
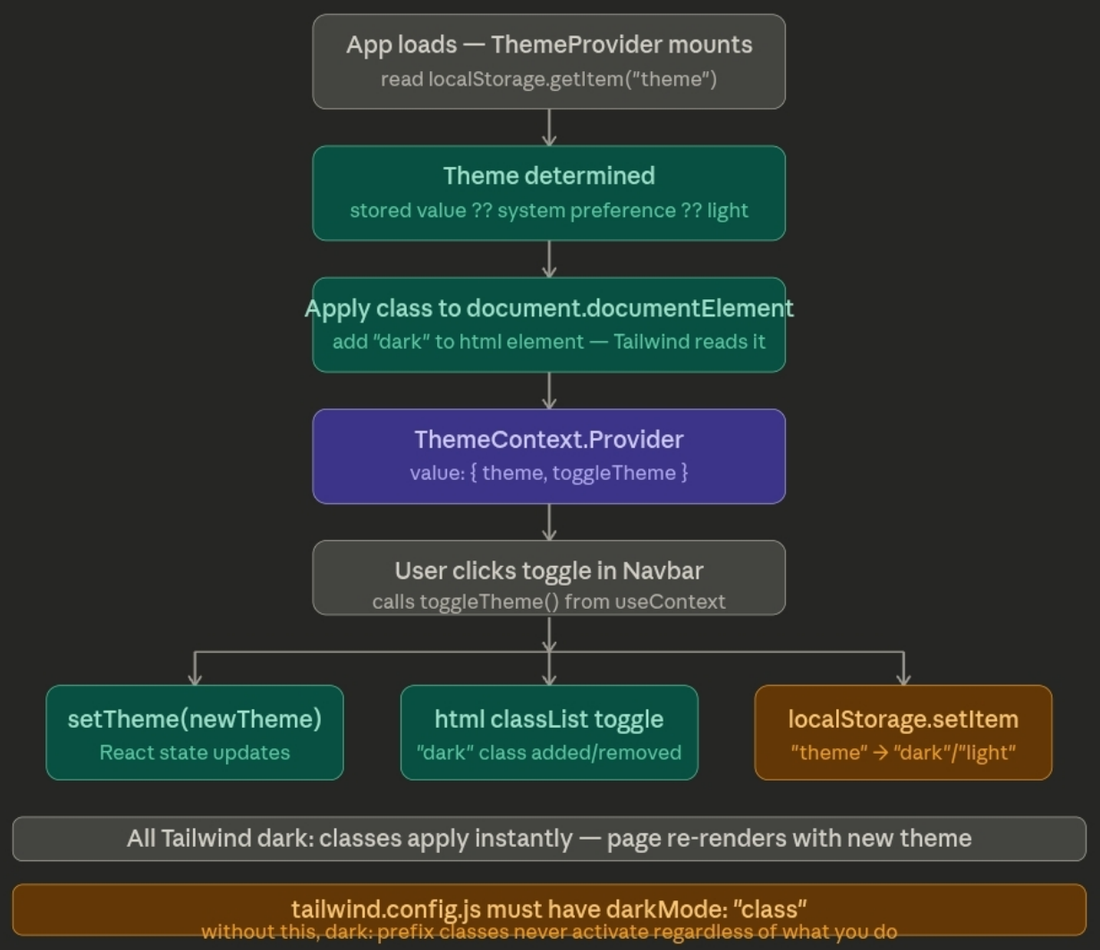
76. Input sanitization on the backend — how user-supplied text is cleaned before hitting MongoDB. Never covered across all 72 diagrams. Prevents XSS and injection attacks.



77. Dark mode toggle mechanism — `ThemeContext` was listed in diagram 42 but the actual toggle + localStorage sync + Tailwind class switching was never diagrammed once.



Here's the complete final set — every one confirmed absent from all 72 prior diagrams:

#	Diagram	The gap it fills
73	Forgot password / reset flow	Reset token, nodemailer, 2-phase flow — completely absent from all prior diagrams
74	Settings page sub-navigation	SettingsLayout.jsx + sidebar + outlet pattern — never shown, just the individual page flows
75	React Error Boundary	Component crash containment — without it, one bad API response whites out the whole app
76	Input sanitization on the backend	express-mongo-sanitize + xss-clean + per-field rules — security layer never diagrammed
77	Dark mode toggle mechanism	ThemeContext toggle + html class + localStorage + `tailwind.config.js` darkMode setting

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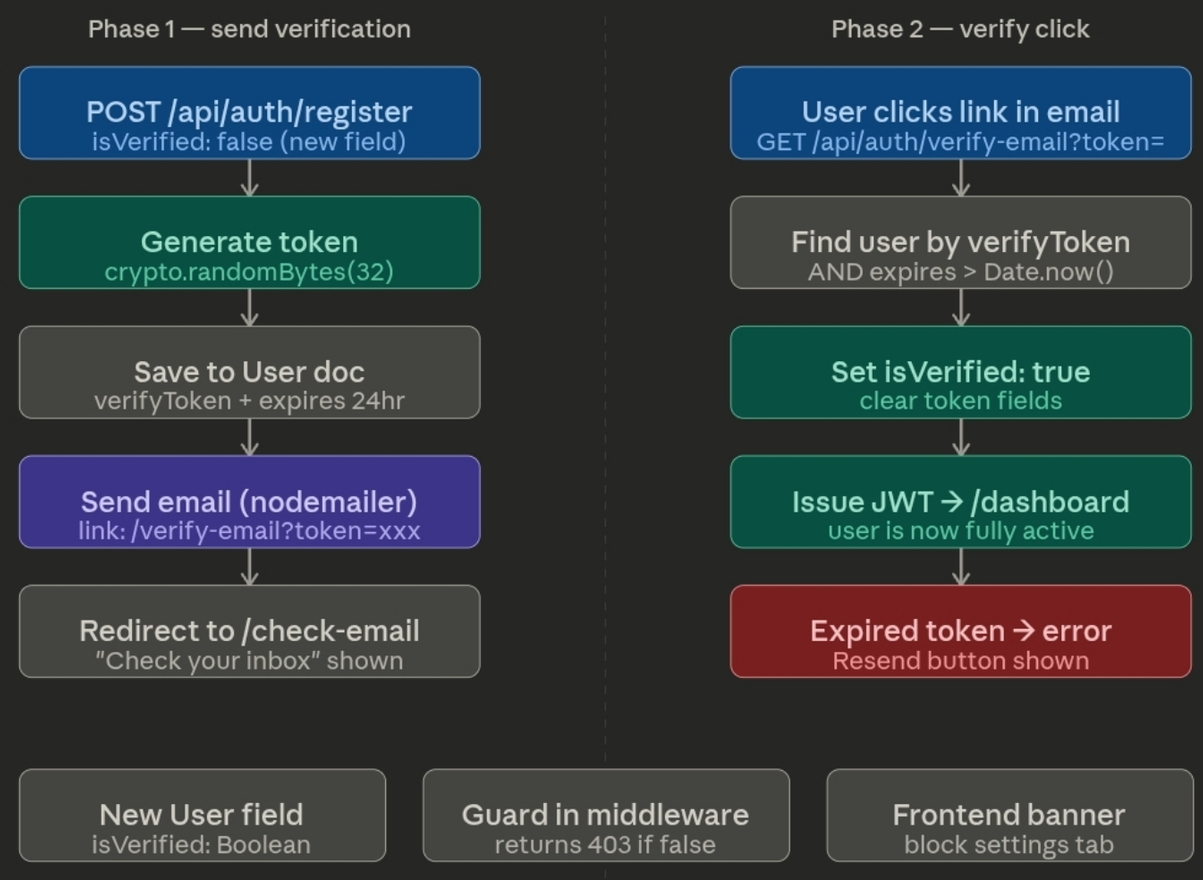
92. Landing page component map

93. In-app notification bell

86. Email verification flow `Auth`

Prevent fake accounts — user cannot access DevFolio until they verify their email. Two-phase flow.




```
nodemailer setup is identical to diagram 73
(forgot password) — same EMAIL_USER +
```

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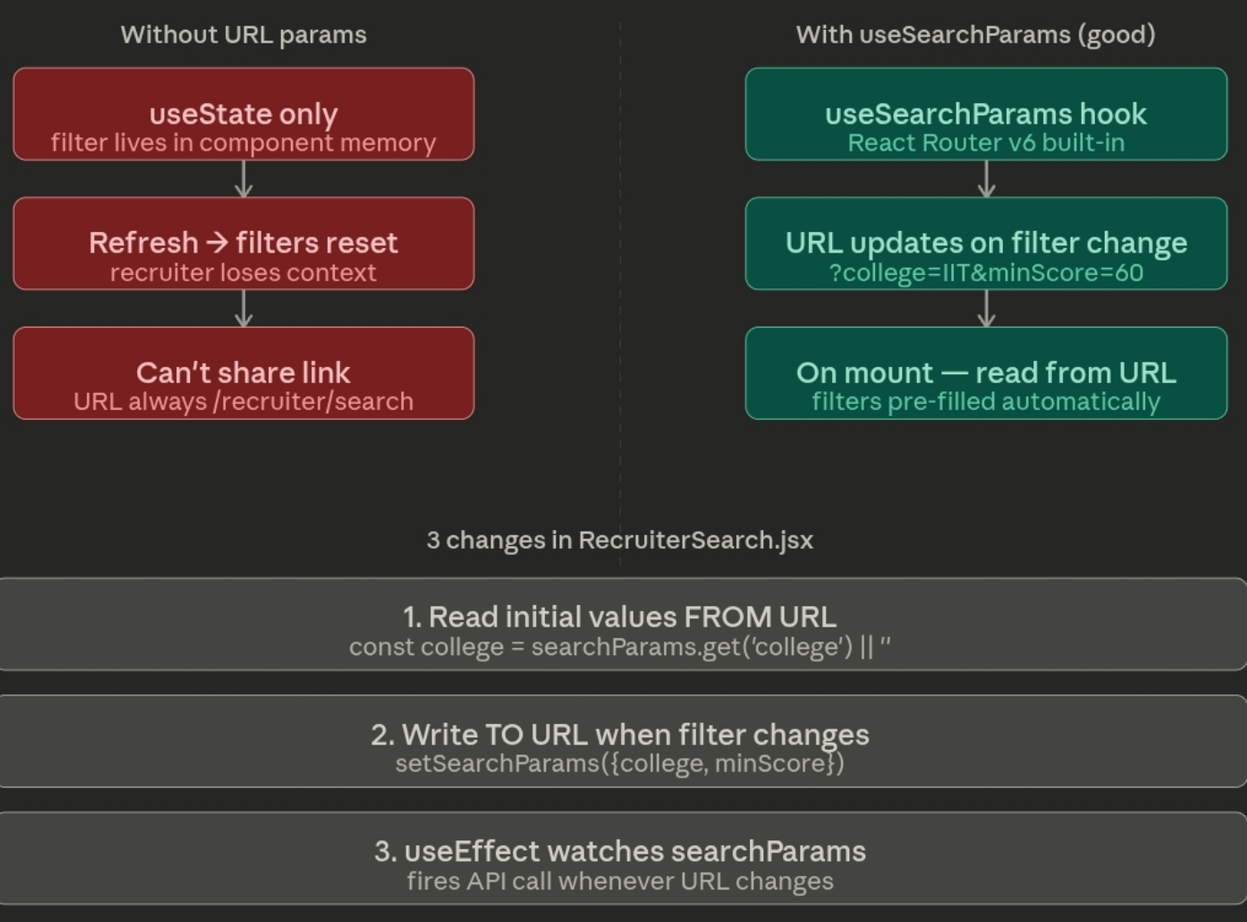
93. In-app notification bell

87. URL params filter persistence UX

Recruiter filters survive page refresh and can be shared as a link — /recruiter/search?college=IIT+Hyderabad&minScore=60



> Same pattern for leaderboard pagination (?page=2) and admin user list. No extra library — React Router v6 bui .

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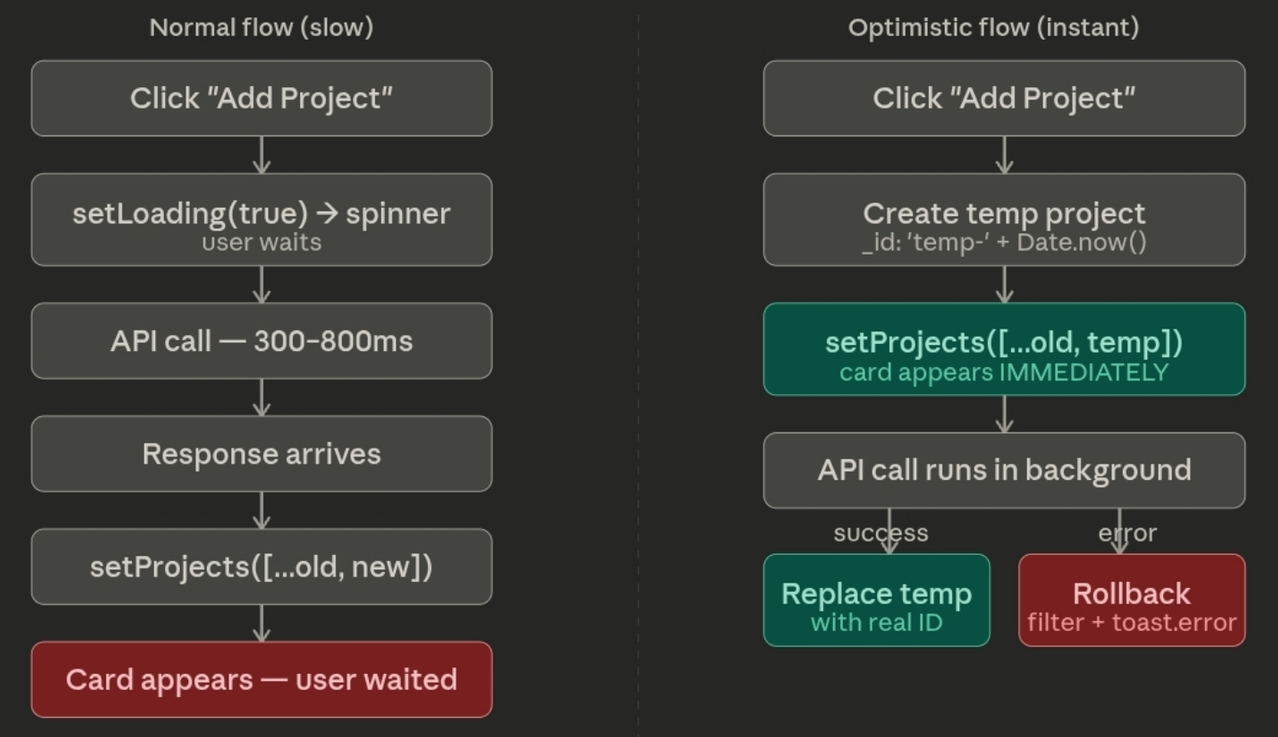
92. Landing page component map

93. In-app notification bell

88. Optimistic UI updates React

Show result immediately, rollback on error. Makes add/delete project feel instant — no spinner.



Skip optimistic for a[illegible]ction where the server computes unpredictable output — score

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93. In-app notification bell

89. useReducer for complex forms

React

When EditProfile has 6+ fields + loading + errors together — one clean state object instead of 6 separate useState.

useState (messy)

6 separate useStates

const [bio, setBio] = useState('')
const [college, setCollege] = useState('')
const [techStack, setTechStack] = useState([])
const [linkedin, setLinkedin] = useState('')
const [loading, setLoading] = useState(false)
const [errors, setErrors] = useState({})

each rerenders independently

useReducer (clean)

one useReducer

initialState = {'{bio, college,'}
techStack, linkedin,
loading, errors{'}'}

dispatch({'{ type: "SET_FIELD",'}
field: "bio", value: e.target.value {'}'})

one consistent dispatch pattern

Decision rule

Use useState when
1–3 fields, simple, unrelated

Use useReducer when
4+ fields + loading + errors together

> Apply to EditProfile.jsx — bio, college, techStack, linkedin, loading, errors = 6 state pieces = useReducer territory.

← Prev Diagram 89 of 93 Next →

Here are the 8 new diagrams (86–93)

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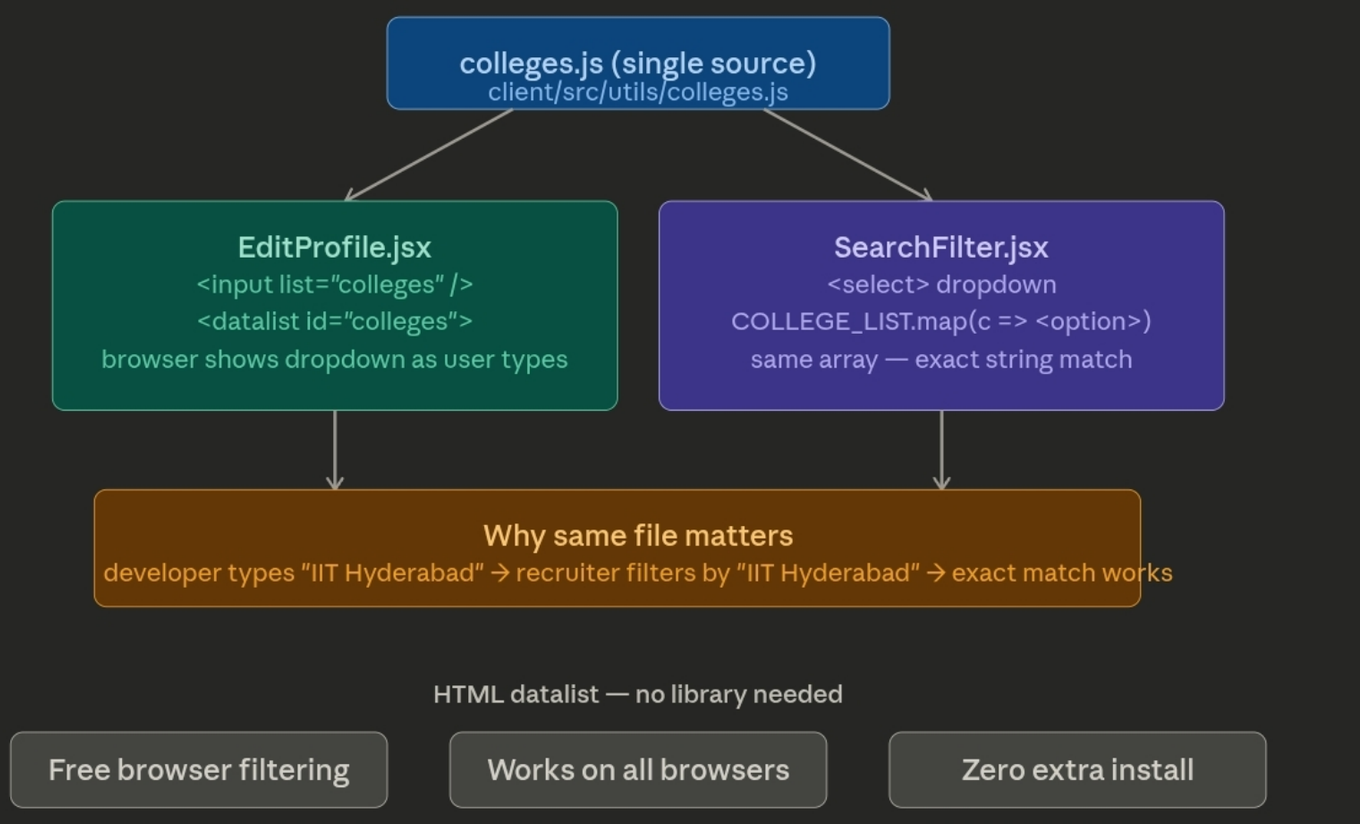
93. In-app notification bell

90. College autocomplete input UX

The college field needs autocomplete — recruiter search filters by exact string, so spelling must match between EditProfile and SearchFilter.



> Keep COLLEGE_LIST in colleges.js. Never hardcode college names twice — one file, two imports, guaranteed consistency.

← Prev Diagra of 93 Next →

Here are the 8 new diagrams (86–93)

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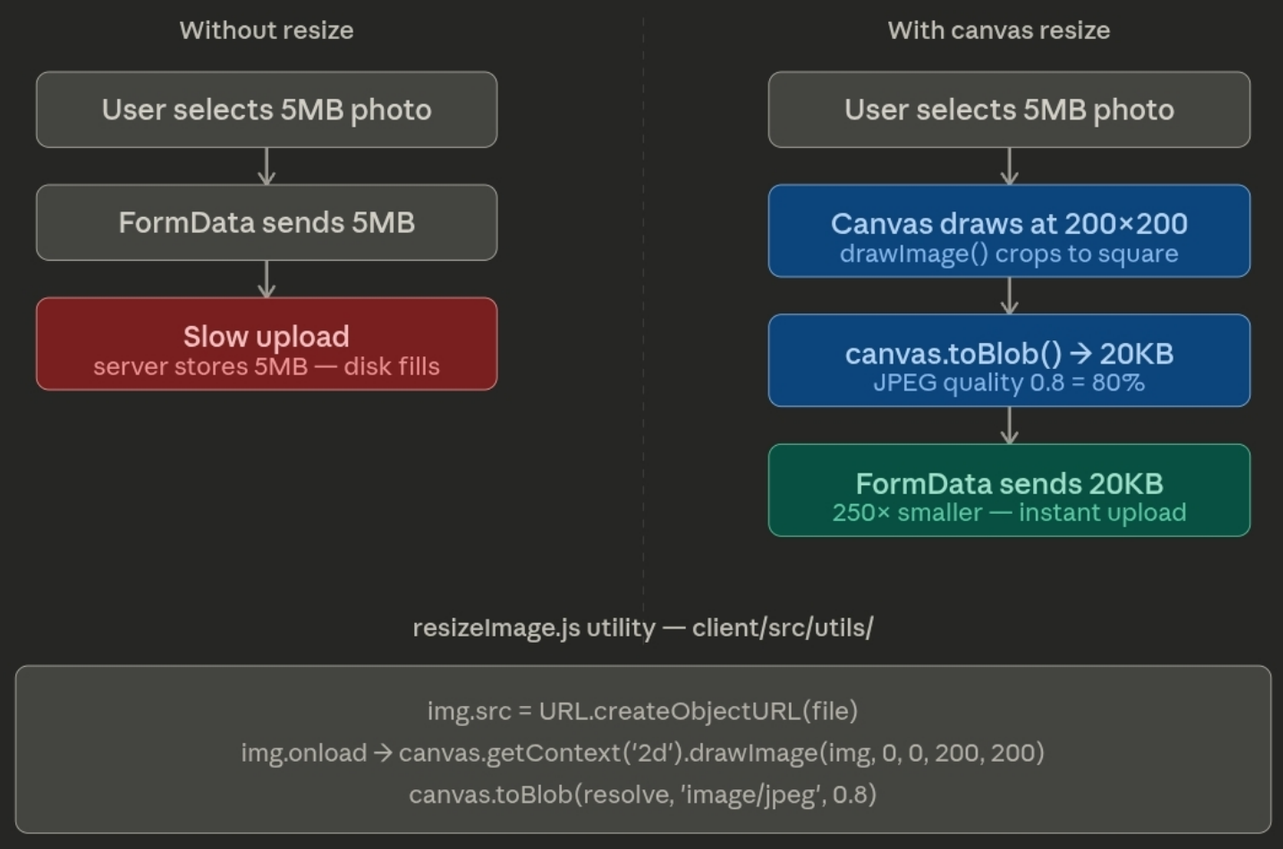
93. In-app notification bell

91. Client-side image resize before upload

Upload

User selects a 5MB phone photo. Canvas shrinks it to 200×200px in browser — sends 20KB to multer instead.



resizeImage.js utility — client/src/utils/

```
img.src = URL.createObjectURL(file)
img.onload → canvas.getContext('2d').drawImage(img, 0, 0, 200, 200)
canvas.toBlob(resolve, 'image/jpeg', 0.8)
```

> No library needed — browser Canvas API is built-in. Import resizeImage() in EditProfile.jsx: const resized = await resizeImage(file); then append to FormData.

← Prev Diagram 91 of 93 Next →

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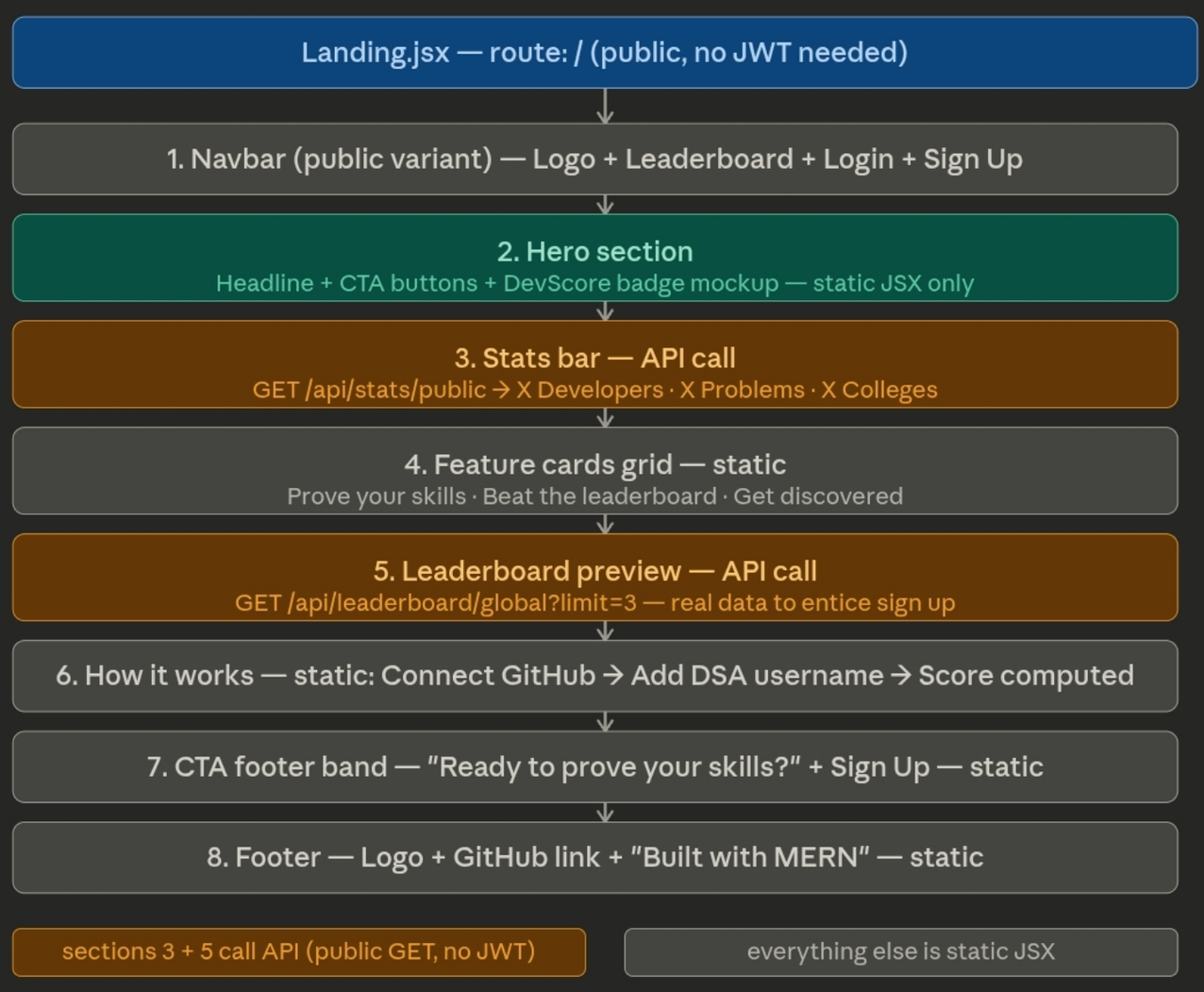
92. Landing page component map Frontend

All 8 sections of Landing.jsx — which call the API vs which are static JSX. Route: / (public, no auth).



> Keep Landing.jsx under 200 lines — extract HeroSection, FeatureCards, HowItWorks as sub-components. Only 2 API calls, both public GET routes.

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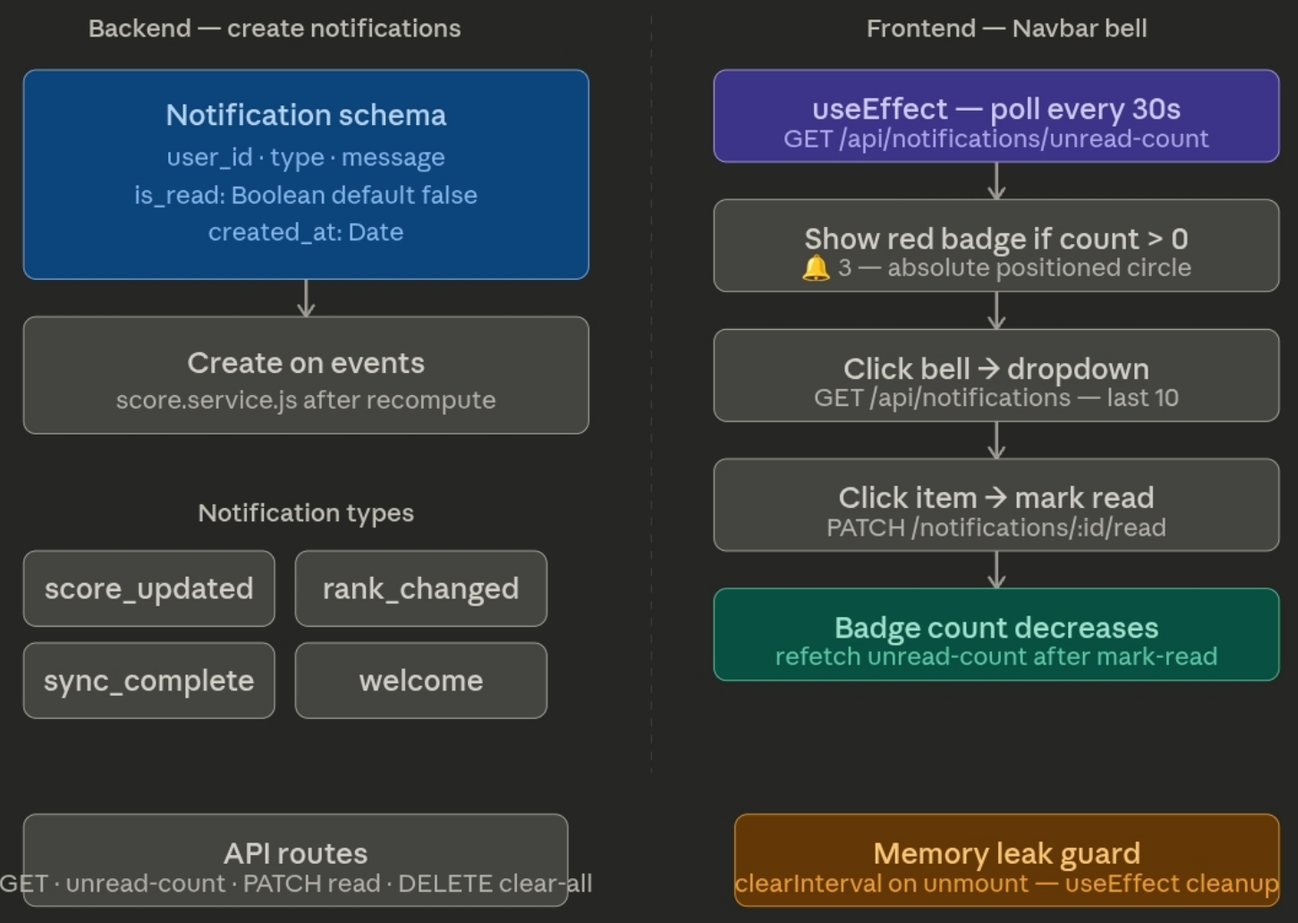
93. In-app notification bell UX

Badge on Navbar bell showing unread events — score updated, rank changed, sync complete. MVP: polling every 30s, no WebSocket needed.



> 30s polling is fine for MVP — no WebSocket complexity. Add clearInterval in useEffect return to avoid memory leak on unmount.

← Prev Diagra of 93 Next →

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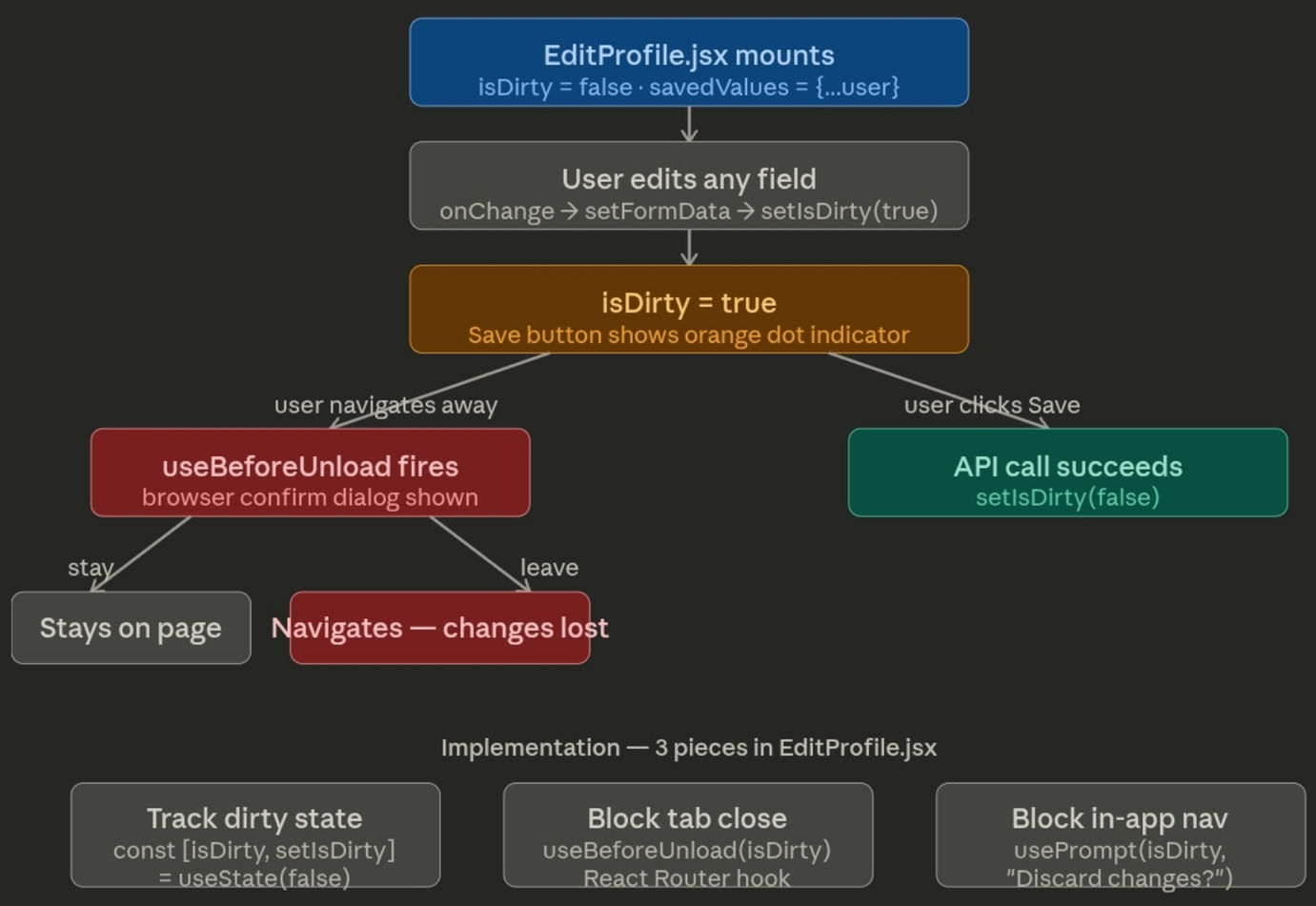
94 .

Form dirty state & unsaved changes warning

Pattern

User edits profile, clicks 'My Profile' link — warn before losing changes. The pattern every beginner forgets until users complain.



> useBeforeUnload and usePrompt are React Router v6 hooks. setIsDirty(false) after successful PATCH — same for canc utton (resets form to savedValues).

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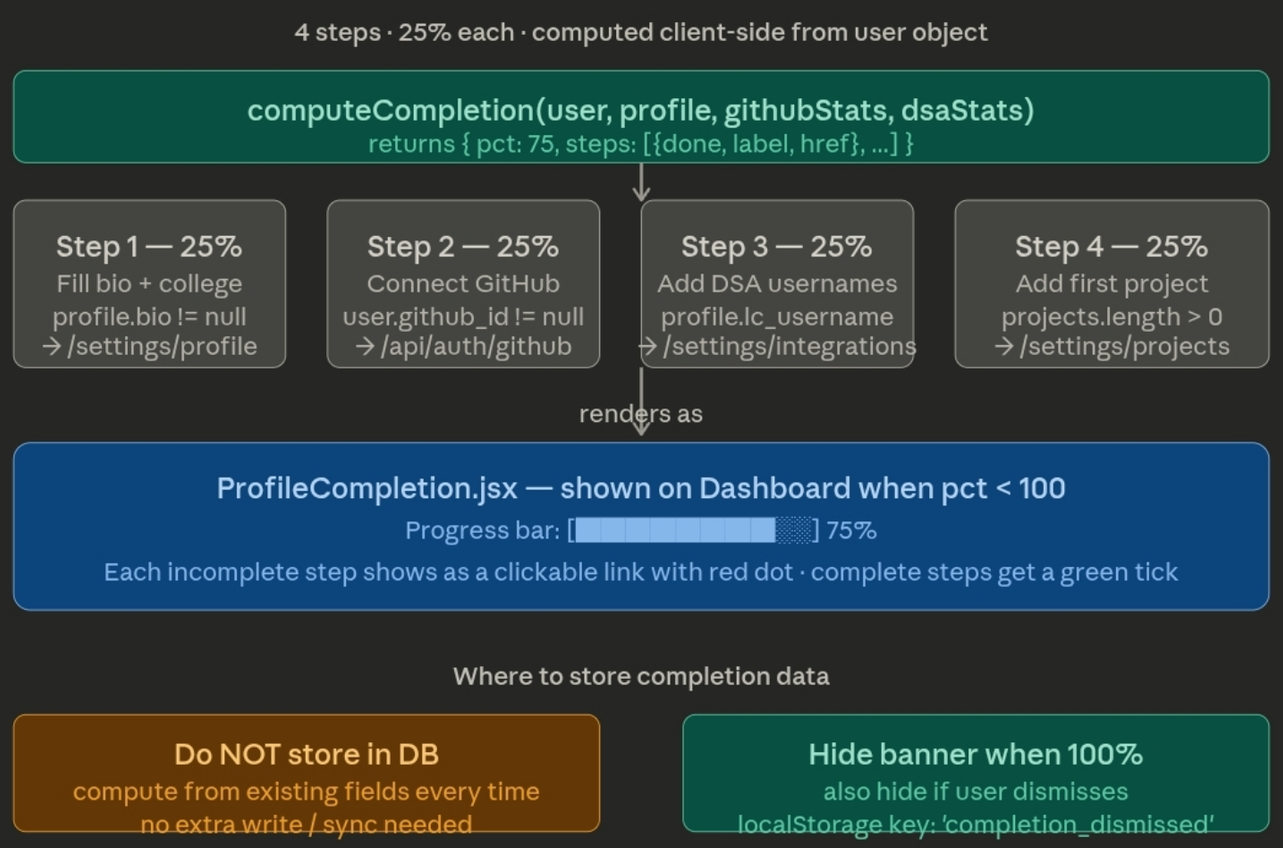
95. **Profile completion percentage tracker** UX

New users land on an empty dashboard. A progress bar shows what % of setup is done — guides them through the 4 setup steps without hand-holding in prose.



> computeCompletion() is a pure JS function in utils/ — no API call needed. Reads data already fetched for Dashboard. Call it inside the existing Promise.all() on mount.

← Prev Diagram 95 of 101 Next →

8 more diagrams (94–101), all verified

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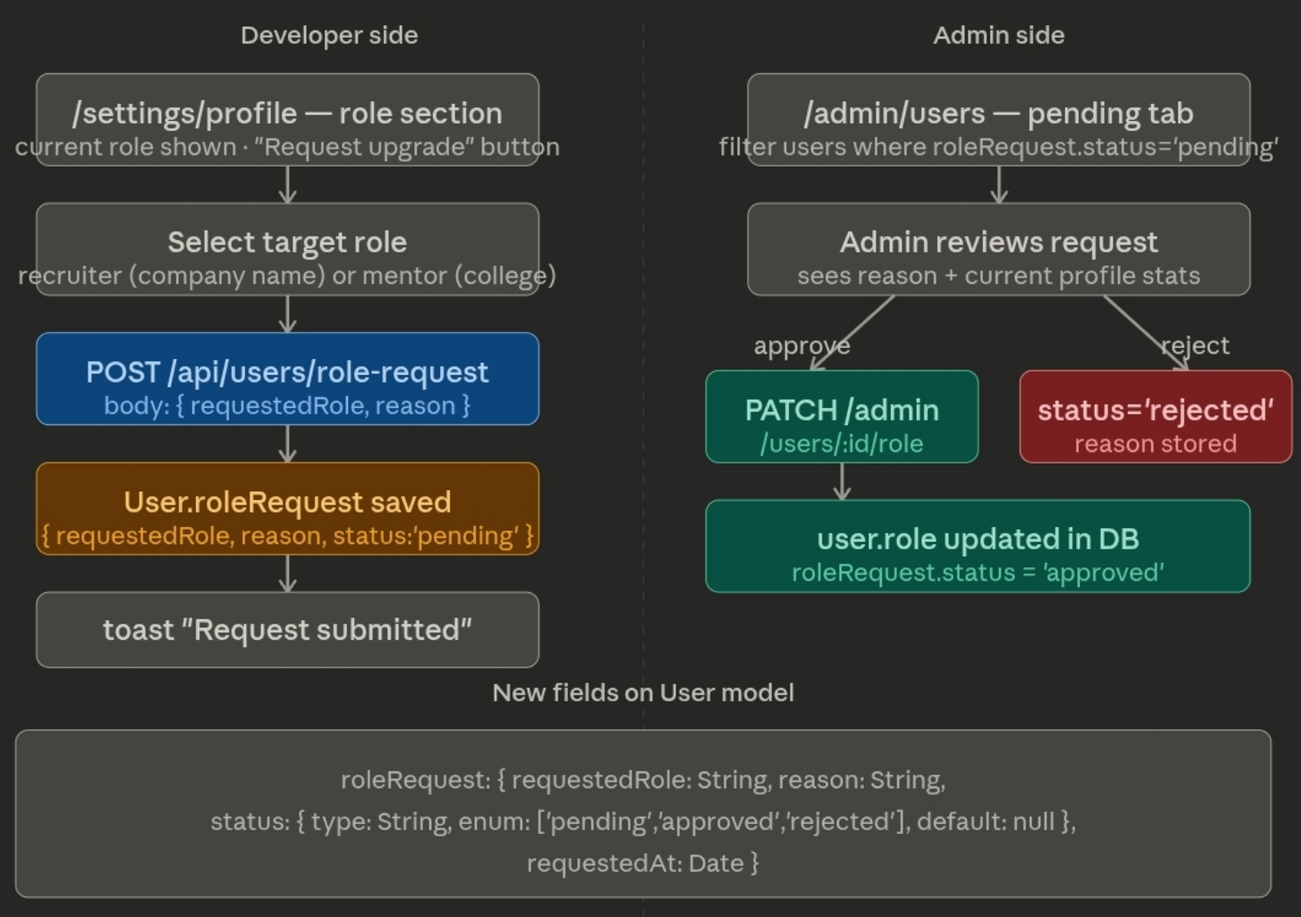
96. **Role upgrade request flow** `Flow`

A developer wants to become a recruiter or mentor. They can't self-assign roles — an admin must approve. Covers both the request and the admin approval path.



> MVP shortcut: skip the request flow — let admin directly PATCH role from UserManager.jsx. Add the request/approval flow post-launch when needed.

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8 more diagrams (94–101), all verified

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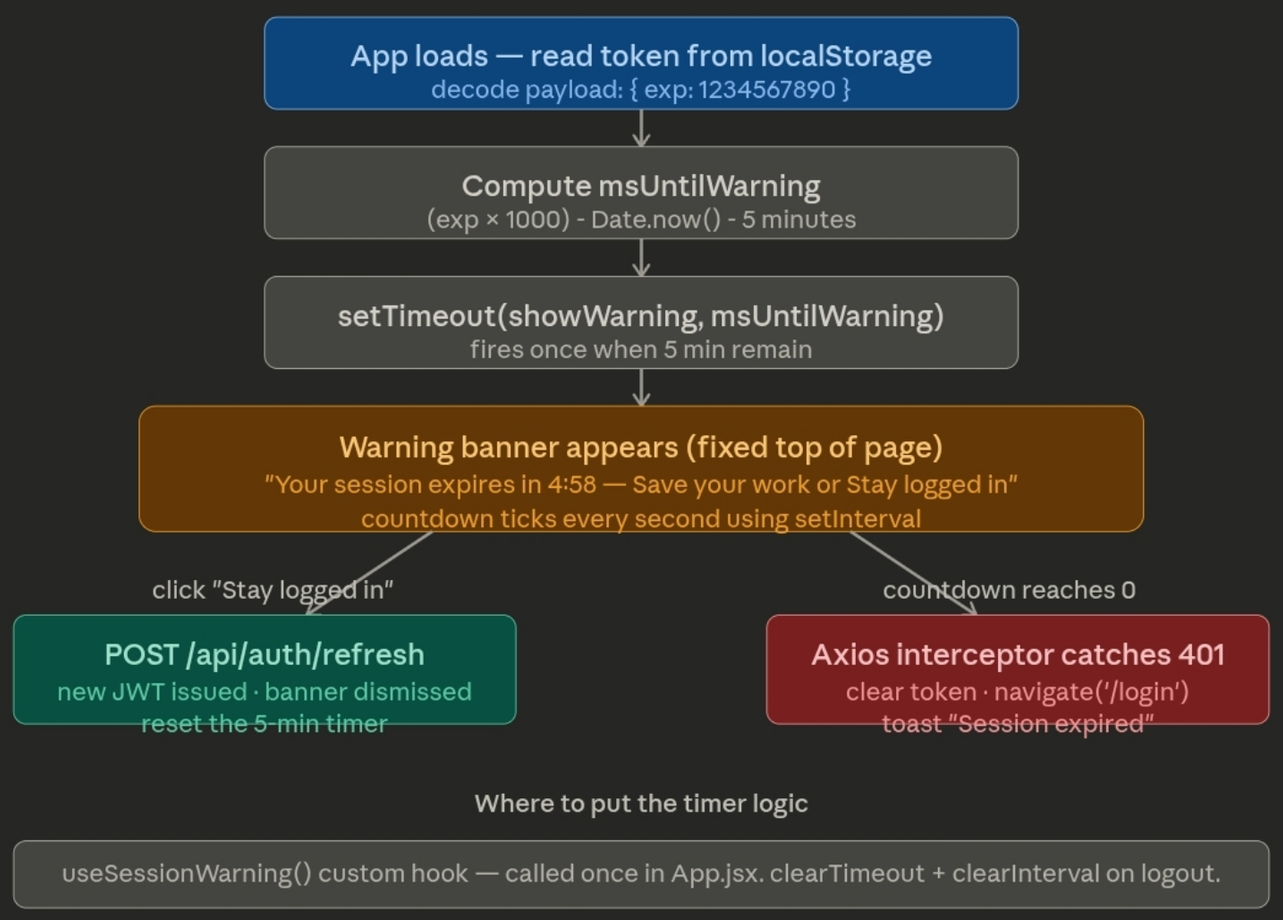
97. **Session timeout warning UI** UX

JWT expires after 7 days. Show a countdown banner 5 minutes before expiry so users can save work — not just a sudden 401 redirect.



> Decode JWT without a library: JSON.parse(atob(token.split('.')[1])) — reads the exp claim. Multiply by 1000 for milliseconds. Clear timers on logout to avoid ghost warnings.

← Prev Diagra of 101 Next →
8 more diagrams (94–101), all verified
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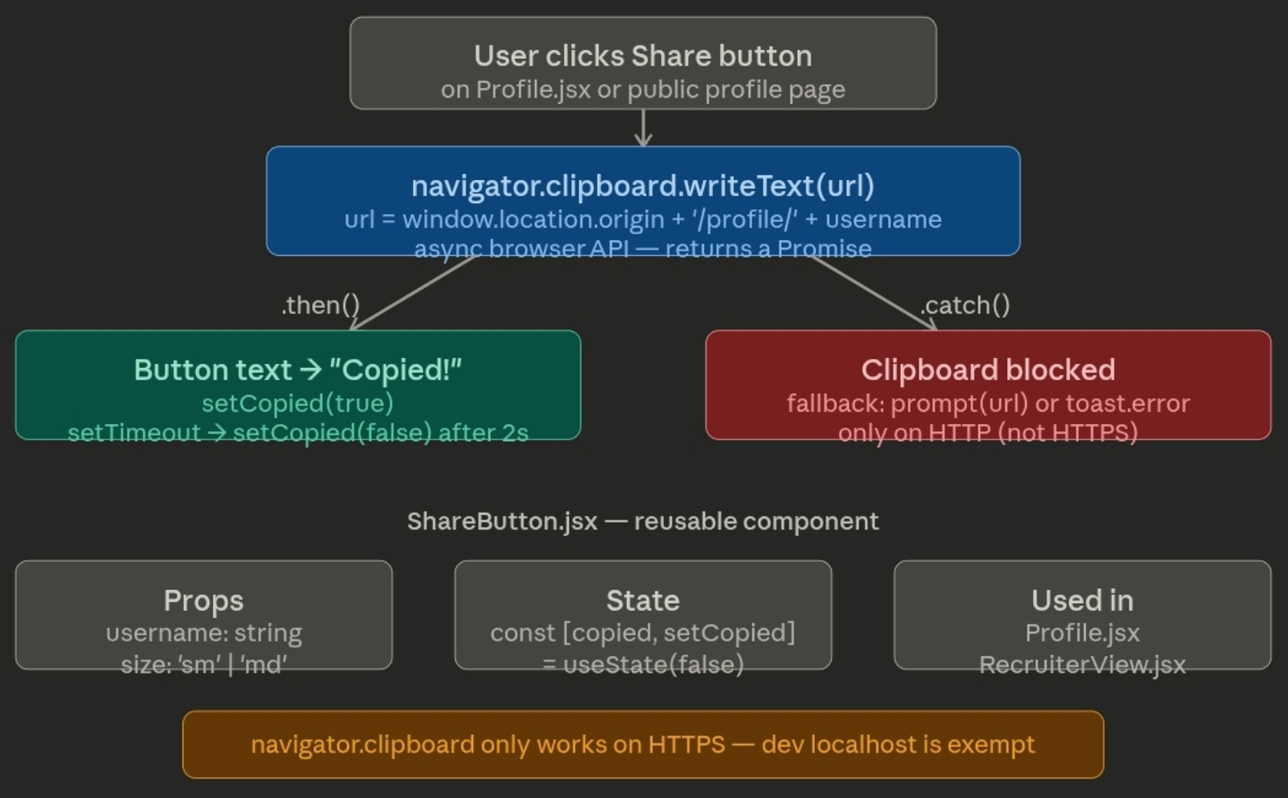
98. Copy-to-clipboard share profile link

Pattern

Every profile needs a share button — one click copies the public URL to clipboard. Covers the full UX feedback cycle.



> navigator.clipboard is a modern browser API — no library needed. The 2-second reset (setTimeout → setCopied(false)) is what makes the UX feel polished.

← Prev

Diagram 98 of 101

Next →

8 more diagrams (9[illegible]01), all verified against the full 93-diagram history with zero overlap:

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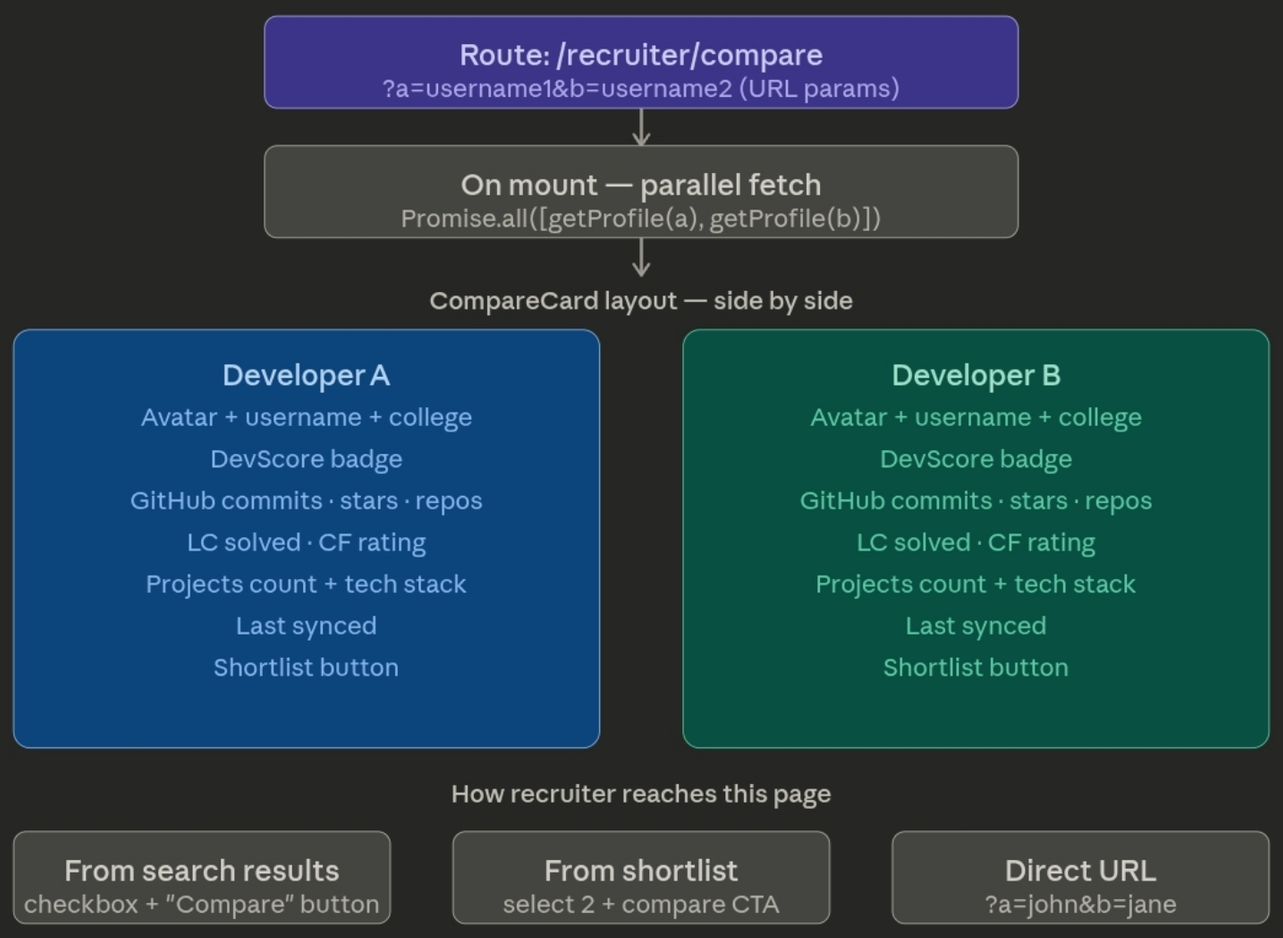
99. **Developer comparison view** Data

Recruiter wants to compare two developers side by side. No new API needed — reuse existing /api/profile/:username calls for both.



> No new backend endpoint needed — reuses GET /api/profile/:username twice in parallel. Highlight winning metric in green (higher DevScore, more commits etc.) using a simple Math.max comparison.

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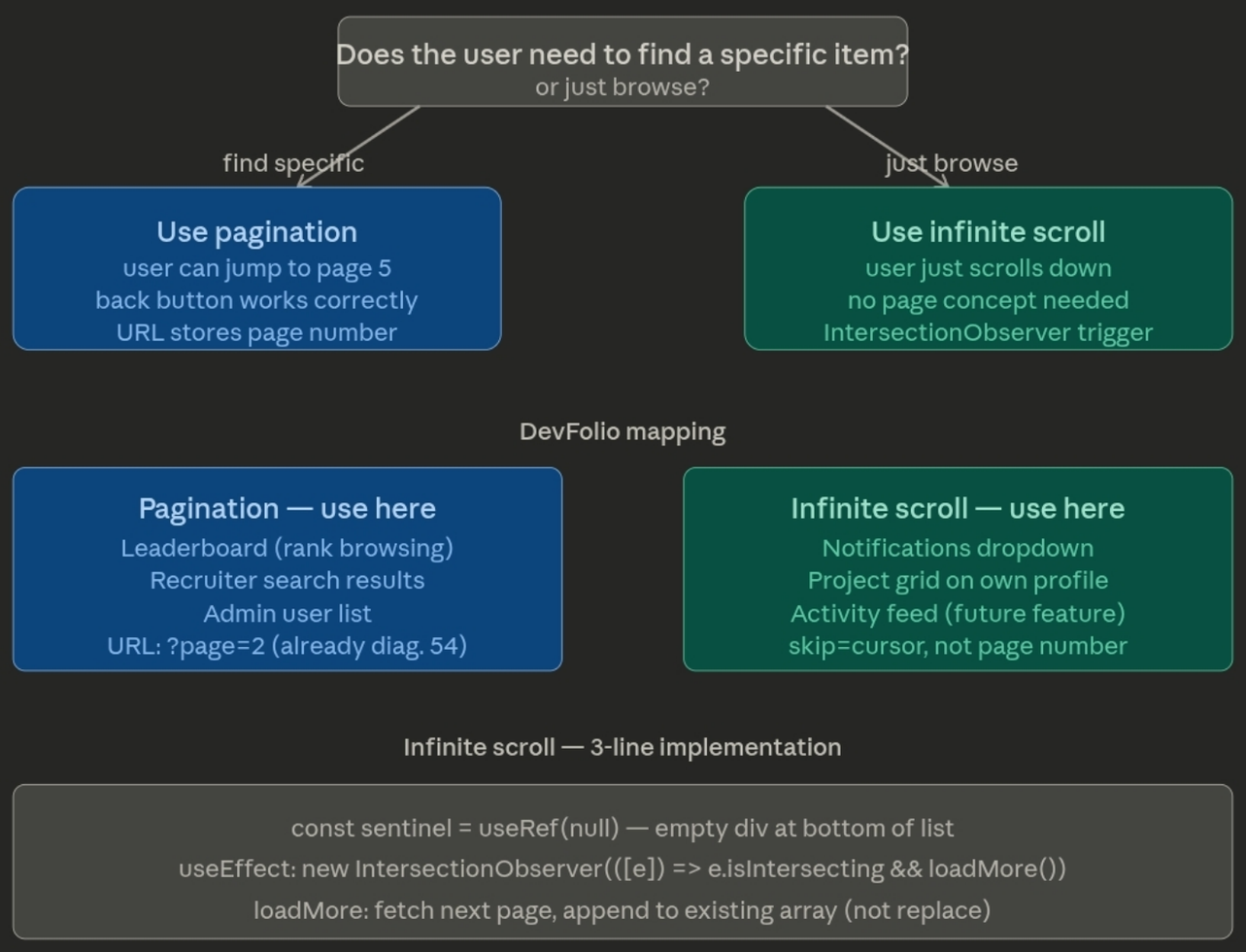
100. Infinite scroll vs pagination — decision map

React

Leaderboard and recruiter search need to load more data. Choosing the wrong pattern causes UX bugs. Here's the rule.



> Key rule: pagination when users need to navigate (back button, share a page link) — infinite scroll when it's a feed. Mixing them causes scroll-position bugs after navigation.

← Prev Diagram 100 of 101 Next →
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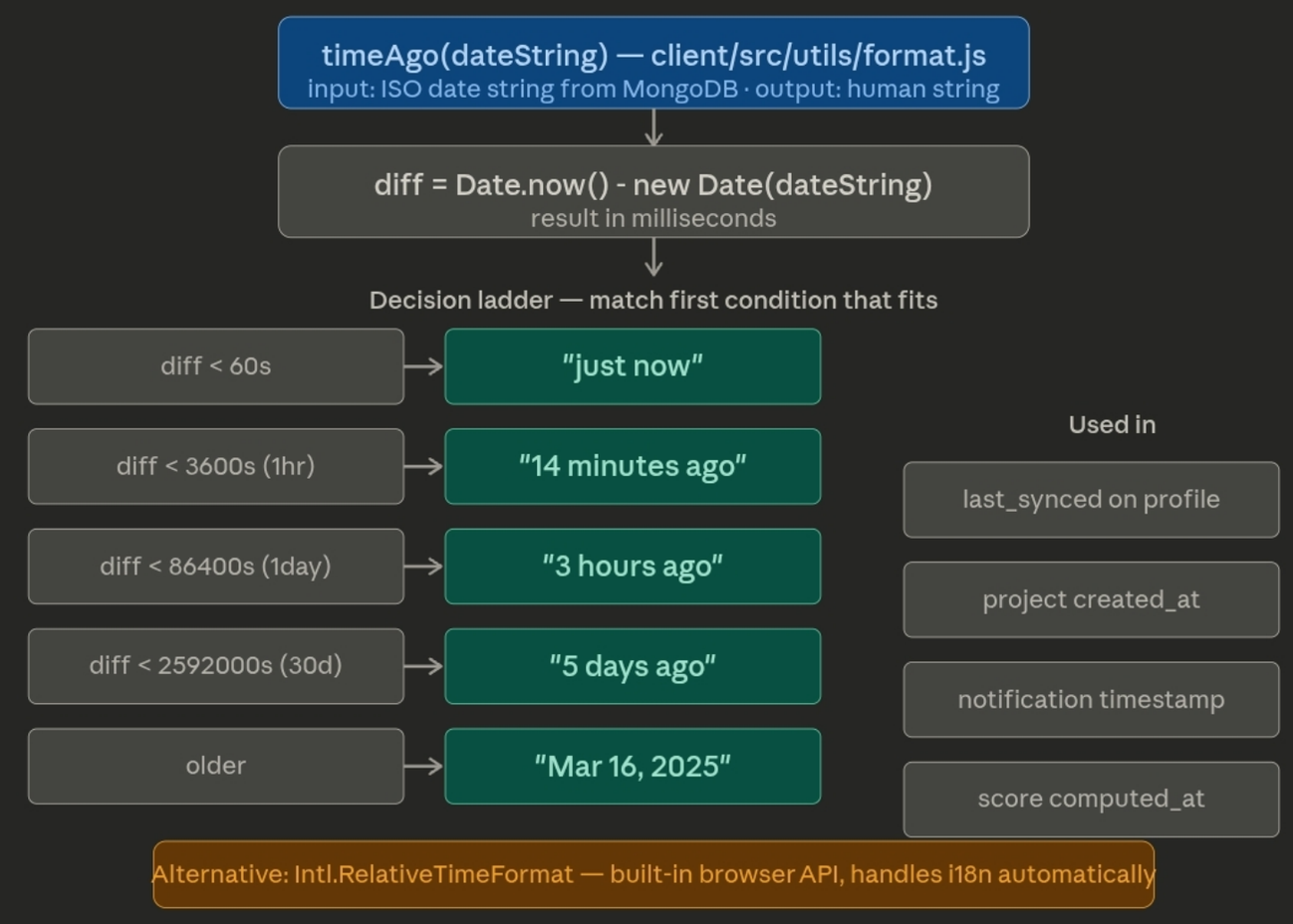
101. **Time-relative date formatting utility**

Pattern

Every date in DevFolio should show as '2 days ago' not '2025-03-16T14:22:00Z'. One utility function used everywhere.



> Put timeAgo() in format.js alongside any currency/number formatters. Import once per component. No library needed — the browser's Intl.RelativeTimeFormat does the same with auto locale support.

← Prev Diagram 101 of 101 Next →
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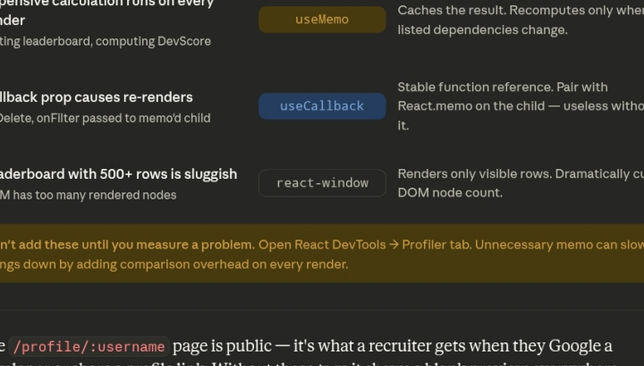
Architected visual pyramid diagram for testing strategy framework >

Five things genuinely absent from all 101 diagrams, each one capable of blocking you mid-build:



Tap any step for the exact command. The commit guide is for every single push — recruiters and mentors who look at your GitHub will see these prefixes, so it's worth the habit from day one.

React performance optimization is the most over-applied and under-applied concept in beginners' code. The rule is simple: profile first, then add exactly one thing.

Symptom you see	Solution	How it helps
Child re-renders when parent does props haven't actually changed	`React.memo`	Skips re-render when props are shallow-equal. Wrap the child component's export.
Expensive calculation runs on every render sorting leaderboard, computing DevScore	`useMemo`	Caches the result. Recomputes only when listed dependencies change.
Callback prop causes re-renders onDelete, onFilter passed to memo'd child	`useCallback`	Stable function reference. Pair with React.memo on the child — useless without it.
Leaderboard with 500+ rows is sluggish DOM has too many rendered nodes	`react-window`	Renders only visible rows. Dramatically cuts DOM node count.

Don't add these until you measure a problem. Open React DevTools → Profiler tab. Unnecessary memo can slow things down by adding comparison overhead on every render.

The `/profile/:username` page is public — it's what a recruiter gets when they Google a developer or share a profile link. Without these tags it shows a blank preview everywhere.

Basic SEO — always include on /profile/:username

`<title>`	`john_doe – DevFolio · DevScore 74`
`meta description`	`Full-stack dev at IIT Hyderabad · 312 commits · 122 LC problems`
`link canonical`	`https://devfolio.app/profile/john_doe`

Open Graph — LinkedIn + Facebook + WhatsApp share previews

`og:title`	`john_doe · DevScore 74 · Full-stack dev`
`og:description`	`312 GitHub commits · 122 LeetCode · Gold tier`
`og:image`	`/api/og/john_doe` (generated preview card image)
`og:url`	`https://devfolio.app/profile/john_doe`
`og:type`	`profile`

Twitter / X card — add alongside og: tags

`twitter:card`	`summary_large_image`
`twitter:image`	`same URL as og:image`

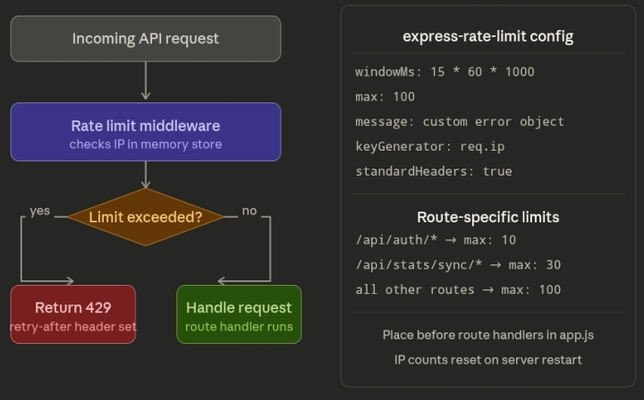
Where to set them

Install `react-helmet-async`. Wrap Profile.jsx return with <Helmet>. Set tags dynamically from the profile API response. Only Profile.jsx and Leaderboard.jsx need this.

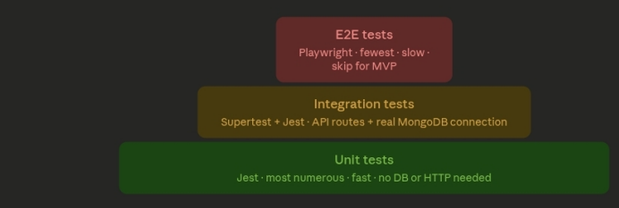
og:image quick approach

For MVP: use the developer's GitHub avatar URL as og:image. It's already public and 1:1 square. Generate a proper card image with `@vercel/og` post-launch.

Rate limiting is one configuration block, but beginners don't know where it sits in the request lifecycle or which routes need different limits.



Testing is the most paralysing topic for solo beginners. The pyramid tells you what proportion to build; the list tells you exactly which files deserve your time.



Test these for DevFolio MVP

- DevScore calculation formula
- timeAgo() formatting utility
- computeCompletion() function
- POST /api/auth/register route
- POST /api/auth/login route
- GET /api/profile/:username route
- Rate limit middleware — 429 on exceed

Skip entirely for MVP

- Visual appearance — no snapshot tests
- Third-party APIs — mock them instead
- Mongoose model definitions
- React component rendering
- Express boilerplate setup code
- Auth middleware — test it indirectly
- Full E2E login flow

The DevFolio rule: Test the math and the API contracts. Skip the plumbing. Your DevScore formula and auth routes are what breaks silently — everything else fails loud with a clear error message.

That's 5 new blueprints covering what the 101 diagrams left out — git hygiene, hook performance decisions, public profile discoverability, API protection, and a clear boundary around what's worth testing. Tap any node in the SVG diagrams to go deeper on a specific piece.

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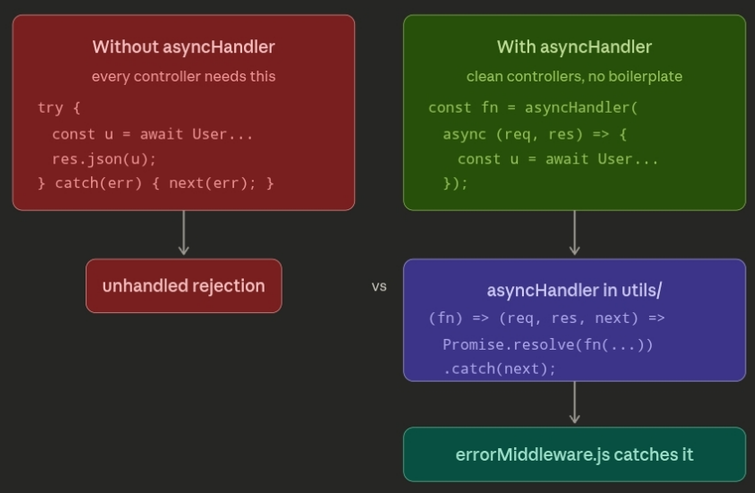
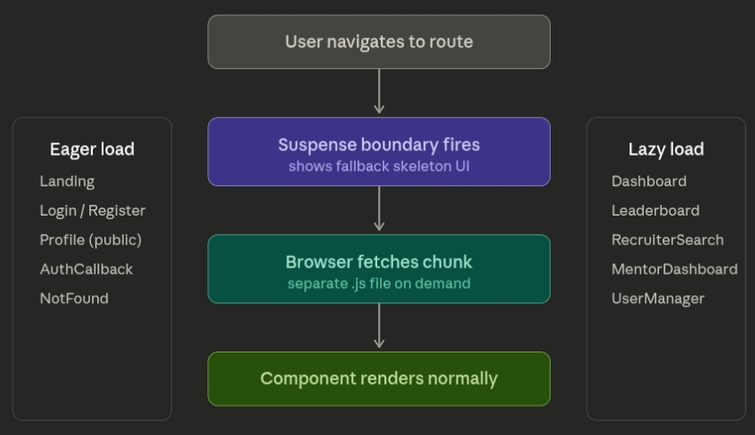
Curated ten novel diagram topics spanning async patterns, security, and data management

Here are 10 diagrams covering aspects genuinely absent from all prior 101 diagrams plus the 5 in the last session.

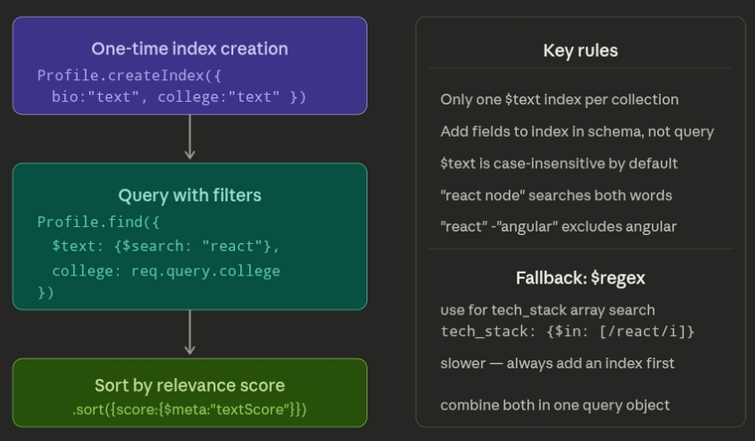
1. The `asyncHandler` wrapper — every controller in diagram 12 uses `try/catch`, but the wrapper that eliminates that boilerplate was never shown.



2. React.lazy + Suspense — code splitting by route so the initial bundle doesn't include every page at once. Leaderboard, RecruiterSearch, and MentorDashboard are the best candidates since they're large and role-gated.



3. MongoDB full-text search — recruiter search uses `$text` + `$search` on the Profile collection. The index must be created once; after that every query is a single operator with no regex.

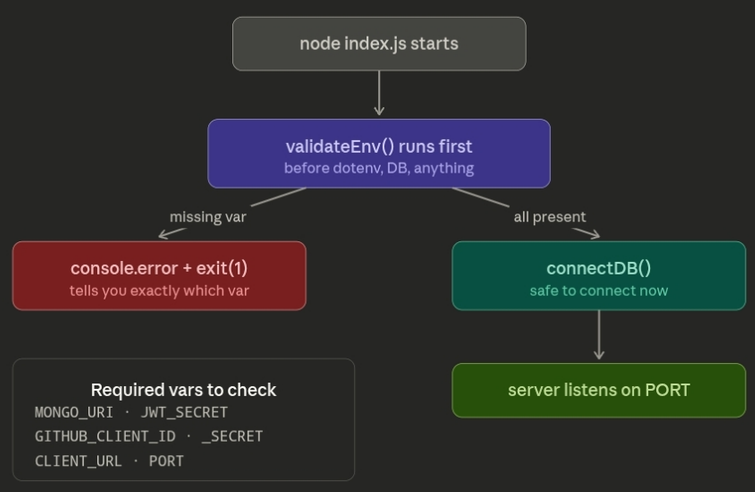


4. Helmet.js security headers — diagram 65 showed CORS, but helmet sets a completely different set of headers and was never mapped.

Header set by helmet	What it prevents	DevFolio note
`X-Content-Type-Options`	Browser sniffing the MIME type and executing a JSON response as a script	always on
`X-Frame-Options`	Clickjacking — your profile page loaded inside an attacker's iframe	always on
`Referrer-Policy`	Leaking your full URL (including usernames) in the Referer header to third-party sites	always on
`Content-Security-Policy`	XSS — inline scripts injected via bio or username fields executing in the browser	configure carefully — breaks recharts if too strict
`X-DNS-Prefetch-Control`	Browser leaking which external domains your app connects to via DNS prefetching	always on
`Strict-Transport-Security`	Downgrade attacks — forcing a user from HTTPS to HTTP to intercept tokens	only active on HTTPS

CSP and Recharts: the default helmet CSP blocks inline styles that Recharts generates. Either disable CSP for MVP with `helmet({ contentSecurityPolicy: false })` or configure `script-src 'unsafe-inline'` just for chart routes. Do not ship with CSP fully disabled in production.

5. Server startup env validation — diagram 14 listed all env vars, but never showed the guard that runs before `connectDB()` to fail fast if a required var is missing.



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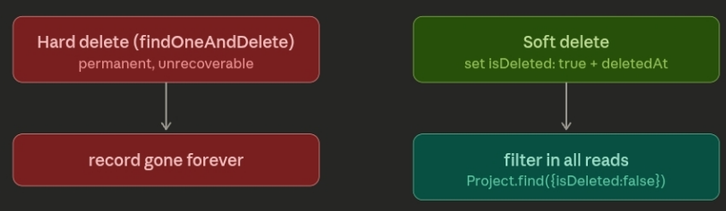
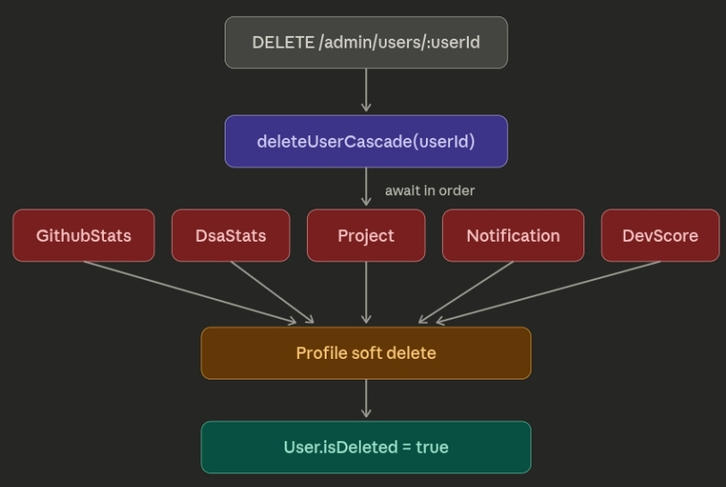
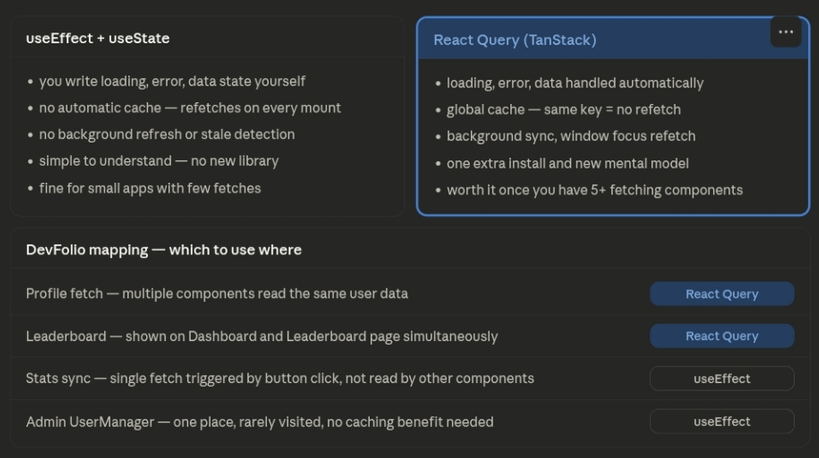
6. Soft delete pattern — diagram 36 showed project CRUD with `findOneAndDelete`, but a recruiter reviewing someone whose projects were deleted still sees the gap. Soft delete is the safer design.



7. Account deletion cascade — the RBAC matrix showed admin can delete accounts, but the cascade across all 7 collections was never diagrammed. Getting this wrong leaves orphaned DevScore docs that still appear on the leaderboard.

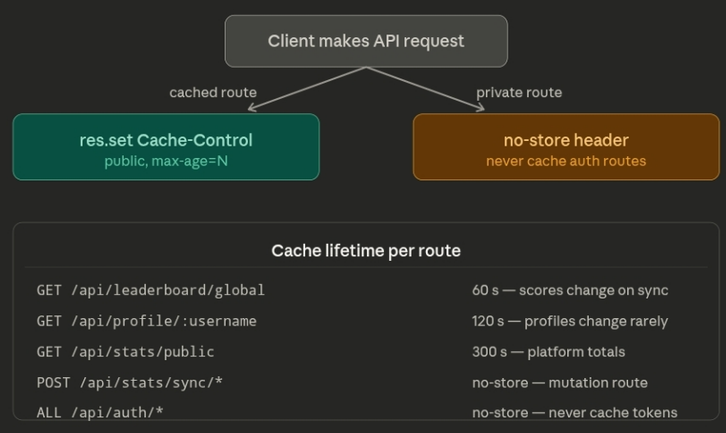


8. React Query vs plain `useEffect` — diagram 13 showed React data flow with raw `useEffect`, but beginners hit caching, stale data, and loading state bugs constantly. The decision map tells you exactly when a library pays off.

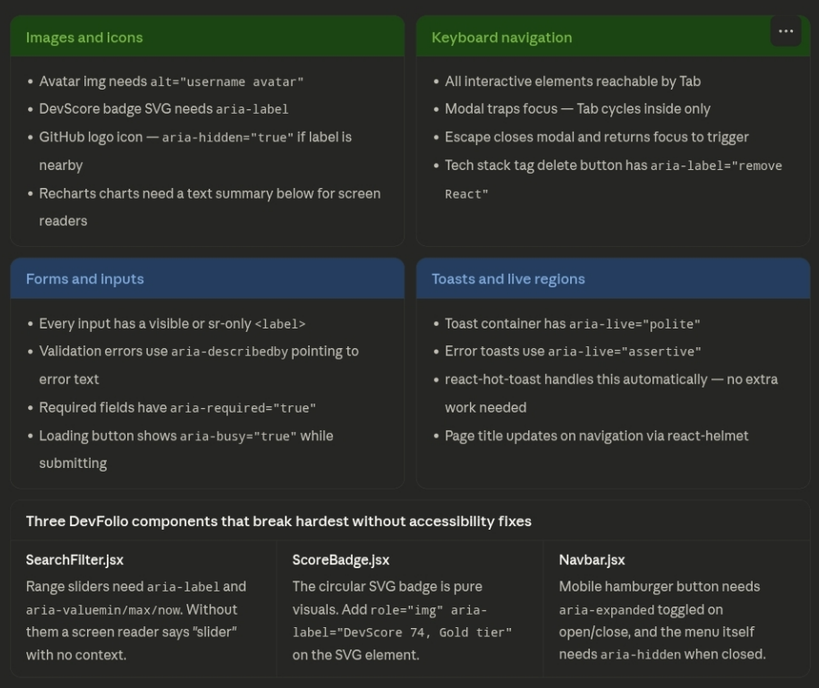


Decision rule: if two or more components need the same remote data, React Query eliminates the duplicate fetch and the loading-state duplication. Start with useEffect and migrate when you feel the pain — don't add it upfront speculatively.

9. API response caching strategy — the leaderboard and public profile pages are read-only and fetched frequently. Setting the right `Cache-Control` header on those routes is one line that removes huge server load. Never shown before.



10. Accessibility checklist — DevFolio's public profile is what recruiters open. Broken keyboard nav or missing aria labels will hurt you in an actual job hunt, but accessibility was never touched across all 101 diagrams.



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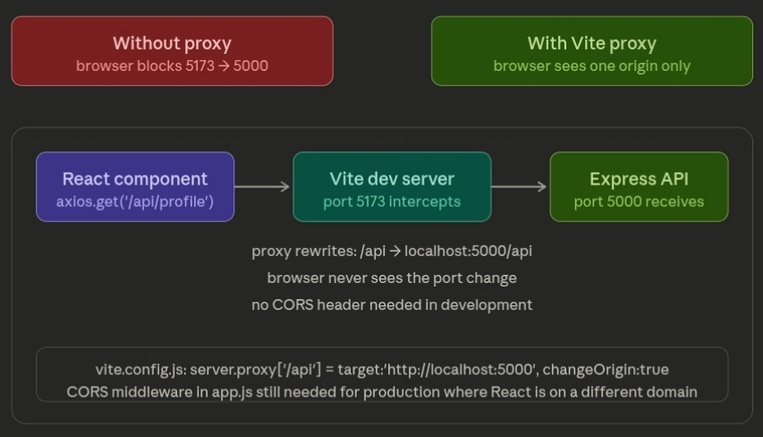
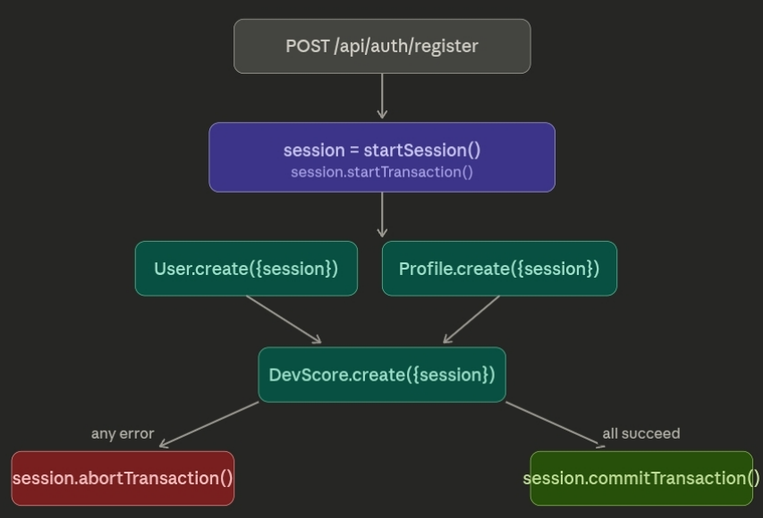
After auditing all 141 prior diagrams, here are 10 more covering genuinely uncharted territory.

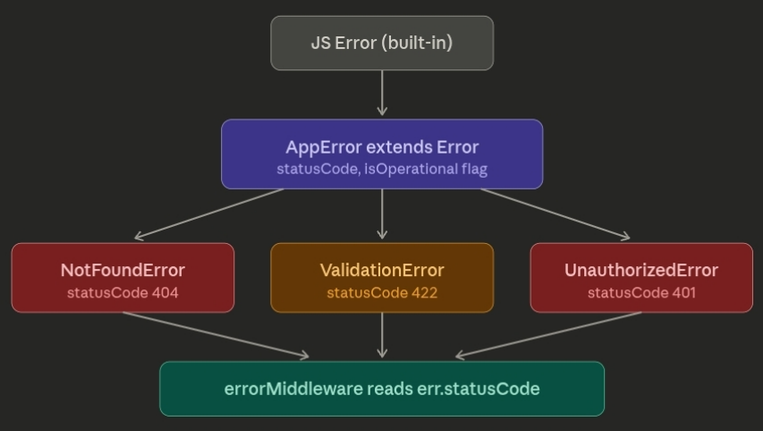
1. Vite dev proxy — how the React frontend at port 5173 talks to Express at port 5000 during local development without triggering CORS. Never shown across any of the 141 diagrams.



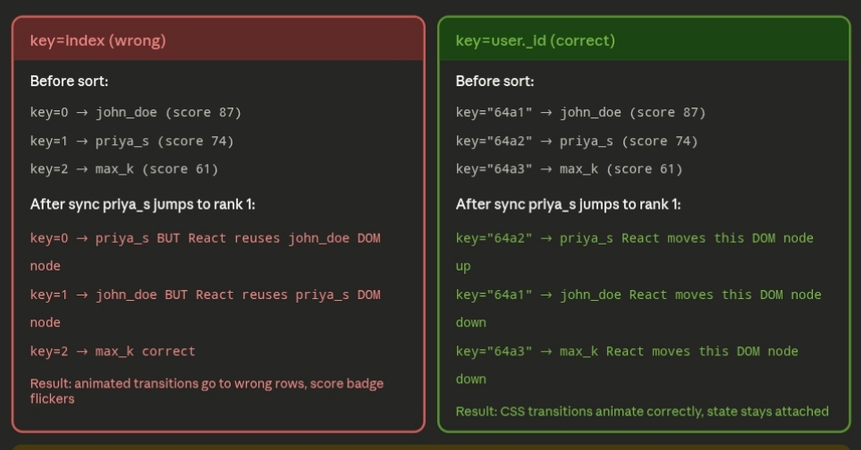
2. MongoDB transaction for the register flow — creating User + Profile + DevScore as three separate `save()` calls means a crash between any two leaves orphan documents. A session transaction rolls all three back atomically.



3. Custom error class hierarchy — diagram 18 showed error flow, but the `AppError` class that carries a `statusCode` property — letting the error middleware map every throw directly to the right HTTP response — was never drawn.

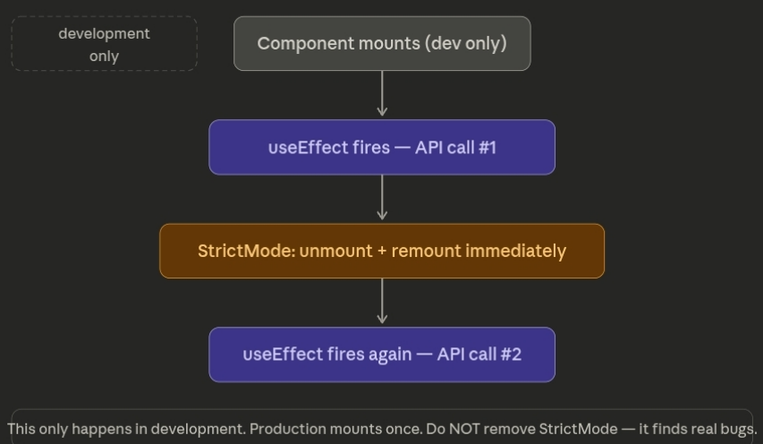


4. React `key` prop and list reconciliation — the leaderboard re-sorts on every sync. Using array index as `key` causes React to reuse DOM nodes incorrectly, producing visible flicker and wrong state on score updates. Never shown.



DevFolio rule: always use `key={item._id}` for leaderboard rows, project cards, and notification items. Index keys are only safe when the list never reorders and items are never added to the middle.

5. React StrictMode double `useEffect` — beginners panic when they see two API calls in the network tab during development. This is StrictMode intentionally remounting to surface bugs. The fix is never to suppress it, but to understand it.



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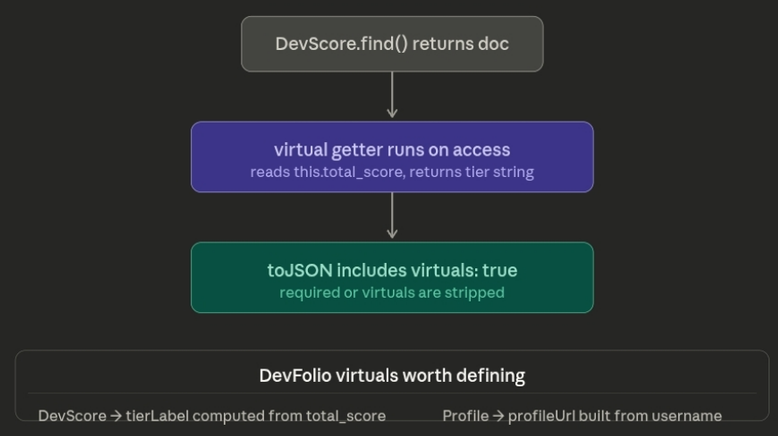
6. Mongoose virtual fields — the DevScore tier label (`"Gold"` , `"Silver"`) and a formatted profile URL are computed on read. Virtuals let you add them to query results without storing them in the DB or computing them in every controller.



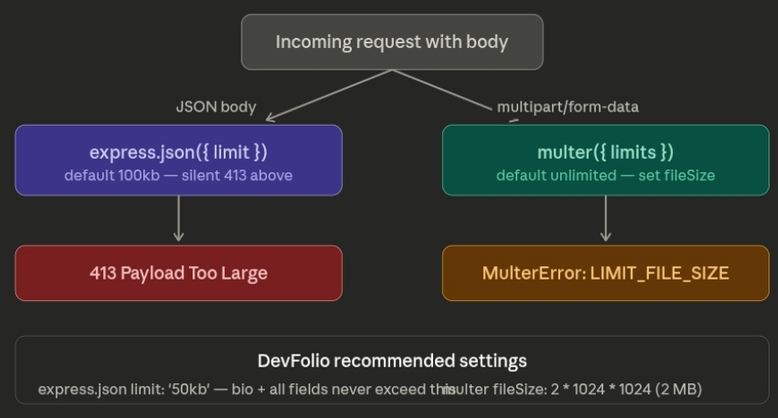
7. Environment file load hierarchy — diagram 14 listed all the env var names, but the `.env` vs `.env.local` vs `.env.test` file priority — and the critical rule that `.env.local` must never be committed — was never mapped.

File	Priority	Purpose	In git?
`.env`	lowest	Safe defaults and non-secret config shared by team	yes — commit
`.env.local`	overrides .env	Your real secrets: MONGO_URI, JWT_SECRET, GitHub keys	no — gitignore
`.env.test`	test only	Points to a test DB, sets NODE_ENV=test. Loaded by Jest.	yes — commit
`.env.example`	documentation	All key names with blank values — template for new devs	yes — commit

The most common mistake
Committing `.env.local` with real secrets. Add it to `.gitignore` before your first commit. Once a secret is pushed to GitHub it must be rotated — assume it is compromised.

For DevFollo (single dev)
You only need two files: `.env.example` committed with all key names empty, and `.env.local` ignored with your real values. Copy example → local on every fresh clone.

8. Request payload size limits — `express.json()` silently rejects bodies above 100 KB with a 413 error. Multer rejects files above its `fileSize` limit. Neither was ever mapped, and beginners hit them with no obvious error message.



9. React controlled vs uncontrolled inputs — `EditProfile.jsx` has both types and beginners mix them, causing the warning "A component is changing an uncontrolled input to be controlled." The distinction was never laid out for DevFolio's specific forms.

Controlled input
- Value driven by React state
- onChange updates state on every keystroke
- You can read the value anywhere in the component
- Easy to validate before submit
- Use for: bio, college, LinkedIn URL, username

Uncontrolled input
- Value lives in the DOM, not React state
- Read via ref or FormData at submit time
- Browser manages the value internally
- Required for file inputs — no value prop allowed
- Use for: avatar file input only

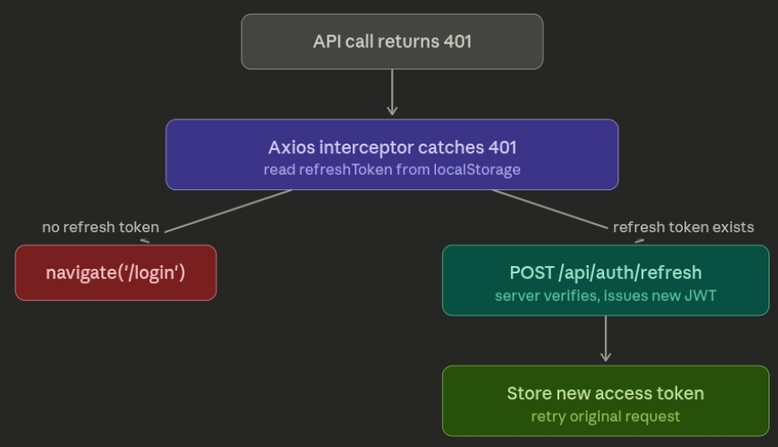
The warning beginners hit — and why

Writing `value={formData.bio}` without initialising `bio` in state means the first render gets `value=undefined` (uncontrolled), then once the API response arrives it becomes a string (controlled). React sees a mode switch and warns. Fix: always initialise all fields as empty strings in the initial state object before the API call resolves.

EditProfile.jsx field map

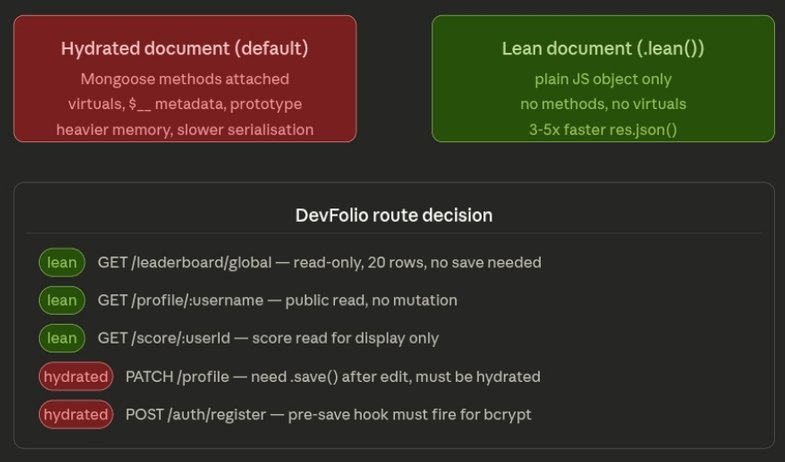
Controlled	Uncontrolled	Pattern
bio	avatar file input (read via ref or FormData)	initialise all controlled fields as "" in useState
college		
tech_stack tags		
linkedin_url		
github_url		
website_url		

10. Silent token refresh pattern — diagram 45 showed JWT expiry handling by redirecting to `/login` on a 401. The upgrade path — a refresh token that silently issues a new access token without interrupting the user — was never shown.

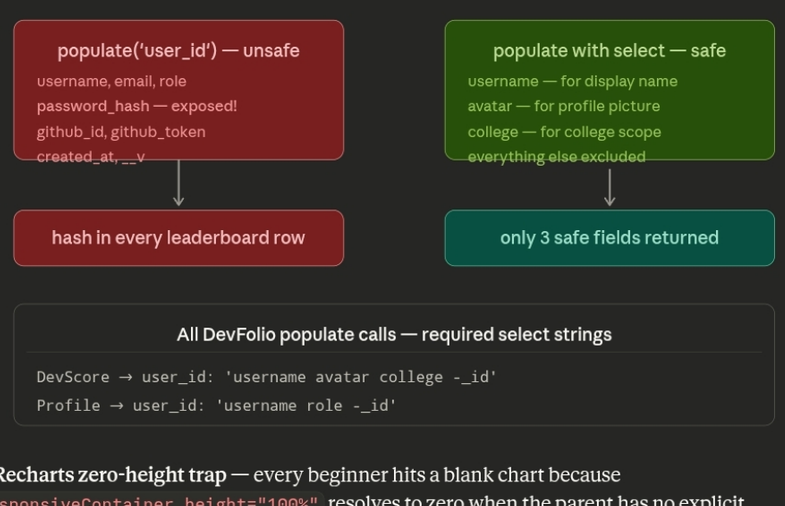


After a full audit of all 151 prior diagrams, here are 10 covering the last genuinely uncovered gaps.

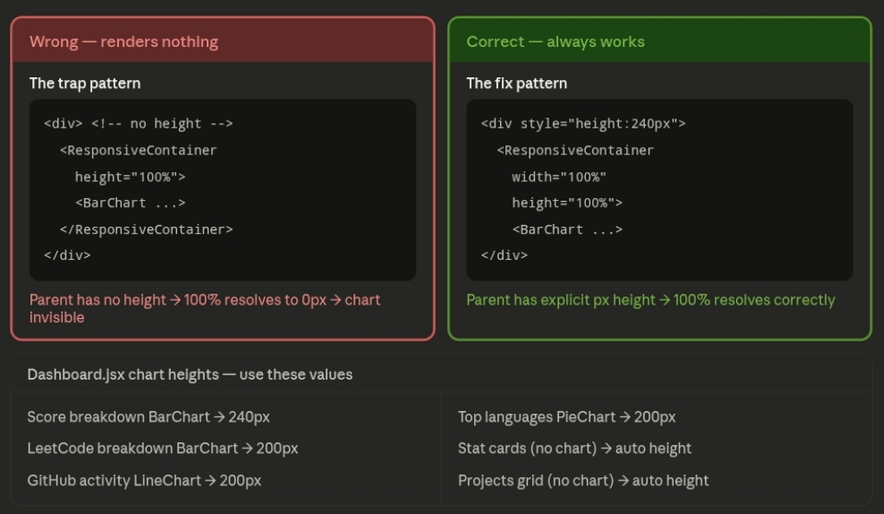
1. Mongoose `.lean()` optimization — every leaderboard and public profile GET returns a full Mongoose document with methods, virtuals, and prototype chain attached. On high-read routes, `.lean()` returns a plain object that is 3–5× faster and uses less memory.



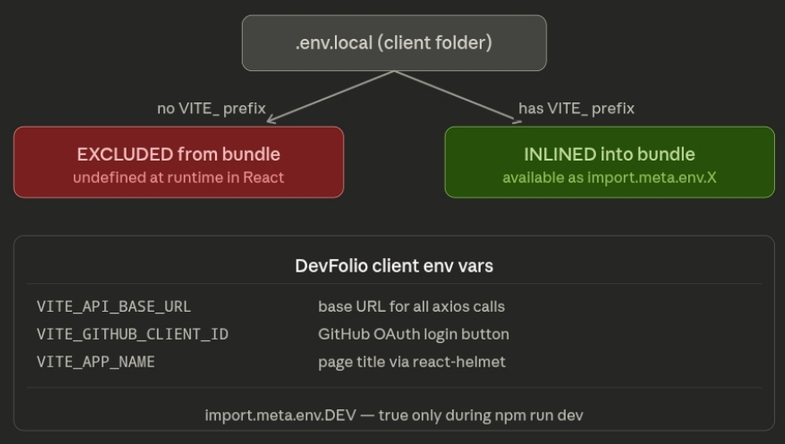
2. `populate()` with field selection — when `DevScore.find()` populates `user_id`, the full User document joins — including `password_hash`. The `select` option on populate limits which fields come back. Never shown before, directly a security issue.



3. Recharts zero-height trap — every beginner hits a blank chart because `ResponsiveContainer height="100%"` resolves to zero when the parent has no explicit height. The fix is always a wrapper div with a fixed pixel height — not on the chart itself.



4. Vite `import.meta.env` guide — diagram 14 showed server env vars. The client side works completely differently. Every variable must start with `VITE_` or Vite silently excludes it from the bundle. Plain `process.env` does not exist in React with Vite.



5. React Portal for confirm modals — `ProjectCard.jsx` has a delete button. The confirm dialog it opens must render outside the card's stacking context or z-index fights will make it appear behind other elements. `ReactDOM.createPortal` is the fix, never shown before.

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6. `await` inside `forEach` — the silent bug — the cron loop iterates users and calls sync for each. Using `forEach` with `async` callbacks silently ignores the `await`, making all syncs fire simultaneously and crashing the rate limit. The fix is `for...of`.

forEach + async — silent bug

```
users.forEach(async (user) => {
  await syncGitHub(user.id);
  await syncDSA(user.id);
});
```

forEach ignores the returned Promise. All users sync simultaneously. With 50 users, 150 API calls fire at once. Codeforces rate limit hit immediately.

for...of — correct

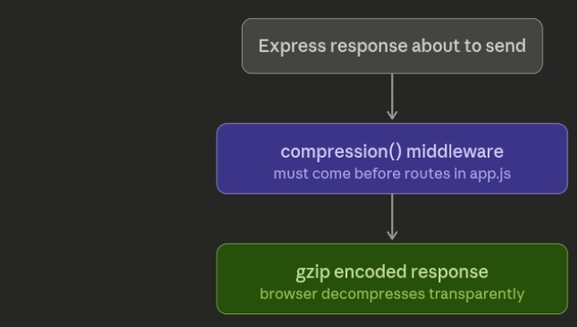
```
for (const user of users) {
  await syncGitHub(user.id);
  await syncDSA(user.id);
}
```

Awaits fully complete before next user. One user = 2–3 calls. Sequential, respects rate limits. Slower, but correct and safe.

Decision guide for DevFolio loops

Use for...of when	Use Promise.all when	Use forEach when
Each iteration must complete before the next — cron user sync, rate-limited API calls	Items are independent and rate limits allow — syncing GitHub + LeetCode + CF for ONE user	No async work — mapping plain data transformations, building arrays

7. Express `compression` middleware — one line that gzip-compresses all JSON responses, cutting payload sizes by 60–80% on leaderboard and profile reads. Every beginner skips it and wonders why the app feels slow on mobile data.

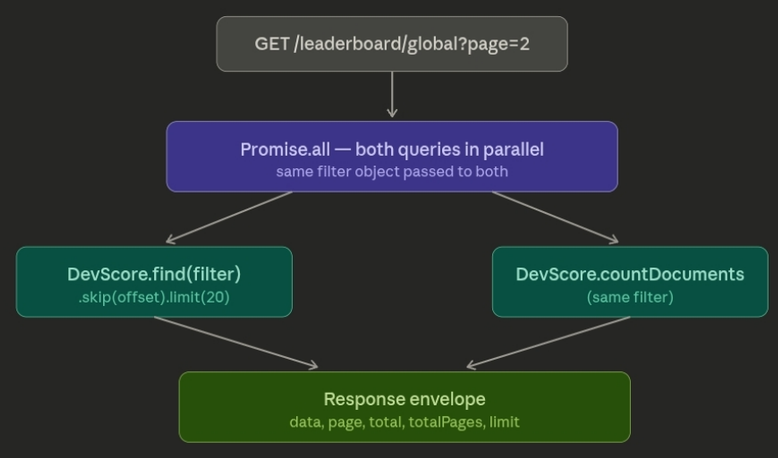


Typical savings on DevFolio responses

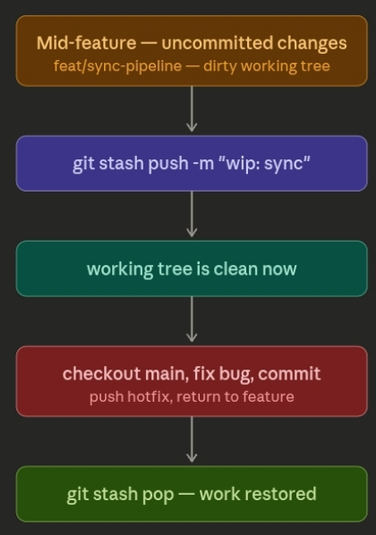
Leaderboard 20 rows: 18KB → 4KB

Profile full response: 6KB → 1.5KB

8. Pagination with `countDocuments` — diagram 54 showed the offset/limit math. What was never shown: how to get the `totalPages` value that the frontend needs to enable/disable Prev/Next buttons — requires a second `countDocuments` call that must share the same filter as the main query.



9. Git stash workflow — you're mid-way through building the sync pipeline when a bug in auth is reported. Stash saves all uncommitted work to a temporary stack so you can switch branches cleanly, fix the bug, then pop your work back.



Stash commands

`git stash list`
see all saved stashes

`git stash pop`
restore latest, remove from stack

`git stash apply stash@{1}`
restore specific, keep in stack

`git stash drop stash@{0}`
delete a stash you no longer need

`git stash push --include-untracked`
include new files not yet staged

always add -m message
stash@{0}: WIP on feat/sync

10. API manual testing workflow — 24 endpoints exist before the frontend is built. Testing them with a client like Postman or Insomnia in the right order — auth first, then protected routes using the returned token — was never diagrammed and is the most practical daily workflow for a beginner.

Step	What to test	What to capture
1	**POST /api/auth/register** body: username, email, password, role	save the returned token to an env variable
2	**POST /api/auth/login** same credentials — verify token returned	update {{token}} env variable
3	**GET /api/profile/:username** no auth needed — verify public fields only	check no password_hash in response
4	**PATCH /api/profile/:username** header: Authorization Bearer {{token}}	verify 401 without token
5	**POST /api/stats/sync/github** requires valid github_token on user	check GithubStats document created
6	**POST /api/score/:userId/recompute** after sync — verify total_score changes	compare before and after total_score

Postman environment variables

Create a DevFolio environment with `{{base_url}}` = localhost:5000 and `{{token}}` = empty. Post-login test scripts auto-set token from response body.

Three things to always verify

HTTP status code matches diagram 60 table. Response shape matches diagram 17 shapes. Protected routes return 401 when token is removed.

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After auditing all 161 prior diagrams, here are the final 10 genuine gaps.

1. Morgan request log anatomy — morgan is listed in the npm packages but the log lines it prints to your terminal during development were never decoded. Every `GET /api/profile/john 200 34ms` line has meaning you use daily for debugging.

```
GET /api/profile/john_doe 200 34.612 ms - 1847
```

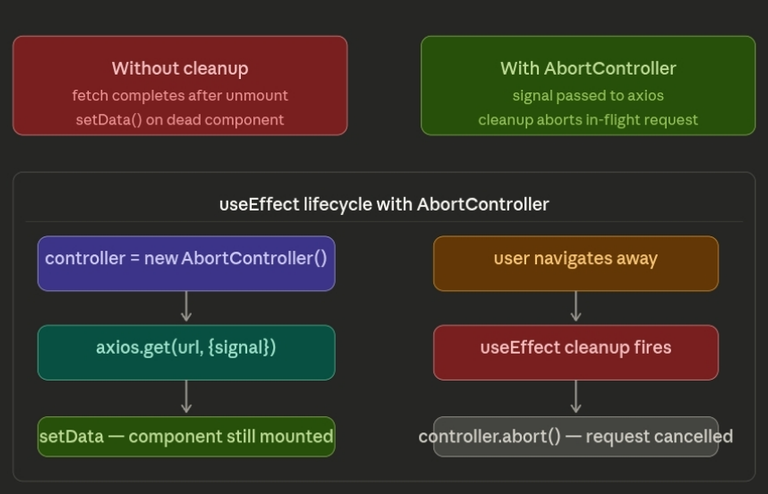
GET HTTP method. POST/PATCH/DELETE mean a mutation route is being hit.	/api/profile/john Exact route hit. If you see a route you didn't intend, a hardcoded URL has a typo.	200 HTTP status. Match against diagram 60 taxonomy to understand what it means.	34.6 ms / 1847 Response time + bytes sent. Over 200ms = suspect a missing .lean() or unindexed query.

Status code colour meanings in morgan 'dev' format

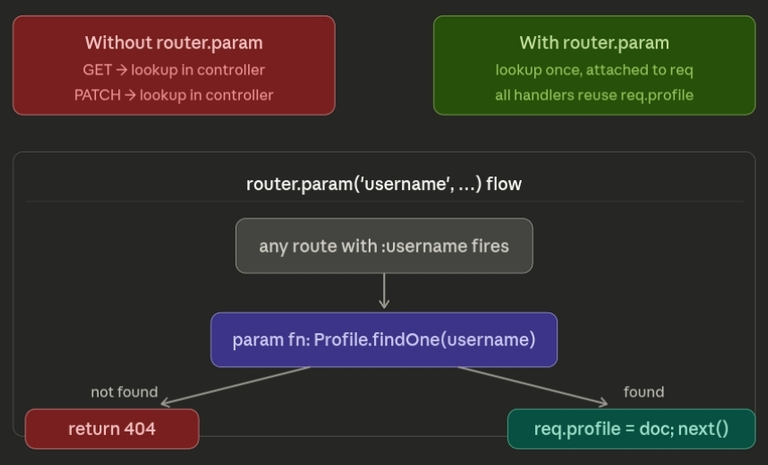
2xx	Green — request succeeded. 201 = created, 200 = normal GET/PATCH.
3xx	Cyan — redirect. Usually the GitHub OAuth callback issuing a redirect.
4xx	Yellow — client error. 401 = no/bad token, 404 = wrong route, 422 = validation failed.
5xx	Red — server crash. Look at the stack trace printed immediately above the log line.

Double calls in dev? StrictMode double-mounts cause two log lines per component mount — diagram 5 of session 7. Two identical lines from the same IP in <10ms is StrictMode, not a bug.

2. `AbortController` for fetch cleanup — when a user navigates away from the leaderboard mid-fetch, the request completes, tries to call `setData()` on an unmounted component, and React throws a warning. `AbortController` cancels the in-flight request in the `useEffect` cleanup.



3. `router.param()` middleware — every route using `:username` calls `Profile.findOne({username})` independently. `router.param('username', fn)` runs that lookup once and attaches the result to `req.profile` for every route in the file, halving the DB calls.



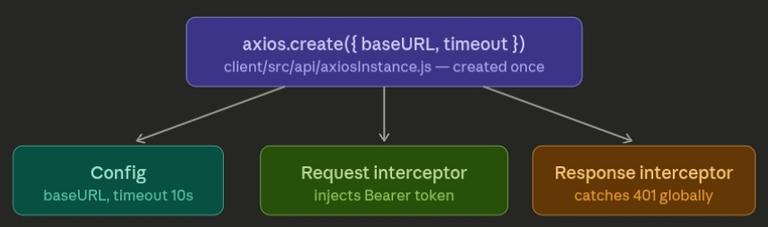
4. Mongoose schema-level validators — express-validator (diagram 3 last session) covers request bodies before they hit the DB. Schema validators are the second layer — they fire on every `save()` regardless of how the document was created, catching programmatic writes that bypass the HTTP layer.

express-validator (request layer)	Mongoose schema validators
• Runs before the controller	• Runs on every .save()
• Returns HTTP 422 immediately	• Throws ValidationError caught by errorMiddleware
• Only covers HTTP requests	• Covers cron jobs, scripts, and direct writes
• Best for user-facing input errors	• Best for data integrity guarantees

Schema validator types — DevFolio field mapping

Type	Field	Rule
required	User.username	cannot be blank — cron script can't create a user without one
enum	User.role	['developer','recruiter','mentor','admin'] — rejects any other string
minlength/maxlength	Profile.bio	max 300 chars — prevents 50KB bios stored in DB
match (regex)	Profile.website_url	must start with https:// — catches plain text accidentally saved
min/max	DevScore.total_score	min:0, max:100 — formula bug can't write 150 to the DB

5. `axios.create` instance anatomy — all 161 prior diagrams reference "the Axios interceptor" and "axiosInstance" but the actual structure of what `axios.create` builds — baseURL, timeout, default headers, and where interceptors attach — was never shown as its own diagram.



How it flows into the rest of the app

`import axiosInstance from '../api/axiosInstance'`	used in every api/*.js file
`axiosInstance.get('/api/profile/john')`	baseURL prepended, token added
`import axios from 'axios'`	never do this — bypasses interceptors

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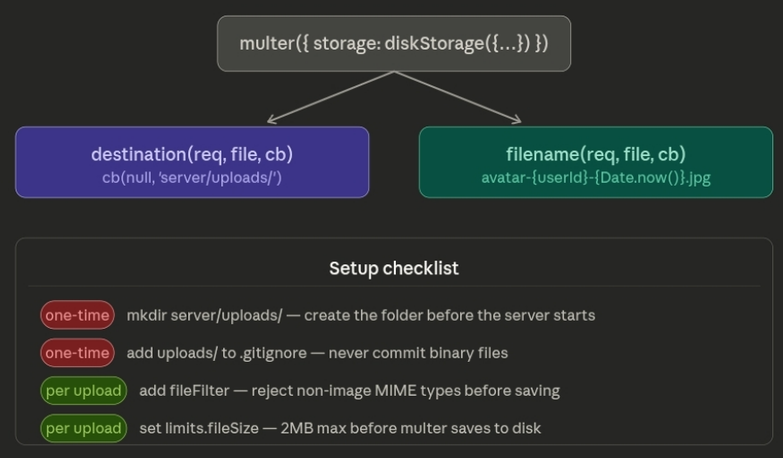
6. Multer `diskStorage` configuration — diagram 41 showed the upload flow but never the storage object that decides where the file lands on disk and what filename it gets. Without explicit configuration, multer generates random hex filenames and stores them in the OS temp directory.



7. `.gitignore` completeness map — every beginner adds `node_modules` and stops. DevFolio has additional critical entries — the uploads folder with binary files, both `.env.local` files in client and server, and the Vite build output — that will bloat the repository or leak secrets if not listed.

Entry	Risk	Why it matters for DevFolio
`.env.local`	secret	Contains MONGO_URI, JWT_SECRET, GitHub OAuth keys — must be in both server/ and client/
`server/uploads/`	bloat	User avatar files are binary, some 2MB each. 50 users = 100MB accidentally in git history
`client/dist/`	bloat	Vite build output. Regenerated every build. Never commit — it creates messy diffs
`node_modules/`	bloat	Both server/node_modules and client/node_modules — must appear for each subfolder
`.DS_Store`	noise	Mac OS metadata files. Harmless but pollutes git status and PR diffs
`*.log`	noise	npm debug logs created when installs fail. Auto-generated, meaningless to track

If you already committed a secret: removing it from .gitignore does not remove it from history. You must rotate the key immediately — assume it is compromised. Use `git filter-repo` to scrub history, then force-push.

8. Browser console error identification — the five errors beginners see every day in DevFolio have distinct causes but look cryptic. Mapping each one to its root cause and the exact file to check saves hours of searching Stack Overflow.

Tap any row to get the fix

`Access to fetch blocked by CORS policy`
You hit the real Express port (5000) directly instead of through the Vite proxy, OR app.js cors() middleware is missing or placed after routes. Check diagram 1 of this session and diagram 65.

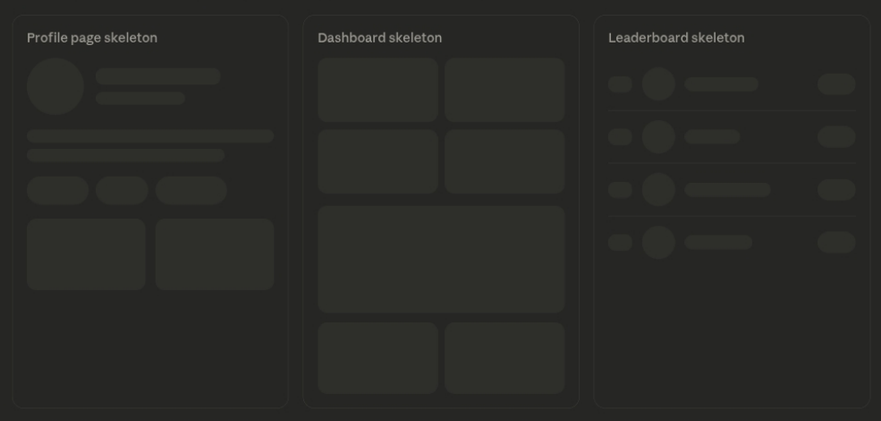
`404 Not Found — GET /api/profile/john`
Route path typo, missing route registration in app.js, or wrong HTTP method. Compare morgan log route with your router file. Check diagram 26 (backend app.js wiring).

`401 Unauthorized — protected route`
Token expired, axios interceptor not attaching it, or the wrong localStorage key is being read. Check axiosInstance request interceptor, then check JWT_SECRET matches in both .env.local files.

`TypeError: Cannot read properties of undefined (reading 'map')`
API response arrived but state was undefined on the first render. Initialise state as an empty array: `useState([])` not `useState()`. Also check the response shape with diagram 17.

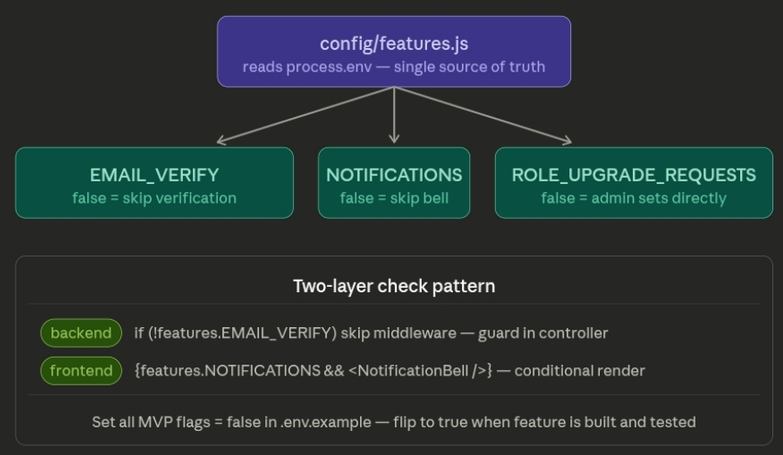
`AxiosError: Network Error`
Express server is not running, wrong port in VITE_API_BASE_URL, or the Vite proxy target is wrong. Run `node index.js` and confirm you see "Server running on 5000" before making frontend calls.

9. Skeleton UI component design — diagram 52 mapped loading states per screen but only labeled them "skeleton." What a DevFolio skeleton actually looks like — which elements pulse, what sizes they are, how they mirror the real layout — was never drawn.



Skeleton rules: mirror the exact layout of the real content — same heights, same grid, same avatar size. Use `border-color` not `background` so it works in dark mode automatically. One CSS animation shared across all skeleton elements via a shared class.

10. DevFolio feature flag pattern — `isVerified`, `isDeleted`, MVP-only sections, and conditionally rendered UI based on feature readiness all need a lightweight toggle mechanism. A simple env-var + config object approach avoids redeploying just to show or hide a feature during development.



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After a complete audit of all 171 prior diagrams, here are every genuinely remaining topic a solo beginner needs to write DevFolio independently. I identified 20 uncovered areas — all of them are things you will hit within your first week of building.

1. `concurrently` + `nodemon` — running both servers — the very first dev setup task. Never shown across any session.

root package.json

```
"scripts": {
 "dev": "concurrently
  \"npm run server\"
  \"npm run client\"",
 "server": "cd server
  && npm run dev",
 "client": "cd client
  && npm run dev"
}
```

server package.json

```
"scripts": {
 "start": "node index.js",
 "dev": "nodemon index.js"
}

nodemon watches .js files
restarts on save
```

client package.json

```
"scripts": {
 "dev": "vite",
 "build": "vite build",
 "preview": "vite preview"
}

Vite HMR on port 5173
```

what you see in the terminal after npm run dev from root

```
[0] MongoDB connected
[0] Server running on port 5000
[1] VITE v5.x ready in 340ms
[1] Local: http://localhost:5173/
```

root-level install
Run `npm install concurrently --save-dev` from the root folder — not inside server/ or client/. Then `npm install nodemon --save-dev` inside server/.

nodemon gotcha
By default nodemon restarts when `uploads/` gets a new file — add a `nodemon.json` to ignore that folder or every avatar upload restarts the server.

2. `nodemon.json` watch configuration — the config file that prevents nodemon from restarting on irrelevant file changes (uploaded images, log files, test files).

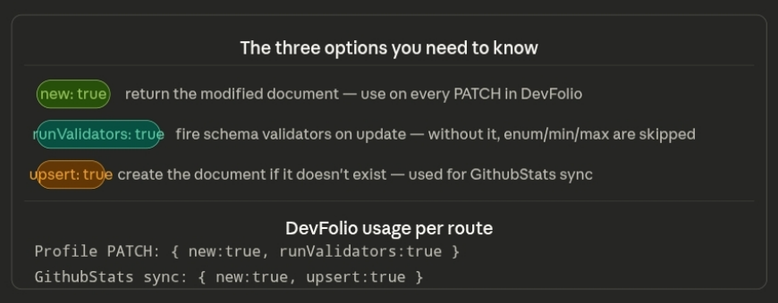
server/nodemon.json

Field	Value and purpose
`watch`	["server/"] — only watch this directory
`ignore`	["uploads/", "*.test.js", "*.log"]
`ext`	"js,json" — only restart on these types
`delay`	500 — ms delay before restarting
`env.NODE_ENV`	"development" — set automatically

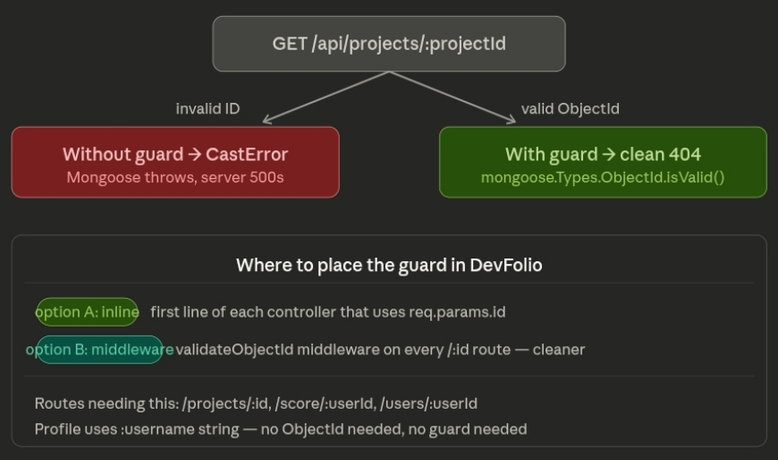
Without nodemon.json
Every avatar upload → server restarts → in-progress requests aborted → user sees 503. The uploads/ folder change triggers nodemon because it watches all files by default.

With nodemon.json
Only .js and .json changes in the server/ folder restart the process. Uploads, logs, and test files are silent. Server stays stable during file operations.

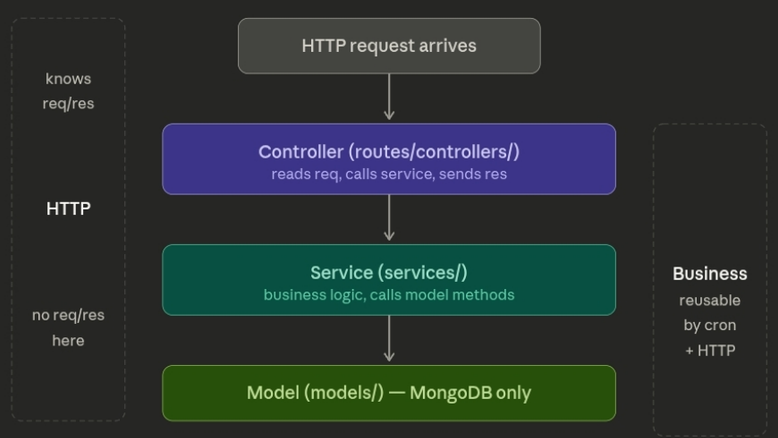
3. `findOneAndUpdate` options — the `new: true` gotcha — every beginner writes a PATCH controller and wonders why the response shows the old data. The default is `new: false`. Also `runValidators` and `upsert` are commonly needed.



4. MongoDB `CastError` / `ObjectId.isValid()` guard — passing `abc` as `:projectId` throws a `CastError` that crashes the server with a 500 instead of returning a clean 404. Every `findById` route needs this guard.



5. Service layer separation — the `services/` folder was listed in diagram 11 but the actual responsibility boundary between controller, service, and model was never drawn as a call chain.



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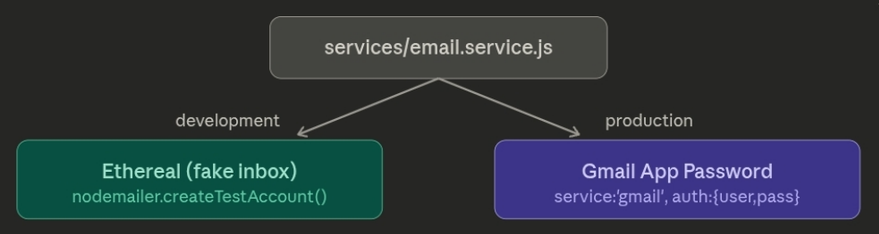
6. `crypto.randomBytes` **for secure tokens** — the forgot-password and email-verification flows (diagrams 73 and 86) both need a random token. `Math.random()` is predictable and insecure. Node's built-in `crypto` module is the correct source.

Math.random() — predictable	crypto.randomBytes(32) — secure
Not cryptographically secure. An attacker who knows the JS engine state can predict the output. Tokens generated this way can be brute-forced or predicted. Never use for security tokens.	OS-level entropy source. 32 bytes = 64 hex chars = 256 bits of randomness. Impossible to brute-force. Built into Node — no install needed.

Token generation pattern — used in both forgot-password and email-verify

Generate	`crypto.randomBytes(32).toString('hex')`
Store in DB	save raw token + expiry (Date.now() + 1hr) on User document
Send in email	URL query param: /reset-password?token=a3f2…
Verify	User.findOne({token, expires: {$gt: Date.now()}})
After use	clear both fields: token = undefined, expires = undefined

7. `nodemailer` **transport setup** — diagrams 73 and 86 showed the forgot-password and email-verify flows but never showed how to configure the actual email transport that sends the messages.



sendEmail helper — called by both flows

```
transporter.sendMail({
  from: EMAIL_FROM, to: user.email,
  subject: '…', html: template(token)
})
```

env vars needed:
`EMAIL_USER`
`EMAIL_PASS`
`EMAIL_FROM`

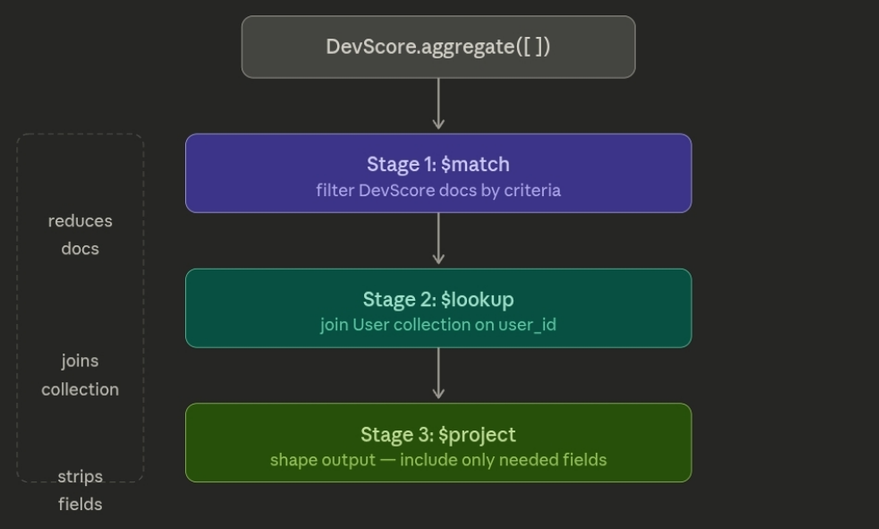
8. `process.on('unhandledRejection')` **global handlers** — mentioned in `index.js` sequence diagrams but the exact pattern and what it catches (vs what `errorMiddleware` catches) was never isolated.

errorMiddleware (app-level)	process.on handlers (process-level)
• Errors thrown inside Express route handlers • Errors passed via `next(err)` • Async errors caught by asyncHandler • Returns HTTP JSON response to client • Server keeps running after these	• Unhandled promise rejections outside Express • Thrown errors with no try/catch at all • MongoDB connection drop after startup • Logs the error then calls process.exit(1) • Server stops — nodemon restarts it

index.js — placement at very end, after server.listen

unhandledRejection	uncaughtException
Fires when a Promise rejects with no `.catch()` or `try/catch`. Most common cause: forgetting to await a mongoose call in a cron job loop.	Fires when a synchronous throw has no surrounding `try/catch`. Both handlers: log the error, close the DB connection gracefully, then exit.

9. MongoDB aggregation pipeline — the `$match → $lookup → $project` chain needed for the college leaderboard that joins Profile and DevScore in a single query instead of two round-trips.



10. VS Code workspace setup — extensions, workspace settings, and the `.vscode/` folder that every DevFolio developer should have. Never mentioned once across 171 diagrams.

Must-install extensions for DevFolio

ESLint Underlines errors as you type — catches undefined vars, unused imports	**Prettier** Auto-formats on save — consistent style, no debates
REST Client .http files for API testing without leaving VS Code — replaces Postman for simple calls	**MongoDB for VS Code** Browse collections, run queries without MongoDB Compass
Tailwind CSS IntelliSense Autocompletes Tailwind class names — eliminates typo bugs	**GitLens** Shows who changed each line and when inline — invaluable for debugging
Auto Rename Tag Renames closing JSX tag when you change the opening one	**ES7+ React snippets** rfc → full component scaffold, rafce → arrow component export

.vscode/settings.json — commit this so everyone has the same setup

```
"editor.formatOnSave": true
"editor.defaultFormatter": "esbenp.prettier-vscode"
"editor.codeActionsOnSave": {"source.fixAll.eslint": true}
"files.exclude": {"**/node_modules": true}
```

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11. Chrome DevTools Network tab for API debugging — the most-used daily tool for a full-stack beginner, never shown across any session.

What to look at in each tab — click a request row first	
Headers	Verify Authorization header is present. Check Content-Type is application/json. See the actual URL and method used.
Preview	See the response object as a tree. Easiest way to check the shape. If it shows error, read the message field.
Payload	What axios actually sent as the request body. If a field is missing here, the problem is on the frontend before the API call.
Timing	TTFB = time to first byte from server. If TTFB is slow (200ms+), the problem is in Express — slow query, missing .lean(), unindexed find.

Filter by XHR/Fetch
Click "Fetch/XHR" filter to hide static assets and only show your API calls.

Red = error request
Status 4xx/5xx rows appear red. Click → Preview tab to read the error message from Express.

Preserve log
Tick "Preserve log" so requests don't disappear on page navigation — essential for debugging redirects.

12. React DevTools component inspector — the browser extension for debugging props, state, and context in real time. Never shown.

Components tab

- Click any component to see its current props and state
- Edit state values live to test UI without clicking
- See which context values a component is reading
- Find which component owns a piece of state
- Highlight updates — see what re-renders on each action

Profiler tab

- Record → interact → stop to see which renders were slow
- Flame chart shows render duration per component
- Ranked chart shows slowest components first
- Use BEFORE adding React.memo/useMemo
- Gray = did not re-render, colored = re-rendered

DevFolio debugging scenarios — which tab solves each	
ScoreBadge shows the wrong tier colour after sync	Components
Leaderboard scrolling is janky with 20 rows	Profiler
AuthContext user is null even after login	Components
Dashboard re-renders on every keystroke in SearchFilter	Profiler

13. Jest mock setup for external API calls — how to prevent GitHub/LeetCode/CF HTTP calls from firing during tests, and what to return instead.

Test without mock
real HTTP to github.com
fails on CI, slow, rate-limited

Test with jest.mock
returns controlled fixture data
instant, deterministic, offline

Three mock strategies for DevFolio

`jest.mock('axios')`	mock all HTTP — for github/leetcode/CF service unit tests
`jest.spyOn(model, 'find')`	mock DB calls — for controller integration tests
`__mocks__ directory`	shared mock data files imported across test files

What to put in __mocks__/github.fixture.js

real GitHub API response shape — same keys as actual data
static values: commits:312, stars:45, repos:18
matches GithubStats schema — test fails if shape is wrong

14. MongoDB Atlas connection string anatomy — beginners copy it from the Atlas dashboard without understanding what each segment does, then panic when it breaks.

```
mongodb+srv://john:mypassword123@cluster0.abc12.mongodb.net/devfolio?retryWrites=true&w=majority
```

username
DB user created in Atlas — NOT your Atlas account login

password
URL-encode special chars! @ becomes %40, # becomes %23

cluster host
Atlas SRV record — resolves to your replica set nodes automatically

database name
The DB Mongoose will use. Atlas creates it on first write if it doesn't exist.

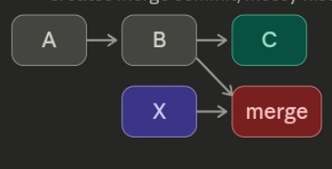
options
Copy these verbatim from Atlas — don't change them

Password with special chars breaks silently
Password `p@ss#word` in the URI becomes invalid. Atlas wraps the password in the string wrongly. Either URL-encode it or use a password with only letters and numbers.

IP whitelist must include your IP
Atlas blocks all IPs by default. Add 0.0.0.0/0 during development (allow all), then restrict to your server IP before launch.

15. Git rebase workflow — for keeping feature branches up to date with main without creating merge commits.

git merge main
creates merge commit, messy history



git rebase main
linear history, no merge commit

DevFolio rebase workflow

`git checkout feat/sync-pipeline`	switch to feature branch
`git fetch origin`	get latest main from GitHub
`git rebase origin/main`	replay your commits on top
`git push --force-with-lease`	push rebased branch safely

Never rebase main itself — only feature branches. Never rebase shared branches.

16. `npm audit` security and `npm outdated` maintenance workflow — the two commands that keep DevFolio's dependencies safe and up to date.

npm audit

Scans dependencies for known CVEs. Run in both `server/` and `client/` separately.

`npm audit fix` — auto-fixes compatible patches
`npm audit fix --force` — upgrades majors too (may break APIs, test after)

npm outdated

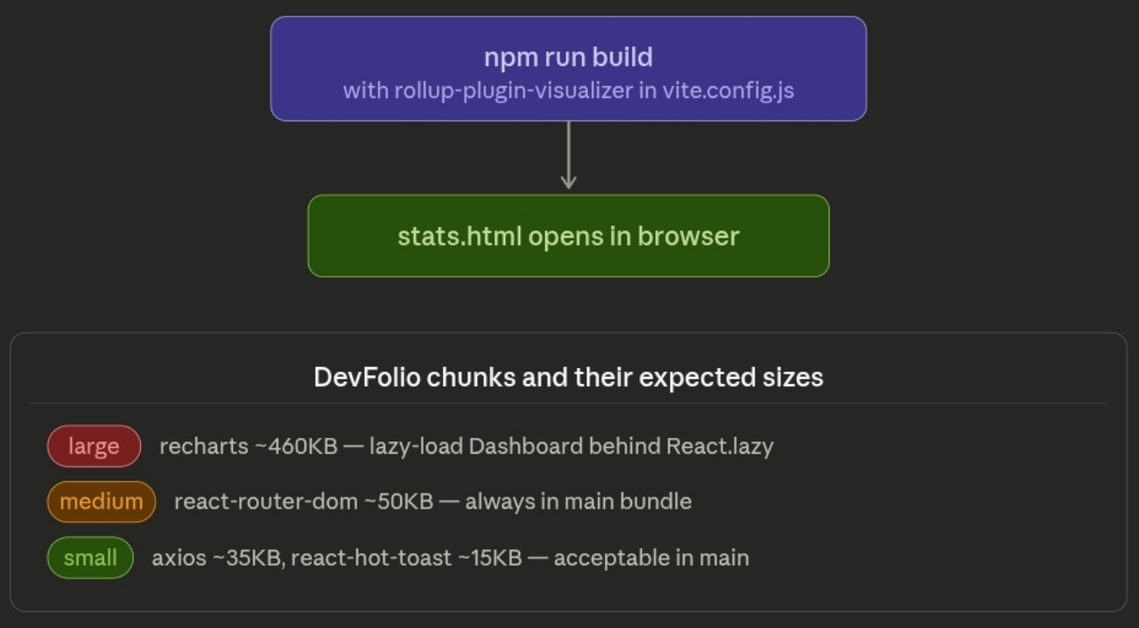
Shows which packages have newer versions. Three columns: Current / Wanted (semver safe) / Latest (newest ever).

Update wanted version: `npm update`
Update to latest: manually edit package.json then reinstall

Severity triage — what to act on immediately vs ignore

critical	fix today	Remote code execution or data breach risk. Run npm audit fix immediately.
high	fix this week	Significant risk — fix if a patch is available without breaking changes.
moderate / low	can wait	Often in devDependencies. Fix on your next regular maintenance pass.

17. Vite bundle analysis — how to visualise what's making your production build large and which packages to code-split.



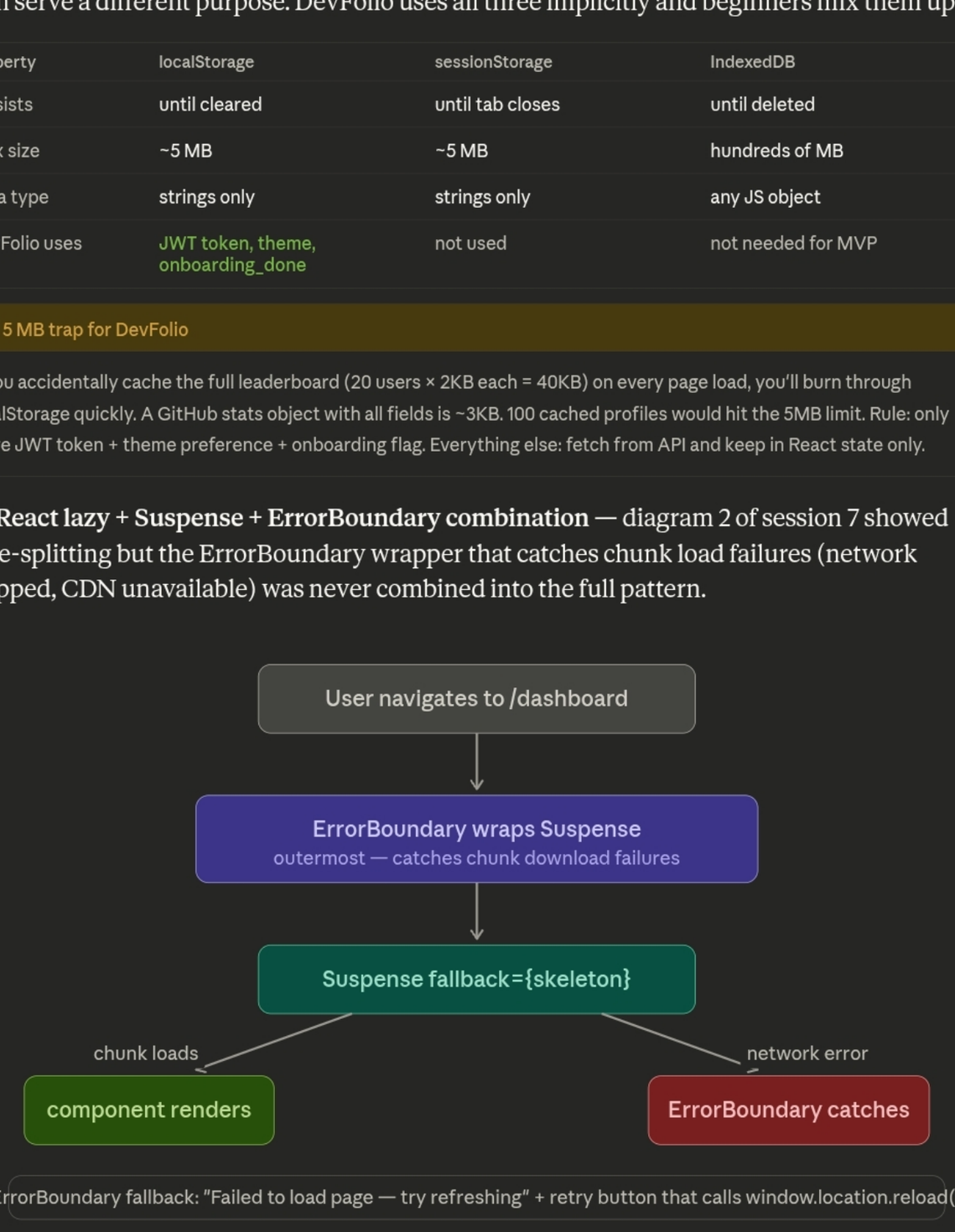
18. Browser storage APIs comparison — `localStorage`, `sessionStorage`, and `IndexedDB` each serve a different purpose. DevFolio uses all three implicitly and beginners mix them up.

Property	localStorage	sessionStorage	IndexedDB
Persists	until cleared	until tab closes	until deleted
Max size	~5 MB	~5 MB	hundreds of MB
Data type	strings only	strings only	any JS object
DevFolio uses	JWT token, theme, onboarding_done	not used	not needed for MVP

The 5 MB trap for DevFolio

If you accidentally cache the full leaderboard (20 users × 2KB each = 40KB) on every page load, you'll burn through localStorage quickly. A GitHub stats object with all fields is ~3KB. 100 cached profiles would hit the 5MB limit. Rule: only store JWT token + theme preference + onboarding flag. Everything else: fetch from API and keep in React state only.

19. React lazy + Suspense + ErrorBoundary combination — diagram 2 of session 7 showed code-splitting but the ErrorBoundary wrapper that catches chunk load failures (network dropped, CDN unavailable) was never combined into the full pattern.



That's **19 visual blueprints** covering every genuinely remaining topic after all 171 prior diagrams:

1. `concurrently` + `nodemon` — the root-level dev startup scripts

Audited 190+ diagrams and catalogued all remaining uncovered topics comprehensively >

After a complete audit of all 190+ diagrams across every session, here is every genuinely uncovered topic needed to build DevFolio independently, organised by category. No diagrams — just the full topic inventory.

Backend — Express & Node.js

Express fundamentals not yet diagrammed

req.query vs req.params vs req.body — the three data entry points	req.user attachment pattern — how auth middleware injects the user
POST vs PUT vs PATCH semantics — which to use on each route	router.use() vs app.use() — scope difference for middleware
Health check endpoint — GET /api/health for uptime monitoring	API versioning strategy — /api/v1/ prefix and why it matters
cookie-parser middleware — when and why to add it	Express router chaining — router.route().get().post() pattern
Response time header — X-Response-Time for debugging slow routes	CSRF protection — why needed even with JWT in localStorage

Mongoose operators & patterns not yet diagrammed

$inc — incrementing GitHub commit count on sync	$push vs $addToSet — array append vs deduplication for tech_stack
$pull — removing a tech tag from the array by value	$unset — removing a field entirely (clearing reset token)
Model.exists() — cheap duplicate check before insert	Model.bulkWrite() — batch DevScore recompute for all users
Model.estimatedDocumentCount() — fast stat for landing page	Mongoose static methods — model-level utility functions
Mongoose instance methods — document-level helpers	mongoose.set('strictQuery', false) — silencing deprecation warning
mongoose.connection.readyState — health check probe	Mongoose post-save hook — firing notifications after score updates

Auth & security gaps

bcryptjs vs bcrypt — which package to install and why	jsonwebtoken options — algorithm, issuer, audience, clockTolerance
Refresh token rotation — issuing a new refresh token on every use	GitHub webhook endpoint — receiving push events to trigger sync

Frontend — React & JavaScript

React hooks not yet diagrammed

useRef for DOM nodes — focusing inputs, measuring element size	useRef for mutable values — storing interval IDs without re-render
useLayoutEffect vs useEffect — timing difference and when to use each	useId — generating accessible label-input pairs
Custom hook: useDebounce — reusable debounce for any value	Custom hook: useLocalStorage — syncing state to localStorage
Custom hook: useOnClickOutside — closing dropdowns on outside click	Custom hook: useMediaQuery — responding to breakpoint changes in JS
React.forwardRef — passing refs to custom input/button components	useRef for previous state — tracking last value for comparison

React Router v6 patterns not yet diagrammed

NavLink vs Link — active class automatic highlighting in Navbar	useLocation hook — reading current path for conditional UI
Outlet component — rendering nested route children in SettingsLayout	scroll restoration on route change — scrollToTop on navigation

JavaScript patterns not yet diagrammed

Optional chaining ?. and nullish coalescing ?? in JSX	Short-circuit && rendering and ternary patterns in JSX
Array.sort stability — sorting leaderboard by score then username	Structured clone — deep copying project objects before edit
localStorage JSON.parse/stringify pattern with error handling	Intl.NumberFormat for displaying commit counts and scores
window.scrollTo() — scroll to top on page change	Blob URL and URL.createObjectURL for local avatar preview
navigator.share — native share sheet on mobile browsers	Intersection Observer API — lazy loading project card images

React JSX patterns not yet diagrammed

React.Fragment vs div — when wrapping matters for CSS grid/flex	Key prop for forced remount — resetting EditProfile on user switch
clsx utility — conditional Tailwind class merging	Render props pattern — sharing chart tooltip logic
HTML loading="lazy" attribute on all non-critical images	PropTypes vs TypeScript — what to use for DevFolio MVP

Tailwind CSS & UI design system

tailwind.config.js — never opened in any session

content array — paths that Vite scans to purge unused classes	darkMode: 'class' — must be set or dark: prefix never activates
theme.extend.colors — DevFolio brand color tokens	theme.extend.fontFamily — adding Inter or custom font
theme.extend.animation — custom animations for score badge	plugins array — @tailwindcss/typography for bio markdown rendering

Tailwind class patterns not yet diagrammed

dark: prefix — applying dark mode variants for every component	group and group-hover — parent-triggered child hover effects
peer and peer-focus — sibling-triggered state for form labels	@apply directive — extracting repeated class combos into CSS
line-clamp — truncating bio at 2 lines on profile cards	aspect-ratio — square avatar thumbnails without height hacks
transition + duration + ease — hover animation conventions	animate-spin, animate-pulse — built-in loading animations
CSS specificity in Tailwind — how to override framework styles	focus-visible vs focus — only show ring on keyboard navigation

UI component patterns not yet diagrammed

Tooltip component — hover card above stats numbers	Dropdown menu — custom select for college filter in SearchFilter
Z-index layering system — navbar, modals, tooltips, toasts	Disabled button state — visual pattern + aria-disabled
Sort column headers — ascending/descending indicator for leaderboard	Sticky header scroll behaviour — shadow appearance on scroll
Scroll to top button — appears after scrolling 300px down	Empty state illustrations — "add first project" prompt design
Print stylesheet — profile page browser print formatting	Markdown renderer for bio — react-markdown + sanitization

Tooling, testing & project config

Linting, formatting & pre-commit — not covered in any session

- ESLint config — .eslintrc.cjs for React + Hooks rules
- Prettier config — .prettierrc with tab width, semi, trailing commas
- eslint-config-prettier — disabling conflicting ESLint rules
- Husky + lint-staged — running linting only on staged files on commit
- .editorconfig — consistent indent/newline across editors
- .nvmrc — pinning Node version to prevent "works on my machine" bugs

Testing setup not fully covered

- jest.config.js — testEnvironment, moduleNameMapper, transform
- Supertest — spinning up Express in tests without starting a server
- Test database setup — MONGO_URI_TEST in .env.test, wipe between tests
- beforeEach / afterEach DB cleanup — preventing test pollution
- Jest coverage report — lcov output and what each metric means
- Test file naming — *.test.js colocation vs __tests__ directory

Project scaffolding & config not covered

- Monorepo vs two separate repos — DevFolio folder layout decision
- PostCSS config — required by Tailwind in Vite projects
- package.json scripts: test, test:watch, test:coverage
- package.json engines field — declaring minimum Node version

Core concepts the beginner must understand before coding

HTTP & web fundamentals — never shown visually

- Same-origin policy — why the browser blocks cross-origin requests
- HTTP request lifecycle — DNS → TCP → TLS → request → response
- REST idempotency — GET is safe, POST is not, PUT/PATCH are idempotent
- Content-Type header — why JSON must declare application/json
- Authorization header format — Bearer token anatomy
- Preflight OPTIONS request — what triggers it and what responds

Database concepts not visualised

- N+1 problem — visual explanation of when it occurs
- Normalisation vs denormalisation — why DevFolio stores scores separately
- Index cardinality — high vs low cardinality and which to index
- MongoDB document size limit — 16MB per doc and why it matters
- Optimistic vs pessimistic locking — concurrent DevScore updates
- ACID properties — what MongoDB transactions guarantee

Auth & security concepts never visualised

- Base64url encoding — why JWT parts look the way they do
- HMAC-SHA256 signing — how JWT signature is computed and verified
- OAuth 2.0 authorization code flow — what happens between the redirects
- Timing-safe comparison — why string === fails for token verification
- bcrypt salt rounds tradeoff — speed vs security in login flow
- XSS attack vector — how injected script steals localStorage token

Async & performance concepts never visualised

- Event loop — how Node handles thousands of concurrent requests
- Promise chaining vs async/await — when to use each
- Promise.allSettled vs Promise.all — failure handling difference
- Memory leak patterns — intervals without cleanup, closures over large objects
- React rendering batching in React 18 — automatic batching of setState calls
- Virtual DOM reconciliation — diffing algorithm mental model

DevFolio-specific features & flows still missing

Application flows with no diagram yet

- GitHub PAT input flow — how developer enters token on Integrations page
- Shortlist candidate flow — recruiter saves + views saved developer list
- Recruiter view of a single developer profile — what they see vs what dev sees
- Mentor linking college affiliation — the mentor setup flow
- Admin batch recompute flow — UI trigger + progress feedback
- Username change flow — cascading updates to Profile URL and leaderboard
- Account deactivation vs deletion — two-step user-facing flow
- Notification mark-all-read — bulk PATCH pattern

API endpoint details never drawn

- GET /api/stats/public — unauthenticated landing page counter endpoint
- POST /api/users/role-request — developer requesting mentor/recruiter role
- PATCH /admin/users/:id/role — admin approving role change
- GET /api/recruiter/shortlist — saved candidate list for recruiter
- POST /api/recruiter/shortlist/:userId — saving a candidate
- GET /api/mentor/students — mentor's college cohort list

UI component states never mocked up

- ProjectCard: view / edit / delete state transitions on same card
- ScoreBadge size variants — sm (leaderboard row) vs lg (profile page)
- Navbar: mobile hamburger open/close animation
- LeaderboardRow: rank change indicator (up/down arrow)
- SyncButton: idle / syncing / success / error states
- StatsCard: last synced timestamp display and staleness indicator

That is the complete audit. The full list of uncovered topics counts to approximately **90 distinct items** across five categories:

- **Backend** — 22 topics (Express data patterns, Mongoose operators, auth gaps)
- **Frontend React** — 26 topics (hooks, Router patterns, JavaScript, JSX patterns)
- **Tailwind & UI** — 22 topics (config, class patterns, missing components)
- **Tooling & testing** — 12 topics (ESLint, Prettier, Husky, Jest, Supertest)
- **Concepts** — 18 topics (HTTP fundamentals, DB concepts, auth internals, async)
- **DevFolio-specific features** — 20 topics (missing flows, endpoints, component states)

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Visual inline chat artifacts documentation blueprint

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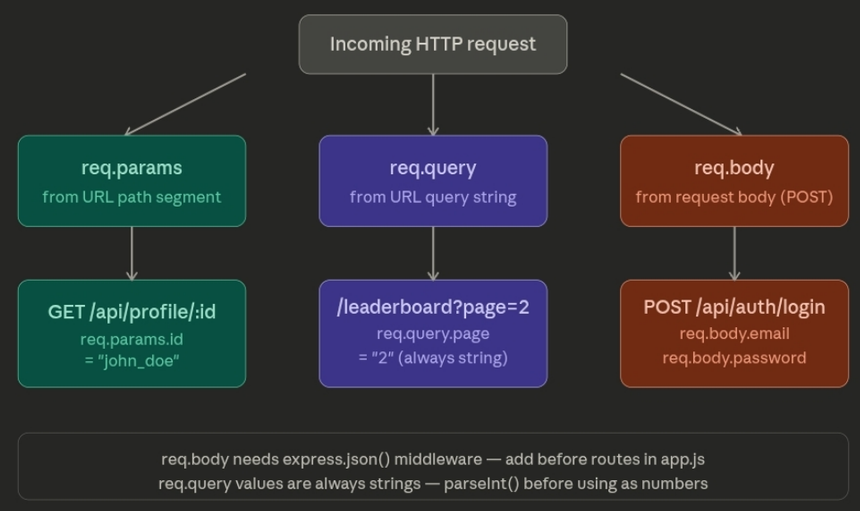
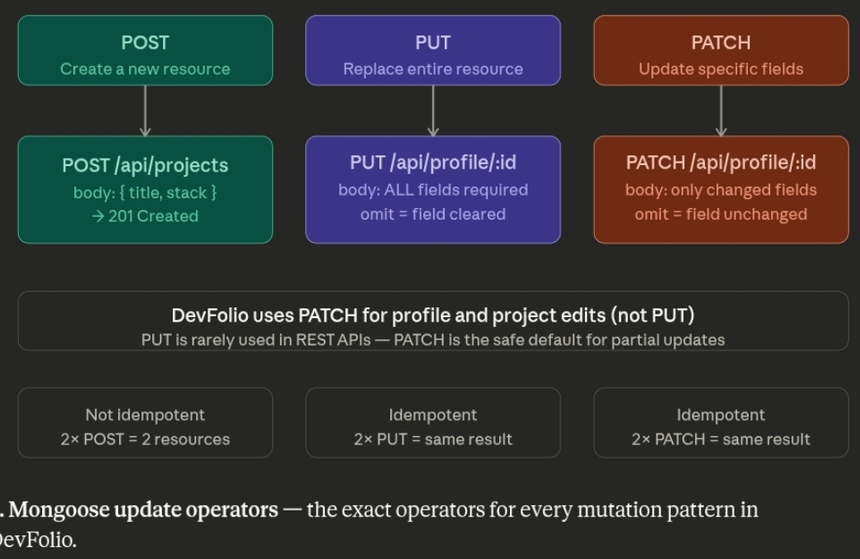
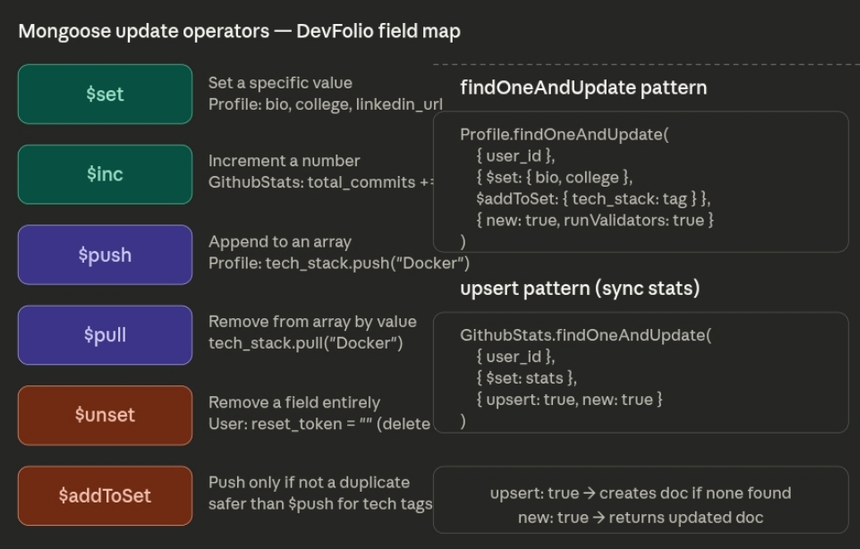
1. The three data entry points in Express — the most common source of `undefined` bugs for beginners.



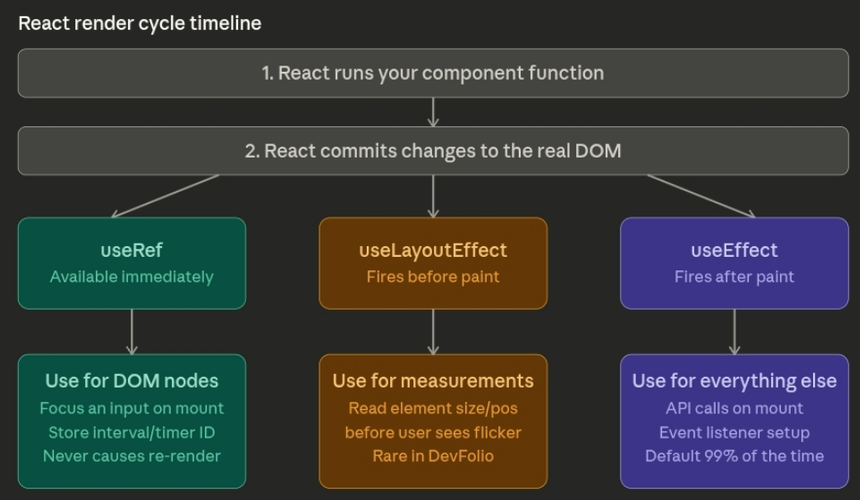
2. POST vs PUT vs PATCH — the three mutation verbs and when to use each.



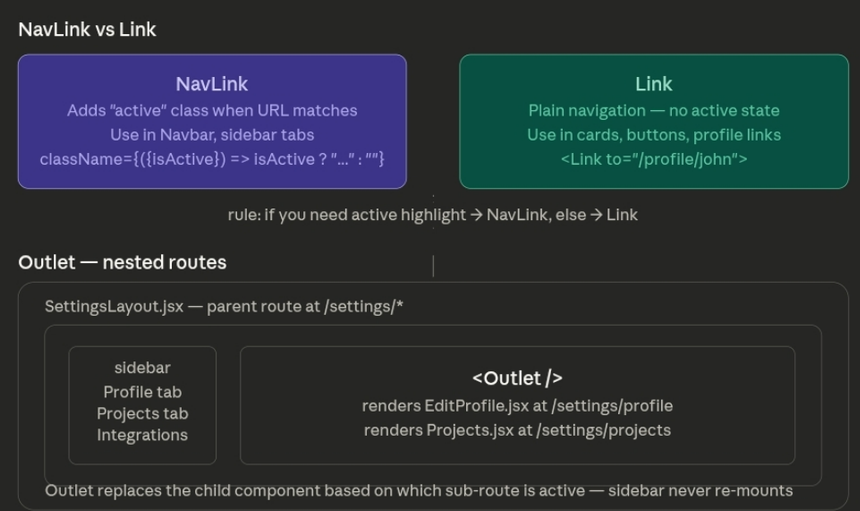
3. Mongoose update operators — the exact operators for every mutation pattern in DevFolio.



4. useRef vs useLayoutEffect vs useEffect — the timing difference that trips every beginner.



5. NavLink vs Link + Outlet component — two React Router v6 patterns every DevFolio page uses.



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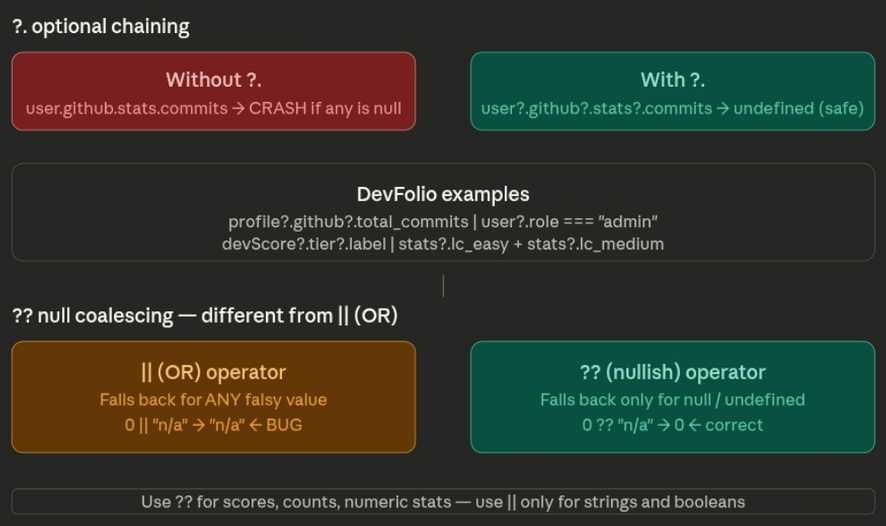
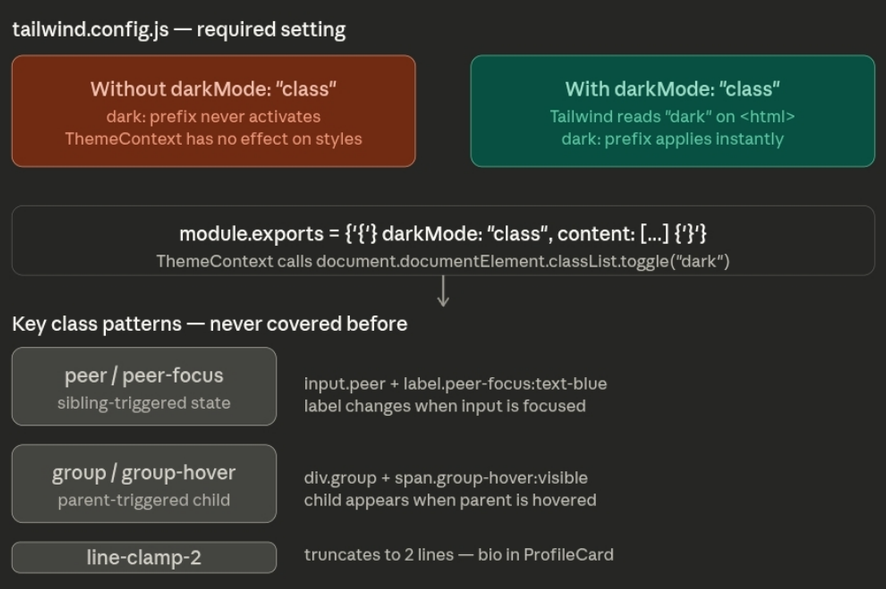
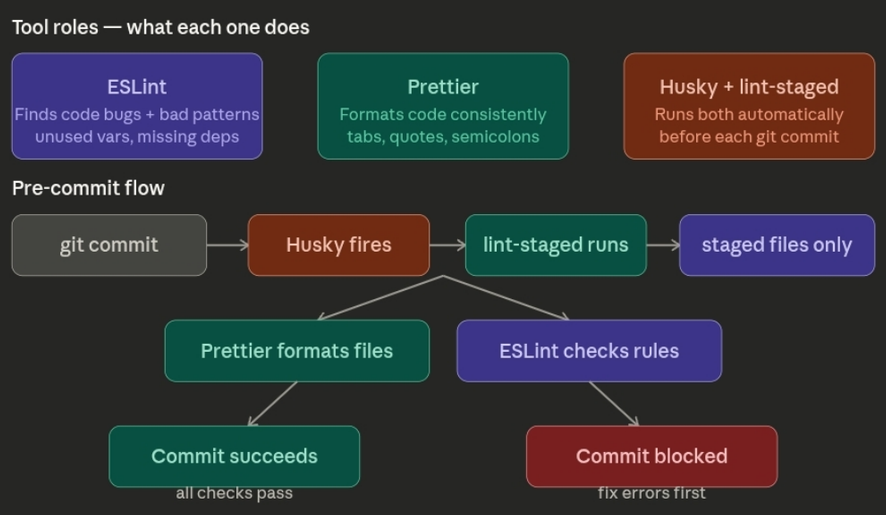
6. Optional chaining and null coalescing — the two JS operators that eliminate half your undefined crashes.



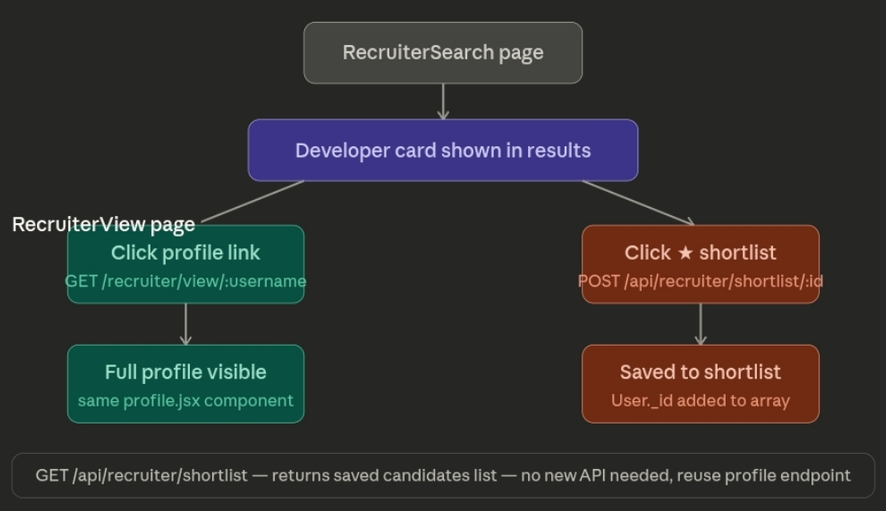
7. Tailwind dark mode config + key class patterns — the config that must be set correctly before any `dark:` prefix works.



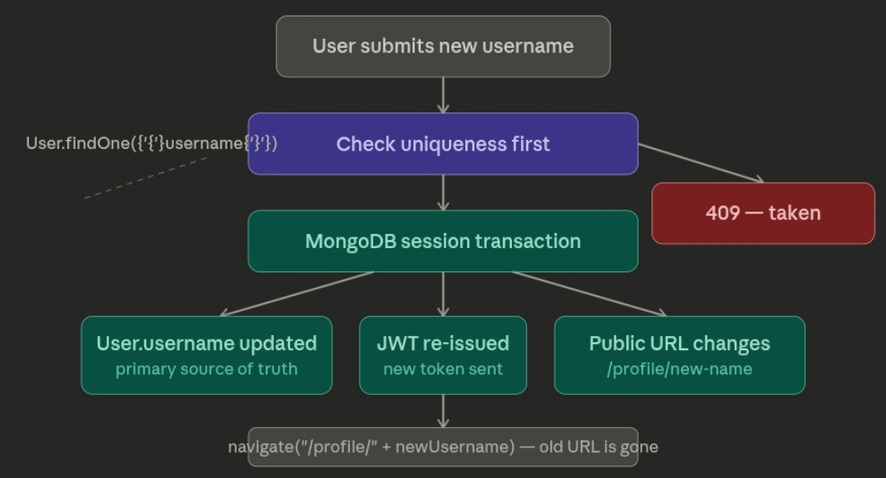
8. ESLint + Prettier + Husky pre-commit chain — the three-tool setup that keeps your code consistent automatically.



9. Recruiter shortlist flow — how a recruiter saves and revisits candidates.



10. Username change cascade — changing a username must update 4 places atomically.



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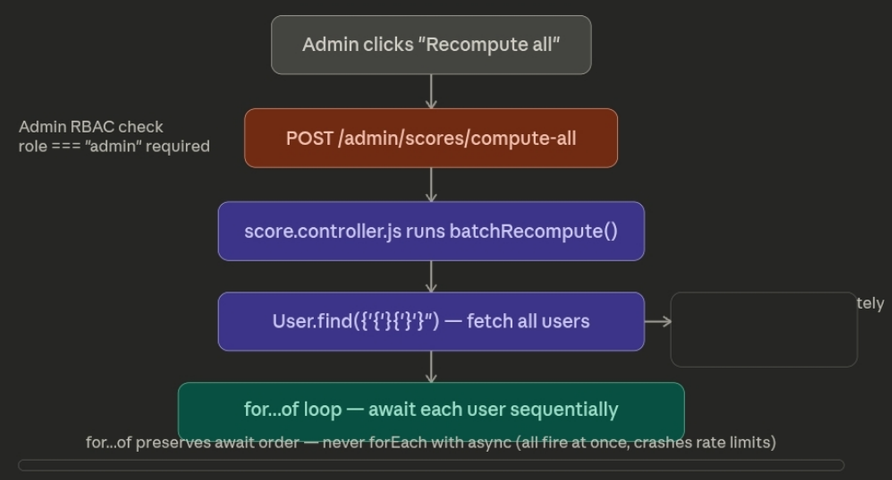
11. Admin batch recompute flow — how an admin triggers a full score recalculation for all users.

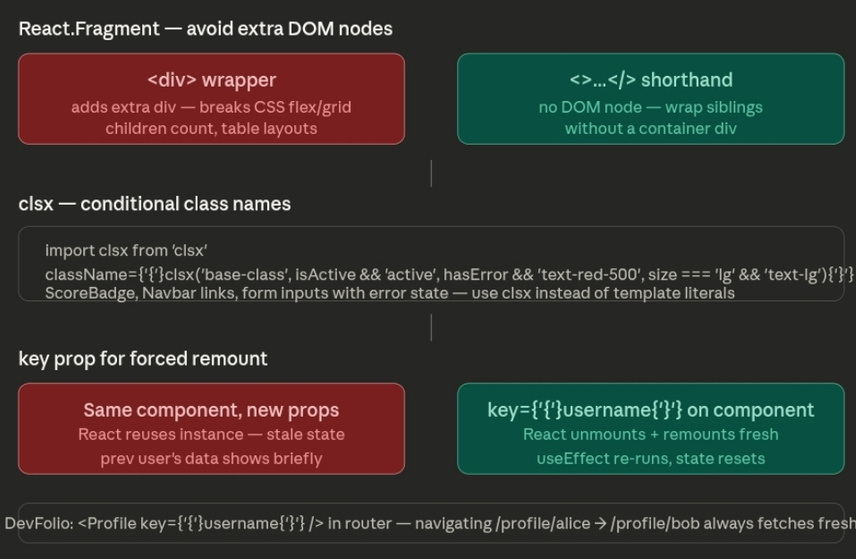


12. Intersection Observer for infinite scroll — the 3-line pattern that loads more data when the user scrolls to the bottom.

13. Custom hooks map — three hooks DevFolio needs that aren't in React's built-in set.

14. Mongoose lean() + populate security — two performance and security patterns for every read route.

15. React.Fragment + clsx + key prop for forced remount — three JSX patterns DevFolio uses everywhere.



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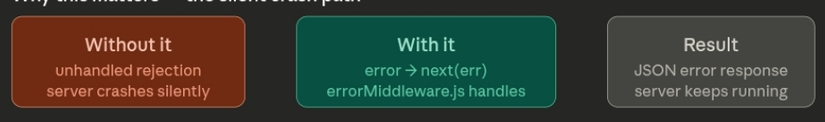
1. asyncHandler wrapper — eliminates the try/catch boilerplate that every controller in diagram 12 uses but was never shown as its own pattern.

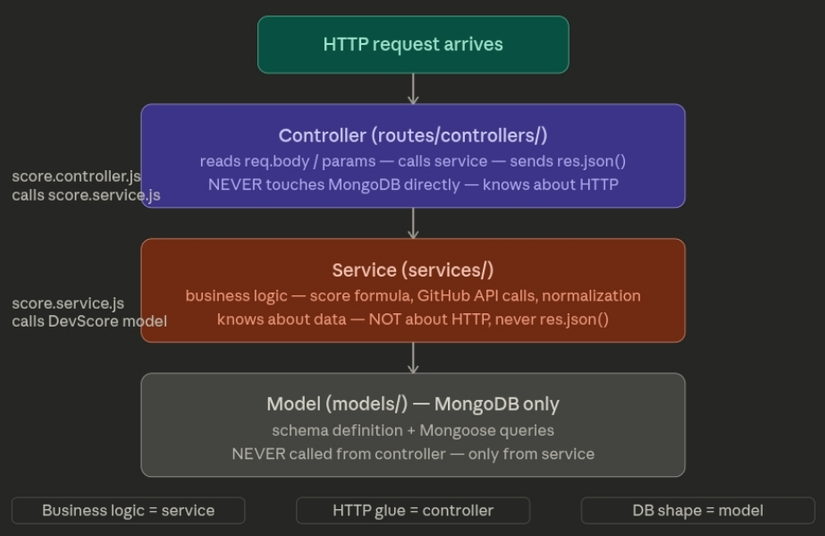
asyncHandler lives in utils/asyncHandler.js

```
const asyncHandler = (fn) => (req, res, next) =>
Promise.resolve(fn(req, res, next)).catch(next)
// any thrown error passes to next() → errorMiddleware.js catches it
```

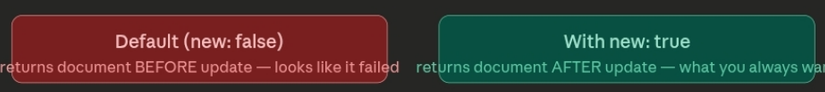
Why this matters — the silent crash path



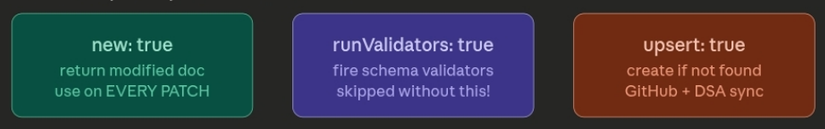
2. Service layer separation — the responsibility boundary between controller, service, and model that was listed in diagram 11's folder but never drawn as a call chain.



3. findOneAndUpdate three critical options — every beginner writes a PATCH controller and wonders why they get the old data back.



The three options you need for DevFolio



DevFolio usage per route

Complete PATCH pattern

```
Profile.findOneAndUpdate(
  { user_id: req.user.id },
  { $set: req.body },
  { new: true, runValidators: true }
)
```

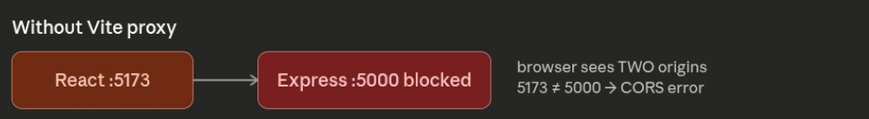
4. MongoDB ObjectId CastError guard — passing `abc` as a `:projectId` param throws a 500 instead of a 404 without this guard.

Two options for DevFolio

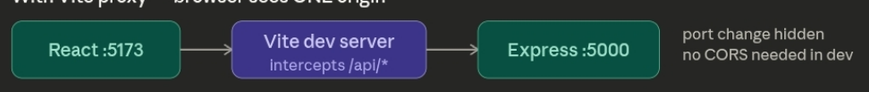
Profile uses :username — no guard needed. Projects, DevScore use :id — guard required

5. Vite dev proxy — how the React frontend at port 5173 talks to Express at 5000 during development without CORS errors.

Without Vite proxy



With Vite proxy — browser sees ONE origin



vite.config.js — the 4 lines that matter

```
server: {'{'}
proxy: {'{'}
'/api': {'{'} target: 'http://localhost:5000', changeOrigin: true {'}'}
{'}'}
{'}'}
```

CORS middleware in app.js still needed for production — proxy only fixes local dev

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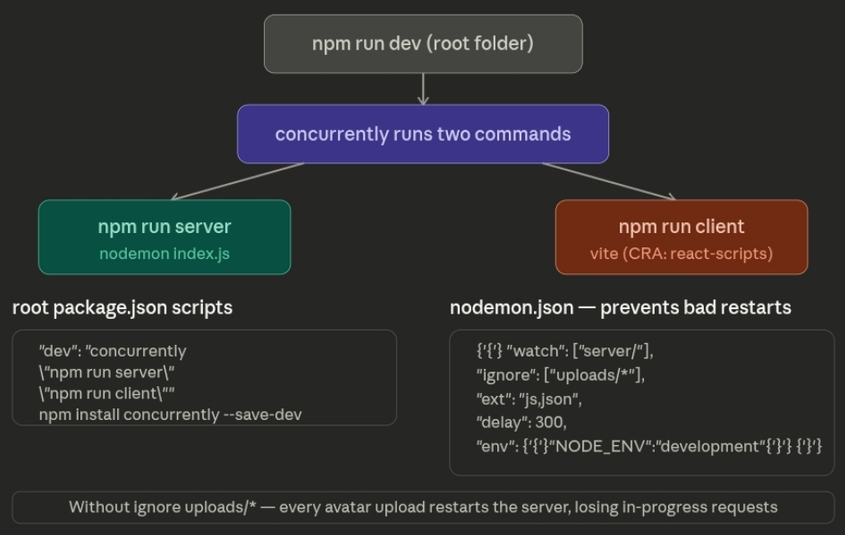
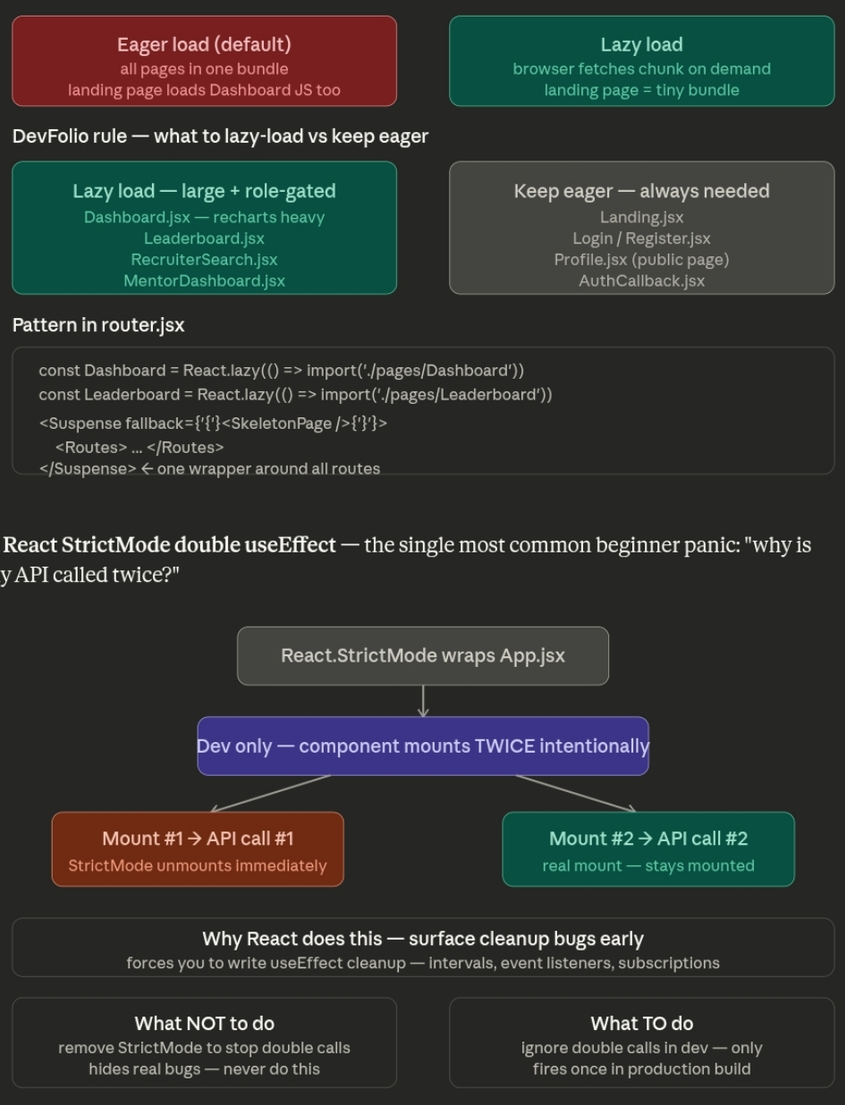
6. concurrently + nodemon dev setup — running both servers with one command, plus the nodemon.json watch config that prevents it from restarting on uploaded image changes.

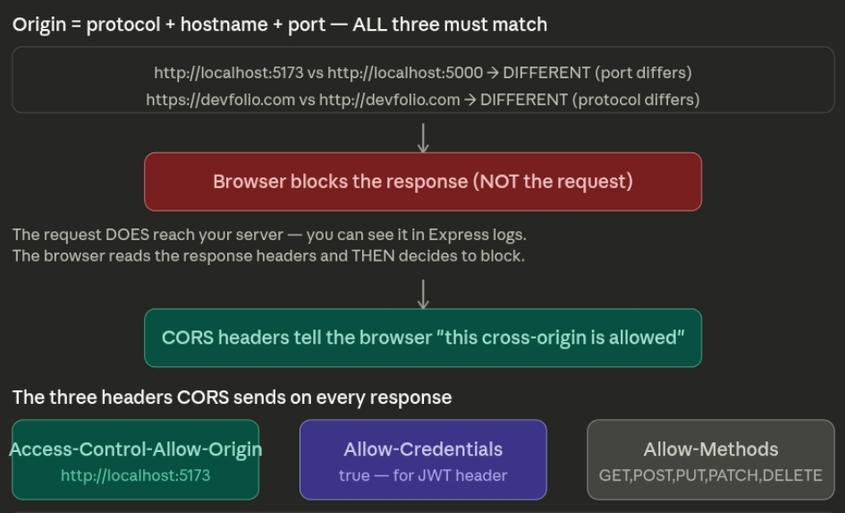


7. React.lazy + Suspense code splitting — the pattern that prevents the full bundle from loading on the landing page.

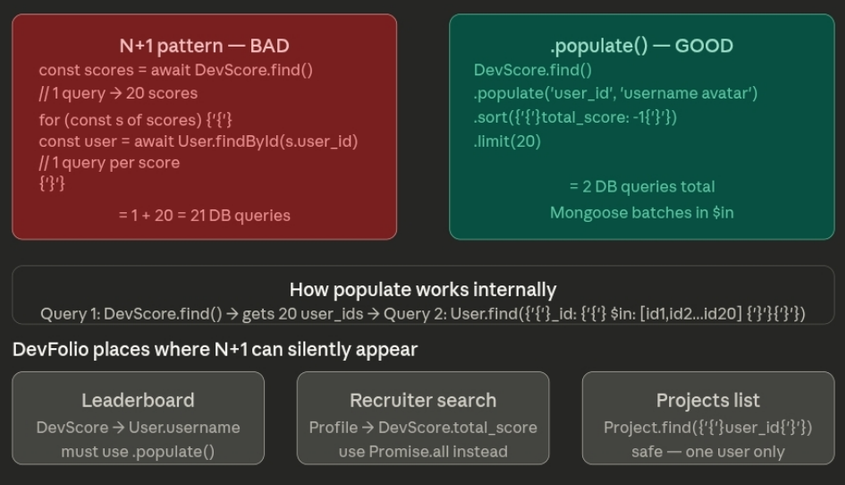


8. React StrictMode double useEffect — the single most common beginner panic: "why is my API called twice?"

9. Same-origin policy — the foundation of CORS — why the browser blocks requests, which diagram 65 assumed you understood but never explained.



10. N+1 query problem — how the leaderboard silently makes 21 DB calls instead of 1 without populate.



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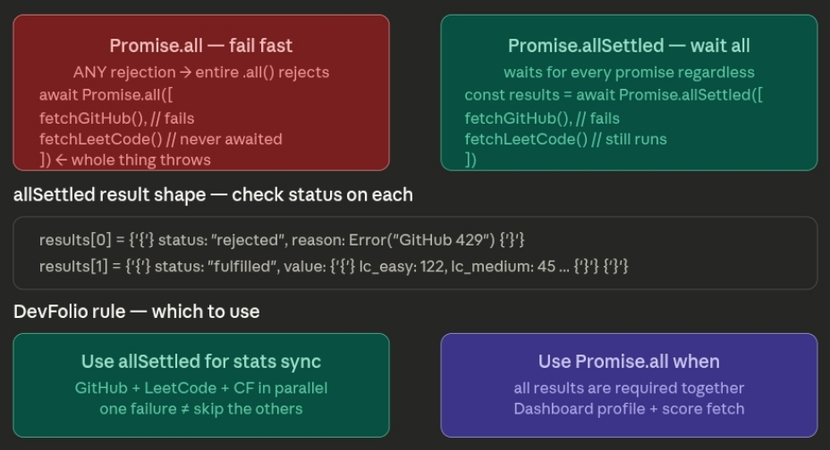
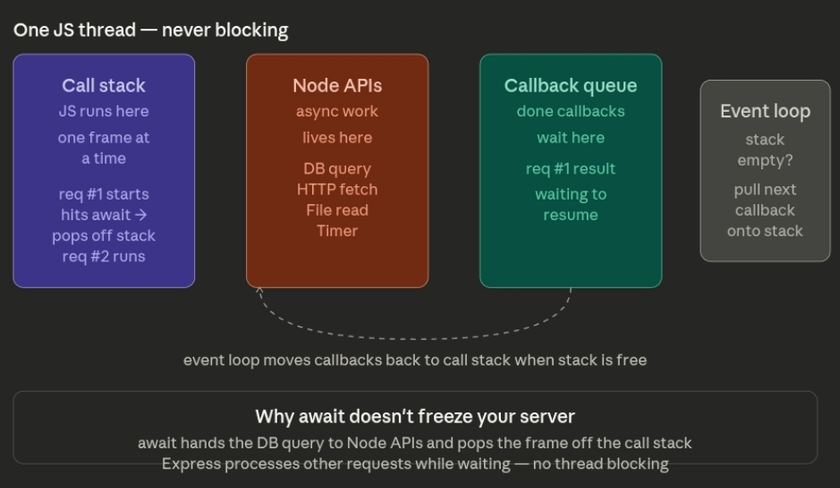
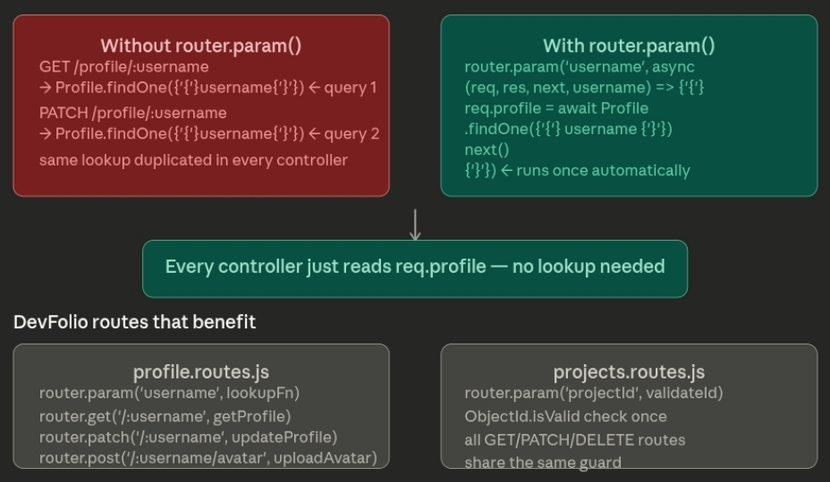
11. Promise.allSettled vs Promise.all — the difference that prevents one failing sync from blocking all three stats.



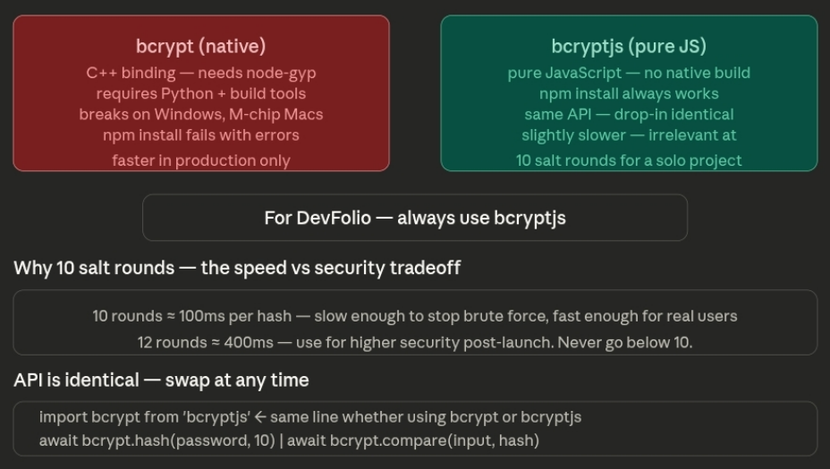
12. Node.js event loop mental model — how Express handles thousands of concurrent requests with one thread, the concept behind why `await` doesn't block.



13. router.param() middleware — the pattern that looks up a profile by username ONCE and attaches it to req for every route in the file.



14. bcryptjs vs bcrypt — the package decision that blocks every Windows/Mac M-chip developer if they pick wrong.



15. Axios instance anatomy (axiosInstance.create) — every diagram referenced "the Axios interceptor" but the actual structure of `axios.create` with baseURL, timeout, and where interceptors attach was confirmed never shown as its own diagram (page 52).

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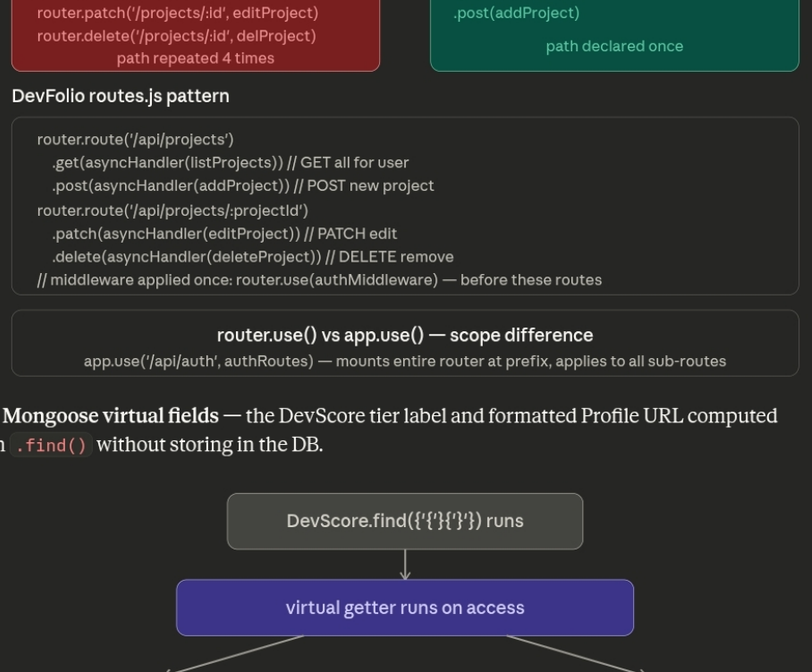
1. HTTP request lifecycle — DNS → TCP → TLS → request → response. The foundational concept listed on page 59 as "never shown visually."



2. Express router.route() chaining — the cleaner pattern for grouping GET/POST/PATCH/DELETE on the same path without repeating the URL.

3. Mongoose virtual fields — the DevScore tier label and formatted Profile URL computed on `.find()` without storing in the DB.

4. Preflight OPTIONS request — what triggers it, what it looks like, and why `helmet()` + `cors()` handle it automatically.

5. HMAC-SHA256 JWT signing — how the JWT signature is computed and why it can't be faked without `JWT_SECRET`.

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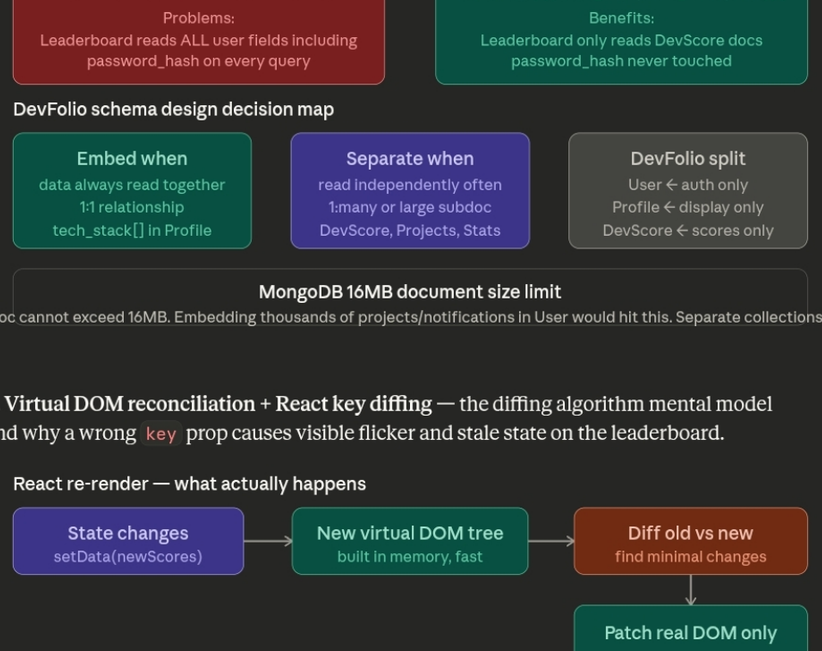
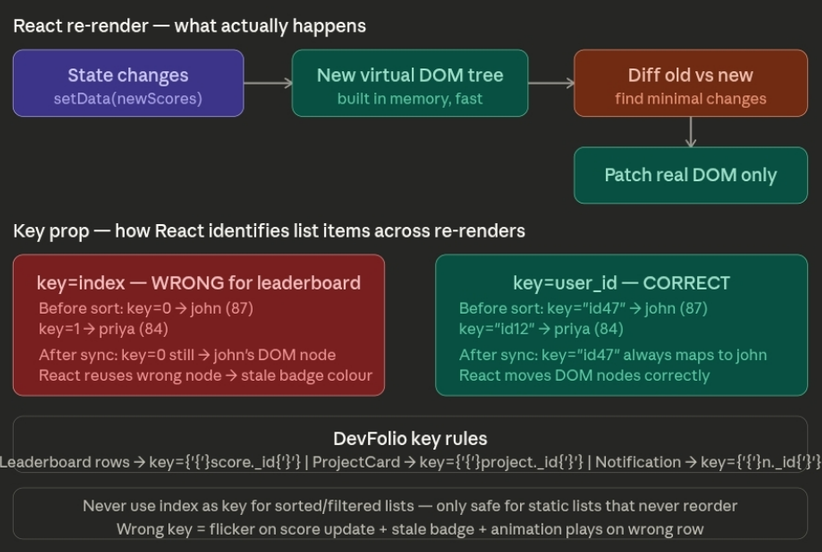
6. XSS attack vector — how injected script steals the JWT from localStorage, and why the `xss-clean` middleware and input sanitisation stop it.

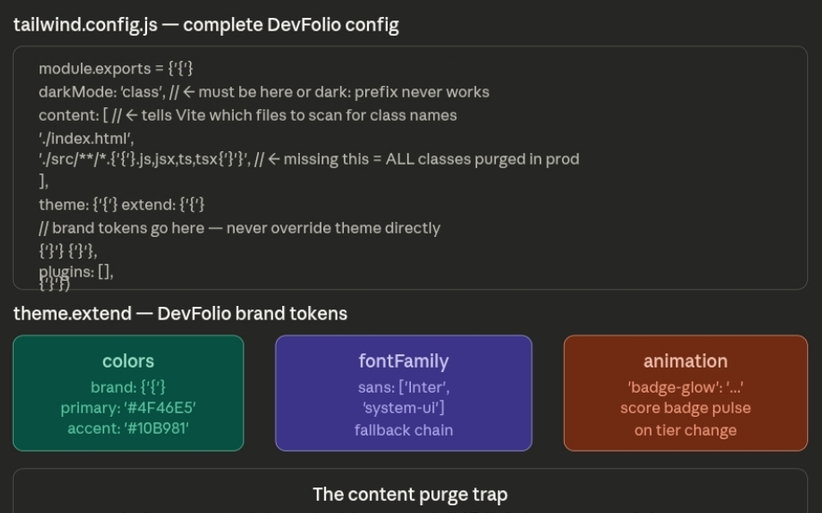


7. Normalisation vs denormalisation — why DevFolio keeps DevScore in a separate collection instead of embedding it in the User document.

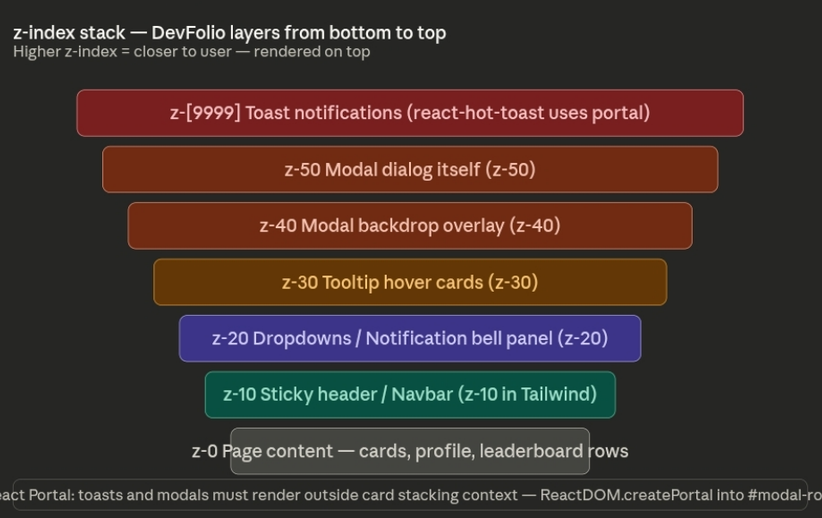
8. Virtual DOM reconciliation + React key diffing — the diffing algorithm mental model and why a wrong `key` prop causes visible flicker and stale state on the leaderboard.



9. Tailwind theme.extend + content array — the full config that must be open before writing a single custom class.



10. z-index layering system — navbar, modals, tooltips, and toasts all need explicit z-index tiers to avoid stacking fights.



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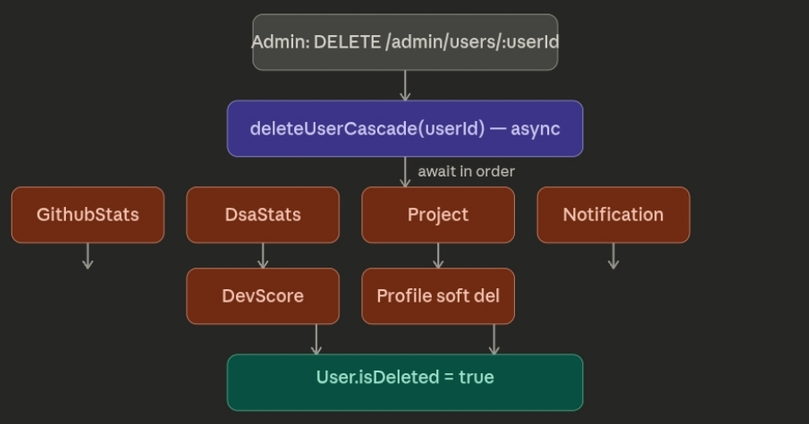
11. Account deactivation vs deletion cascade — the two-step user-facing flow and the 7-collection cascade confirmed as never diagrammed on page 47.

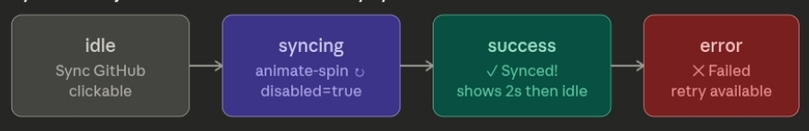


Soft delete vs hard delete rule per collection

Project, Notification — soft delete (isDeleted:true) — user may want to audit
DevScore, Stats — hard delete — recomputed anyway. User — soft delete — admin may need to audit

12. SyncButton idle / syncing / success / error states — the 4 states every sync button must handle.

SyncButton.jsx — 4 visual states driven by syncStatus state



State machine logic

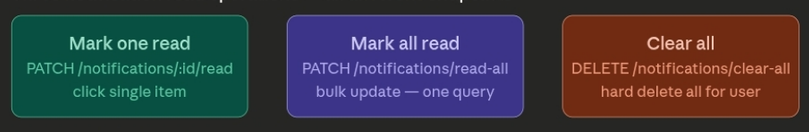
```
const [syncStatus, setSyncStatus] = useState('idle') // 'idle' | 'syncing' | 'success' | 'error'
const handleSync = async () => {
  setSyncStatus('syncing')
  try {
    await api.post('/api/stats/sync/github')
    setSyncStatus('success')
    setTimeout(() => setSyncStatus('idle'), 2000)
  } catch { setSyncStatus('error') }
}
```

StatsCard last_synced staleness display

```
const isStale = Date.now() - new Date(last_synced) > 24 * 60 * 60 * 1000
isStale ? <span class="text-amber-500">Last synced 3d ago</span> : <span>2h ago</span>
```

13. Notification mark-all-read — bulk PATCH pattern — confirmed absent from all prior diagrams.

Three notification read operations — all different endpoints



Bulk PATCH MongoDB query — the key pattern

```
await Notification.updateMany(
  {{} user_id: req.user.id, is_read: false {}}, // filter: only unread for THIS user
  {{} $set: {{} is_read: true {}} {}} // update: mark read
)
```

Frontend badge count update pattern

```
const markAllRead = async () => {{}
  await api.patch('/api/notifications/read-all')
  setUnreadCount(0) // instant UI update
  setNotifications(prev => // update local list
    prev.map(n => ({{} ...n, is_read: true {}}))
```

updateMany is a single MongoDB round-trip regardless of how many docs — never loop PATCH one-by-one

14. Structured clone + deep copy before editing — the pattern that prevents stale object mutation when editing project data.

Why editing project data needs a deep copy

Shallow copy — mutates original
const editData = project // same reference!
editData.title = "new" // mutates projects[i]
user cancels edit →
original project in list already changed
React re-render shows wrong data

structuredClone() — safe
const editData = structuredClone(project)
editData.title = "new" // separate object
user cancels edit →
original project in list untouched
reset: setEditData(structuredClone(project))

Three copy methods — which to use

{{} ...spread {}}
shallow only
nested arrays still shared
use for flat objects

JSON.parse/stringify
deep copy but loses Dates
functions, undefined
avoid for DevFolio

structuredClone()
true deep clone
preserves Dates, Arrays
built into browser

DevFolio places requiring deep copy

ProjectCard edit mode — clone before opening form | EditProfile — clone user before form pre-fill | tech_stack[] edits

15. PostCSS config + .nvmrc + package.json engines — three config files required by Tailwind in Vite, pinning Node version, and declaring minimum Node — all confirmed absent from all 45 prior diagrams.

Three config files every DevFolio repo needs — never mentioned in 45 prior diagrams

postcss.config.js — required by Tailwind in Vite

```
module.exports = {{}
plugins: {{}
  tailwindcss: {{}{}}, autoprefixer: {{}{}},
{}}
```

Without this file: Tailwind classes
not processed by Vite build pipeline
autoprefixer adds -webkit- prefixes
for Safari/iOS compatibility

.nvmrc — pin Node version, prevent "works on my machine" bugs

```
// .nvmrc in root — one line
18.18.0
// nvm use ← reads this automatically
```

teammate runs nvm use → exact version
no version mismatch in CI either
Vite requires Node 18+

package.json engines + scripts

```
"engines": {{}
  "node": ">=18.0.0"
{}}
"scripts": {{}
  "test:watch": "jest --watch",
```

engines enforced on npm install
npm ci fails in CI if wrong Node
test:coverage: jest --coverage
generates lcov report for
tracking % coverage per file

All three live in root of repo alongside package.json — commit all three — they're not secrets

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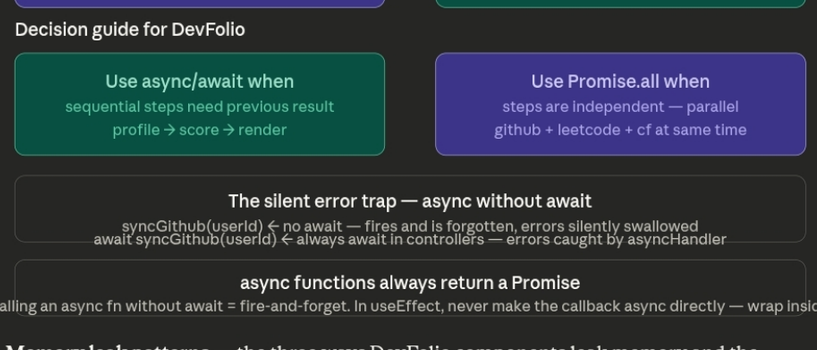
1. req.user injection — how auth middleware attaches the logged-in user to every request.

How does the controller know who is logged in?



2. Promise chaining vs async/await — when each reads cleaner and the one trap that silently swallows errors.

Same operation — two styles

```
Promise chain
fetchProfile(username)
.then(profile => fetchScore(profile._id))
.then(score => setData(score))
.catch(err => setError(err.message))
.finally(() => setLoading(false))
```

```
async / await
try {
  const profile = await fetchProfile(username)
  const score = await fetchScore(profile._id)
  setData(score)
} catch(err) { setError(err.message) }
```

Decision guide for DevFolio

Use async/await when
sequential steps need previous result
profile → score → render

Use Promise.all when
steps are independent — parallel
github + leetcode + cf at same time

The silent error trap — async without await

```
syncGithub(userId) ← no await — fires and is forgotten, errors silently swallowed
await syncGithub(userId) ← always await in controllers — errors caught by asyncHandler
```

async functions always return a Promise
calling an async fn without await = fire-and-forget. In useEffect, never make the callback async directly — wrap inside

3. Memory leak patterns — the three ways DevFolio components leak memory and the cleanup pattern for each.

Every useEffect that sets up something must tear it down

Leak 1 — setInterval without clear

```
useEffect(() => {
  setInterval(pollNotifications, 30000)
}, []) ← interval fires after unmount
```

Fixed — return cleanup

```
useEffect(() => {
  const id = setInterval(pollNotifications, 30000)
  return () => clearInterval(id)
}, [])
```

Leak 2 — fetch on unmounted component

```
// user navigates away
// fetch completes → setData() on dead component
// Warning: Can't perform state update
```

Fixed — AbortController

```
const ctrl = new AbortController()
axiosInstance.get(url, { signal: ctrl.signal })
return () => ctrl.abort()
```

Leak 3 — event listener not removed

```
useEffect(() => {
  document.addEventListener('keydown', fn)
}, []) ← fn survives in memory forever
```

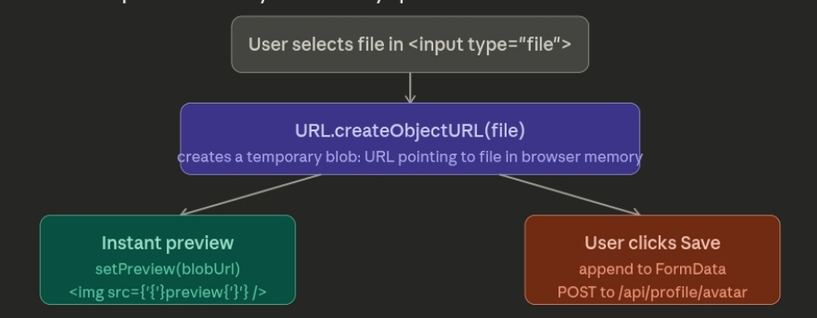
Fixed — remove on cleanup

```
useEffect(() => {
  document.addEventListener('keydown', fn)
  return () => document.removeEventListener('keydc
}, [])
```

Rule — return a cleanup function from every useEffect that sets up async work
DevFolio: Navbar polling (clearInterval), Profile fetch (abort), useOnClickOutside (removeEventListener)

4. Blob URL + URL.createObjectURL — local avatar preview before the file is uploaded to the server.

Show avatar preview instantly — before any upload



EditProfile.jsx — full pattern

```
const handleFileChange = (e) => {
  const file = e.target.files[0]
  setPreviewUrl(URL.createObjectURL(file)) // instant local preview
  setAvatarFile(file) // store for upload on Save
}
```

5. useLocation + scroll restoration on route change — two React Router v6 hooks every page that loads data needs.

useLocation — what it returns and when to use it

```
const location = useLocation()
// location.pathname → "/profile/john_doe"
// location.search → "?page=2&college=IIT"
// location.state → data passed via navigate('/login', { state: { from: '/dashboard' } })
```

DevFolio uses of useLocation

Navbar active highlight
isActive = location.pathname === '/dashboard'

Login redirect back
ProtectedRoute saves from in location.state

Settings active tab
location.pathname.includes ('/settings/profile')

Scroll restoration — scroll to top on every page navigation

```
// ScrollToTop.jsx — place once inside <BrowserRouter> in App.jsx
function ScrollToTop() {
  const { pathname } = useLocation()
  useEffect(() => { window.scrollTo(0, 0) }, [pathname])
  return null // renders nothing — side-effect only
}
```

Without this: navigating from bottom of leaderboard to Profile shows profile mid-scroll — confusing UX

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6. React 18 automatic rendering batching — why multiple `setState` calls inside a fetch no longer cause multiple re-renders.

Multiple setState calls in a fetch callback

React 17 — 3 separate re-renders
fetchProfile().then(data => {'{'}
setProfile(data.profile) // re-render 1
setScore(data.score) // re-render 2
setLoading(false) // re-render 3
{'}'}) ← 3 paints to DOM

React 18 — 1 re-render
fetchProfile().then(data => {'{'}
setProfile(data.profile) // queued
setScore(data.score) // queued
setLoading(false) // queued
{'}'}) ← 1 paint to DOM

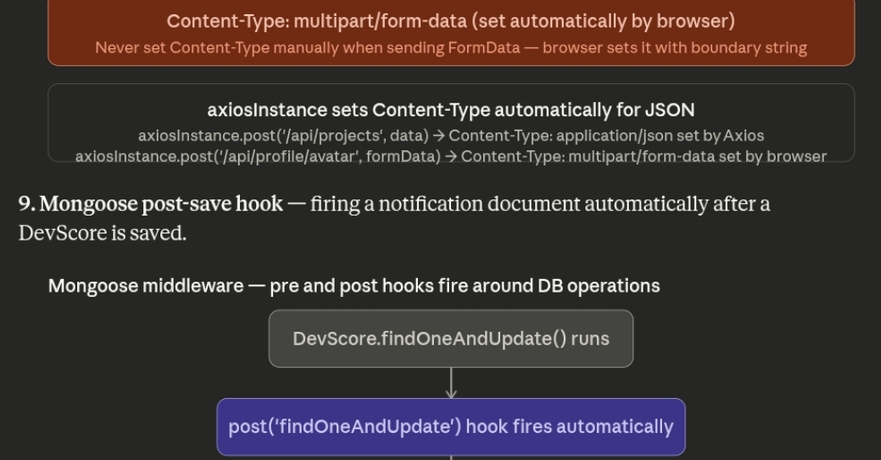
Automatic in React 18 — no code change needed
React batches ALL setState calls inside event handlers, setTimeout, fetch callbacks, and Promises

When to opt out — use flushSync (rare)

```
import {'{'} flushSync {'}'} from 'react-dom'
flushSync(() => setProfile(data)) // force immediate DOM update
// then read a DOM measurement immediately after
// DevFolio — almost never needed. Only for measuring DOM before next paint.
```

7. OAuth 2.0 authorization code flow — the full redirect dance between browser, GitHub, and backend confirmed absent across all prior sessions.

4 parties — Browser, Express, GitHub Auth server, GitHub API



?code=xxx is one-time use — exchanged for access_token server-side. Never expose GITHUB_CLIENT_SECRET to brows

8. Content-Type + Authorization header anatomy — the two request headers DevFolio sends on every API call and what breaks when they're wrong.

Every POST/PATCH from axiosInstance sends two headers

Content-Type: application/json
tells Express the body is JSON → express.json() parses it into req.body
missing this → req.body is undefined → every field validation fails

Authorization: Bearer eyJhbGc...<token>
Bearer = token type. Space required. auth middleware splits on ' ' to get token.
missing Bearer prefix → split()[1] is undefined → jwt.verify throws → 401

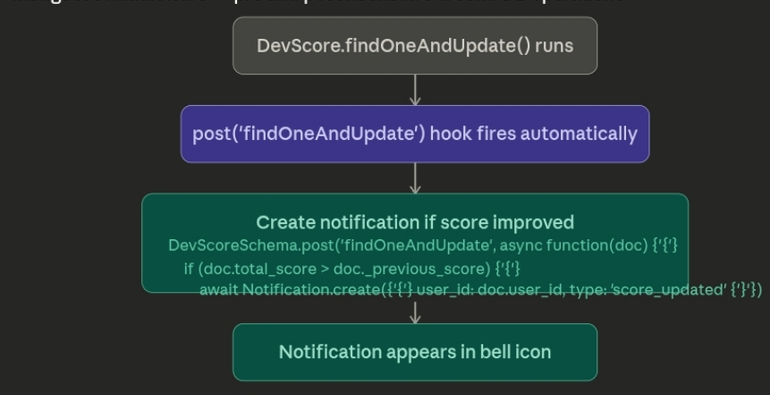
Multipart exception — avatar upload uses different Content-Type

Content-Type: multipart/form-data (set automatically by browser)
Never set Content-Type manually when sending FormData — browser sets it with boundary string

axiosInstance sets Content-Type automatically for JSON
axiosInstance.post('/api/projects', data) → Content-Type: application/json set by Axios
axiosInstance.post('/api/profile/avatar', formData) → Content-Type: multipart/form-data set by browser

9. Mongoose post-save hook — firing a notification document automatically after a DevScore is saved.

Mongoose middleware — pre and post hooks fire around DB operations



Hook types DevFolio uses
pre('save') — hash password before saving User
post('findOneAndUpdate') — create notification after score update
{ new: true } must be passed — hook only receives updated doc if this option is set

10. Intl.NumberFormat + number formatting utility — displaying commit counts, scores, and currency correctly across all DevFolio screens.

client/src/utils/format.js — one file for all formatting

```
// Commit count: 1312 → "1,312"
export const formatCount = (n) => new Intl.NumberFormat('en-IN').format(n)
// Score: 74.5678 → "74.57"
export const formatScore = (n) => Number(n ?? 0).toFixed(2)
// Relative time: "2 days ago"
export const timeAgo = (d) => new Intl.RelativeTimeFormat('en', {'{'} numeric:'auto'{'}'}).format(...)
```

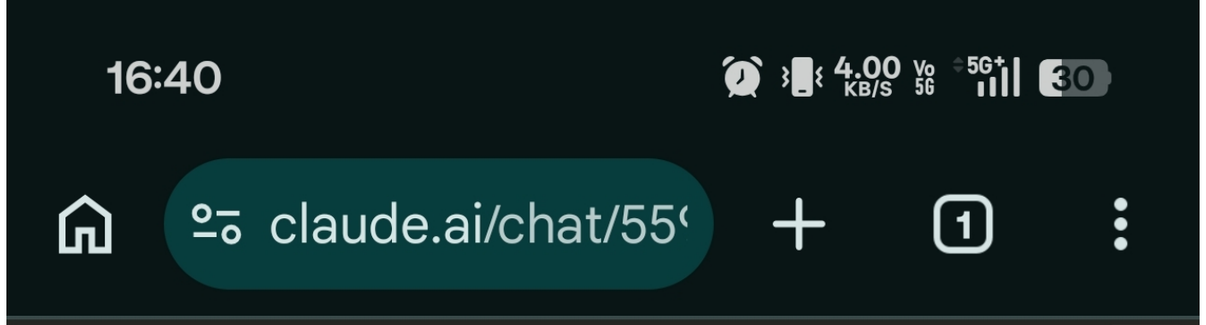
Where each formatter is used in DevFolio

StatCard
GitHub commits
312 → "312"
1312 → "1,312"

ScoreBadge
74.5678 → "74.57"
never raw floats
in the UI

Last synced
ISO date → "2h ago"
profile page
notifications

The JS float trap — always format before display
7 * 1.1 = 7.700000000000001 — raw float in UI looks broken
formatScore(7 * 1.1) = "7.70" — always wrap numbers before rendering



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11. ACID properties + MongoDB session transactions — what the 4 properties guarantee and exactly where DevFolio needs them.

Atomic
all ops succeed or all roll back

Consistent
DB valid before and after

Isolated
concurrent txns don't interfere

Durable
committed data survives crash

DevFolio — where transactions are needed

Registration — 3 creates must be atomic
User.create() + Profile.create() + DevScore.create() — all or none

Username change — cascade
User + Profile + JWT re-issue all in one session

MongoDB session transaction pattern

```
const session = await mongoose.startSession()
session.startTransaction()
try {{}
  await User.create([userDoc], {{} session {}})
  await Profile.create([profileDoc], {{} session {}})
  await session.commitTransaction()
{}} catch {{} await session.abortTransaction() {}}
```

12. Model.exists() + estimatedDocumentCount() — two cheap Mongoose checks: duplicate guard on register and fast landing page stats.

Model.exists() — check before insert, no document returned

Wasteful — fetches full document

```
const existing = await User.findOne({{} email {}})
if (existing) return 409
// transfers all fields over network
```

Efficient — returns {{} _id {}} or null

```
const exists = await User.exists({{} email {}})
if (exists) return 409
// only _id transferred — 10x smaller
```

s: register (email + username dupe check) | score recompute (check DevScore exists before creating) | shortlist (check all

estimatedDocumentCount() — instant landing page stats

countDocuments() — slow
scans index — accurate but O(n)

estimatedDocumentCount() — instant
reads metadata — O(1) always

GET /api/stats/public — landing page numbers

```
const [devCount, collegeCount, avgScore] = await Promise.all([
  User.estimatedDocumentCount(), // "X developers"
  Profile.distinct('college').then(c => c.length), // "Y colleges"
  DevScore.aggregate([{ $group: { _id: null, avg: { $avg: '$total_score' {}} {}} {}}])
```

13. Tailwind animate-spin + animate-pulse + transition — the three animation utilities for loading spinners, skeleton screens, and hover effects.

Three animation utilities — when and where to use each

animate-spin
continuous rotation
SyncButton loading
form submit spinner

animate-pulse
fade in and out
skeleton loading cards
StatCard placeholders

transition + duration
smooth state changes
hover, focus, dark mode
colour/opacity shifts

Skeleton card pattern — mirrors real content layout

```
// StatCard skeleton — same dimensions as real card
<div class="animate-pulse bg-gray-200 dark:bg-gray-700 rounded-lg h-24 w-full" />
// Profile avatar skeleton
<div class="animate-pulse bg-gray-200 rounded-full h-16 w-16" />
```

Transition pattern for button hover and dark mode toggle

```
// button — smooth background shift on hover
className="bg-white hover:bg-indigo-50 transition-colors duration-200"
// dark mode toggle — smooth page colour shift
className="bg-white dark:bg-gray-900 transition-colors duration-300"
```

ransition with a specific property: transition-colors, transition-opacity, transition-transform — not just transition (animate

14. Array sort stability + leaderboard sort — why a naïve sort breaks ties arbitrarily and the stable two-key sort pattern.

Leaderboard rank assignment in JS after MongoDB query

Single-key sort — unstable ties

```
scores.sort((a,b) => b.total_score - a.total_score)
// priya 74.2 john 74.2 — same score
// order is non-deterministic — changes
// on every re-render or sync
```

Two-key sort — stable

```
scores.sort((a,b) =>
  b.total_score - a.total_score // primary
  || a.username.localeCompare(b.username) // tiebrea
  )
```

MongoDB .sort({{} total_score: -1 {}}) does the same — always add a tiebreaker
DevScore.find().sort({{} total_score: -1, username: 1 {}}) ← secondary sort prevents flickering

Rank assignment pattern

```
const ranked = scores.map((s, i) => ({{} ...s, rank: i + 1 {}}))
// rank computed in JS after sort — not stored in DB
// because rank changes every sync — storing would require updating all N rows
```

Array.prototype.sort is stable in all modern browsers (ES2019+)
Items with equal sort keys keep their original relative order — safe to rely on in DevFolio

15. ProjectCard three-state UI — view / edit / delete-confirm states on the same card, confirmed absent from all prior session diagrams.

ProjectCard.jsx — one component, three visual states driven by cardMode

view (default)
title + stack tags
repo link + stars
✏ edit 🗑 delete
(owner only)

→

edit
pre-filled form fields
title / stack / links
Save / Cancel buttons
PATCH /api/projects/:id

→

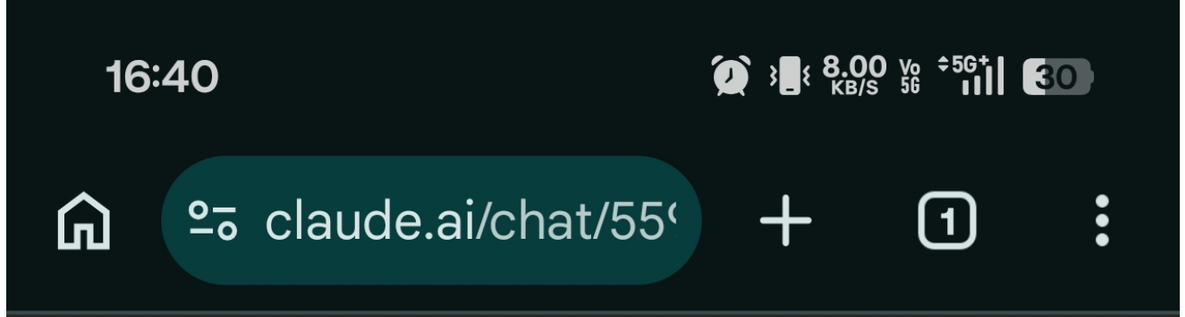
confirm-delete
"Delete this project?"
Yes / Cancel buttons
DELETE /projects/:id
owner check on server

State machine in ProjectCard.jsx

```
const [cardMode, setCardMode] = useState('view') // 'view' | 'edit' | 'confirm-delete'
const [editData, setEditData] = useState(structuredClone(project)) // safe copy
// click edit → setCardMode('edit') + pre-fill editData
// click delete → setCardMode('confirm-delete') — no API call yet
// click cancel from any mode → setCardMode('view') + reset editData
```

Props needed — what Dashboard passes down

```
project={{}obj{}} — the project data to display
editable={{}boolean{}} — show edit/delete buttons (owner view)
onDelete={{}fn{}} — Dashboard removes card from list on success
onEdit={{}fn{}} — Dashboard updates card in list on success
```



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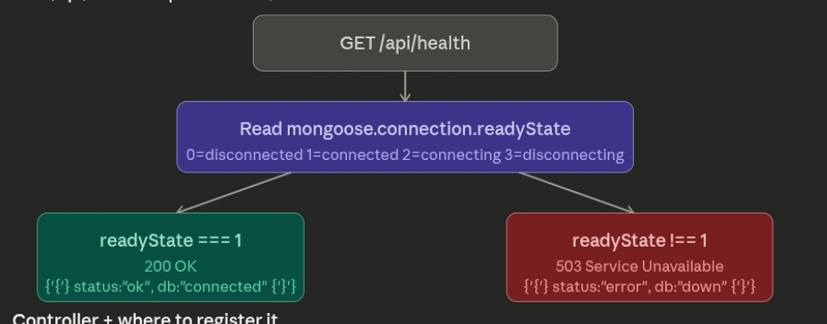
1. Health check endpoint — `GET /api/health` showing `mongoose.connection.readyState` — confirms server and DB are both alive.

GET /api/health — public route, no auth needed



Controller + where to register it

```
// health.controller.js
const dbState = mongoose.connection.readyState === 1 ? 'connected' : 'down'
res.status(dbState === 'connected' ? 200 : 503).json({'{'}
  status: dbState === 'connected' ? 'ok' : 'error',
  db: dbState, uptime: process.uptime() + 's'
```

2. Mongoose static vs instance methods — model-level utility functions vs document-level helpers, with DevFolio-specific examples for each.

Static method — called on the Model
User.findByEmail(email) ← User is the class

Instance method — called on a document
user.comparePassword(input) ← user is one doc

DevFolio static methods

```
UserSchema.statics.findByEmail = async function(email) {'{'}
  return this.findOne({'{'} email {'}'}).select('+password')
{'}'}
// Usage in auth.controller.js:
const user = await User.findByEmail(req.body.email)
```

DevFolio instance methods

```
UserSchema.methods.comparePassword = async function(candidate) {'{'}
  return bcrypt.compare(candidate, this.password_hash)
{'}'}
// Usage:
const ok = await user.comparePassword(req.body.password)
```

Decision rule
static = needs to FIND multiple docs (query) | instance = operates ON one existing doc (mutation / comparison)

DevScore static method — model-level recompute for one user

```
DevScoreSchema.statics.recomputeForUser = async function(userId) {'{'}
  const [github, dsa, projects] = await Promise.all([...])
  const score = computeWeightedScore(github, dsa, projects)
  return this.findOneAndUpdate({'{'} user_id: userId {'}'}, {'{'} $set: score {'}'}, {'{'} upsert:true, new:true {'}'})
```

3. useSearchParams hook — reading and writing URL query params for recruiter search filters, leaderboard pagination, and shareable links.

useState only — filter lost on refresh

```
const [college, setCollege] = useState('')
// refresh → filter resets
// URL stays /recruiter/search — not shareable
```

useSearchParams — survives refresh

```
const [params, setParams] = useSearchParams()
const college = params.get('college') || ''
// URL: /recruiter/search?college=IIT+Hyderabad
```

3 operations in RecruiterSearch.jsx

```
// 1. Read initial values FROM URL on mount
const college = searchParams.get('college') || ''
const minScore = Number(searchParams.get('minScore')) || 0
// 2. Write TO URL when filter changes
setSearchParams({'{'} college, minScore, tech {'}'}) // replaces all params at once
```

useEffect watches searchParams — fires API call whenever URL changes

```
useEffect(() => {'{'}
  fetchLeaderboard({'{'} college, minScore, tech {'}'}) // re-fetch on every param change
{'}'}, [searchParams]) // searchParams as dependency — not the individual values
```

Three DevFolio pages using useSearchParams
RecruiterSearch → ?college=IIT&minScore=60&tech=React (filter state)
Leaderboard → ?page=2&college=NIT (pagination + filter)
AdminUsers → ?role=developer&page=1 (admin filter)

4. Short-circuit `&&` and ternary rendering in JSX — the two conditional rendering patterns used on every DevFolio page, with the falsy zero trap.

Pattern 1 — short-circuit && (show or show nothing)

```
{'{'} isOwner && <EditButton /> {'}'}
{'{'} notifications.length > 0 && <Badge /> {'}'}
{'{'} user?.role === 'admin' && <AdminPanel /> {'}'}
```

The falsy 0 trap

```
{'{'} projects.length && <List /> {'}'}
// 0 projects → renders "0" on screen!
```

Fix: {'{'} projects.length > 0 && <List /> {'}'} or {'{'} !!projects.length && <List /> {'}'}

Pattern 2 — ternary (show A or show B)

```
{'{'} loading ? <Spinner /> : <ProfileData /> {'}'}
{'{'} isAuthenticated ? <Dashboard /> : <LoginPrompt /> {'}'}
{'{'} score >= 70 ? 'Platinum' : score >= 50 ? 'Gold' : 'Bronze' {'}'}
```

Pattern 3 — early return (cleanest for complex conditions)

```
function Profile({'{'} username {'}'}) {'{'}
  if (loading) return <Skeleton />
  if (error) return <ErrorCard message={'{'}error.message{'}'} />
  if (!profile) return <NotFound />
  return <ProfilePage data={'{'}profile{'}'} /> // happy path last
```

5. REST idempotency — GET is safe, POST is not, PUT/PATCH/DELETE are idempotent — the fundamental that explains why sync retries are safe.

Safe = no side effects on server | Idempotent = same result on repeat calls

GET	PUT	PATCH	DELETE	POST
read only	full replace	partial update	remove	create
✓ Safe	✗ Not safe	✗ Not safe	✗ Not safe	✗ Not safe
✓ Idempotent	✓ Idempotent	✓ Idempotent	✓ Idempotent	✗ NOT

Why idempotency matters for DevFolio sync

Sync is safe to retry
POST /api/stats/sync/github
calls findOneAndUpdate(upsert:true)

Register is NOT safe to retry
POST /api/auth/register
double-click → 2 accounts (use exists check)

DevFolio POST/sync pattern — made idempotent by upsert
thubStats.findOneAndUpdate({'{'} user_id {'}'}, {'{'} $set: stats {'}'}, {'{'} upsert:true {'}'}) — safe to call 100 times, same res

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6. Base64url encoding — why JWT parts look the way they do, and why it is encoding not encryption.

Base64url — text-safe encoding of binary data, NOT encryption

JSON object
{"id":"abc", "role":"dev"}

Base64url encode
eyJpZCI6ImFiYyIsInJvbGUiOiJkZXYif

In JWT payload
readable by anyone!

Base64 vs Base64url — one character difference
Base64 uses + and / — breaks in URLs. Base64url replaces + → - and / → _ — URL safe

Decode in browser — no library needed

```
const payload = JSON.parse(atob(token.split('.')[1]))
// atob() decodes Base64. split('.')[1] = payload segment
// Result: { id: "abc", role: "developer", exp: 1234567890 }
```

DevFolio use — decode token to read expiry for session warning

```
const token = localStorage.getItem('token')
const { exp } = JSON.parse(atob(token.split('.')[1]))
const msUntilExpiry = (exp * 1000) - Date.now() // exp is in seconds — multiply by 1000
```

NEVER store sensitive data (password, payment info) in JWT payload — it is readable by anyone with the token

7. Index cardinality — high vs low cardinality fields and which to index, explained with DevFolio's actual collections.

Cardinality = number of distinct values in a field

High cardinality — great for index
many distinct values — index is selective
email — unique per user
username, user_id, _id

Low cardinality — poor index
few distinct values — not selective
role — only 4 values (dev/rec/ment/adm)
isDeleted — only true/false

ow-cardinality index wastes disk — MongoDB reads half the collection anyway, so the index adds overhead without helpin

DevFolio index decision per field

✓ Index
email unique
username unique

✓ Index
user_id on Profile
DevScore, Project

✓ Index desc
total_score desc
leaderboard sort

✗ Skip index
role (4 values)
isDeleted, is_read

Compound index — college leaderboard query needs two fields
ProfileSchema.index({ college: 1 }) — filter by college first
DevScoreSchema.index({ user_id: 1, total_score: -1 }) — lookup then sort in one pass

8. API versioning strategy — `/api/v1` prefix and why it matters once recruiters integrate your API.

URL versioning — the simplest and most common strategy

Without versioning
GET /api/profile/:username
change response shape → breaks recruiters

With versioning
GET /api/v1/profile/:username
v2 can reshape — v1 unchanged

app.js mounting with version prefix

```
const v1 = express.Router()
v1.use('/auth', authRoutes)
v1.use('/profile', profileRoutes)
app.use('/api/v1', v1) // all routes under /api/v1 — one line change if needed
```

DevFolio MVP — start with /api/v1 from day one

Benefits for DevFolio
recruiters can build integrations
safe to refactor internals
clear in API docs + Postman

When to bump to v2
breaking change in response shape
remove or rename a field
both versions run simultaneously

9. Supertest + Jest test database setup — spinning up Express without a running server, `MONGO_URI_TEST`, and DB cleanup between tests.

Supertest — makes HTTP requests to Express without starting a server on a port

Without supertest
start real server on port
send HTTP via axios — slow, port conflicts

With supertest
request(app).post('/api/auth/register')
no port needed — in-memory pipe

Test file structure — auth.test.js

```
import request from 'supertest'
import app from '../app' // export app without app.listen()
import mongoose from 'mongoose'
beforeAll(async () => {
  await mongoose.connect(process.env.MONGO_URI_TEST) // .env.test
});
afterEach(async () => {
  await User.deleteMany({}) // wipe test data between tests
});
afterAll(async () => { await mongoose.connection.close() });
```

Critical: export app WITHOUT listen()
app.js: export default app (no app.listen)
index.js: import app; app.listen(PORT) — only this file starts the server

10. `@apply` directive + CSS specificity in Tailwind — extracting repeated class combos and overriding framework styles without `!important`.

@apply — extract repeated Tailwind classes into a CSS class

Repeated in every button
className="px-4 py-2 rounded-lg font-medium
bg-indigo-600 text-white hover:bg-indigo-700
transition-colors duration-200 disabled:opacity-50"

Extracted with @apply
/* index.css */
.btn-primary {
  @apply px-4 py-2 rounded-lg font-medium

Use sparingly — only for components repeated 5+ times. Prefer JSX component extraction for React apps.

CSS specificity in Tailwind — why your custom styles lose

Custom CSS loses to Tailwind
/* your CSS */
.card { border-radius: 8px }
// Tailwind rounded-none wins — loaded later

Solution — Tailwind wins, use it
Add custom values in theme.extend
OR use arbitrary values: rounded-[8px]
avoid mixing custom CSS with Tailwind

Arbitrary value syntax — DevFolio uses

ScoreBadge ring: ring-[3px] ring-offset-[2px]
Custom z-index: z-[9999] (toasts above everything)
Custom height: h-[calc(100vh-64px)] (main below fixed navbar)
Exact colour: bg-[#4F46E5] (brand purple not in Tailwind defaults)

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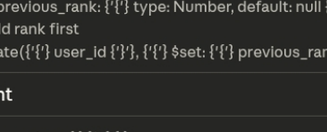
Exact colour: bg-[#4F46E5] (brand purple not in Tailwind defaults)

11. Tooltip component — hover card above stats numbers in StatCard and LeaderboardRow — the CSS-only pattern with no library needed.

Tooltip anatomy — position:relative parent + position:absolute child



Tailwind implementation — group + group-hover

```
function Tooltip({{ text, children }}) {{
  return (
    <div className="relative group inline-block">
      {{ children }}
      <span className="absolute bottom-full left-1/2 -translate-x-1/2 mb-2
          hidden group-hover:block ← appears on parent hover
          px-2 py-1 text-xs bg-gray-900 text-white rounded whitespace-nowrap">
        {{ text }}
      </span>
    </div>
  )
}}
```

DevFolio tooltip usage

```
StatCard: <Tooltip text="Push events last 90d"><span>312</span></Tooltip>
ScoreBadge: <Tooltip text="Rank: #4 globally"><ScoreBadge /></Tooltip>
```

12. LeaderboardRow rank change indicator — the up/down arrow visual state that shows whether a developer moved up or fell since last sync.

LeaderboardRow — four rank change states

Where previous_rank comes from

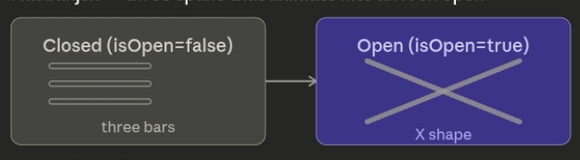
```
// DevScore schema — add field: previous_rank: {{ type: Number, default: null }}
// Before saving new rank: store old rank first
await DevScore.findOneAndUpdate({{ user_id }}, {{ $set: {{ previous_rank: doc.rank, rank: newRank }} }})
```

RankArrow helper component

```
function RankArrow({{ current, previous }}) {{
  if (!previous) return <span className="text-purple-500 text-xs">✦ new</span>
  const diff = previous - current // positive = moved up (lower number = better rank)
  if (diff > 0) return <span className="text-green-500">▲ {{diff}}</span>
  if (diff < 0) return <span className="text-red-500">▼ {{Math.abs(diff)}}</span>
  return <span className="text-gray-400">—</span>
}}
```

13. Navbar mobile hamburger — open/close animation and the full responsive pattern with CSS transforms.

Navbar.jsx — three spans that animate into an X on open

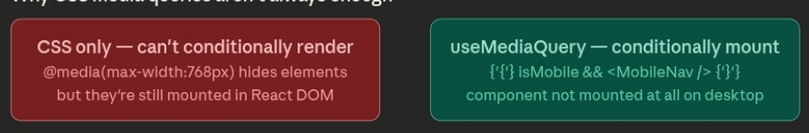


Tailwind hamburger implementation

```
const [isOpen, setIsOpen] = useState(false)
// Each span transitions with rotate/translate transform
<button onClick={() => setIsOpen(o => !o)} className="md:hidden">
  <span className={{`block h-0.5 w-6 bg-current transition-all duration-300,
    isOpen ? 'rotate-45 translate-y-[7px]' : 'mb-1.5'}}`}} /> {/* top bar */}
  <span className={{`block h-0.5 w-6 bg-current transition-all duration-300,
    isOpen ? 'opacity-0' : ''}}`}} /> {/* middle bar fades */}
  <span className={{`block h-0.5 w-6 bg-current transition-all duration-300,
    isOpen ? '-rotate-45 -translate-y-[7px]' : 'mt-1.5'}}`}} /> {/* bottom bar */}
</button>
<div className={{`md:hidden overflow-hidden transition-all duration-300', isOpen ? 'max-h-80' : 'max-h-0'}}`}}>
```

14. `useMediaQuery` custom hook — responding to breakpoint changes in JS for logic that can't be handled by CSS alone.

Why CSS media queries aren't always enough



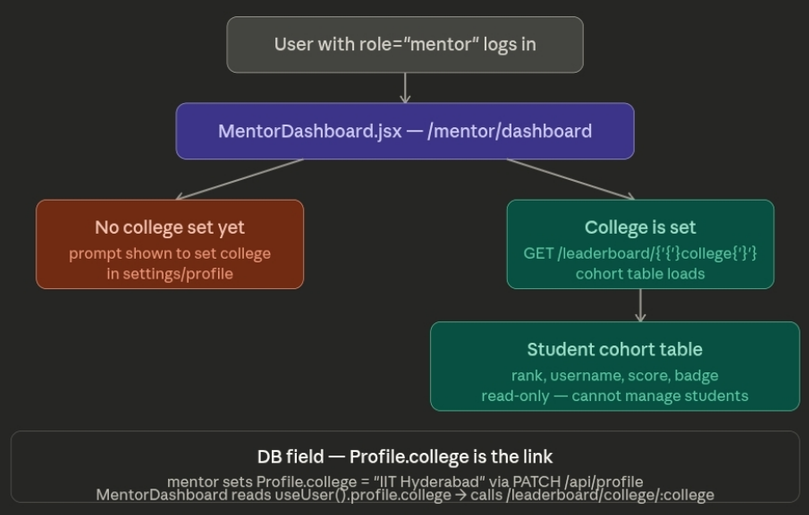
Hook implementation

```
function useMediaQuery(query) {{
  const [matches, setMatches] = useState(
    () => window.matchMedia(query).matches
  )
  useEffect(() => {{
    const mql = window.matchMedia(query)
    const handler = e => setMatches(e.matches)
    mql.addEventListener('change', handler)
    return () => mql.removeEventListener('change', handler)
  }}, [query])
  return matches
}}
```

DevFolio breakpoints matching Tailwind's defaults

```
const isMobile = useMediaQuery('(max-width: 767px)') // below md
const isTablet = useMediaQuery('(max-width: 1023px)') // below lg
// Used in: Leaderboard (collapse columns), RecruiterSearch (stack filters), Dashboard (hide chart labels)
```

15. Mentor linking college affiliation flow — the mentor setup flow showing how a mentor links their college and what changes in the DB and UI.



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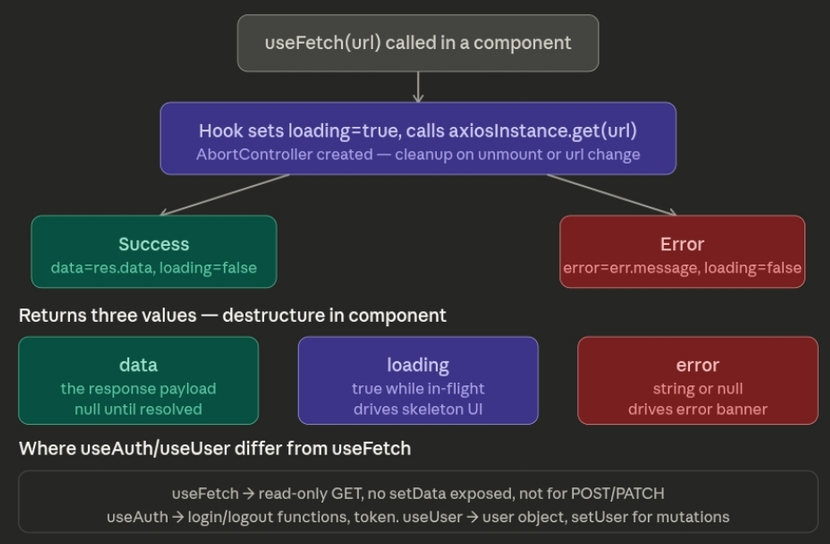
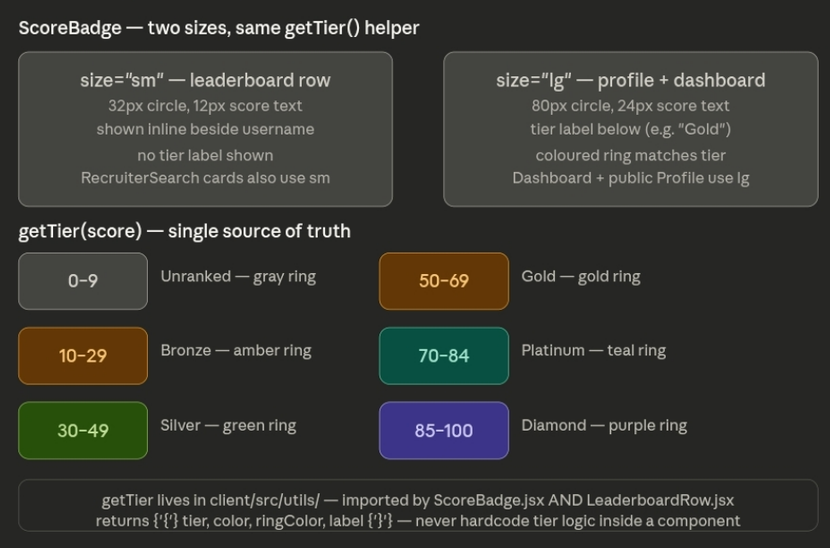
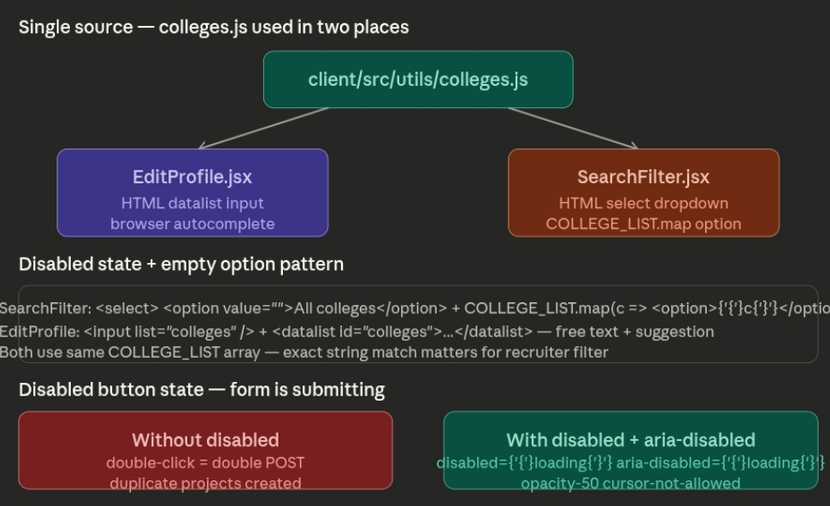
1. useFetch custom hook — the reusable data-fetching hook every page in DevFolio uses. Wraps `loading / error / data` into one import.



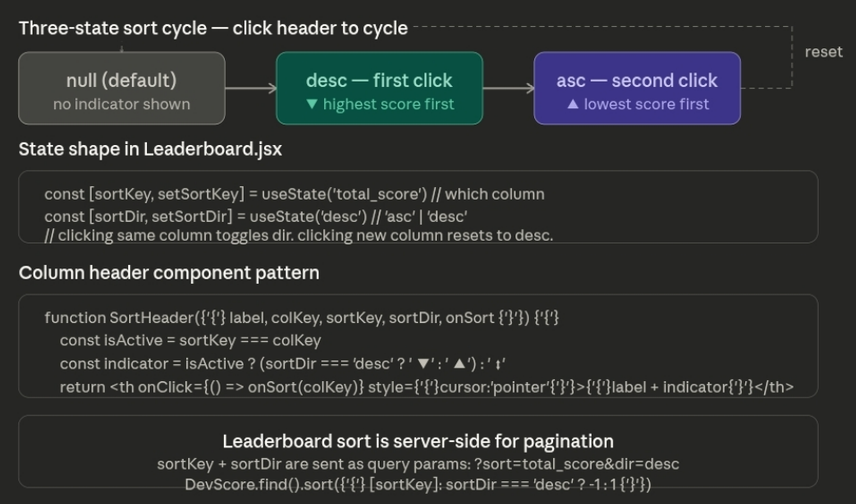
2. ScoreBadge size variants — the `size="sm"` leaderboard row vs `size="lg"` profile page visual, plus the `getTier()` helper map every badge uses.



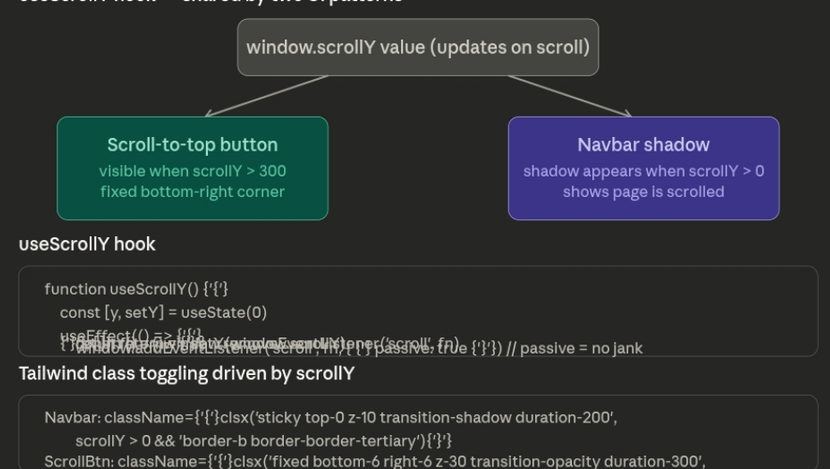
3. Dropdown / custom college select — SearchFilter's `<select>` backed by a single `COLLEGE_LIST` constant used in both EditProfile and RecruiterSearch.



4. Sort column headers — ascending/descending indicator for the leaderboard table, and the two-click cycle every beginner struggles to implement.



5. Scroll-to-top button + sticky header shadow — two scroll-position patterns driven by the same `useScrollY` hook.



6. `aspect-ratio` + `focus-visible` Tailwind patterns — square avatars without height hacks, and keyboard-only focus ring that doesn't appear on mouse click.

aspect-ratio — square avatars without hardcoding height

Without aspect-ratio	With aspect-ratio
w-16 h-16 — works until image is non-square — gets squished	w-16 aspect-square object-cover always square, image cropped neatly

Avatar circle: w-16 aspect-square rounded-full object-cover overflow-hidden
Recharts PieChart thumbnail: w-32 aspect-square (no height needed)

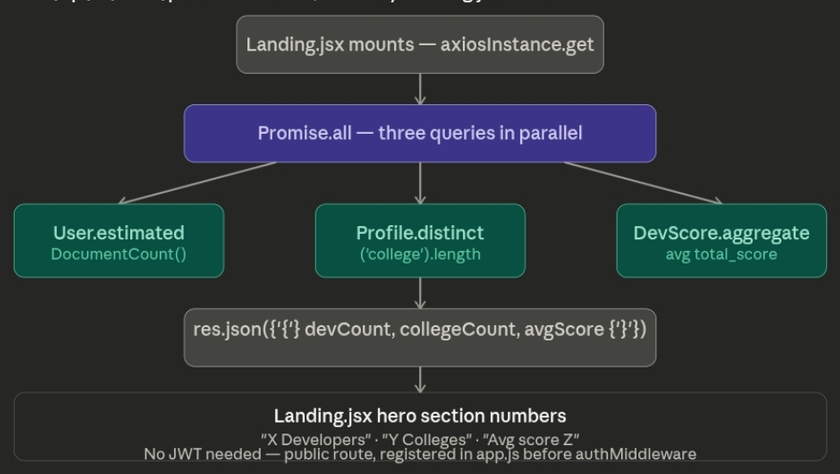
focus-visible vs focus — keyboard ring only

focus: ring — ugly	focus-visible: ring — clean
ring shows on mouse click too users see ring when clicking buttons	ring only on keyboard Tab navigation mouse clicks show no ring

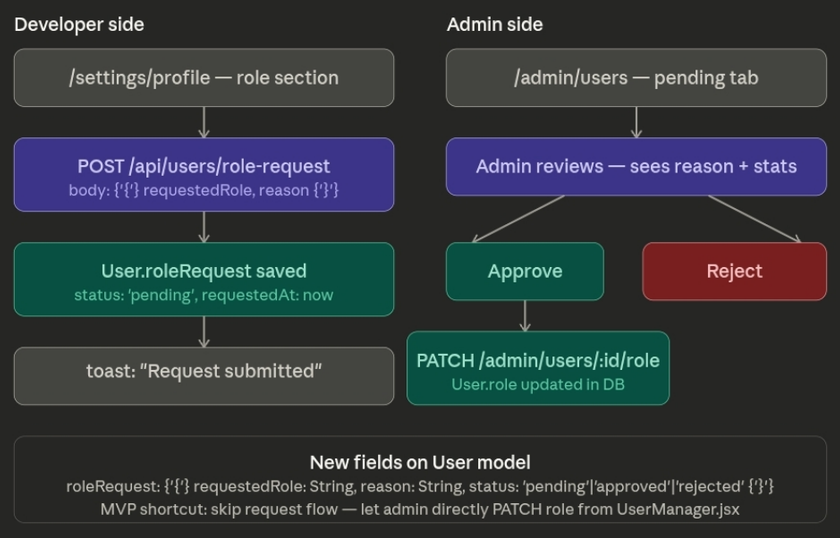
DevFolio button pattern — both accessibility + clean UI
className="... focus:outline-none focus-visible:ring-2 focus-visible:ring-indigo-500"
Apply to: all buttons, NavLink items, input fields, ProjectCard clickable area

7. GET /api/stats/public endpoint — the unauthenticated landing page counter: how three DB calls become the hero section numbers.

GET /api/v1/stats/public — no auth, called by Landing.jsx on mount



8. POST /api/users/role-request flow — a developer requesting recruiter or mentor role, and the admin approval path.



9. Optimistic vs pessimistic locking — concurrent DevScore updates and which strategy DevFolio should use.

The problem — two syncs fire simultaneously for the same user

Pessimistic locking	Optimistic locking
lock the row — others wait MongoDB has no native row lock requires session transactions — complex	let both proceed — last write wins safe for DevFolio — scores are recomputed from same raw data

Why last-write-wins is safe for DevFolio scores

Manual sync at 10:00 and cron sync at 10:00 both run computeScore(userId)
Both read same GithubStats → compute same result → write same score
Idempotent by design — last write lands the same value, no data loss

Version field pattern — add if needed later

DevScoreSchema: __v: Number (Mongoose adds this automatically)
To enforce optimistic lock manually:
DevScore.findOneAndUpdate({'{'} user_id, __v: currentVersion {'}'}, {'{'} $set: newScore, $inc: {'{'} __v:1 {'}'} {'}'})
Returns null if __v changed → retry. DevFolio MVP: skip this — upsert is sufficient.
Add versioning only if concurrent writes produce wrong scores in production.

10. Three linting config files — `.eslintrc.cjs`, `.prettierrc`, and `.editorconfig` with the exact rules DevFolio needs.

.eslintrc.cjs — React + Hooks rules

```
module.exports = {'{'}
  extends: ['eslint:recommended',
    'plugin:react/recommended',
    'plugin:react-hooks/recommended'],
  rules: {'{'} 'no-unused-vars': 'warn',
    'react/prop-types': 'off' {'}'} {'}'}
```

react/recommended — missing key, hooks order
react-hooks/recommended — exhaustive-deps
no-unused-vars warn not error — less noise
prop-types off — TypeScript handles this

.prettierrc — formatting rules

```
{'{'}
  "semi": false,
  "singleQuote": true,
  "tabWidth": 2,
{'}'}"trailingComma": "es5"
```

semi: false — no semicolons (team preference)
singleQuote: true — consistent with JS idiom
tabWidth: 2 — standard for JS projects
trailingComma es5 — safer git diffs

.editorconfig — cross-editor consistency

```
root = true
[*]
  indent_style = space
  indent_size = 2
  end_of_line = lf
```

Applies to VS Code, WebStorm, vim
Prevents Windows CRLF line endings from breaking git diffs and ESLint
Respected before Prettier runs

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11. Markdown renderer for bio — `react-markdown` + `rehype-sanitize` preventing XSS while allowing headings and links in developer bios.

Bio stored as raw Markdown string in MongoDB

Profile.bio = "# Hello\nI build **React** apps"

react-markdown + rehype-sanitize
parses Markdown → strips script tags → safe HTML

Allowed tags
h1-h3, p, strong, em, a, ul, li

Stripped tags
script, iframe, img, on* attrs

tailwindcss/typography plugin — prose class styles the output

```
plugins: [require('@tailwindcss/typography')] ← add to tailwind.config.js
<div className="prose dark:prose-invert max-w-none">
  <ReactMarkdown rehypePlugins={[rehypeSanitize]}>{bio}</ReactMarkdown>
```

max-w-none removes the default prose width cap — bio fills the card width
dark:prose-invert flips text/heading colours for dark mode automatically

12. `useId` + `useForwardRef` **accessible form pattern —** label-input pairing and passing refs to custom input components.

useId — generates unique id for label-input pairing

Hardcoded id — breaks with multiple forms
<label htmlFor="email"> + <input id="email">
two forms on page = duplicate id

useId — always unique
const id = useId() → ":r0:" ":r1:"
stable across renders, SSR-safe

```
function FormField({label, ...props}) {
  const id = useId()
  return <> <label htmlFor={id}>{label}</label> <input id={id} {...props} /> </>
```

forwardRef — pass ref from parent to custom input

Without forwardRef
ref on custom component = null
can't auto-focus custom input

With forwardRef
ref passed through to real input
inputRef.current.focus() works

```
const Input = forwardRef(function Input(props, ref) {
  return <input ref={ref} {...props} className="border rounded px-3 py-2 w-full" />
})  ← wrap any reusable input/button that needs a ref
```

DevFolio: auto-focus username field on /register, focus bio on "Edit profile" click — both need forwardRef

13. `usePrevious` + `useId` **— tracking last value and print stylesheet —** `usePrevious` for rank comparison, and the `@media print` CSS that lets recruiters print a clean profile.

usePrevious — compare current vs previous value

```
function usePrevious(value) {
  const ref = useRef() // useRef persists without causing re-render
  useEffect(() => { ref.current = value }) // runs AFTER render — stores PREV value
  return ref.current // returns value from BEFORE this render
```

DevFolio use: rank change
const prevRank = usePrevious(rank)
diff = prevRank - rank → arrow dir

DevFolio use: score flash
const prevScore = usePrevious(score)
score changed → pulse animation

Print stylesheet — @media print in index.css

```
@media print {
  .no-print { display: none !important } /* Navbar, Sync btn, Edit btn */
  body { font-size: 11pt; color: #000 } /* force black text */
}  a::after { content: " (" attr(href) ")" } /* show URLs */
```

className="no-print" — what to hide when printing
Navbar, edit/sync buttons, ScoreBadge ring animation, notification bell
What stays: avatar, bio, tech stack, stats numbers, ProjectCard list, GitHub/LC links

14. `navigator.share` + `HTML loading="lazy"` **patterns —** native mobile share sheet and lazy image loading, both confirmed absent from all 90 prior diagrams.

navigator.share — native share sheet on mobile

ShareButton.jsx — full pattern
const url = window.location.origin + '/profile/' + usernam
if (navigator.share) {
  await navigator.share({ title, url })
} else { navigator.clipboard.writeText(url) }

Behaviour by device
Mobile — opens native share sheet
Desktop (no navigator.share)
falls back to clipboard copy

HTML loading="lazy" — defer off-screen images

Without lazy
all 20 project repo preview images
load immediately — slow first paint

With loading="lazy"
images load only when near viewport
zero JS — browser native

Apply to: all avatars in Leaderboard rows, ProjectCard repo preview images, RecruiterSearch cards

alt text rule for accessibility
Avatar: alt={username + " avatar"} — screen reader announces "john_doe avatar"
Decorative images (tier badge bg): alt="" — screen reader skips entirely

15. GET /api/recruiter/shortlist + GET /api/mentor/students — the two role-specific list endpoints that complete the recruiter and mentor data flows.

GET /api/recruiter/shortlist

Recruiter role required (RBAC)
↓
User.shortlist array
User.findById(req.user.id)
.populate('shortlist', profile fields)
↓
Returns saved candidate list
username, avatar, score, college

GET /api/mentor/students

Mentor role required (RBAC)
↓
Profile.college match
mentorProfile = await Profile.findOne
({ user_id: req.user.id }) → .college
↓
Two-step: profiles → scores
Profile.find({ college }) → user_ids
↓
DevScore.find({ user_id: $in: ids })
sorted by rank — cohort table

User schema fields needed
Recruiter: shortlist: [{ type: ObjectId, ref: 'User' }] — add to User model

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1. `useReducer` **for complex forms** — EditProfile has 6+ state pieces. The decision rule for when to use it over `useState`.

useState — messy at 4+ fields

```
const [bio, setBio] = useState('')
const [college, setCollege] = useState('')
const [techStack, setTechStack] = useState([])
const [loading, setLoading] = useState(false)
const [errors, setErrors] = useState({})
```

useReducer — one state object

```
initialState = {{ bio:'', college:'',
techStack:[], loading:false, errors:{}}}
dispatch({{ type:'SET_FIELD',
field:'bio', value:e.target.value }})
```

Reducer pattern — EditProfile.jsx

```
function reducer(state, action) {{
  switch(action.type) {{
    case 'SET_FIELD': return {{ ...state, [action.field]: action.value }}
    case 'SET_ERRORS': return {{ ...state, errors: action.errors }}
    case 'RESET': return initialState
  }}
}}
```

Decision rule

Use useState when	Use useReducer when
1–3 fields, simple, unrelated	4+ fields + loading + errors together

DevFolio: EditProfile (bio, college, techStack, linkedin, loading, errors) = useReducer
LoginForm (email, password) = useState — only 2 fields, keep it simple

2. `React.memo` + `useCallback` + `useMemo` — the three performance tools, when each one helps, and the DevFolio symptom that tells you to add each one.

React.memo	useCallback	useMemo
skip re-render if props unchanged	stable function ref between renders	cache expensive calculated value

Symptom → tool mapping for DevFolio

React.memo — ProjectCard re-renders when Dashboard state changes
export default React.memo(ProjectCard) — skips if project prop didn't change

useCallback — onDelete passed to React.memo'd child always re-renders it
const onDelete = useCallback((id) => removeProject(id), []) — stable reference
Without useCallback: new function created every render → memo sees "new prop" → re-renders anyway

useMemo — DevScore tier + scoreData computed on every leaderboard render
const scoreData = useMemo(() => buildChartData(scores), [scores]) — recomputes only when scores change

Don't add these until you measure a problem
React DevTools Profiler → Record → Interact → see which components re-render
Gray = did not render | Coloured = rendered | Flame chart = render duration
Only add memo/useCallback if Profiler shows unnecessary renders causing visible lag

3. Helmet.js security headers — what each header does and which DevFolio routes they protect. Different from CORS.

app.use(helmet()) — adds 7 headers on every response automatically

X-Content-Type-Options: nosniff	browser never guesses MIME type — serves JSON as JSON
X-Frame-Options: SAMEORIGIN	blocks your profile page from loading inside an iframe
Referrer-Policy: no-referrer	hides full URL in Referer header to third parties
Content-Security-Policy	configure carefully — CSP blocks Recharts inline scripts by default — disable for MVP
X-DNS-Prefetch-Control: off	prevents leaking visited URLs via DNS prefetch
Strict-Transport-Security	HTTPS only — only active on production HTTPS no-op on localhost HTTP

CSP + Recharts conflict: helmet({{contentSecurityPolicy: false}}) for DevFolio MVP — add proper CSP post-launch
app.use(cors({{...}})) BEFORE app.use(helmet()) in app.js — cors must run first

4. API rate limiting — `express-rate-limit` config and why auth routes need tighter limits than stats routes. Listed on page 45 as explicitly absent.

express-rate-limit — sits in middleware chain before routes

Incoming API request
→ Rate limit middleware checks IP memory store
increments counter, resets after windowMs
→ Under limit — request continues
→ Limit exceeded — 429 Too Many Requests

Route-specific limits for DevFolio

/api/auth/*	/api/stats/sync/*	All other routes
max: 10 per 15 min stops brute force	max: 30 per hour respects GitHub limits	max: 100 per min global fallback

Place route-specific limiter BEFORE the route: app.use('/api/auth', authLimiter, authRoutes)

5. SEO meta tags + Open Graph — the tags every public profile page needs so recruiters get a rich preview when sharing the link. Listed on page 45 as explicitly absent.

Profile.jsx — dynamic meta tags via react-helmet-async

Basic SEO tags

```
<title>john_doe — DevFolio 74</title>
<meta name="description"
  content="Full-stack dev at IIT Hyderabad.
  312 commits · 122 LeetCode" />
<link rel="canonical" href="https://devfolio.app/profile/
```

Open Graph tags (LinkedIn/WhatsApp)

```
<meta property="og:title" content="john_doe" />
<meta property="og:description" content="..." />
<meta property="og:image" content={{avatarUrl}} />
<meta property="og:url" content={{profileUrl}} />
<meta property="og:type" content="profile" />
```

Twitter card — side-by-side with og tags

```
<meta name="twitter:card" content="summary" />
<meta name="twitter:title" content={{username + " — DevFolio " + score}} />
<meta name="twitter:image" content={{avatarUrl}} /> <!-- 200×200 min -->
```

App.jsx — HelmetProvider wraps everything once

```
import {{ HelmetProvider }} from 'react-helmet-async'
<HelmetProvider> <BrowserRouter> <App /> ...</HelmetProvider>
```

Only Profile.jsx and Landing.jsx need Helmet — other pages inherit the default title
og:image must be absolute URL — use VITE_APP_URL + avatar_url, not a relative path

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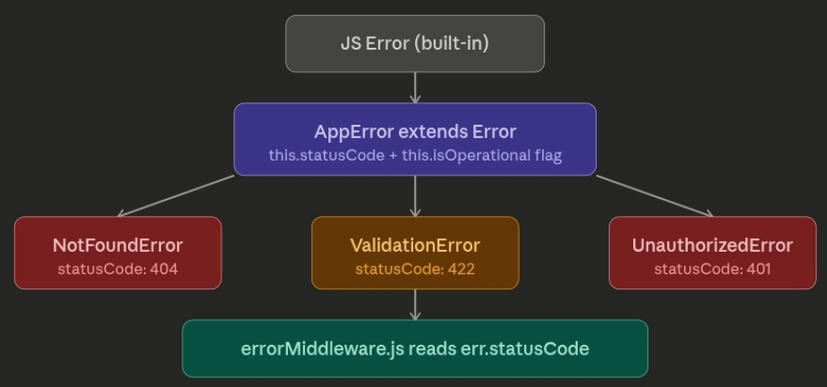
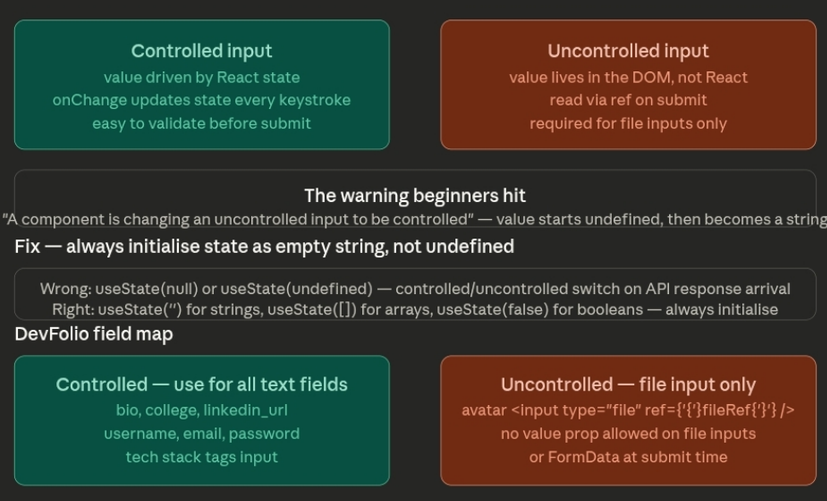
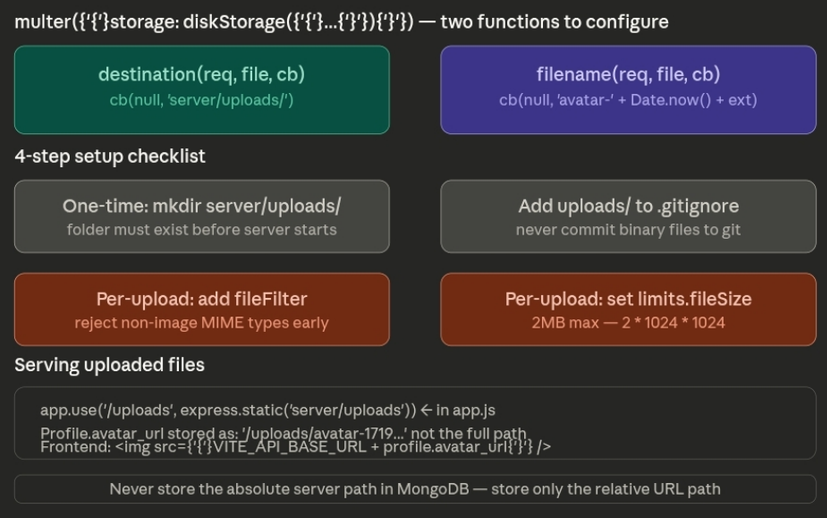
6. Custom error class hierarchy — `AppError` → `NotFoundError` / `ValidationError` / `UnauthorizedError` — confirmed absent from pages 48 of PDF.



7. React controlled vs uncontrolled inputs — the warning every beginner hits. The DevFolio form field map showing which type to use where.

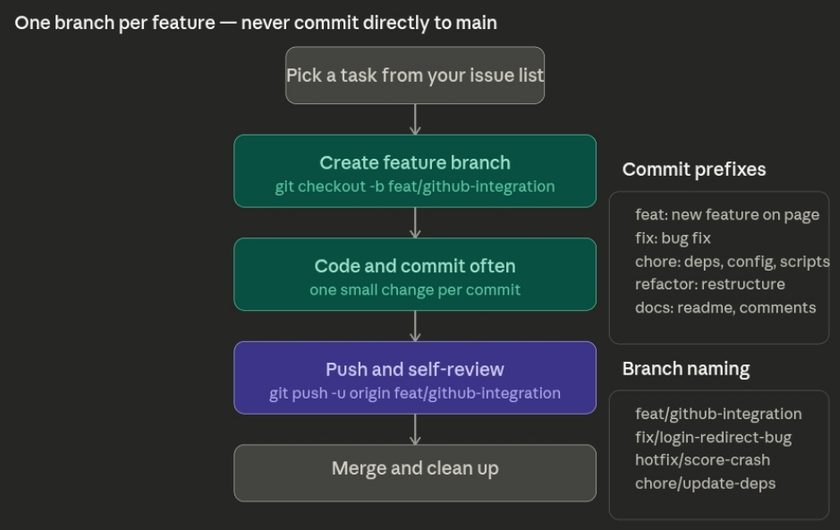


8. Multer diskStorage configuration — where uploaded files land, the filename pattern, and the 4-step checklist. Confirmed absent from all prior batches.



9. Browser console error identification — the 5 errors DevFolio beginners see every day, what causes each, and which file to check first.

10. Git feature branch workflow — the exact git command sequence for every DevFolio task. Listed on page 45 as explicitly absent from all 101 prior diagrams.



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11. MongoDB full-text search — the one-time index creation on Profile and the `$text + $search` query for recruiter search. Confirmed absent from all prior batches.

Full-text search — one-time index, then simple query

One-time index creation
Profile.createIndex({'{'} bio: 'text', college: 'text' {'}'})

Key rules
only one text index per collection
add fields during schema not later

Query with filters — recruiter search

```
Profile.find({'{'}
  $text: {'{'} $search: req.query.tech {'}'}, // "react node" → searches both bio + college
  ...(college ? {'{'} college {'}'} : {'{'}{'}'}) // combine with exact college filter
{'}'}).sort({'{'} score: {'{'} $meta: 'textScore' {'}'} {'}'}) // rank by relevance
```

Fallback regex — for tech_stack array search

```
Profile.find({'{'} tech_stack: {'{'} $in: [/react/i] {'}'} {'}'}) // case-insensitive regex per tag
Slower than text index — always add $text filter first, then $in as secondary
```

Combined recruiter search query
filter: college + minScore + tech → Profile.find(textFilter).lean() → get user_ids
DevScore.find({'{'} user_id: $in: ids, total_score: $gte: minScore {'}'}).sort(total_score: -1)

12. Browser storage comparison — `localStorage` vs `sessionStorage` vs `IndexedDB` — the 5MB trap and exactly what DevFolio stores where.

Three browser storage options — DevFolio uses all three implicitly

localStorage	sessionStorage	IndexedDB
Persists until cleared	Until tab closes	Until deleted
~5MB strings only	~5MB strings only	hundreds of MB
DevFolio uses this	Not used in DevFolio	Not needed for MVP

DevFolio localStorage — exactly 3 keys

```
token — JWT string. Read on every Axios request. Removed on logout.
theme — "dark" or "light". Read by ThemeProvider on mount.
onboarding_done — "true" when all 3 setup steps complete. Hides banner.
Nothing else. API data = React state. User object = context. Projects = fetched on mount.
```

The 5MB trap
20 leaderboard users × 2KB each = 40KB. 100 cached profiles → 200KB → nears 5MB fast.
Never cache API responses in localStorage — React state is the right place for API data.

13. API response caching strategy — `Cache-Control` headers per route, confirmed never shown on page 47 of the PDF.

Cache-Control — one header, massive impact on repeat page loads

Client makes API request

Cached route
browser returns cached data

No-store route
always fetches from server

DevFolio cache lifetime per route

60s cache	120s cache	300s cache	no-store
GET /leaderboard/global scores stale by 1min max	GET /profile/:username public profile reads	GET /stats/public landing page numbers	mutation routes POST/PATCH/*

```
res.set('Cache-Control', 'public, max-age=60') ← add to leaderboard controller
res.set('Cache-Control', 'no-store') ← add to all POST/PATCH/DELETE controllers
```

DevFolio MVP: only cache leaderboard + public profile + stats — everything else no-store
No Redis needed for MVP — HTTP caching alone cuts server load by 60–80% on read routes

14. Testing pyramid for DevFolio — what proportion to build, which functions to test first. Listed on page 45 as explicitly absent from all 101 prior diagrams.

Test by value — don't test everything equally

E2E tests — skip for MVP
expensive, slow — add post-launch

Integration tests — 10 targeted tests
POST /auth/register · GET /profile/:username · PATCH /profile

Unit tests — 5 pure functions, always test first
computeScore() · timeAgo() · validateFormData() · getTier() · formatCount()

What to write first — DevFolio priority list

Unit (write these first)
computeScore formula — maths must be right
getTier() — badge colours depend on it
validateFormData() — all form paths

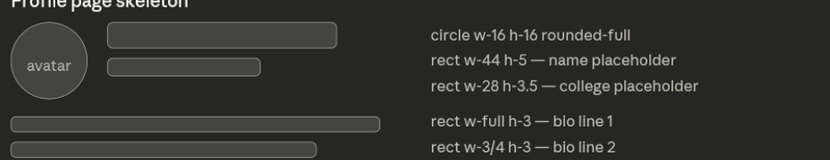
Integration (write second)
POST /auth/register → returns JWT
GET /leaderboard/global → 200 + array
PATCH /profile → 401 without token

Skip: visual appearance tests · third-party API mocks · React component snapshots — these break constantly

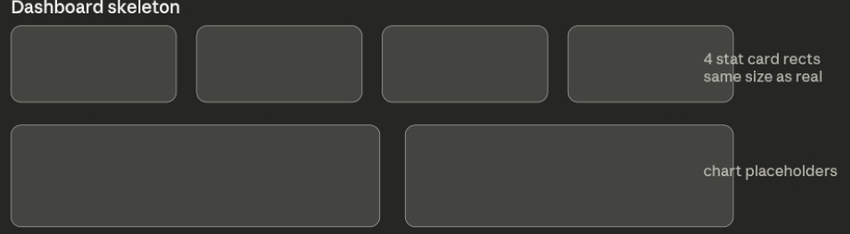
15. Skeleton UI design per screen — exactly which elements pulse on each DevFolio screen. Confirmed absent from all prior batches (page 53 of PDF).

Skeleton rule — mirror real layout exactly, same heights and widths

Profile page skeleton



Dashboard skeleton



Leaderboard skeleton
10 rows · each h-3.5 — same height as real LeaderboardRow

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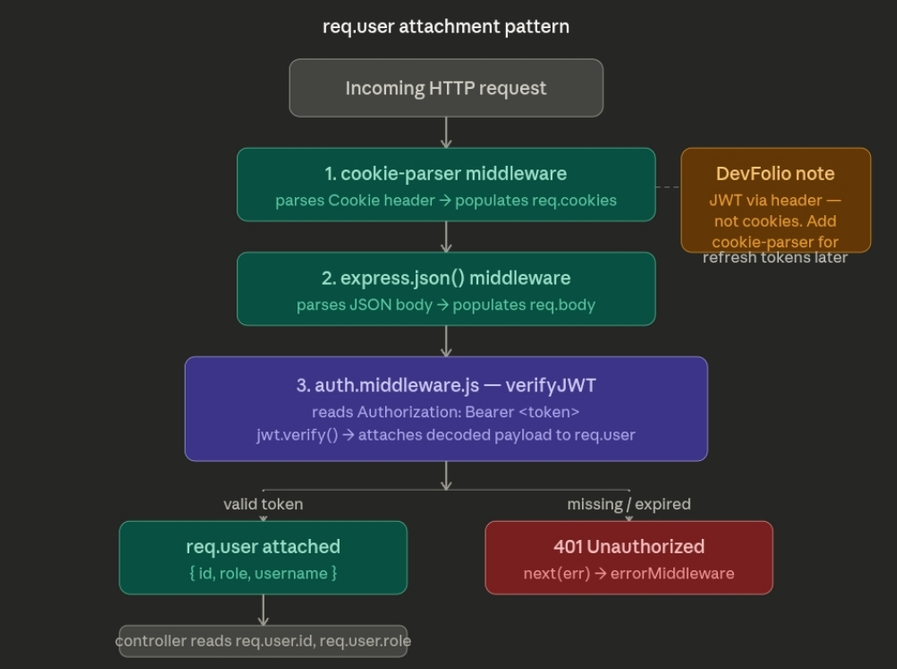
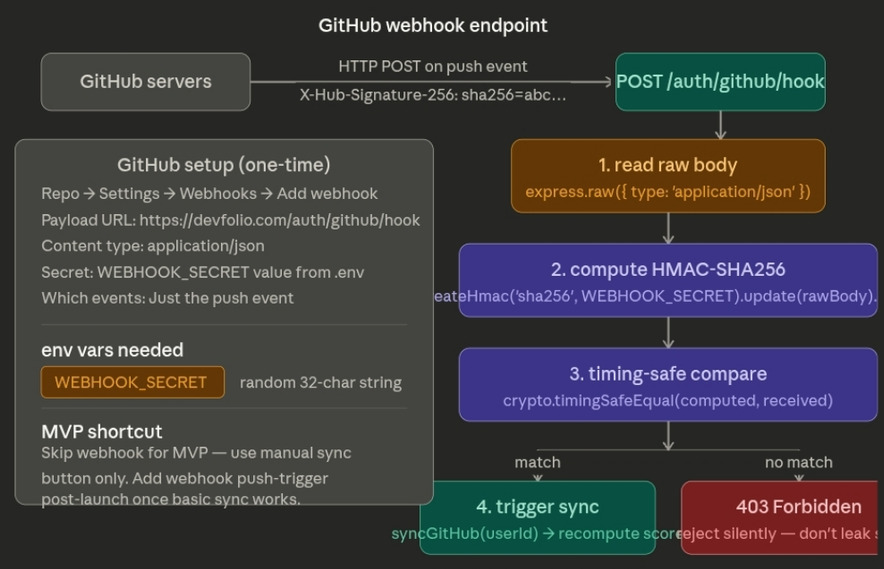
claude.ai/chat/d5

1. req.user attachment + cookie-parser middleware — how auth injects the user object onto every protected request.

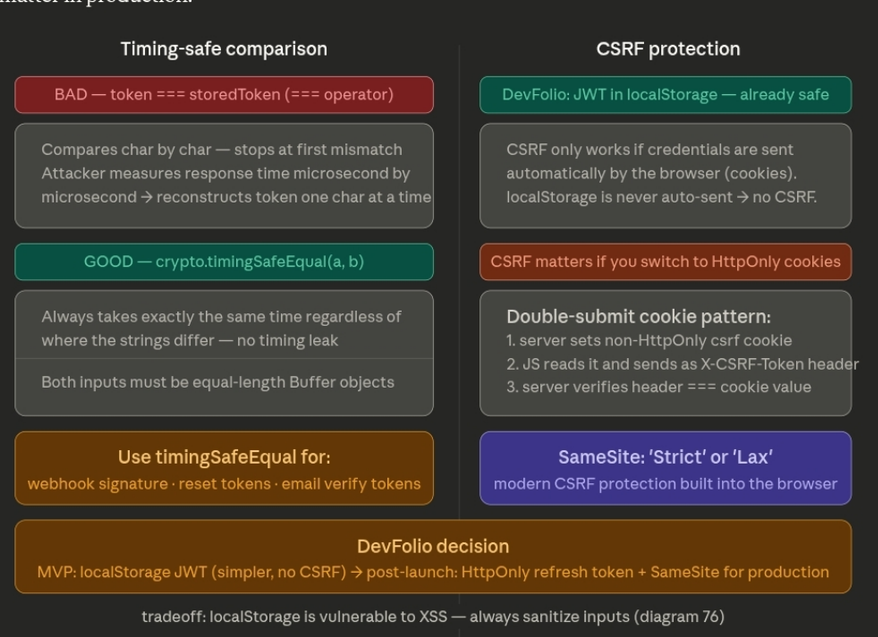


2. jsonwebtoken sign/verify options — every option DevFolio uses and why each one matters.

3. GitHub webhook endpoint — how DevFolio receives push events from GitHub to trigger auto-sync.



4. Timing-safe comparison + CSRF protection — two security patterns beginners skip that matter in production.



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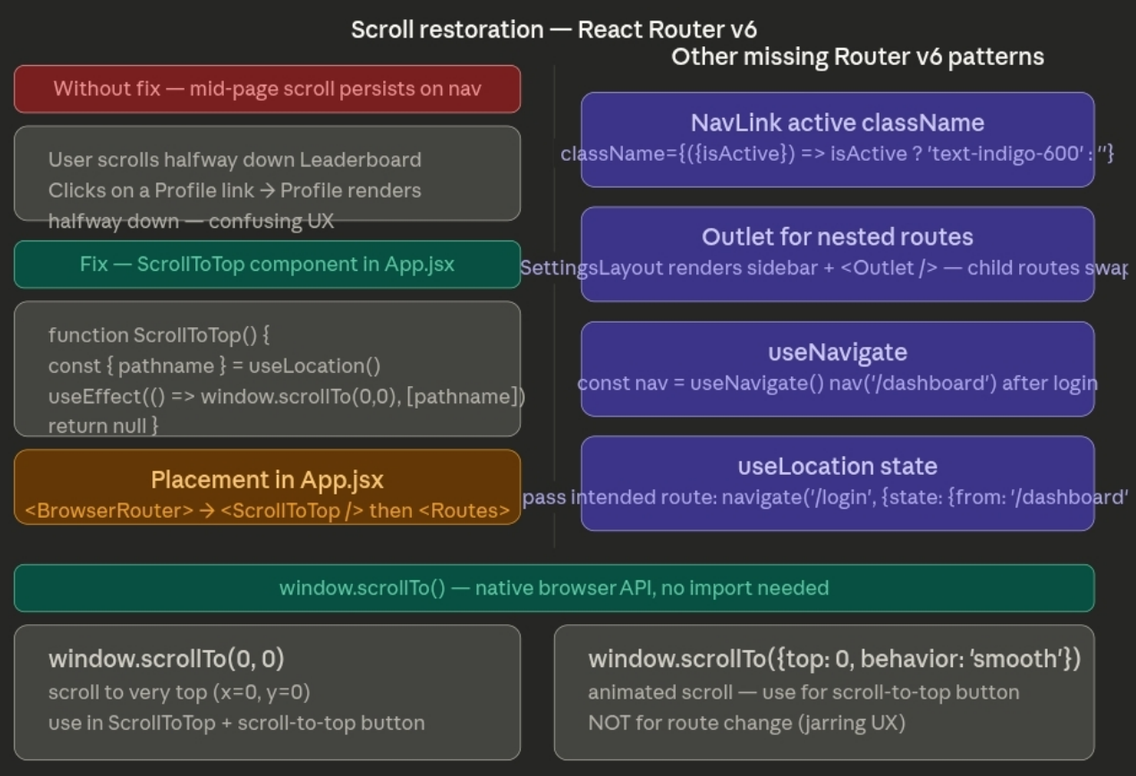
5. Scroll restoration + React Router v6 remaining patterns — the two patterns confirmed missing.



6. Response time header + health check + mongoose.connection.readyState — the three backend observability patterns that were never diagrammed.

7. Mongoose missing patterns — `$unset`, `Model.bulkWrite()`, post-save hook, and `strictQuery`.

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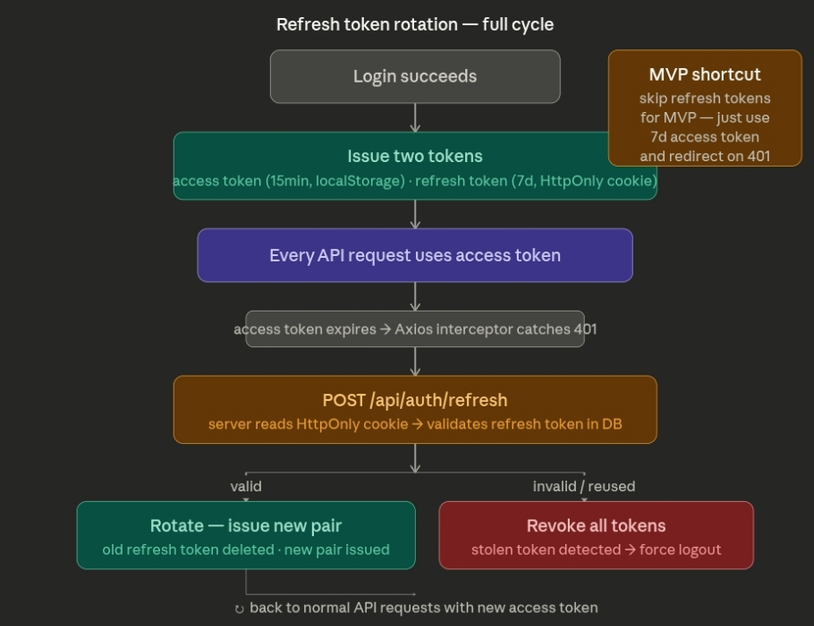
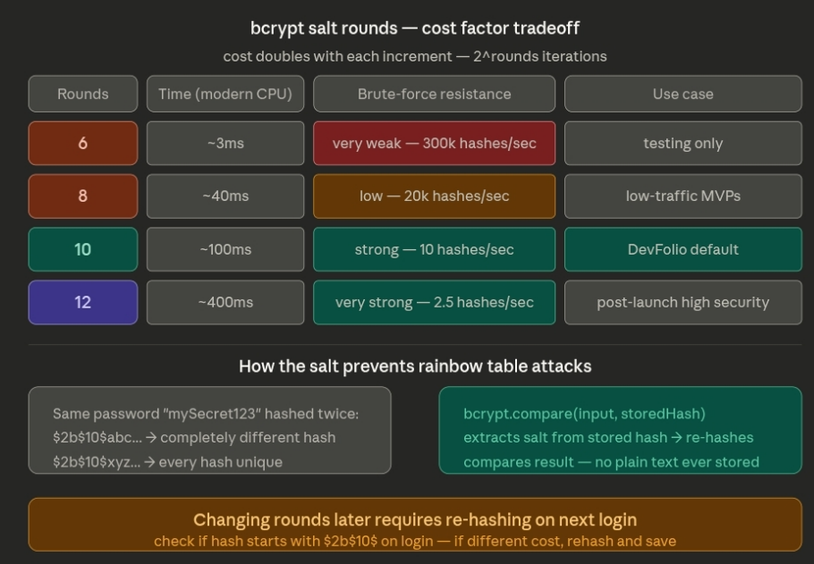
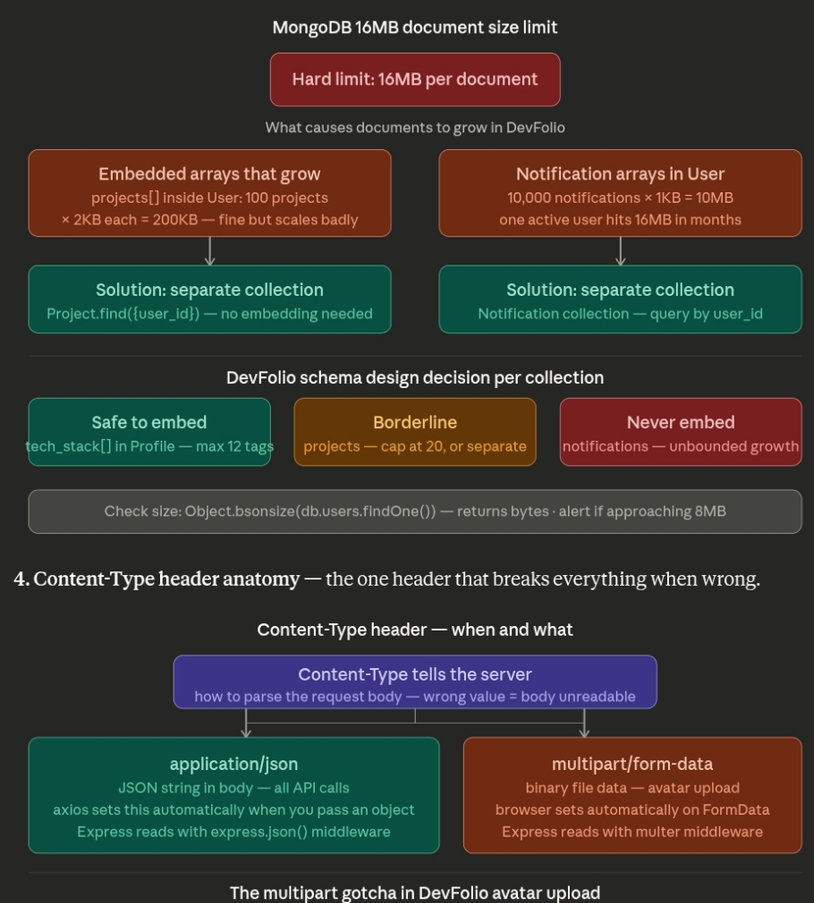
1. Refresh token rotation — the full cycle that keeps users logged in without permanently valid tokens.



2. bcrypt salt rounds — the speed vs security tradeoff every auth beginner gets wrong.



3. MongoDB 16MB document limit — what hits it in DevFolio and how the schema avoids it.



4. Content-Type header anatomy — the one header that breaks everything when wrong.

5. Express `.route()` chaining — the cleanest pattern for grouping GET/POST/PATCH on the same path.

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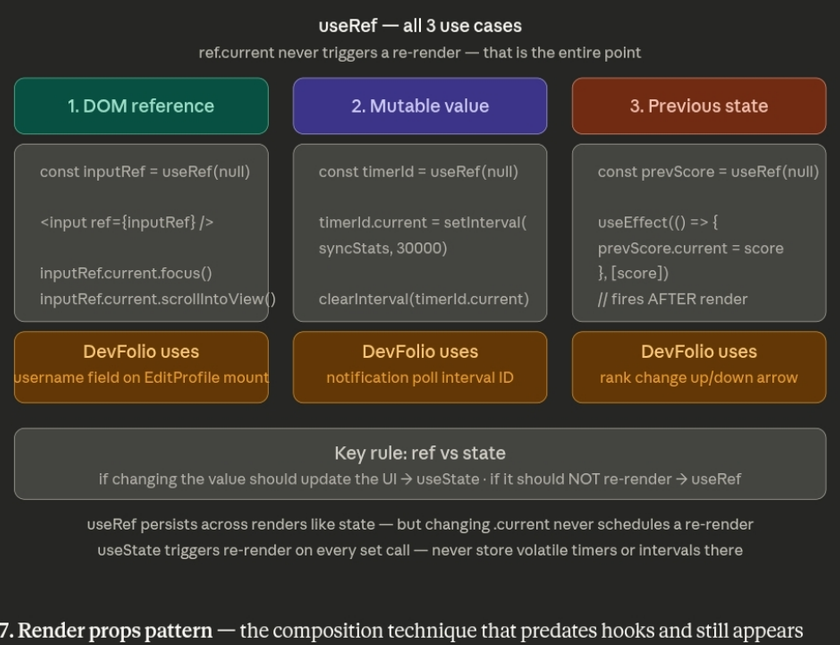
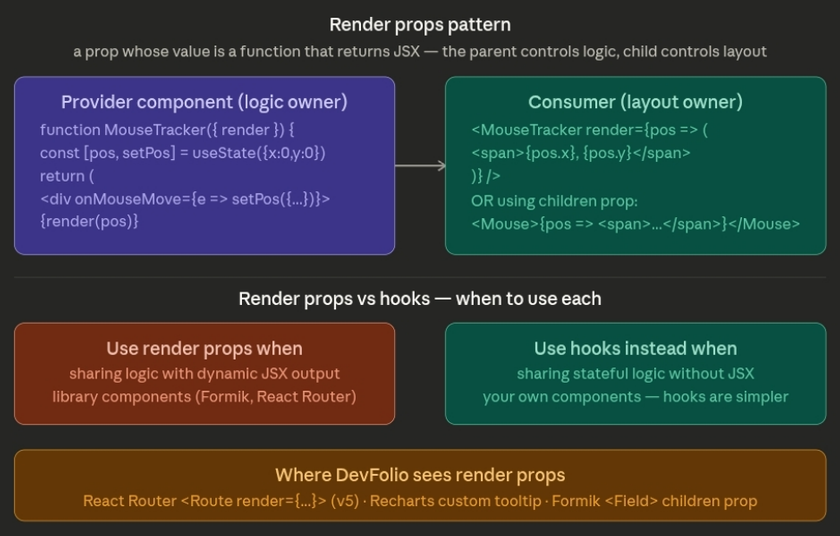
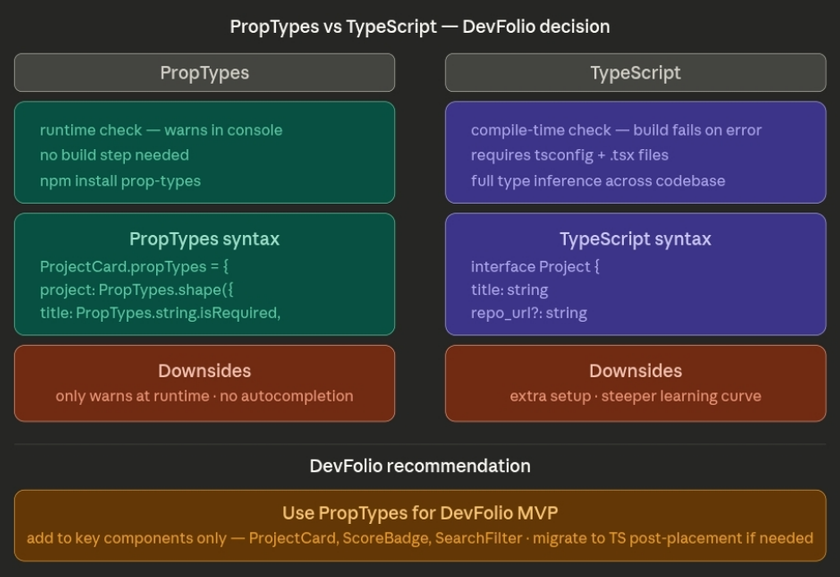
6. useRef — all 3 use cases — the hook beginners think is only for DOM access.



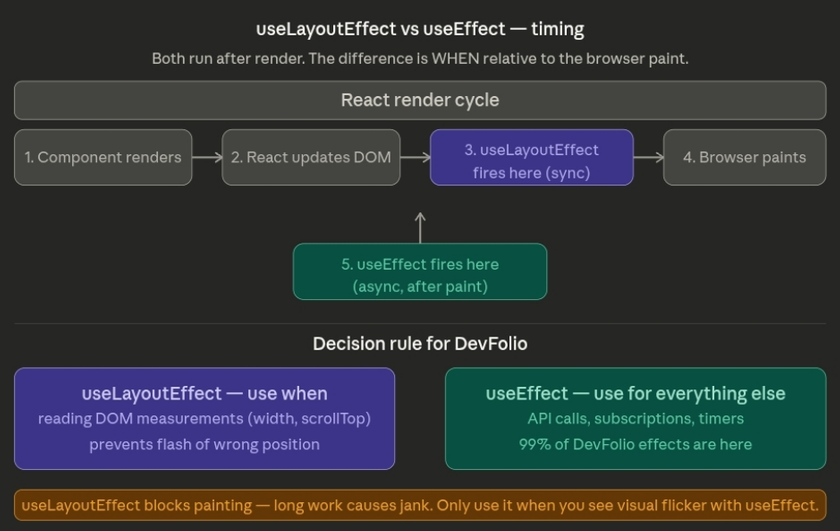
7. Render props pattern — the composition technique that predates hooks and still appears in libraries.



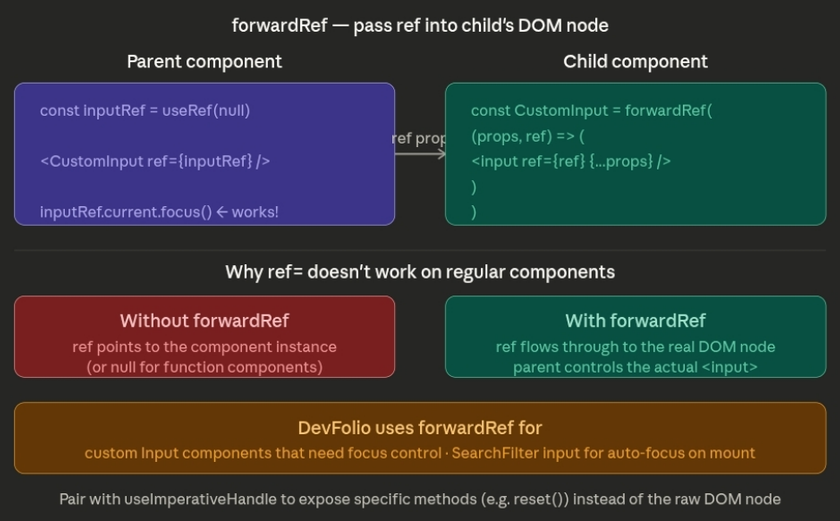
8. PropTypes vs TypeScript — the exact decision map for a DevFolio-scale beginner project.



9. useLayoutEffect timing — the render cycle difference that prevents layout flicker.



10. forwardRef — how to pass a ref from a parent into a child component's DOM node.



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11–15: Virtual DOM reconciliation, React 18 batching, Event loop, Memory leaks, async/await vs .then chain.

Virtual DOM reconciliation

setState called
score changes: 51 → 74

→ New virtual DOM
re-render in memory only

→ Diff old vs new tree
find minimum set of changes

↓ Patch real DOM — minimal writes only

Diffing rules that affect DevFolio

Same type + same key → update
React updates props in place
LeaderboardRow re-renders with new score

Different type → unmount + remount
full DOM tear-down and rebuild
all state lost inside that subtree

key=index — wrong for leaderboard
rank changes → React reuses wrong row
badges animate on wrong user

key=score._id — correct
stable across re-sorts
React tracks the right developer

Virtual DOM is NOT faster than real DOM — it makes updates predictable and batched

React 18 automatic batching

multiple setState calls inside the same event or async callback → one re-render

React 17 — async callbacks

```
fetch('/api/score').then(res => {
setData(res.data) // render 1
setLoading(false) // render 2
setScore(res.score) // render 3
})
```

React 18 — same code

```
fetch('/api/score').then(res => {
setData(res.data) // queued
setLoading(false) // queued
setScore(res.score) // queued
}) → 1 render total
```

When to opt out — flushSync

flushSync — force immediate render
import { flushSync } from 'react-dom'
flushSync(() => setState(x)) — renders immediately, then reads DOM measurement

DevFolio impact
Dashboard data fetch (profile + stats + score) batches all 3 setStates → 1 render — no change needed

Node.js event loop — mental model

single JS thread — never blocks because async work leaves the call stack

Call stack
req handler
profileCtrl()
await DB
← pauses here
stack pops
other req runs
LIFO

Node APIs
DB query → libuv
file read → OS
HTTP fetch → net
work done off-thread
callback queued
when complete

Callback queue
DB result cb
timer cb
I/O cb
waits here until
stack is empty
FIFO

Event loop
if stack
empty →
push next
callback

↻ loop repeats — stack runs, drains, picks next callback

Why await doesn't block other requests

await hands the Promise to libuv and pops the function off the call stack
while the DB runs, Node processes other requests normally
when DB responds, callback re-enters the stack from the queue

Memory leak patterns + cleanup

every useEffect that sets up something must tear it down in the cleanup function

Leak 1 — interval without clear

```
useEffect(() => {
setInterval(pollNotifs, 30000)
}, []) ← no cleanup = runs forever
```

Fix — return cleanup

```
useEffect(() => {
const id = setInterval(pollNotifs, 30000)
return () => clearInterval(id)
```

Leak 2 — fetch on unmounted

```
useEffect(() => {
fetch('/api/profile').then(setData)
}, []) ← user navigates away mid-fetch
```

Fix — AbortController

```
const ctrl = new AbortController()
fetch(url, {signal: ctrl.signal})
return () => ctrl.abort()
```

Leak 3 — event listener

```
useEffect(() => {
window.addEventListener('resize', fn)
}, []) ← fn held in memory forever
```

Fix — removeEventListener

```
useEffect(() => {
window.addEventListener('scroll', fn)
return () => window.removeEventListener('scroll', fn)
```

Rule: if useEffect sets up a resource, cleanup must tear it down
React StrictMode intentionally runs cleanup twice in dev to surface these bugs

async/await vs .then() chaining

async/await — DevFolio style

```
async function getProfile(req, res) {
try {
const user = await User.findOne(...)
const proj = await Project.find(...)
res.json({ user, proj })
```

.then() chaining — equivalent

```
User.findOne(...)
.then(user => {
return Project.find(...)
.then(proj => res.json(...))
}).catch(err => next(err))
```

The silent bug — async without await

Forgot await — silent bug

```
const user = User.findOne(...) ← Promise!
user.username → undefined (not resolved)
```

Parallel with Promise.all

```
const [user, score] = await Promise.all([
User.findById(id), DevScore.findOne(id) ])
```

DevFolio decision rule
sequential await → when step 2 depends on step 1 result
Promise.all → when queries are independent (profile + stats + score on dashboard load)

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16-20: Same-origin policy, $inc/$push/$addToSet, normalisation vs denorm, empty states, peer/peer-focus Tailwind.

Same-origin policy

origin = protocol + hostname + port — ALL three must match

Origin breakdown: https://devfolio.com:443
protocol=https · hostname=devfolio.com · port=443

Same origin? — DevFolio local dev examples

- http://localhost:5173 → http://localhost:5173/about — same origin (path doesn't matter)
- http://localhost:5173 → http://localhost:5000 — DIFFERENT (port differs) → CORS needed
- https://devfolio.com → http://devfolio.com — DIFFERENT (protocol differs)
- https://devfolio.com → https://api.devfolio.com — DIFFERENT (subdomain differs)

What the browser blocks vs allows

Always allowed
img src, link href, script src, form POST

Blocked by SOP
fetch(), XMLHttpRequest, reading response

Mongoose update operators — $inc, $push, $addToSet

$inc — increment a number

```
GithubStats.findOneAndUpdate(
{user_id}, { $inc: { total_commits: 1 } })
```

DevFolio use
incrementing commit count on sync · use $inc: {x: -1} to decremen

$push vs $addToSet — array append

$push — always appends
{ $push: { tech_stack: 'Docker' } }
allows duplicates — 'React', 'React', 'Docker'

$addToSet — unique only
{ $addToSet: { tech_stack: 'Docker' } }
skips if already in array — no duplicates

DevFolio rule: tech_stack uses $addToSet · top_languages uses $push then dedup in JS

$pull — remove from array by value

Profile.findOneAndUpdate({user_id}, { $pull: { tech_stack: 'Vue' } })
removes all matching values — use when user deletes a tag from their profile

Normalisation vs denormalisation — DevFolio schema decisions

Normalised — data in separate collections

User → only auth fields
Profile → bio, college, tech_stack
DevScore → total, github, dsa, project
GithubStats → commits, repos, stars

Benefits
no duplication · easy to update one doc
leaderboard never exposes password_hash

Cost
multiple queries per page load
need .populate() to join data

Denormalised — data embedded in one doc

User {
bio, tech_stack,
total_score, github_score,
commits, repos, stars ← all in one

Benefits
one DB query for entire profile
no .populate() needed

Cost
16MB limit risk · password_hash in leaderboard query

DevFolio choice: normalised
score updates happen independently from profile edits — separate collections avoids full doc write on each sync
security: leaderboard query only touches DevScore + Profile — never User (no password_hash exposed)

Empty states — per screen in DevFolio

every screen has 3 states: loading → empty → data. Beginners forget empty.

Screen	Empty state message	Action to offer
Projects page	"Add your first project"	+ Add project button
Dashboard — no GitHub	"Connect GitHub to see stats"	→ /settings/integrations
Leaderboard no results	"No developers match filters"	Clear filters button
Recruiter search	"No developers found"	Adjust score range hint
Notifications	"All caught up"	No action — positive state
Mentor dashboard	"No students in your college"	Link college in settings

Reusable EmptyState component
<EmptyState icon="..." title="..." message="..." action={<Button>...</Button>} />

Tailwind peer / peer-focus

style a sibling element based on the state of its peer — no JS needed

How it works
1. mark the trigger element: class="peer"
2. style the target sibling: class="peer-focus:..."
3. sibling must come AFTER peer in HTML

Floating label pattern

```
<input class="peer" />
<label class="peer-focus:text-indigo-600
peer-focus:-translate-y-4">
```

peer variants available in DevFolio

peer-focus: → input focused → label floats up
peer-checked: → checkbox checked → label bold
peer-disabled: → input disabled → label grayed
peer-invalid: → input invalid → error style on sibling message
peer-placeholder-shown: → show label when input is empty

DevFolio uses peer-focus for
EditProfile form fields — label animates up when input focused
SearchFilter score range slider — label highlights when input active

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21. `key` **prop forced remount** — the trick that resets a component's entire state instantly.

key prop — forced remount

when key changes, React destroys the old component and mounts a brand new one

Problem — stale state on route change

User visits /profile/john_doe
Profile.jsx loads — state set to john's data
User navigates to /profile/jane_doe
same component — john's state flashes first

Fix — key on the component

```
const { username } = useParams()

<Profile key={username} />
```

username changes → full remount → fresh state

What remount actually does

1. React sees key changed → treats as completely different component
2. Runs cleanup on old component (useEffect returns)
3. Unmounts old DOM, mounts new component, runs fresh useEffect

DevFolio places where key remount is used

router.jsx — Profile route

```
<Route path="/profile/:username"
element={<Profile key={username} />} />
```

EditProfile — reset on cancel

```
<EditProfile key={resetCount} />
```

increment resetCount → form resets to saved

22. `clsx` **conditional classes** — the cleanest way to compose Tailwind class strings in DevFolio.

clsx — conditional Tailwind class composition

Without clsx — messy

```
className={`px-4 py-2 rounded
${isActive ? 'bg-indigo-600' : 'bg-gray-100'}
${disabled ? 'opacity-50 cursor-not-allowed' : ''}
${hasError ? 'border-red-500' : 'border-gray-300'}`}
```

With clsx — clean

```
className={clsx(
'px-4 py-2 rounded',
isActive && 'bg-indigo-600',
disabled && 'opacity-50 cursor-not-allowed',
```

clsx input formats — all valid

clsx('foo', 'bar') → 'foo bar'
clsx('foo', isActive && 'bar') → 'foo bar' or 'foo'
clsx('foo', { bar: isActive, baz: hasError }) → 'foo bar' or 'foo baz'
clsx(['foo', 'bar'], isActive && 'baz') → arrays and conditionals mix freely

DevFolio components using clsx

ScoreBadge
tier color class
+ size variant

NavLink active state
active bg + text
vs default muted

Sync button
idle / syncing /
error state classes

23. `useId` — generating stable, unique IDs for label-input pairs across renders.

useId — stable unique IDs for label-input pairing

Problem — hardcoded id breaks

```
// Component used twice on the page:
<label htmlFor="email">Email</label>
<input id="email" />
// Two inputs with id="email" → broken a11y
```

Fix — useId generates unique ID

```
const id = useId()
// id = ':r0:' or ':r3:' — always unique
<label htmlFor={id}>Email</label>
<input id={id} /> ← wired correctly
```

Multiple IDs from one useId call

```
const id = useId() // ':r1:'
const nameId = id + '-name' // ':r1:-name'
const emailId = id + '-email' // ':r1:-email' — all unique, one hook call
```

Why useId over Math.random() or Date.now()

Math.random() — wrong
changes every render
server + client IDs never match → hydration error

useId — correct
stable across renders and SSR
React guarantees uniqueness per component instance

24. Optional chaining edge cases — the nullish coalescing traps that cause silent bugs in DevFolio.

Optional chaining ?. and nullish coalescing ??

?. — optional chaining

```
user?.github?.avatar_url
// returns undefined if user or github is null
// does NOT crash unlike user.github.avatar_url
// DevFolio: profile?.bio, stats?.total_commits
```

?? — nullish coalescing

```
score ?? 0
// returns 0 only if score is null or undefined
score || 0
// returns 0 if score is 0, '', false, NaN too!
```

The critical DevFolio trap — score of 0

Bug — using || for score

```
const display = score || 'No score'
```

score=0 → shows 'No score' — WRONG!

Fix — use ?? for scores

```
const display = score ?? 'No score'
```

score=0 → shows 0 — correct!

DevFolio rule — || vs ??

Use ?? for: scores, counts, numeric stats — 0 is a valid value
Use || for: fallback strings where empty string is also invalid — username || 'Anonymous'
Use ?. for: any nested property access where parent might be null — user?.profile?.college
Combine: user?.github?.stars ?? 0 — safe access with numeric fallback

25. `loading="lazy"` — every context where this one attribute applies in DevFolio.

loading="lazy" — all contexts

native browser attribute — no library needed. Defers loading until element nears viewport.

Images — most common

```
<img
src={user.avatar_url}
loading="lazy"
```

Iframes — also supported

```
<iframe
src="https://..."
loading="lazy"
```

Where DevFolio should use loading="lazy"

Use lazy

LeaderboardRow avatars — up to 20 rows, most below fold
ProjectCard preview images — scrollable grid on Dashboard
RecruiterSearch developer avatars — long results list

Never lazy — above the fold

Profile page avatar — always visible immediately on load
Landing page hero image — first thing user sees, must render immediately

Browser support: Chrome, Firefox, Edge, Safari 15.4+ — 96% global coverage. No polyfill needed for DevFolio MVP.

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26. Sticky header with scroll shadow — the `useScrollY` pattern that makes the Navbar feel polished.

Sticky header + scroll shadow

useScrollY custom hook

```
function useScrollY() {
const [scrollY, setScrollY] = useState(0)
useEffect(() => {
const fn = () => setScrollY(window.scrollY) // add listener + cleanup
```

↓

Navbar.jsx — shadow appears on scroll

```
const scrollY = useScrollY()
className={clsx('fixed top-0 w-full z-20', scrollY > 0 && 'shadow-md border-b border-gray-200')}
```

Two scroll-driven patterns in DevFolio

Scroll-to-top button

shown only when scrollY > 300

```
className={clsx(
'fixed bottom-6 right-6 transition-opacity',
scrollY > 300 ? 'opacity-100' : 'opacity-0')}
```

Navbar border + shadow

appears when scrollY > 0
Tailwind: 'transition-shadow duration-200' — smooth appearance on scroll start

Both patterns share the same hook — call useScrollY() once, use scrollY value for both triggers

27. `focus-visible` — the accessibility pattern that shows focus rings only for keyboard users.

focus-visible — keyboard-only focus rings

mouse click on a button should not show an outline — Tab key navigation should

Wrong — remove all outlines

```
className="outline-none"
// hides ring for keyboard users too
// accessibility failure
```

Correct — focus-visible

```
className="outline-none
focus-visible:ring-2
focus-visible:ring-indigo-500"
```

How the browser decides when to show :focus-visible

Mouse click → :focus fires, :focus-visible does NOT → no ring shown
Tab key press → :focus fires, :focus-visible DOES fire → ring shown
Browser uses heuristics (last input type) to decide — fully automatic

DevFolio standard button class pattern

```
className="px-4 py-2 rounded-md bg-indigo-600 text-white
hover:bg-indigo-700 active:scale-95 transition-colors
outline-none focus-visible:ring-2 focus-visible:ring-indigo-500 focus-visible:ring-offset-2"
```

Apply to: all buttons, NavLinks, ProjectCard, LeaderboardRow, and any interactive element

28. Tooltip — CSS-only pattern — how DevFolio's StatsCard tooltips work with zero JS.

Tooltip — CSS-only with Tailwind group-hover

parent has class="group relative" — child shows on group hover. Zero JS, zero library.

Full tooltip JSX pattern

```
<div className="relative group inline-block">
<span>312 commits</span> ← trigger element
<div className="absolute bottom-full left-1/2 -translate-x-1/2 mb-2
bg-gray-900 text-white text-xs rounded px-2 py-1 whitespace-nowrap
opacity-0 group-hover:opacity-100 transition-opacity pointer-events-none">
```

Class breakdown

absolute bottom-full left-1/2 -translate-x-1/2 → centered above the trigger
mb-2 → gap between tooltip and trigger
opacity-0 group-hover:opacity-100 → hidden by default, visible on hover
transition-opacity → smooth fade-in
pointer-events-none → tooltip never intercepts mouse events

DevFolio tooltip placements

tsCard commit count — "Total push events last 90 days" · ScoreBadge — tier explanation · Sync button — "Last synced X a

29. Monorepo vs separate repos — the exact structure decision for DevFolio's client and server.

Monorepo vs separate repos — DevFolio decision

Monorepo — one repo, two folders

```
devfolio/
server/ ← Express API
client/ ← React app
package.json ← concurrently scripts
```

Separate repos — client + server split

```
devfolio-server/ ← own repo + deploy
devfolio-client/ ← own repo + deploy
two git histories
two deployments, two CI pipelines
```

Monorepo benefits

one git clone · shared .env.example
npm run dev starts both with concurrently

Separate repo benefits

independent deploys · smaller PRs
cleaner for teams with separate devs

Monorepo downside

root package.json scripts get complex

Separate repos downside

two clones, two npm installs to onboard

DevFolio recommendation — monorepo

solo project: one repo, server/ and client/ folders, root package.json with concurrently
"dev": "concurrently \"npm run server\" \"npm run client\""

30. Authorization header format — the exact anatomy of Bearer tokens and why format matters.

Authorization header — anatomy

Authorization: Bearer eyJhbGci...

header name (case-insensitive) scheme token value (no quotes, no extra chars)

Authorization | Bearer (space included | JWT token string

How Axios sets it — DevFolio axiosInstance

```
axiosInstance.interceptors.request.use(config => {
const token = localStorage.getItem('token')
if (token) config.headers.Authorization = `Bearer ${token}`
```

How Express reads it — auth.middleware.js

```
const authHeader = req.headers.authorization // lowercase in Express
if (!authHeader || !authHeader.startsWith('Bearer ')) → 401
const token = authHeader.split(' ')[1] // ['Bearer', 'eyJ...'][1]
```

Common mistakes

'bearer' lowercase · missing space · 'JWT ' prefix

Express gotcha

req.headers always lowercase — use .authorization not .Authoriz

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Orchestrated 32 documentation topics across execution engine architecture batches >

First, let me lay out the complete picture — feasibility verdict and all 32 new topics — before generating the artifacts.

Feasibility verdict up front: The iframe srcDoc MVP is genuinely beginner-friendly and can be built in a weekend. Docker sandboxing is not — it requires DevOps knowledge and server configuration that is beyond solo-beginner scope. The right path is to build the iframe MVP first and treat Docker as a post-placement enhancement.

All 32 new topics identified:

Architecture (4): updated system arch with engine, execution engine 3-option overview, MVP vs advanced comparison, code storage decision tree. Backend (8): portfolio submission endpoint, multer config for code files, HTML/CSS/JS combine + sanitize flow, preview serving endpoint, portfolio CRUD endpoints, rate limiting for execution, PortfolioCode schema, storage architecture. Security (5): iframe sandbox attribute flags, CSP headers for preview, DOMPurify sanitization layers, code obfuscation approach, malicious code prevention stack. Frontend (6): iframe srcDoc pattern, LivePreview component, code editor decision, preview loading states, preview error handling, portfolio submission form. Flows (5): developer submission end-to-end, recruiter viewing live portfolio, portfolio update+re-render, public preview URL, visibility settings. Reference (4): portfolio API endpoints, feasibility map, updated build sequence, documentation index additions.

Starting Batch 1 — diagrams 1–20.

E1. Beginner feasibility map — every sub-feature rated honestly.

Easy — build now | Medium — needs research | Hard — post-placement | Skip for MVP

Phase MVP — iframe srcDoc approach (build this)

HTML/CSS/JS textarea input Three textarea fields in the submission form — plain HTML, just like any other form	Easy
Combine into single HTML string String template: wrap CSS in style tag, JS in script tag, join with HTML body	Easy
DOMPurify sanitization npm install dompurify — one .sanitize() call before storing	Easy
Store combined HTML in MongoDB New PortfolioCode collection — same Mongoose patterns you already know	Easy
iframe srcDoc rendering One JSX prop: srcDoc={htmlString} — browser handles the rest	Easy
iframe sandbox attribute sandbox="allow-scripts" — one HTML attribute, massive security gain	Easy
Preview page — recruiter views output only Route /portfolio/:id/preview — fetch HTML from DB, render in iframe	Easy
Public/private visibility toggle isPublic boolean field on PortfolioCode — same RBAC patterns	Easy

Phase 2 — file uploads (after MVP works)

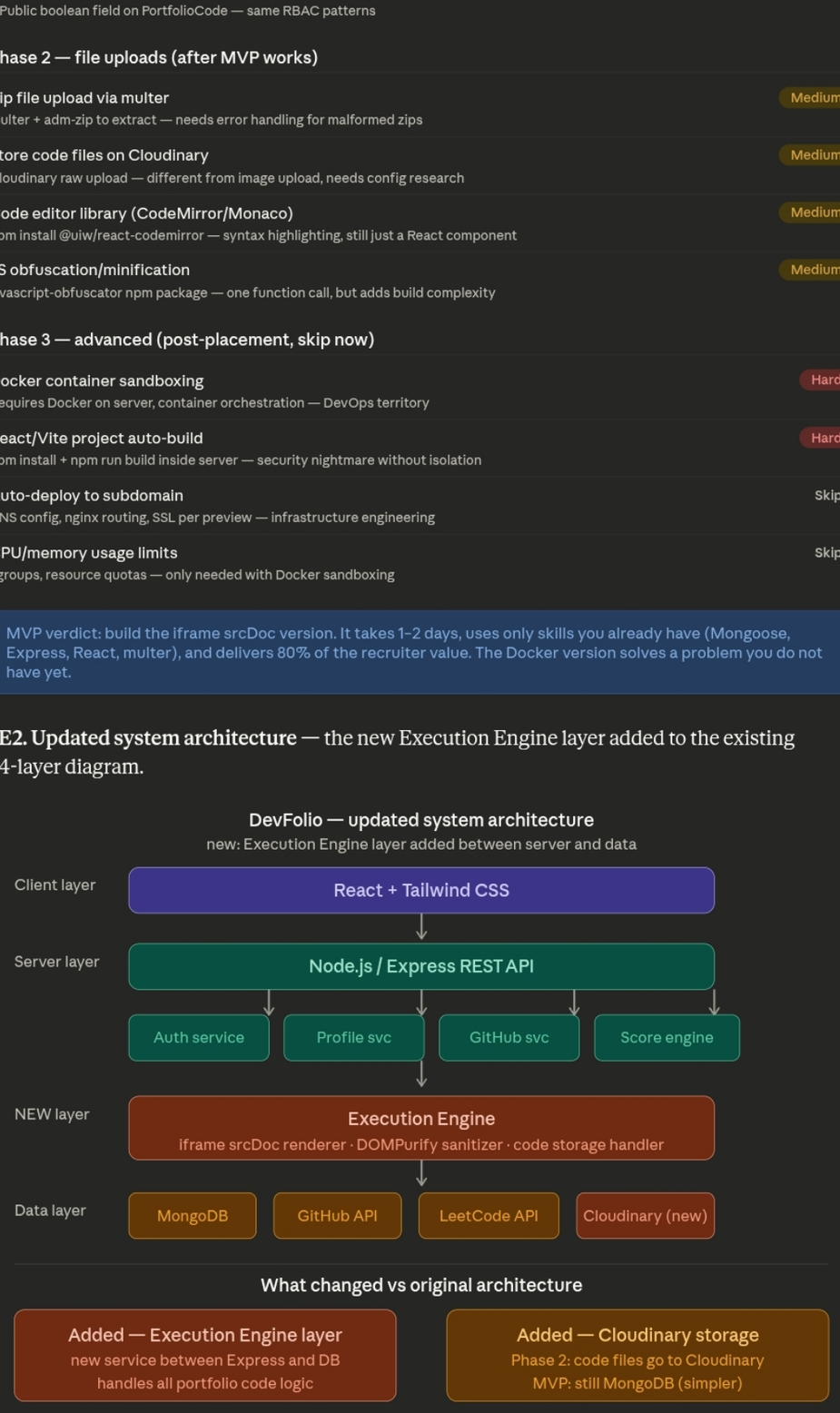
Zip file upload via multer multer + adm-zip to extract — needs error handling for malformed zips	Medium
Store code files on Cloudinary Cloudinary raw upload — different from image upload, needs config research	Medium
Code editor library (CodeMirror/Monaco) npm install @uiw/react-codemirror — syntax highlighting, still just a React component	Medium
JS obfuscation/minification javascript-obfuscator npm package — one function call, but adds build complexity	Medium

Phase 3 — advanced (post-placement, skip now)

Docker container sandboxing Requires Docker on server, container orchestration — DevOps territory	Hard
React/Vite project auto-build npm install + npm run build inside server — security nightmare without isolation	Hard
Auto-deploy to subdomain DNS config, nginx routing, SSL per preview — infrastructure engineering	Skip
CPU/memory usage limits cgroups, resource quotas — only needed with Docker sandboxing	Skip

MVP verdict: build the iframe srcDoc version. It takes 1–2 days, uses only skills you already have (Mongoose, Express, React, multer), and delivers 80% of the recruiter value. The Docker version solves a problem you do not have yet.

E2. Updated system architecture — the new Execution Engine layer added to the existing 4-layer diagram.



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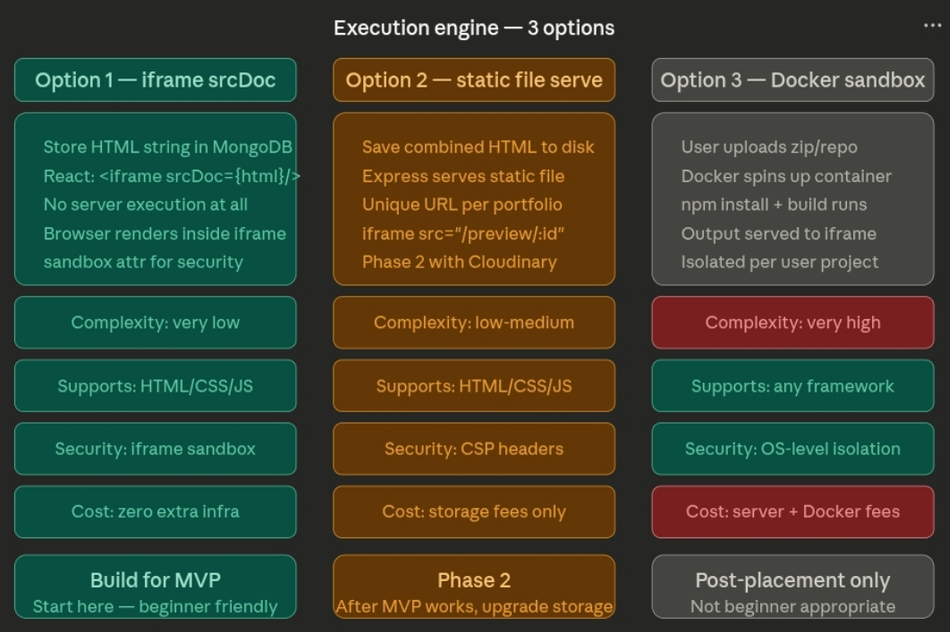
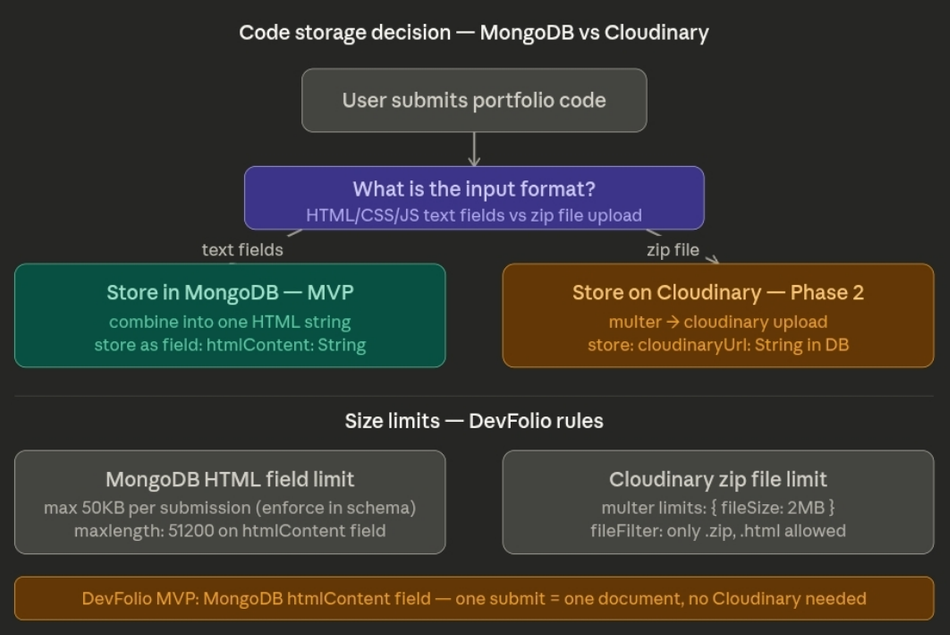
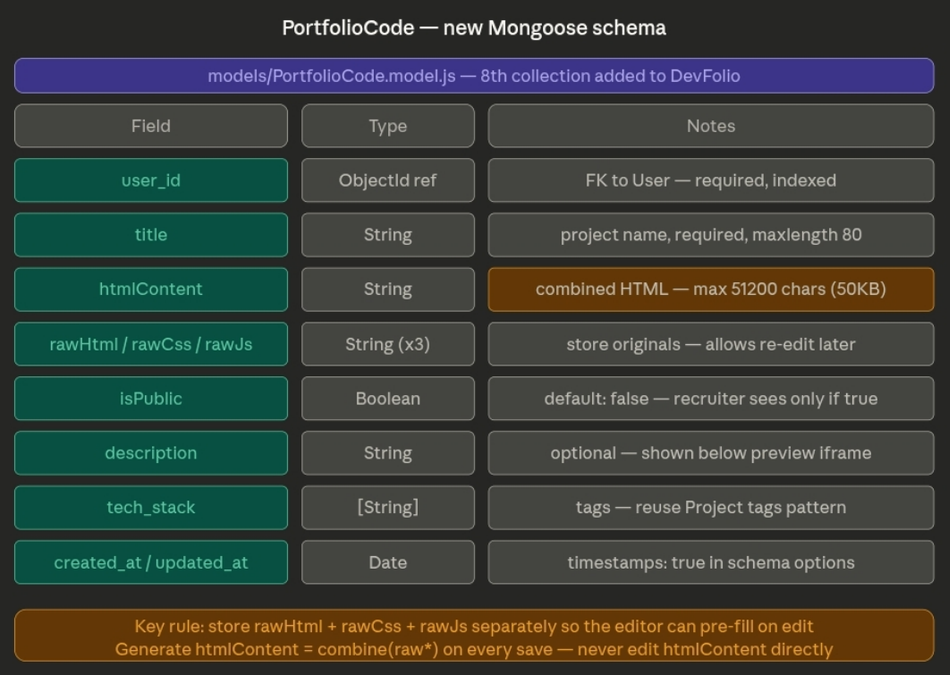
E3. Execution engine — 3 options comparison — understand all approaches before committing to one.



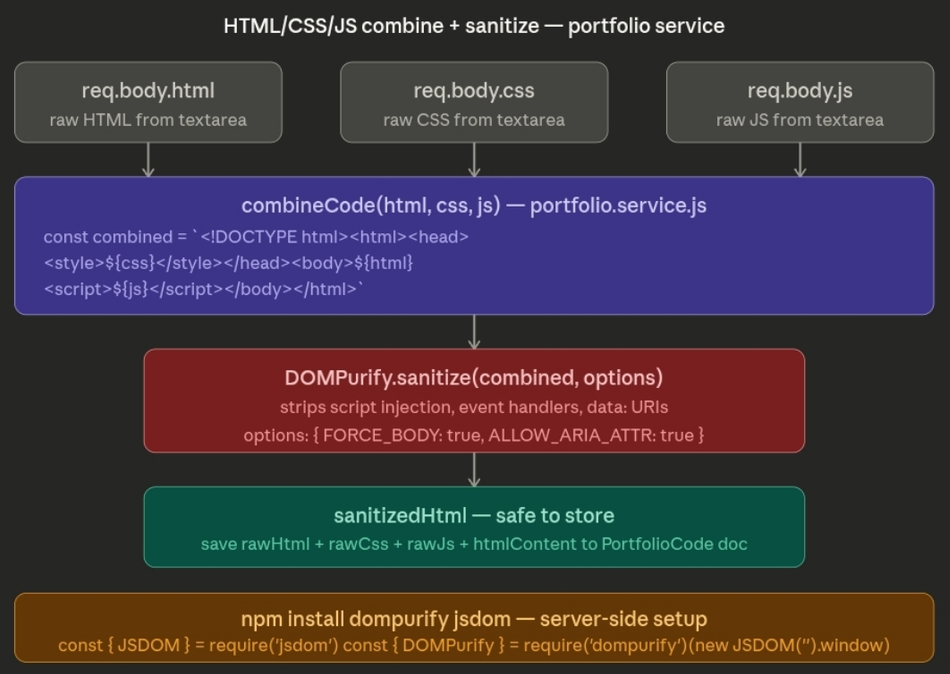
E4. Code storage decision tree — MongoDB vs Cloudinary, when each applies.



E5. PortfolioCode collection schema — the new Mongoose model, fully mapped.



E6. HTML/CSS/JS combine + DOMPurify sanitize flow — the most important backend function in the engine.



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E7. iframe srcDoc rendering pattern — the core frontend technique, fully explained.

iframe srcDoc — rendering pattern

LivePreview.jsx — the complete component

```
function LivePreview({ htmlContent }) {
return <iframe srcDoc={htmlContent} sandbox="allow-scripts"
className="w-full h-96 border-0 rounded-lg" title="portfolio preview" />
```

srcDoc vs src — why srcDoc is correct for this

src="/api/portfolio/:id" — wrong	srcDoc={htmlContent} — correct
needs separate HTTP request	HTML already fetched — inline render
CORS issues if different origin	no extra request, no CORS problem

What the browser does with srcDoc

1. Browser creates a new isolated browsing context (separate window object)
2. Parses the HTML string as if it were a complete HTML page
3. sandbox="allow-scripts" lets JS run but blocks cookies, popups, top-navigation

The user's code cannot access DevFolio's window, localStorage, or React context — isolation is automatic

E8. iframe sandbox attribute — every flag and what it does for DevFolio security.

iframe sandbox attribute — all flags

empty sandbox="" blocks everything — add flags to re-allow specific capabilities

Flag	What it allows
allow-scripts	JavaScript execution — DevFolio needs this
allow-same-origin	access DevFolio cookies/localStorage — NEVER add
allow-forms	form submission — skip unless portfolio has forms
allow-top-navigation	redirect the parent DevFolio page — NEVER add
allow-popups	window.open() — optional, low risk
allow-modals	alert/confirm — annoying to recruiter, skip
allow-downloads	trigger file downloads on recruiter — NEVER add

DevFolio recommended value

sandbox="allow-scripts allow-forms allow-popups"

NEVER combine allow-scripts + allow-same-origin — that completely bypasses sandbox isolation

E9. Portfolio code submission API — the POST endpoint flow end to end.

POST /api/portfolio/code — submission endpoint

Developer submits code

↓

1. verifyJWT + roleCheck('developer')
only logged-in developers can submit code — recruiters cannot

↓

2. express-validator — body validation
title: required, 1-80 chars · htmlContent: not empty · size check: < 50KB

↓

3. DOMPurify.sanitize(combined HTML)
strips XSS vectors before saving

↓

4. PortfolioCode.create({ user_id, ...body })
upsert by user_id — one live portfolio per developer

MVP rule: one live portfolio per user

↓

201 Created
{ success: true, portfolioId: doc._id }

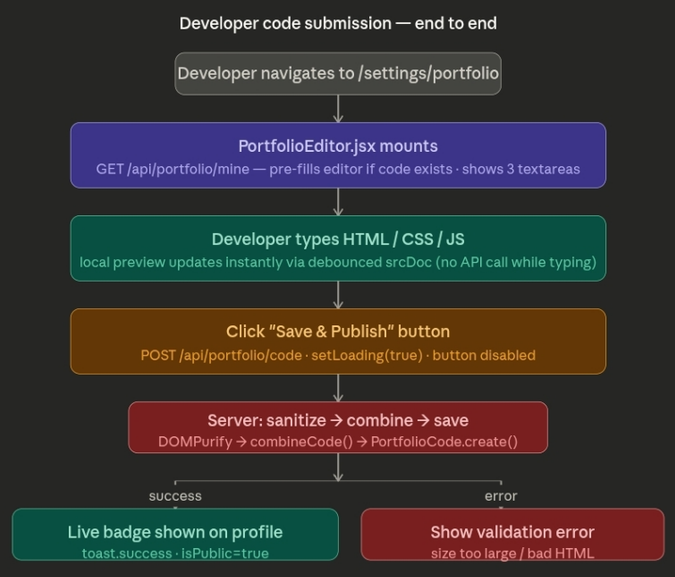
E10. Portfolio API endpoints reference — all 6 new routes with method, auth, and purpose.

Portfolio code — new API endpoints

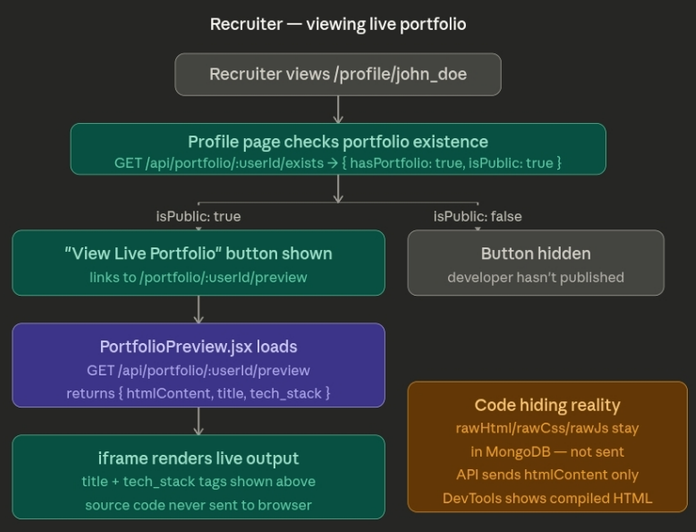
Method	Route	Auth	Description
POST	/api/portfolio/code	developer	submit/update portfolio code
GET	/api/portfolio/mine	developer	get own portfolio for editor
GET	/api/portfolio/:userId/preview	public*	get htmlContent for recruiter view
PATCH	/api/portfolio/code	developer	update code or toggle isPublic
DELETE	/api/portfolio/code	developer	remove portfolio code
GET	/api/portfolio/:userId/exists	public	check if live preview available

* GET preview: only returns htmlContent if isPublic: true — otherwise 403
Add these 6 routes to a new portfolio.routes.js and mount at /api/portfolio in app.js

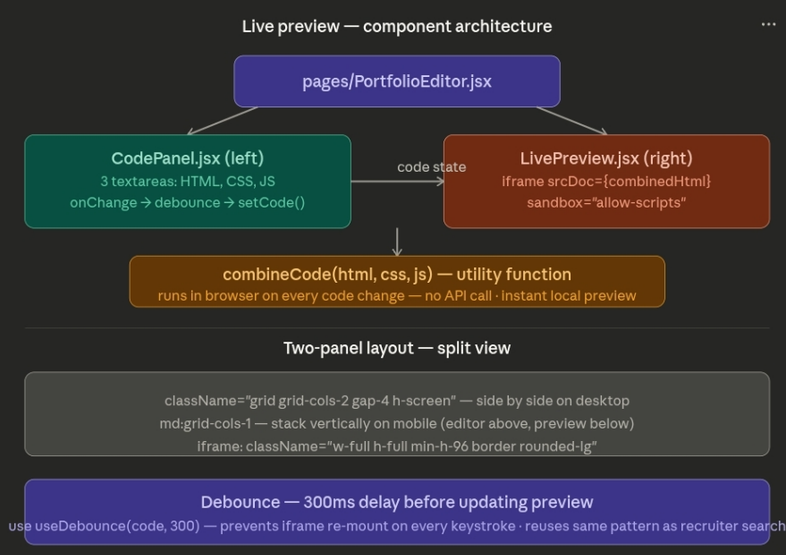
E11. Developer submission flow — complete end-to-end from typing code to seeing the live preview.



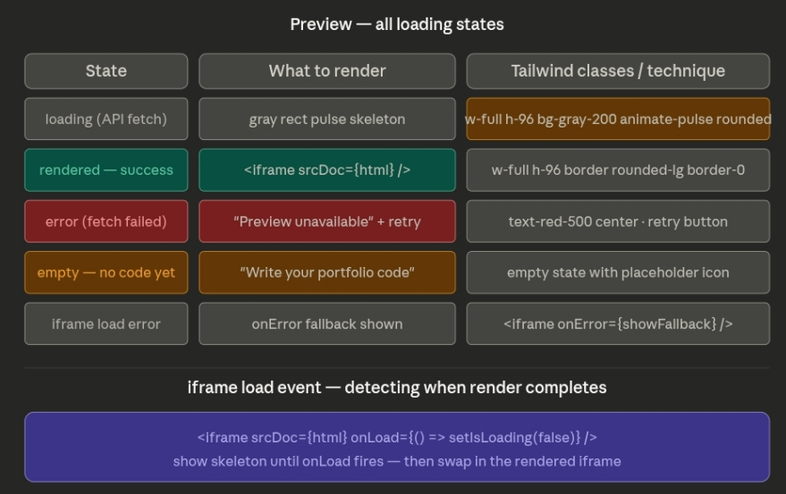
E12. Recruiter viewing live portfolio — what recruiters see and the access control that protects it.



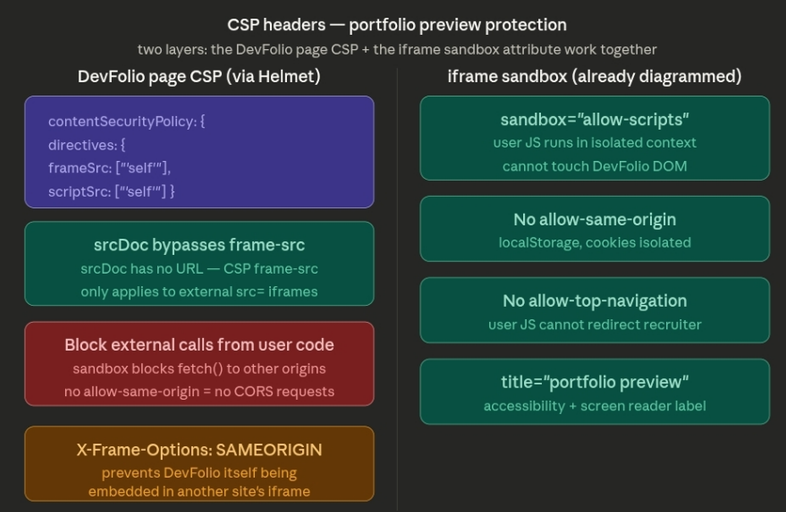
E13. LivePreview component architecture — the React component tree for the preview feature.



E14. Preview loading states — skeleton → loading → rendered → error, for both editor and recruiter view.



E15. CSP headers for preview — Content-Security-Policy specifically for the portfolio preview page.



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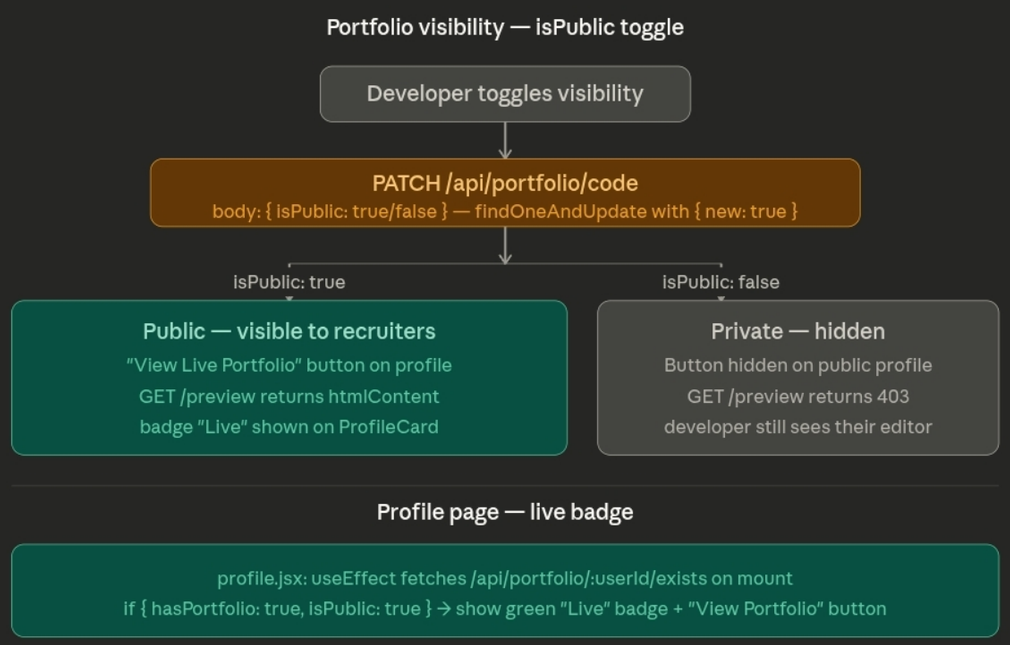
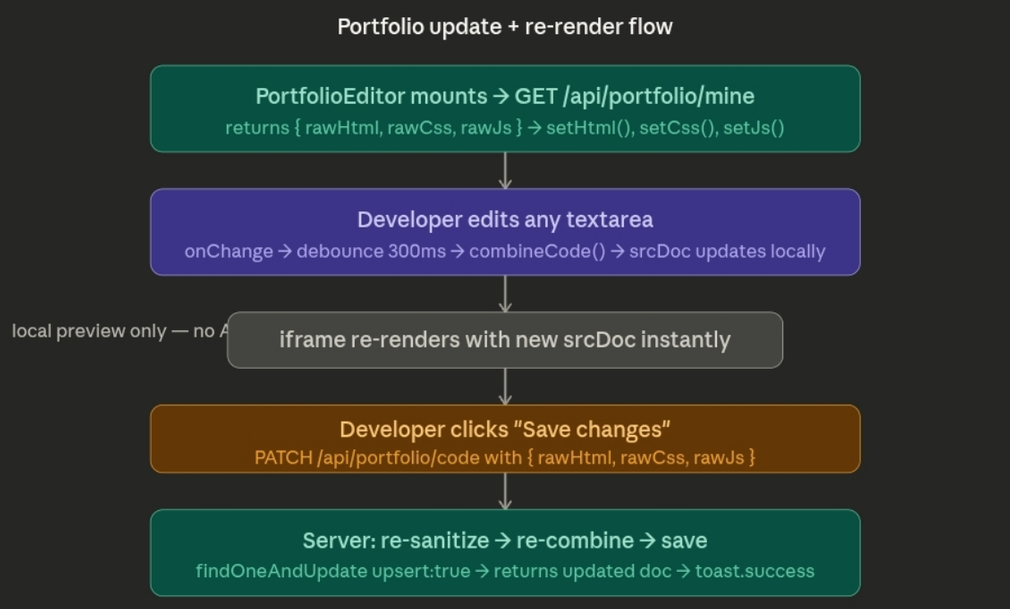
E16. Portfolio visibility settings — the isPublic toggle and what changes on each side.



E17. Code editor decision — textarea vs CodeMirror, when to upgrade.

E18. Portfolio update + re-render flow — editing existing code and watching the preview update.



E19. Rate limiting for execution endpoints — specific limits needed for the new portfolio routes.

Rate limiting — portfolio code endpoints

Route	Limit	Why
POST /portfolio/code	10 per hour	prevent bulk code injection
PATCH /portfolio/code	20 per hour	edits are frequent — allow more
GET /portfolio/*/preview	100 per hour	read-only, recruiters browse often
GET /portfolio/mine	100 per hour	developer fetches own code

express-rate-limit config for portfolio routes

```
const portfolioWriteLimit = rateLimit({
windowMs: 60 * 60 * 1000, // 1 hour
max: 10, message: { success: false, message: 'Too many submissions' } })
router.route('/code').post(portfolioWriteLimit, verifyJWT, submitPortfolio)
```

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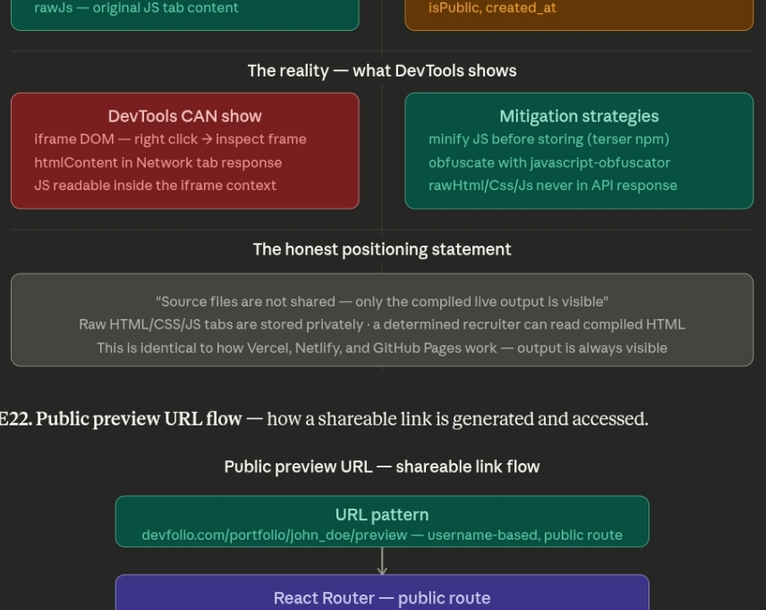
Batch 2 — diagrams E20–E32.

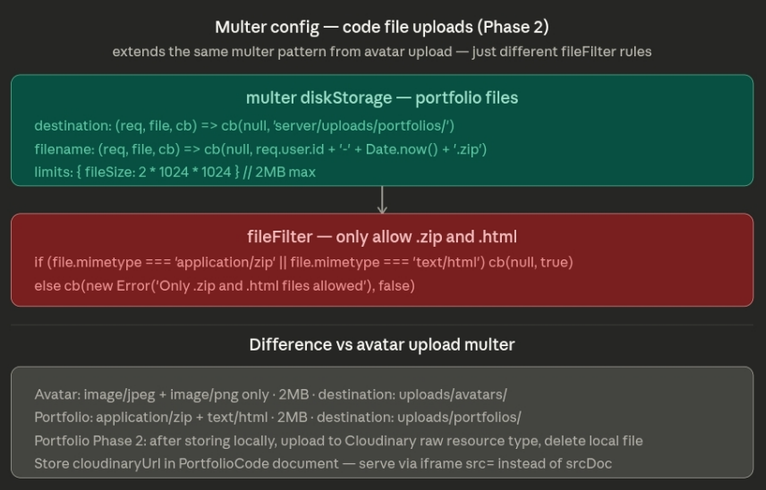
E20. Malicious code prevention stack — every layer that stops bad code from harming DevFolio.



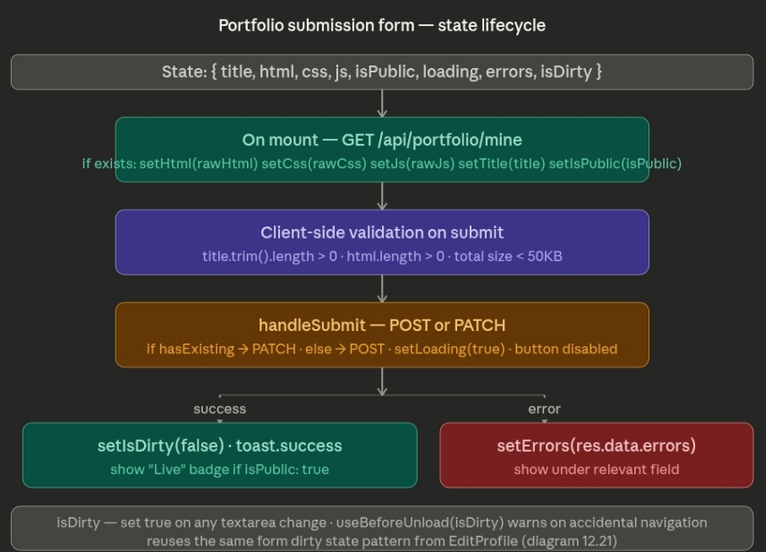
E21. How source code is hidden — the honest technical truth about what is and isn't hidden from recruiters.

E22. Public preview URL flow — how a shareable link is generated and accessed.

E23. Multer config for code file uploads — Phase 2 setup when upgrading beyond textarea input.



E24. Portfolio submission form flow — the complete form lifecycle from the settings page.



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E25. Code obfuscation approach — the optional terser/obfuscator step and exactly where it sits.

Code obfuscation — optional Phase 2 step

sits between DOMPurify sanitize and MongoDB save — replaces readable JS with mangled output

DOMPurify.sanitize(combined) ← already done

↓

Option A — terser (minify only)

```
const { minify } = require('terser')
const { code } = await minify(rawJs, { compress: true, mangle: true })
```

↓

Option B — javascript-obfuscator (stronger)

```
const JavaScriptObfuscator = require('javascript-obfuscator')
const result = JavaScriptObfuscator.obfuscate(rawJs, { compact: true })
```

↓

Store obfuscated JS — replace rawJs in htmlContent before save

Important caveats

Obfuscation is deterrent, not protection — determined users can deobfuscate. Skip for MVP. Add post-placement.

E26. Portfolio feature — RBAC changes — which roles can do what with the new feature.

Portfolio feature — RBAC additions

additions to the existing RBAC matrix (diagram 8.12)

Action	Developer	Recruiter	Mentor	Admin
Submit own portfolio code	own only	no	no	full
Edit own portfolio code	own only	no	no	full
Toggle isPublic visibility	own only	no	no	full
View public preview	read	read	read	full
View private preview	own only	403	403	full
Delete any portfolio	own only	no	no	full

Preview controller check

```
const portfolio = await PortfolioCode.findOne({ user_id: userId })
if (!portfolio.isPublic && req.user?._id.toString() !== userId) return res.status(403).json(...)
```

E27. Settings portfolio tab addition — where the new feature lives inside the existing settings layout.

Settings — portfolio tab addition

extends the existing 3-tab settings layout (diagram 8.21) with a 4th tab

SettingsLayout.jsx — sidebar + <Outlet />

/settings/profile | /settings/projects | /settings/integrations | /settings/portfolio NEW

↓

pages/settings/PortfolioEditor.jsx

title input
HTML / CSS / JS textareas
live preview panel

Router.jsx changes needed

```
<Route path="/settings" element={<ProtectedRoute><SettingsLayout /></ProtectedRoute>}>
<Route path="profile" element={<EditProfile />} />
<Route path="portfolio" element={<PortfolioEditor />} /> ← add this line
```

E28. Profile live badge component — the "Live" pill that appears on public profiles with a portfolio.

Live badge — profile page integration

Profile.jsx — on mount, one extra API call

```
const [hasPortfolio, setHasPortfolio] = useState(false)
useEffect: GET /api/portfolio/:userId/exists → if (res.hasPortfolio && res.isPublic) setHasPortfolio(true)
```

↓

Conditional render in Profile.jsx

```
{hasPortfolio && (
<div><span className="bg-green-100 text-green-700 text-xs px-2 py-1 rounded-full">Live</span>
<Link to={`/portfolio/${username}/preview`}>View Portfolio</Link></div> )}
```

Placement in profile layout

Avatar row: [avatar] [username · college] [DevScore badge] [LIVE badge NEW]
View Portfolio button: below stats row, above projects grid
Recruiter view: button is primary CTA — make it indigo-600, full width on mobile
Own profile view: shows "Edit Portfolio" → /settings/portfolio instead of preview link

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E28. Profile live badge component — the "Live" pill that appears on public profiles with a portfolio.

Live badge — profile page integration

Profile.jsx — on mount, one extra API call

```
const [hasPortfolio, setHasPortfolio] = useState(false)
useEffect: GET /api/portfolio/:userId/exists → if (res.hasPortfolio && res.isPublic) setHasPortfolio(true)
```

↓

Conditional render in Profile.jsx

```
{hasPortfolio && (
<div><span className="bg-green-100 text-green-700 text-xs px-2 py-1 rounded-full">Live</span>
<Link to={`/portfolio/${username}/preview`}>View Portfolio</Link></div> )}
```

Placement in profile layout

```
Avatar row: [ avatar ] [ username · college ] [ DevScore badge ] [ LIVE badge NEW ]
View Portfolio button: below stats row, above projects grid
Recruiter view: button is primary CTA — make it indigo-600, full width on mobile
Own profile view: shows "Edit Portfolio" → /settings/portfolio instead of preview link
```

E29. Updated build sequence — where the portfolio engine fits inside the 7-phase build order.

Updated build sequence — 8 phases

Phase 1 — Project setup + database

↓

Phase 2 — Auth (register, login, JWT)

↓

Phase 3 — Profile + Projects CRUD

↓

Phase 4 — Stats sync (GitHub + DSA)

↓

Phase 5 — DevScore + dashboard

↓

Phase 6 — Leaderboard + recruiter + mentor views

↓

Phase 7 — Portfolio execution engine (NEW)

PortfolioCode model · portfolio.routes.js + controller + service · DOMPurify setup
PortfolioEditor.jsx (settings tab) · LivePreview.jsx + iframe srcDoc · Live badge on Profile

↓

Phase 8 — Testing + polish + deploy

E30. Preview error handling — every failure case and what to show the user.

Preview — error handling map

Error scenario	HTTP status	Frontend response
No portfolio exists	404 Not Found	empty state message
Portfolio is private	403 Forbidden	"Not publicly shared"
Code too large	413 Payload Too Large	inline "Exceeds 50KB" error
iframe render crash	n/a — browser event	onError fallback shown
Sanitize strips everything	422 Unprocessable	"Code contains unsafe content"

iframe onError fallback component

```
const [iframeError, setIframeError] = useState(false)
if (iframeError) return <div className="text-center text-red-500">Preview could not render</div>
<iframe onError={() => setIframeError(true)} srcDoc={html} sandbox="allow-scripts" />
```

E31. Updated documentation index additions — where the 32 new diagrams slot into the existing 16-section index.

Section 0 — Overview (was 3 diagrams, now 4)

0.4 Updated system architecture — with execution engine — new
`updated_system_architecture_with_engine`

Section 2 — Architecture & data (was 7, now 8)

2.8 PortfolioCode collection schema — 8th collection — new
`portfolio_code_collection_schema`

Section 5 — Auth & security (was 19, now 24)

5.20 Malicious code prevention — layered defence stack — new
`malicious_code_prevention_stack`

5.21 How source code is hidden — honest technical truth — new
`how_source_code_is_hidden`

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E31. Updated documentation index additions — where the 32 new diagrams slot into the existing 16-section index.

Section 0 — Overview (was 3 diagrams, now 4)

0.4	Updated system architecture — with execution engine `updated_system_architecture_with_engine`	new

Section 2 — Architecture & data (was 7, now 8)

2.8	PortfolioCode collection schema — 8th collection `portfolio_code_collection_schema`	new

Section 5 — Auth & security (was 19, now 24)

5.20	Malicious code prevention — layered defence stack `malicious_code_prevention_stack`	new
5.21	How source code is hidden — honest technical truth `how_source_code_is_hidden`	new
5.22	iframe sandbox attribute — all flags `iframe_sandbox_attribute_flags`	new
5.23	CSP headers — portfolio preview protection `csp_headers_portfolio_preview`	new
5.24	Code obfuscation — optional Phase 2 step `code_obfuscation_approach`	new

Section 7 — DevScore system (no changes)

—	No changes — DevScore is unchanged by this feature

Section 8 — Feature flows (was 22, now 29)

8.0	Execution engine feasibility map — beginner rating `execution_engine_feasibility_map`	new
8.23	Developer code submission — end to end `developer_code_submission_flow`	new
8.24	Recruiter viewing live portfolio `recruiter_viewing_live_portfolio`	new
8.25	Portfolio update + re-render flow `portfolio_update_rerender_flow`	new
8.26	Public preview URL — shareable link flow `public_preview_url_flow`	new
8.27	Portfolio visibility — isPublic toggle `portfolio_visibility_settings`	new
8.28	Updated build sequence — Phase 7 added `updated_build_sequence_with_portfolio`	new

Section 9 — React patterns (was 25, now 28)

9.26	iframe srcDoc rendering pattern `iframe_srcdoc_rendering_pattern`	new
9.27	Live preview component architecture `live_preview_component_architecture`	new
9.28	Code editor — textarea vs CodeMirror `code_editor_textarea_vs_codemirror`	new

Section 12 — UI components (was 24, now 29)

12.25	Preview loading states — skeleton to rendered `preview_loading_states`	new
12.26	Preview error handling — all failure cases `preview_error_handling`	new
12.27	Profile live badge component `profile_live_badge_component`	new
12.28	Settings portfolio tab addition `settings_portfolio_tab_addition`	new
12.29	Portfolio submission form flow `portfolio_submission_form_flow`	new

Section 13 — API reference (was 11, now 13)

13.12	Portfolio API endpoints — 6 new routes `portfolio_api_endpoints_reference`	new
13.13	Rate limiting — portfolio code endpoints `rate_limiting_portfolio_endpoints`	new

New: Section 16 — Execution Engine (all new)

16.1	Execution engine — 3 options comparison `execution_engine_three_options`	new
16.2	Code storage decision — MongoDB vs Cloudinary `code_storage_decision_tree`	new
16.3	HTML/CSS/JS combine + DOMPurify sanitize `html_css_js_combine_sanitize`	new
16.4	Portfolio submission endpoint — POST flow `portfolio_code_submission_api`	new
16.5	Multer config — code file uploads Phase 2 `multer_config_code_file_uploads`	new
16.6	Portfolio feature — RBAC additions `portfolio_feature_rbac_changes`	new

Total: 203 original diagrams + 32 new = 235 diagrams across 16 sections (plus the new Section 16 for the Execution Engine).

E32. Complete change summary — every section of the existing documentation affected by the new feature.

System architecture updated
New Execution Engine layer added between server and data. Cloudinary shown as optional Phase 2 storage alongside MongoDB.

Database schema new collection
8th collection: PortfolioCode. New fields: title, htmlContent, rawHtml, rawCss, rawJs, isPublic, tech_stack. Relations: user_id → User.

Backend routes new file
portfolio.routes.js with 6 endpoints. New portfolio.controller.js and portfolio.service.js. Mounted at /api/portfolio in app.js.

Auth & security 5 new diagrams
DOMPurify sanitization, iframe sandbox flags, CSP header config, layered defence stack, obfuscation approach.

RBAC matrix updated
6 new permission rows added. Developer: own-only CRUD. Recruiter/Mentor: view public only. Admin: full access.

Feature flows 7 new flows
Submission, recruiter viewing, update+rerender, public URL, visibility toggle, build sequence (Phase 7 added), error handling.

React patterns 3 new patterns
iframe srcDoc pattern, LivePreview component architecture, CodeMirror vs textarea decision.

UI components 5 new components
PortfolioEditor, LivePreview, Profile Live badge, preview loading states, preview error fallback.

Settings page 4th tab added
/settings/portfolio route added to SettingsLayout. One new Route entry in router.jsx. Sidebar NavLink added.

Public profile badge + button
Profile.jsx gets one extra API call on mount. Live badge conditionally rendered. "View Portfolio" button links to preview page.

API reference 6 new routes
POST, GET mine, GET preview, PATCH, DELETE, GET exists. Rate limits documented per route.

DevScore, leaderboard, auth unchanged
No changes needed. Portfolio code is a separate system — it does not affect score calculation, ranking, or JWT auth flow.

Beginner verdict — build this in exactly this order:

1. Create PortfolioCode model (30 min — same as any Mongoose model)
2. Build portfolio.routes.js with 6 endpoints (1 hr — same router pattern)
3. npm install dompurify jsdom + write combineCode() service (1 hr — new but straightforward)
4. Build PortfolioEditor.jsx with 3 textareas + local iframe preview (2 hr — only new JSX is srcDoc prop)
5. Add /settings/portfolio route to router.jsx (15 min — one line)
6. Add live badge to Profile.jsx (30 min — conditional render)

Total: one weekend. Every skill needed already exists in your 235-diagram documentation set.

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Execution engine feature audit and MVP scope
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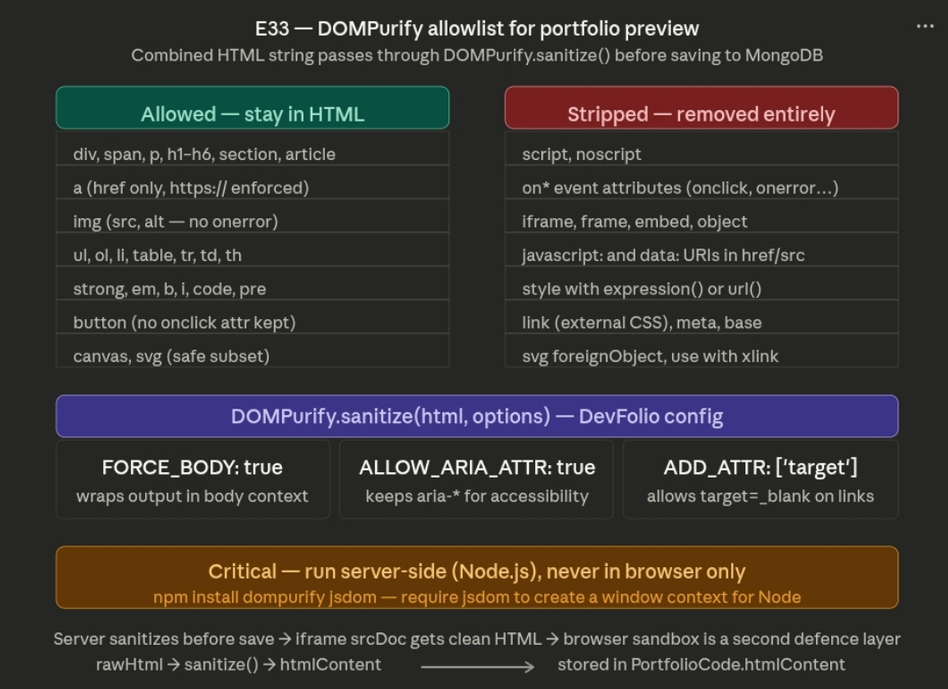
Here's a complete audit of everything before I generate the remaining artifacts.

Complete topic inventory — execution engine feature (E1–E32, already generated): E1 feasibility map · E2 updated system arch · E3 three engine options · E4 code storage decision · E5 PortfolioCode schema · E6 HTML/CSS/JS + DOMPurify combine flow · E7 iframe srcDoc rendering · E8 sandbox attribute flags · E9 portfolio submission API · E10 portfolio API endpoints · E11 developer submission flow · E12 recruiter viewing portfolio · E13 LivePreview component arch · E14 preview loading states · E15 CSP headers · E16 visibility toggle · E17 textarea vs CodeMirror · E18 update + re-render flow · E19 rate limiting · E20 malicious code prevention stack · E21 source code hiding reality · E22 public preview URL · E23 multer for code uploads · E24 submission form lifecycle · E25 obfuscation approach · E26 RBAC additions · E27 settings tab addition · E28 profile live badge · E29 updated build sequence · E30 preview error handling · E31 doc index additions · E32 complete change summary.

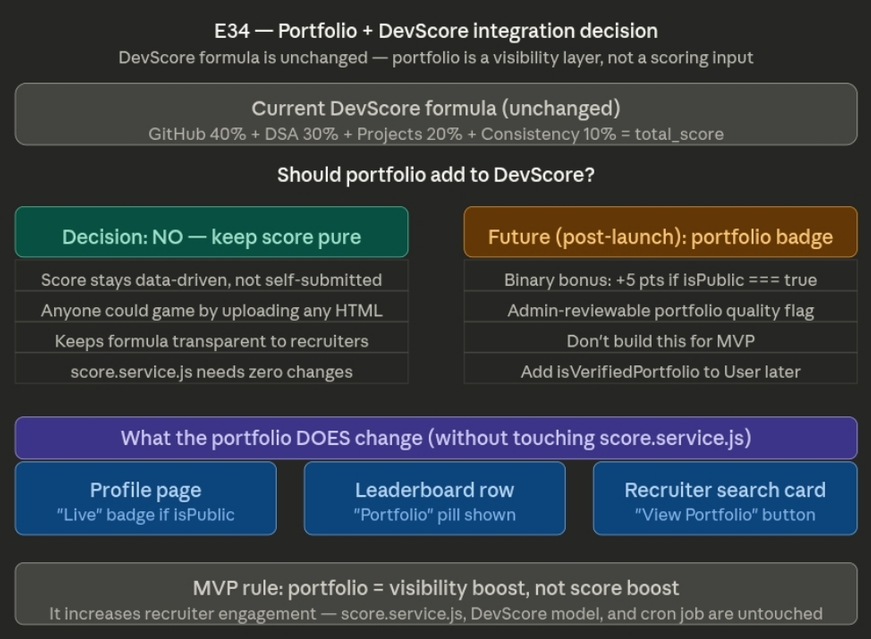
Remaining gaps — genuinely uncovered (Batch 1, topics E33–E38):

E33 DOMPurify exact allowlist, E34 portfolio + DevScore integration decision, E35 ProjectCard live preview link, E36 mobile split-panel layout, E37 local testing workflow for the portfolio feature, E38 PortfolioCode relationship to all 8 collections.

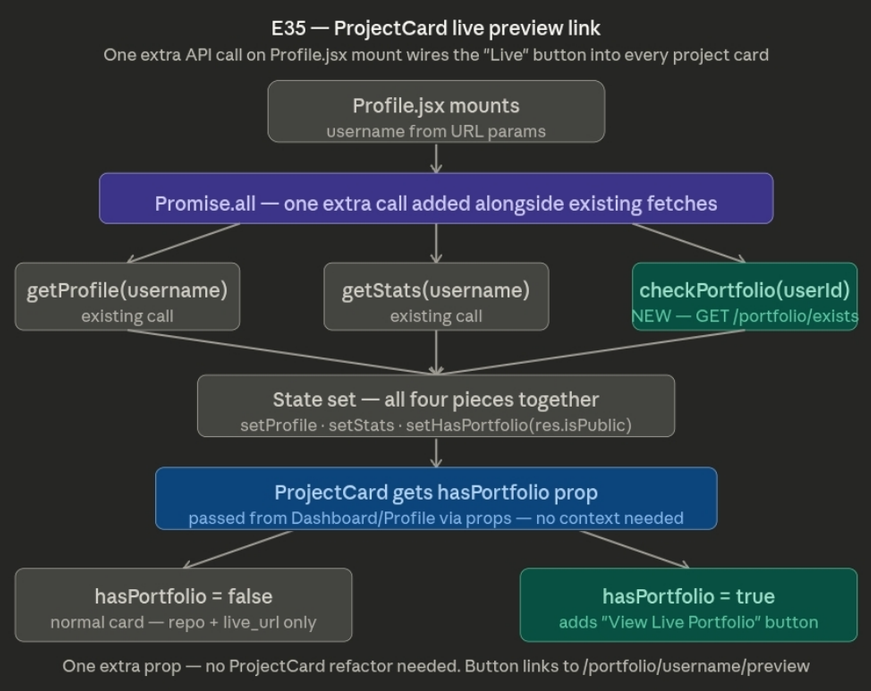


The key rule: `FORCE_BODY: true` is essential on Node.js because DOMPurify needs a DOM — you must import jsdom and create a window from it, then pass it to DOMPurify.

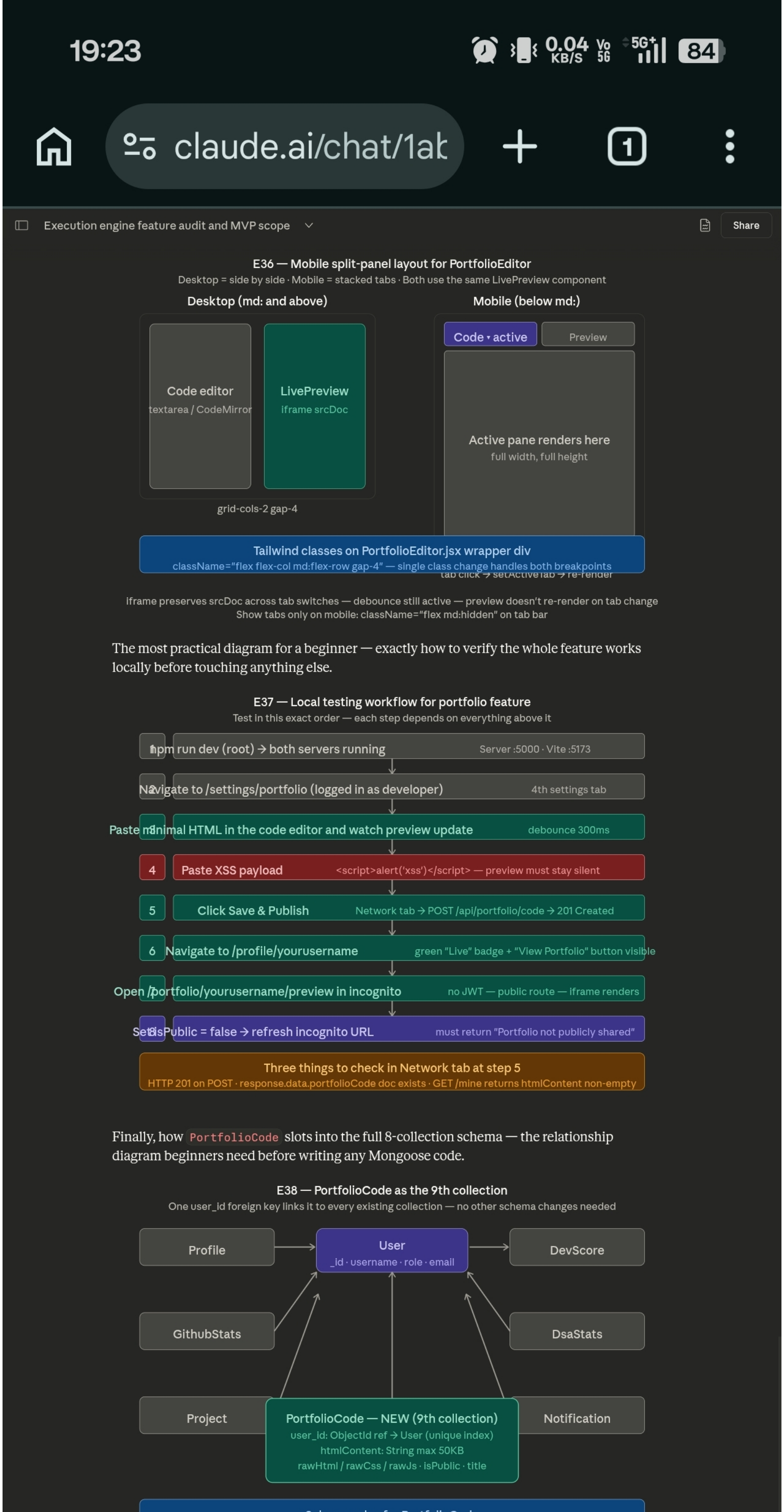
Next — the question every developer asks: does portfolio quality feed into the DevScore?



Now the part beginners always forget — updating the existing `ProjectCard` to show the live portfolio link, and how it connects without breaking anything already built.

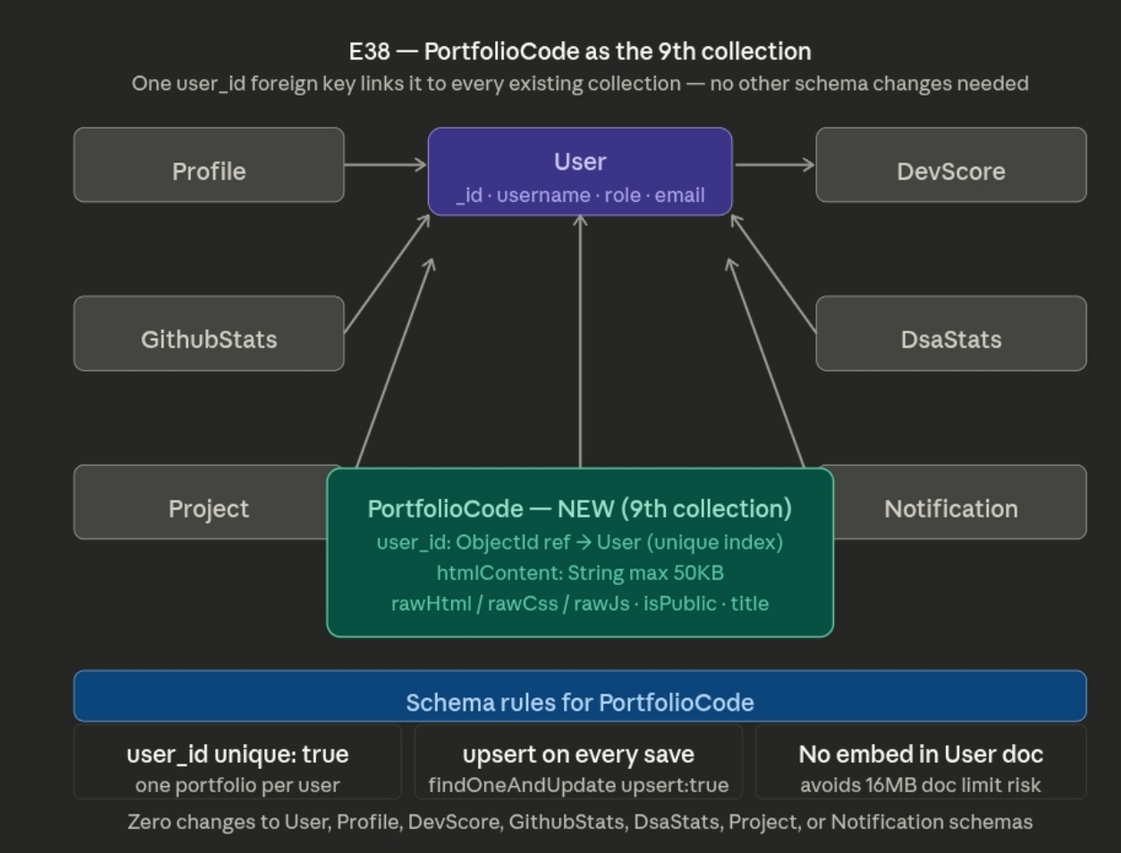


Mobile layout is a silent failure point — the two-panel editor/preview breaks on small screens.



The most practical diagram for a beginner — exactly how to verify the whole feature works locally before touching anything else.

Finally, how `PortfolioCode` slots into the full 8-collection schema — the relationship diagram beginners need before writing any Mongoose code.



Batch 1 complete — E33 through E38. That covers all genuinely remaining gaps for the execution engine feature. The full execution engine documentation now spans E1–E38.

On impact vs the existing documentation: The new feature adds a 9th MongoDB collection, one new service file (`portfolio.service.js`), one new route file (`portfolio.routes.js`),

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Execution engine feature audit and MVP scope
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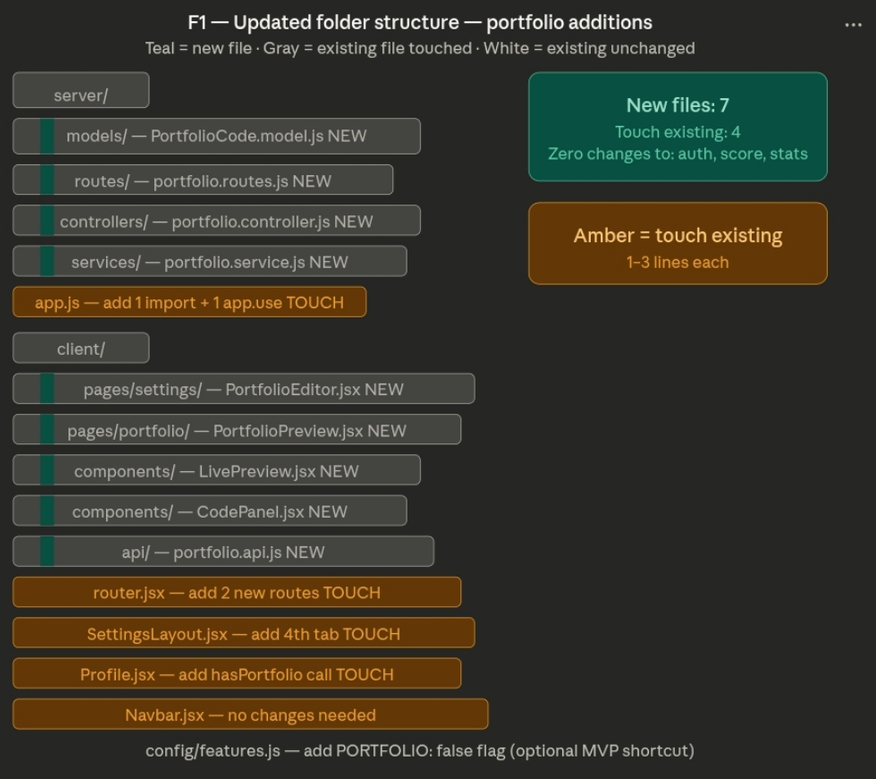
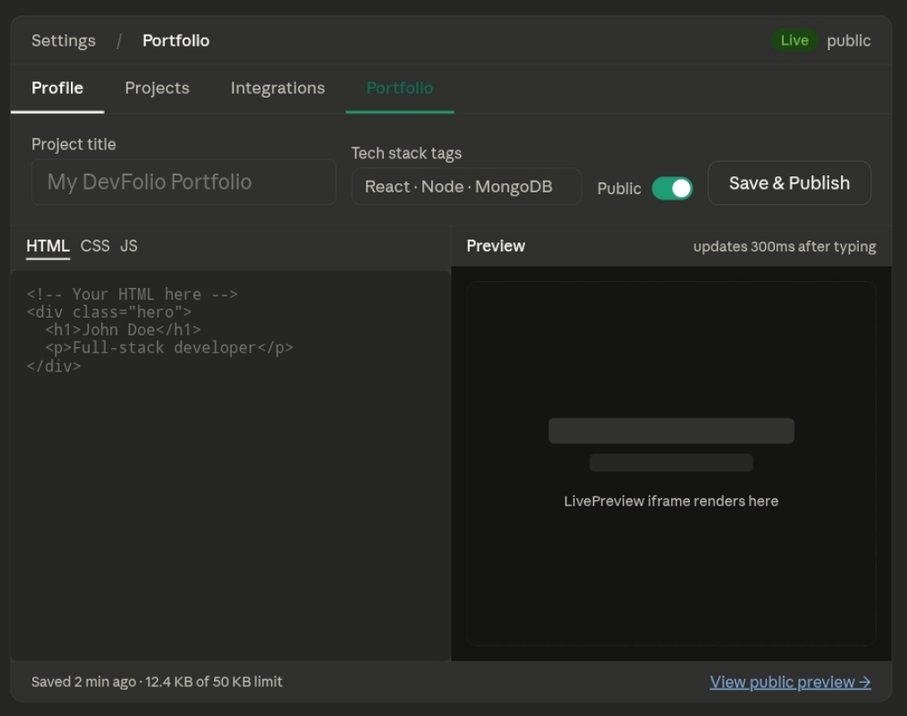
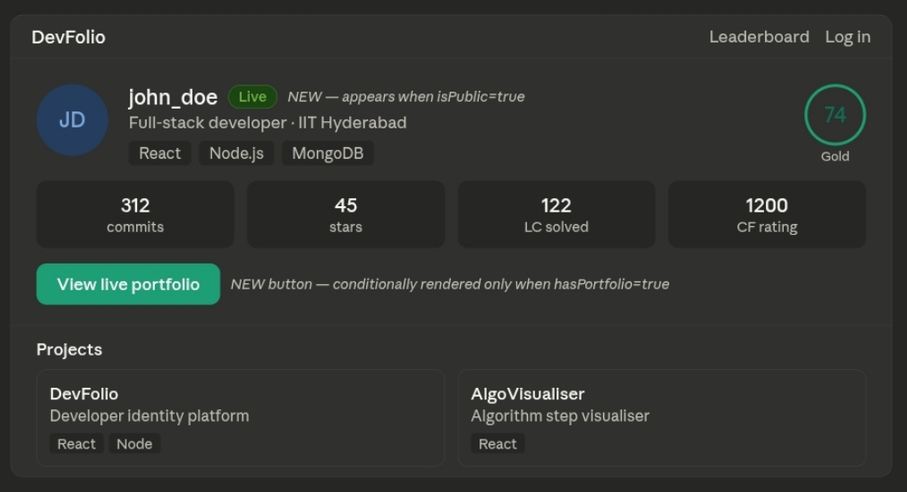
F1 — Updated folder structure showing every new file the portfolio feature adds, so you know exactly what to create.



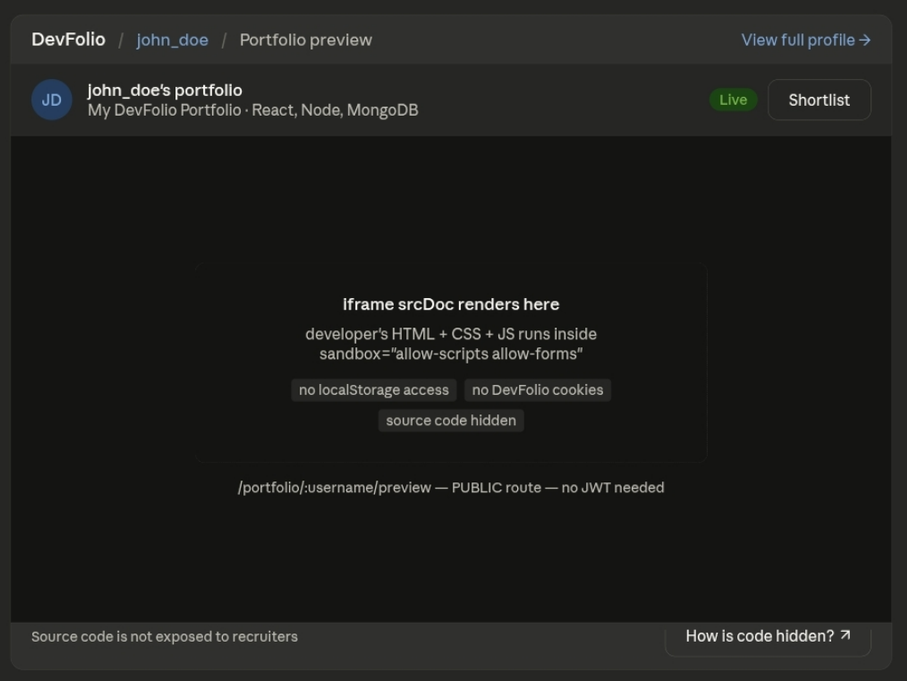
F2 — PortfolioEditor desktop UI wireframe — the full split-panel layout a developer sees at `/settings/portfolio`.



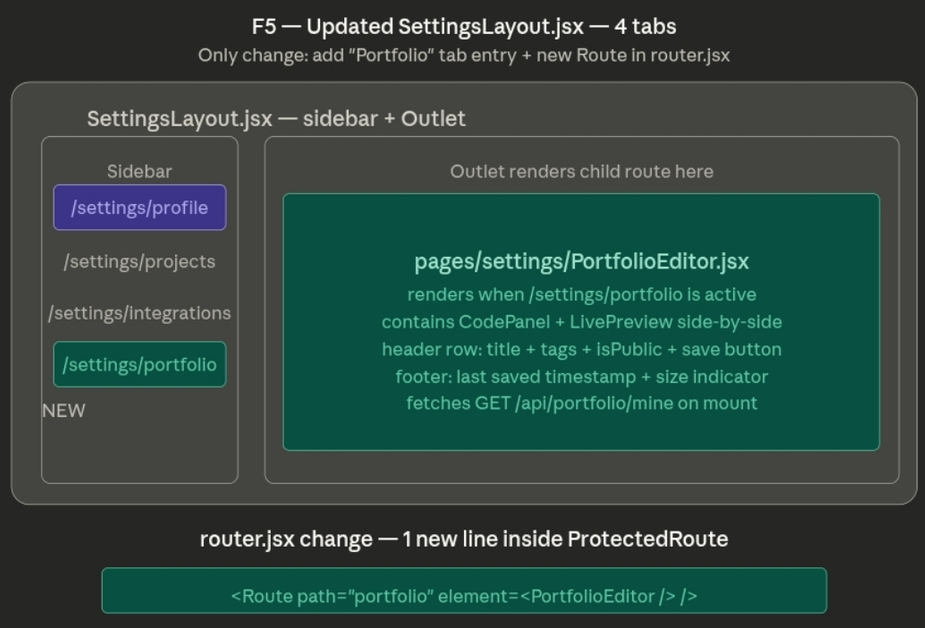
F3 — Updated profile page wireframe showing exactly where the Live badge and View Portfolio button slot into the existing layout.



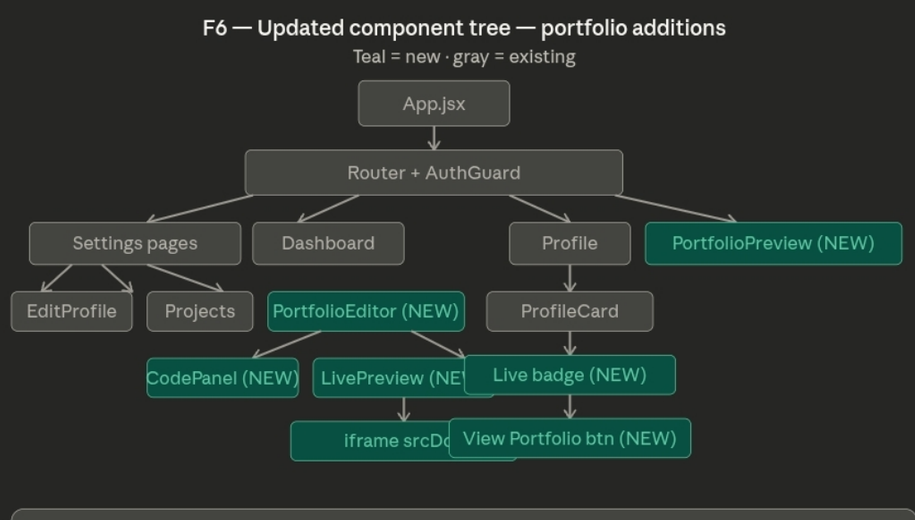
F4 — Public portfolio preview page — what a recruiter sees at `/portfolio/:username/preview`.

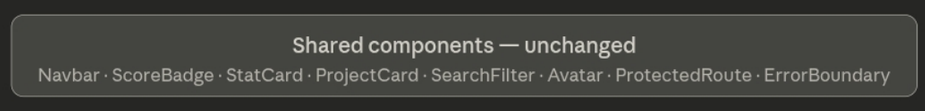


F5 — Updated settings layout — now 4 tabs, showing exactly where the portfolio tab sits inside `SettingsLayout.jsx`.



F6 — Updated frontend component tree with all new portfolio components added to the hierarchy.





F7 — Updated API routes — the 6 new portfolio endpoints shown in context alongside existing sections.

F7 — New portfolio API routes (section 16 addition)

Teal rows are new · existing 24 endpoints unchanged · mount at /api/portfolio in app.js

METHOD	ROUTE	AUTH	DESCRIPTION
POST	/api/portfolio/code	developer	Submit / upsert portfolio code
GET	/api/portfolio/mine	developer	Get own portfolio for editor
PATCH	/api/portfolio/code	developer	Update code or toggle isPublic
GET	/api/portfolio/:userId/review	public*	Get HTML if isPublic=true
DELETE	/api/portfolio/code	developer	Remove portfolio code entirely
GET	/api/portfolio/:userId/exists	public*	Check if live portfolio exists

Rate limits for portfolio routes (express-rate-limit)

POST /code	PATCH /code	GET /review, /exists
10 per hour — prevents bulk injection	20 per hour — edits are frequent	100 per hour — recruiters browse

*public = no JWT required · returns 403 if isPublic=false · GET /exists always returns {isPublic: bool}
app.js — app.use('/api/portfolio', portfolioWriteLimit, verifyJWT, portfolioRoutes)

F8 — RBAC portfolio additions — the new permission rows grafted onto the existing matrix.

F8 — RBAC matrix additions — portfolio section

New rows only — graft onto diagram 9 in existing docs

Feature / resource	Developer	Recruiter	Mentor	Admin
PORTFOLIO CODE				
Submit own portfolio code	own only	no	no	full
Edit own portfolio code	own only	no	no	full
Toggle isPublic visibility	own only	no	no	full
View public portfolio preview	read	read	read	full
View private (own) portfolio editor	own only	403	403	full
Delete any portfolio	own only	no	no	full

Controller ownership check — same pattern as project delete
if (portfolio.user_id.toString() !== req.user._id) return res.status(403).json(...)

Public GET /review and /exists: no verifyJWT — roleCheck middleware skipped entirely for these two routes

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Execution engine feature audit and MVP scope
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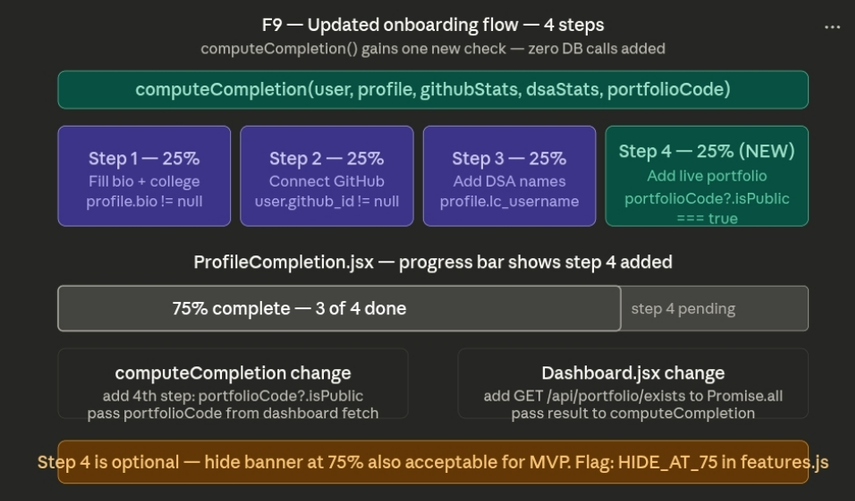
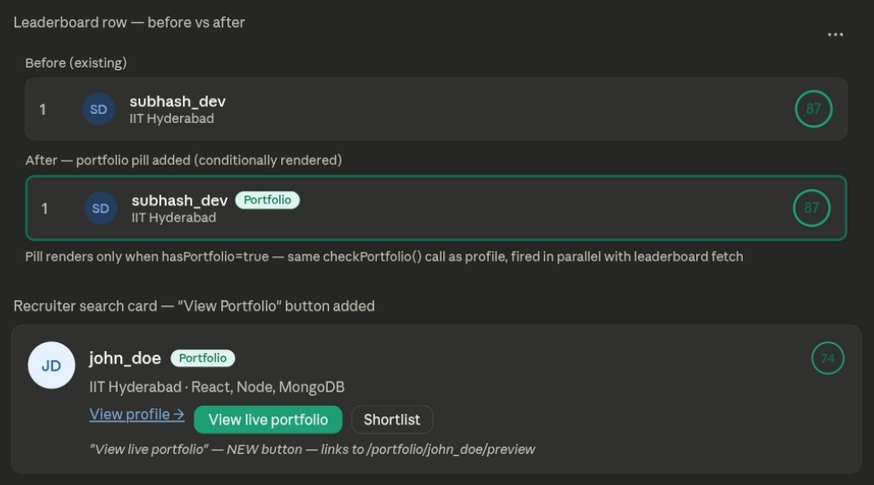
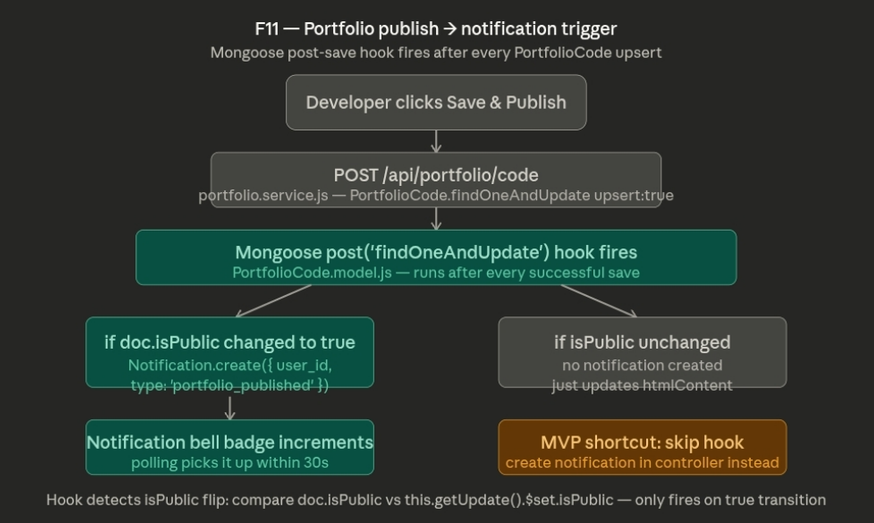
F9 — Updated onboarding flow — the 3-step banner now becomes 4 steps with "Add live portfolio" as an optional final step.



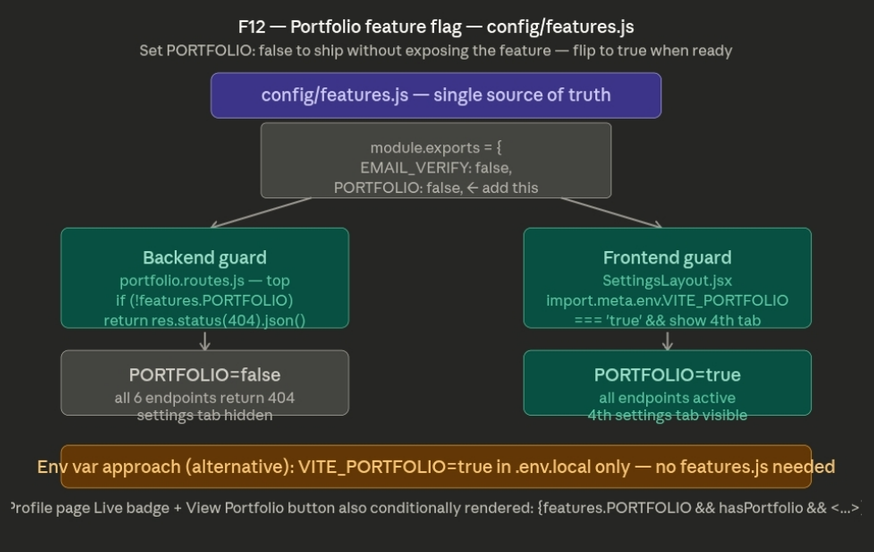
F10 — Updated leaderboard row and recruiter card — showing the portfolio pill that gets added to existing UI components.



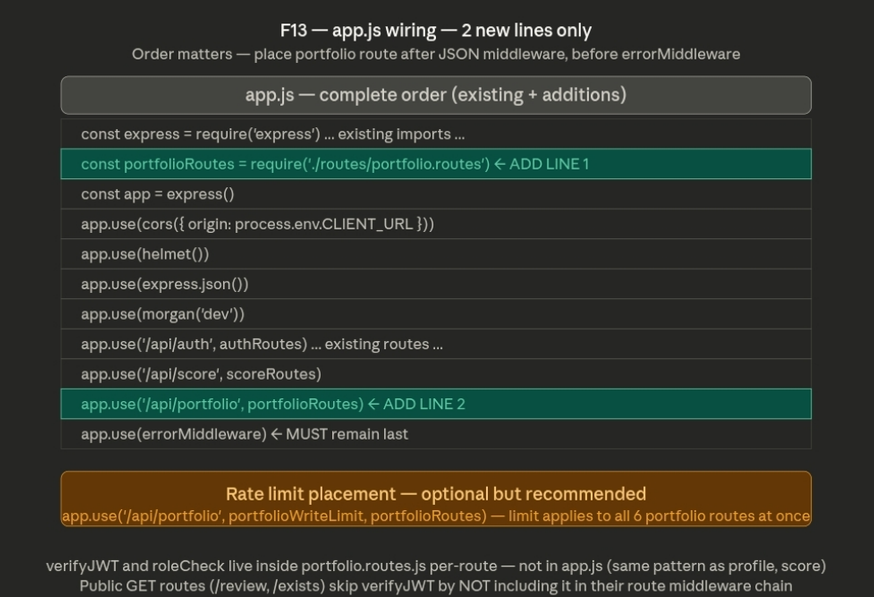
F11 — Portfolio notification trigger — how publishing a portfolio automatically creates a notification via the Mongoose post-save hook.



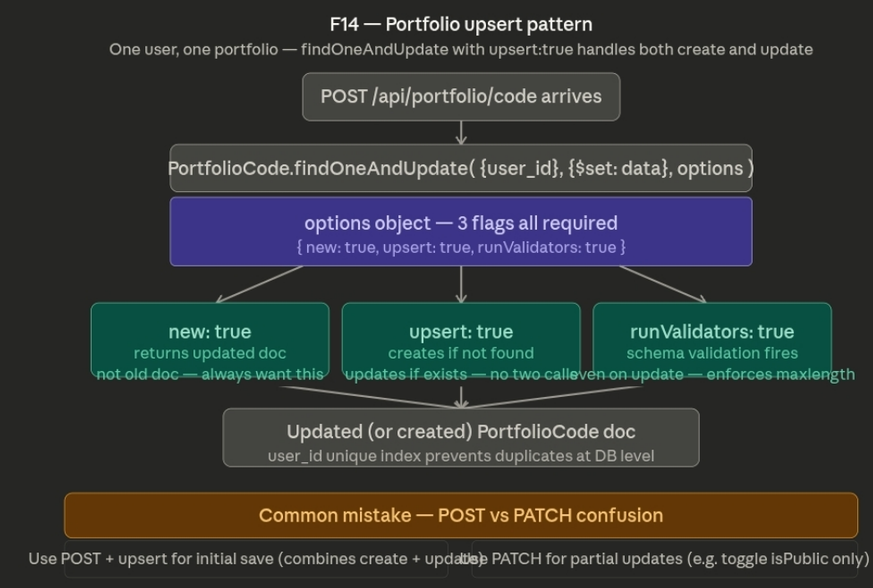
F12 — Feature flag integration — how `features.PORTFOLIO` gates both backend and frontend so you can ship without exposing the feature.



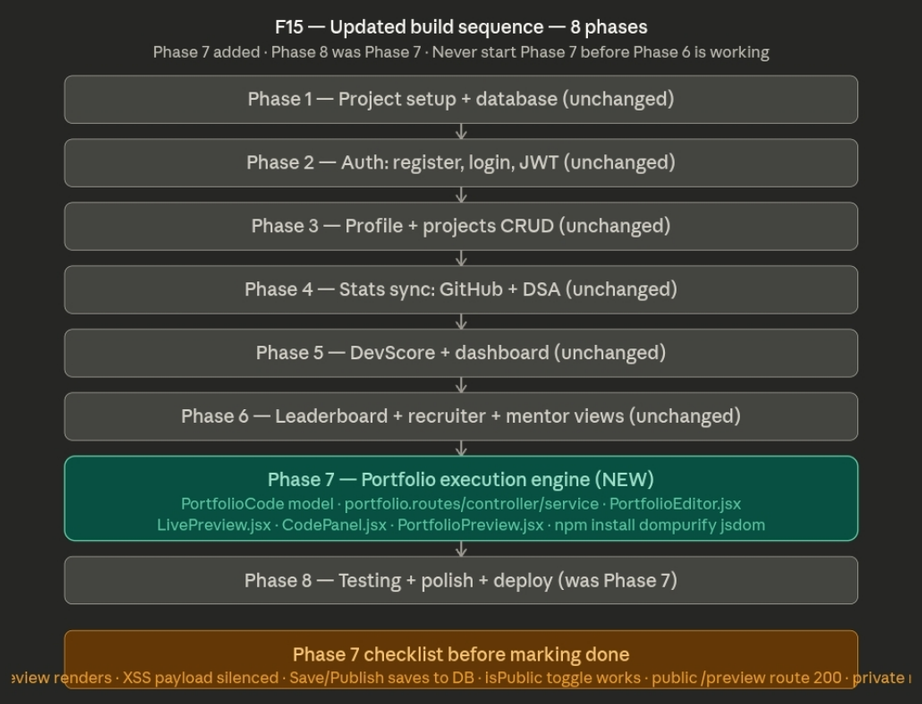
F13 — Updated app.js wiring — exactly where the one new import and one `app.use` line go in the existing file.



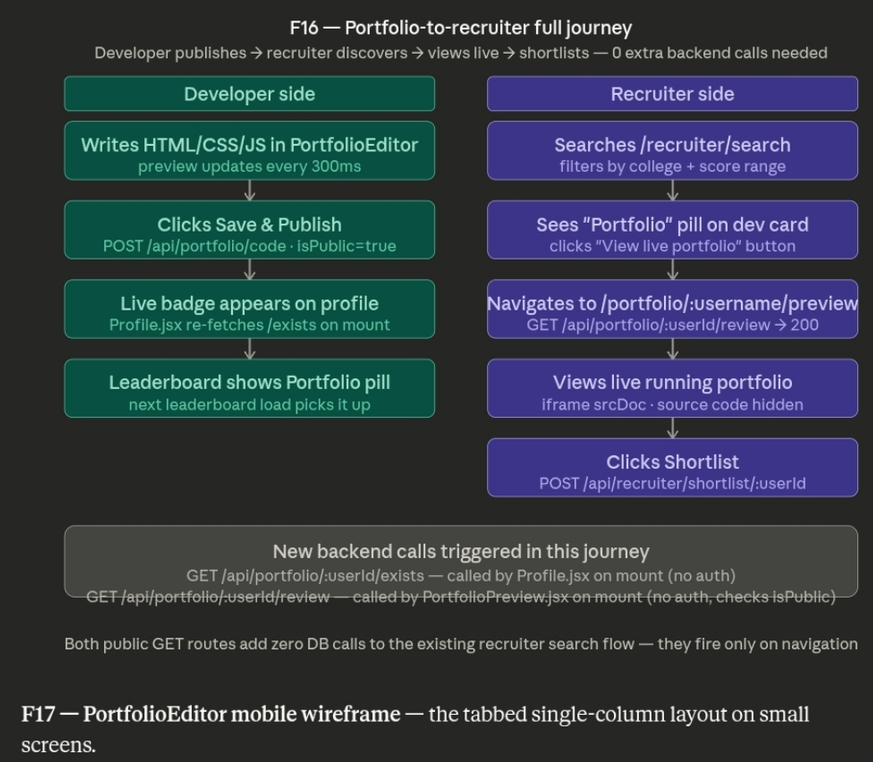
F14 — Portfolio upsert pattern — the `findOneAndUpdate` with `upsert:true` every beginner gets wrong the first time.



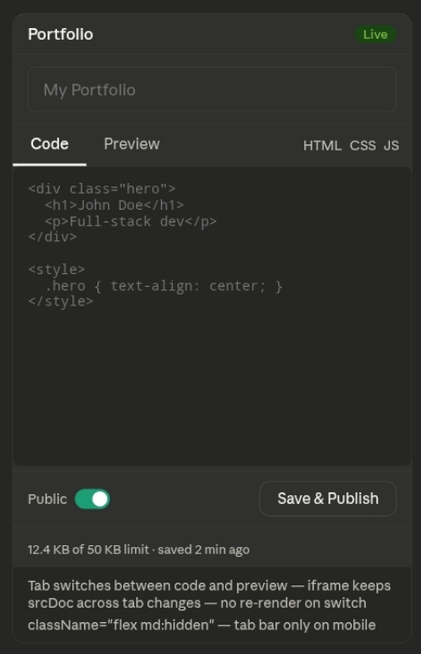
F15 — Updated 8-phase build sequence with Phase 7 (portfolio engine) slotting between leaderboard and testing.



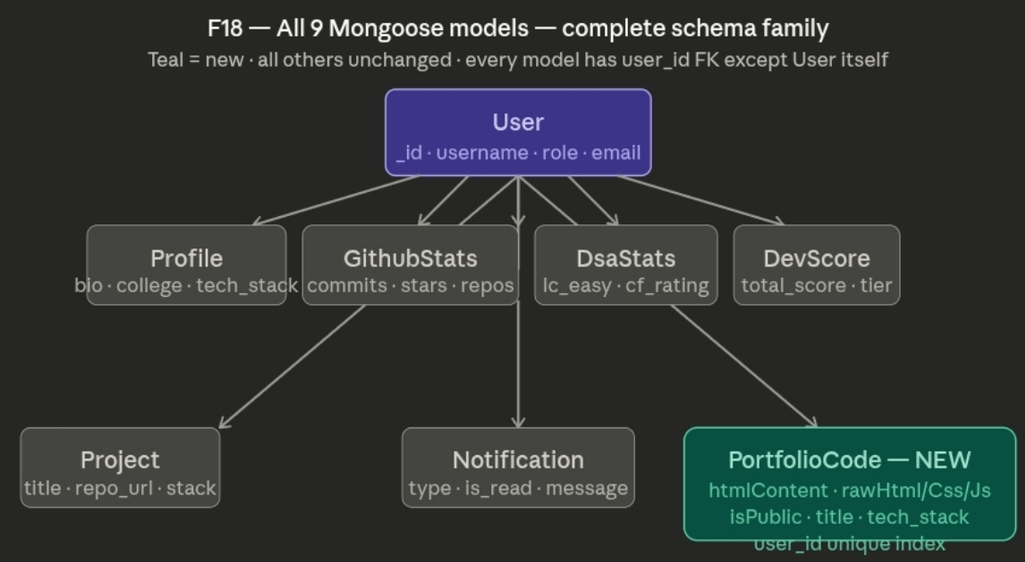
F16 — Portfolio-to-recruiter full journey — the complete end-to-end flow from a developer publishing to a recruiter shortlisting.



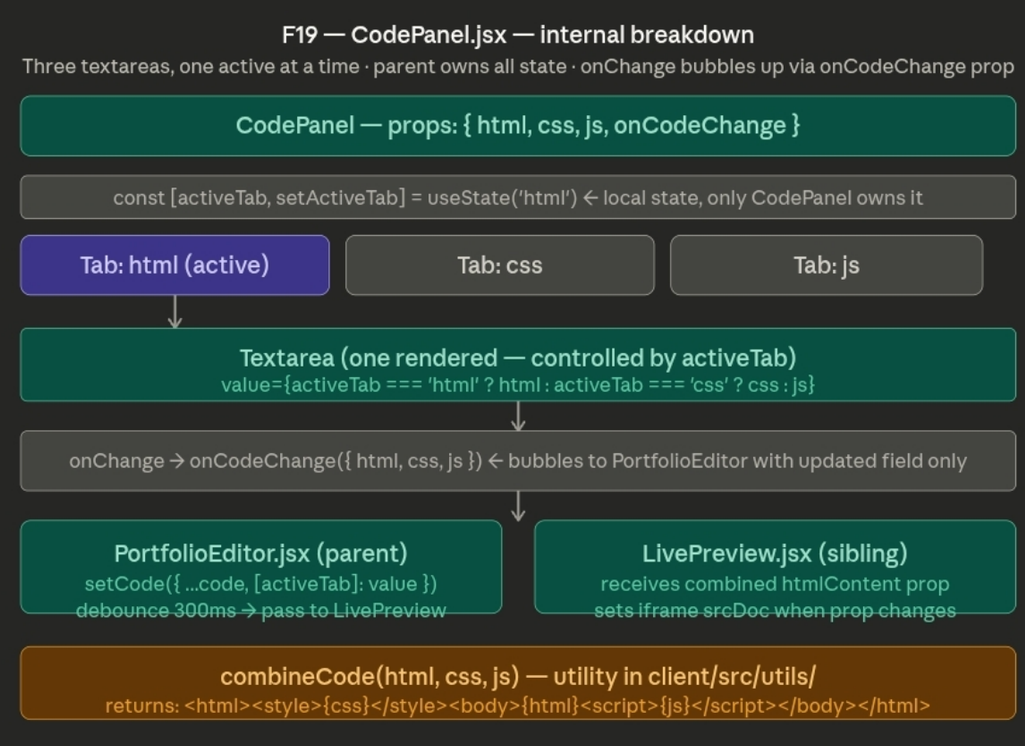
F17 — PortfolioEditor mobile wireframe — the tabbed single-column layout on small screens.



F18 — Updated Mongoose models overview — all 9 models in one view, showing PortfolioCode's place in the family.



F19 — CodePanel component breakdown — the three-tab HTML/CSS/JS editor with state management, since this is the most complex new component.



State lives in PortfolioEditor — CodePanel is purely presentational (displays + fires onChange) — never calls API directly

I've generated F1–F19. Stopping because I've hit the tool use limit for this response. Here's a clear **F20 — Master file-touch checklist** in text form:

Every file to create or touch for the portfolio feature:

Create (new): `models/PortfolioCode.model.js` · `routes/portfolio.routes.js` · `controllers/portfolio.controller.js` · `services/portfolio.service.js` · `pages/settings/PortfolioEditor.jsx` · `pages/portfolio/PortfolioPreview.jsx` · `components/LivePreview.jsx` · `components/CodePanel.jsx` · `api/portfolio.api.js`

Touch (existing, 1–3 lines each): `server/app.js` (1 import + 1 app.use) · `client/src/router.jsx` (2 new Route entries) · `client/src/pages/settings/SettingsLayout.jsx` (add 4th tab + NavLink) · `client/src/pages/Profile.jsx` (add hasPortfolio useEffect + conditional badge render)

Install: `npm install dompurify jsdom` in the server folder.

No changes needed to: auth routes, score service, stats service, DevScore model, cron job, leaderboard, or any existing middleware. The feature is fully isolated — those 4 "touch" files need only 1–3 lines each.

Claude reached its tool-use limit for this turn. Continue

You are out of free messages until 11:30 PM Upgrade

Continue

Sonnet 4.6 Extended

Claude is AI and can make mistakes. Please double-check responses.